Rollback, Compensate, or Rescope? Repairing Deployed Logs of Rank-One Model Edits (Technical Report)

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ABSTRACT

This report accompanies the paper of the same title and contains the proofs of its statements. Appendix A treats the linear repairs, rollback and compensation; Appendix B treats rescoping; Appendix C holds the tables and figures of the experiments that left the body of the paper, and the table of closest works and the comparisons with related work that left its Sec. 6. Notation, definitions and the numbering of the statements of the paper are those of the paper.

Use of AI tools. The first author initiated and supervised this work and set its direction, scope and constraints. Under this supervision, generative AI was used extensively: an AI assistant (Claude, Anthropic) to develop the approach, draft proofs, code and text, and run the experiments and their independent verifications (separate instances with their own code); a second language model as a critic; an image model for the draft of Fig. 1 of the paper. The authors reviewed the work and are responsible for its content. Wherever this report or the repository use the words "referee", "referee pass", "prover", "audit" or "verification" for a check of a proof or of an experiment, they denote separate automated passes of such assistants, not reviews by other people. "Registered" means written into the authors' protocol file before the run, with the time stamps of their machine; no external registry was used.

A LINEAR REPAIRS: ROLLBACK AND COMPENSATION

This appendix proves the statements of the paper on rollback and on linear repairs. Table 1 lists, for both appendices, where each result of the paper is proved, and which statements are kept in this report only.

A.1 Conventions

Numbers are rationals written in binary. The size of an instance is its total bit length. A Karp reduction is a polynomial-time many-one reduction. A problem is *strongly* NP-hard if it remains NP-hard on instances whose numerators and denominators are bounded by a polynomial in the instance size [26]; such a problem has no pseudo-polynomial algorithm unless P = NP. X3C (exact cover by 3-sets) asks whether $U = [3h]$ is partitioned by h of the given 3-sets $C_1, \dots, C_H$. It has no numbers and is NP-complete by the standard map from 3-dimensional matching [37], so it is strongly NP-complete.

For contributions $w_1, \dots, w_T \in \mathbb{Q}^n$ (Def. 2.6) we write $r(R) = \sum_{i \notin R} w_i$ and $D = r(\emptyset)$, and $\chi(C)$ for the indicator vector of a set C.

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Table 1: Results of the paper and the statements of this report that prove them.

Paper	Result	Proved in
Thm. 3.1; the table that Sec. 3 cites	the three repairs under one criterion; granularity	Sec. B.9: Props. B.69, B.74, B.72, B.75, B.77; Thm. B.76; Table 6
Thm. 3.2	capacity of linear restoration; unreported keys; stored records	Thm. A.15; Cor. A.16; Props. A.18, A.19; Sec. A.8.4
Prop. 3.3	locality certificate and frame rule	Prop. B.1; Examples B.2–B.4
Thm. 3.4	least repair; union-closed classes	Thm. B.8; Prop. B.12; Remark B.11
Remark 3.5	several edited modules: what extends and what does not	Sec. B.10: Thms. B.82, B.85; Cor. B.90; Sec. B.10.1
Thm. 3.6	two-block rule: marginal and calibration-conditional; sample sizes	Prop. B.58; Thms. B.61, B.62, B.63; Thms. B.45, B.50
Thm. 3.7	rollback under a tolerance; Scope-Cut as one minimum cut; in its status: two detectors per edit	Thm. A.5; Thm. B.14, Prop. B.15; Sec. B.5: Props. B.35, B.36; Thm. B.40
Prop. 4.2 Sec. 4	rebase lemma and retrofit top-1 selection; stored threshold, guards and capped edits; log-only thresholds and refusals; registered variants; complaint operators beyond the least repair (extensions)	Lemma B.65; Thm. B.66; Prop. B.67 Remarks B.7, B.48, B.60; Secs. B.7.6, B.7.7; Sec. B.11, Algorithm 1
in this report only	conditioning; greedy and smoothed rollback; compensation is not local; capacity law of exact protection; nested frontier; hardness frontier; lineage size	Prop. A.8; Thm. A.5; Lemmas A.9, A.10; Thm. A.12; Thm. B.16; Table 4; Prop. B.42

A deletion set R is *feasible* at tolerance c if $\|r(R)\|_p \le c\,\|D\|_p$; for $p = 2$ we compare squared norms, so every test is exact rational arithmetic. $R = [T]$ is always feasible. A *perturbation* of an instance is a family $\xi = (\xi_i)_{i \in [T]}$ of rational vectors; the perturbed instance is $w'_i := w_i + \xi_i$, with residuals $r'(R)$ and damage D', and the perturbation has *mass* $|\xi| := \sum_i \|\xi_i\|_p$. An instance is *zero-sum-free* if $r(R) \neq 0$ for every $R \neq [T]$. For $c \le c'$, an algorithm is a (κ, c')*-bicriteria approximation* for tolerance c if it returns R with $\|r(R)\|_p \le c'\|D\|_p$ and $|R| \le \kappa\,\mathrm{OPT}_c$.

A.2 What a correction of the weight guarantees

In this subsection models differ only in the weight of the edited module, and we use (H1) and (H2) of Sec. 2 in the following form: the key of a position depends on the tokens up to it and not on the weight of the edited module, and the logits at a position are determined by these tokens and by the module outputs at the positions

up to it. Arithmetic is exact. For a prompt followed by an answer of L tokens, the *needed set* N consists of the positions of the prompt and of the first $L-1$ answer tokens; it is prefix-closed and contains every position at which a token of the answer is predicted under teacher forcing.

Theorem A.1 (Exact preservation). *Let (q_i, y_i) be the request of an edit and N the needed set of the sequence q_i followed by y_i. If a correction $X \in \mathbb{R}^{n\times d}$ satisfies $Xk_t = 0$ for all $t \in N$, then the logits of $\mathcal{M}[W_E + X]$ and of $\mathcal{M}[W_E]$ on this sequence agree at every $t \in N$.*

Proof. The key at every position is the same under both weights. At $t \in N$ the module output is $(W_E + X)k_t = W_E k_t$. The logits at $t \in N$ are determined by the tokens and by the module outputs at the positions up to t, all of which lie in N because N is prefix-closed. □

By induction over the generation steps, if X annihilates the keys of the greedy continuation of $\mathcal{M}[W_E]$, the greedy output is preserved. For sampling the guarantee is weaker: the per-step distributions coincide only on prefixes whose keys are all protected. Protecting only the last prompt key guarantees nothing beyond the first generated token.

Proposition A.2 (Restoration). *Let q be a query and N the needed set of its pre-edit trace $x^0(q)$. If a weight W' satisfies $(W' - W_0)k_t = 0$ for every $t \in N$, then the logits of $\mathcal{M}[W']$ equal those of $\mathcal{M}[W_0]$ there, so the teacher-forced answer of $\mathcal{M}[W']$ is $y_0(q)$; if the hypothesis holds at every position of $x^0(q)$, the greedily decoded answer is $y_0(q)$ as well. For $W' = W_E + X$ the hypothesis reads $Xk_t = -dWk_t$. Restoring a proper subset of the needed set gives no such guarantee, even at the restored positions.*

Proof. The first claim is the argument of Theorem A.1 with W_0 as the reference model. Under greedy decoding the two models choose the same token at every step, by induction over the steps, and stop at the same step. For the last claim, consider one block after the edited module whose attention at position 2 averages positions 1 and 2 with the identity value map. Its output at position 2 is the mean of the two module outputs. Restoring the module output at position 2 exactly leaves the error $(W' - W_0)k_1/2$, which is nonzero in general, and a linear read-out of this error can change the largest logit. □

Proposition A.2 is the weight form of the certificate criterion of Def. 2.5: a query that is certified for a weight W' is resolved. A tolerance at one complaint key is a surrogate: it is not necessary for the pre-edit answer and, short of exact restoration along the trace, not sufficient.

Lemma A.3 (Shared keys). *If a key k is both a complaint key and a protected key, an admissible correction exists only if $dWk = 0$. In general the obstruction is dW restricted to $\operatorname{col} K_q \cap \operatorname{col} K_{\text{prot}}$ (Lemma A.9). Identical token prefixes produce identical keys, so a chat template puts the same prefix keys into every needed set; these keys have to be excluded from the complaint keys.*

Proof. An admissible X needs $Xk = 0$ and $Xk = -dWk$ at the same time. □

A.3 Realisability and membership

Lemma A.4 (Realisability). *Let $w_1, \dots, w_T \in \mathbb{Q}^n$ be arbitrary. Some edit log with $d = T + 1$ has $\Delta_i k_q = w_i$ for all i at one complaint key k_q, and on it the own keys $k_1, \dots, k_T$ and k_q are linearly independent.*

Proof. Let $e_1, \dots, e_{T+1}$ be the standard basis. Set $k_q = e_1$, $k_i = e_{i+1}$, $u_i = w_i$ and $\psi_i = e_1 + e_{i+1}$. Then $s_i = \langle \psi_i, k_i \rangle = 1 > 0$, $\Delta_i k_q = u_i \langle \psi_i, e_1 \rangle = w_i$ and $\Delta_j k_i = \mathbf{1}[i = j]\, u_i$. With the targets $z_i^\star := (W_0 + \sum_{j<i} \Delta_j) k_i + u_i$ every write vector is the residual required by Def. 2.1, for any order of the batches. On these logs a rollback preserves the module output of every kept edit at its own key. □

So every instance of Min-Rollback$_{p,c}$, perturbed or not, arises from an edit log, and the results below hold for edit logs whose own keys and complaint key are linearly independent. They are worst-case statements about the module-level tolerance criterion, not statements about the logs that an editor produces.

Membership. For $p \in \{1, 2, \infty\}$ (and every integer p) the decision version of Min-Rollback$_{p,c}$ is in NP: the certificate is R, and $\|r(R)\|_p^p$ is compared with $c^p \|D\|_p^p$ exactly.

A.4 Rollback is hard at every tolerance

Two parameters describe the cancellation among the contributions. The *negative-interaction graph* joins i and j when $\langle w_i, w_j \rangle < 0$, and ν is its vertex cover number. For $\theta \in [0, 1]$, all pairwise cosines are at least $-\theta$ if $\langle w_i, w_j \rangle \geq -\theta \|w_i\|_2 \|w_j\|_2$ for all $i \neq j$. For $p = 2$, *FO-greedy* deletes an edit $i \notin R$ that maximises the first-order score $\langle r(R), w_i \rangle$, and *BI-greedy* an edit that minimises $\|r(R) - w_i\|_2$; both stop as soon as $\|r(R)\|_2 \leq c\|D\|_2$.

Theorem A.5 (Rollback is hard at every tolerance, robustly; Thm. 3.7(a) of the paper). *Fix $p \in [1, \infty]$ and rationals $0 \leq c \leq c' < 1$.*

(a) Min-Rollback$_{p,c}$ *is strongly NP-hard.*

(b) For every $\varepsilon > 0$, unless P = NP, *no polynomial-time algorithm returns R with $\|r(R)\|_p \leq c'\|D\|_p$ and $|R| \leq T^{1-\varepsilon}\mathrm{OPT}_c$. It is NP-hard to distinguish $\mathrm{OPT}_c \leq T^\varepsilon$ from $\mathrm{OPT}_{c'} = T$. Deleting everything has ratio T.*

(c) Let $c > 0$. Part (b) holds on every instance obtained from the hard instances by a perturbation of mass less than $\rho(c, c')\|D\|_p$, where $\rho(c, c') := \min(c, 1 - c')/(1 + c')$, and on instances on which exact restoration is trivial ($\mathrm{OPT}_0 = T$). Consequently, for $p = 2$, no algorithm whose expected running time on σ-perturbed instances is polynomial in T, n and $1/\sigma$ solves or $T^{1-\varepsilon}$-approximates Min-Rollback$_{2,c}$, *unless* NP $\subseteq$ BPP. *Here the instance is scaled to entries in $[-1, 1]$ and every entry receives independent noise of mean square at most σ^2, drawn from a distribution that can be sampled in polynomial time (for instance, Gaussian noise rounded to polynomially many bits).*

(d) (Greedy.) *For every $c \in [0, 1)$ and $L \geq 1$ there are $T = 2L + 2$ contributions with $\mathrm{OPT}_c = 2$ (for $p = 2$) on which FO-greedy and BI-greedy delete all T contributions under every tie-breaking rule.*

(e) (Small angles.) *The hard instances of* (a)–(c) *can be chosen with* $\nu \le 2$ *and, for any fixed* $\theta > 0$*, with all pairwise cosines at least* $-\theta$.

The paper displays parts (a) and (b) and, as its part (c), the greedy part (d); it states part (c), the robustness of the hardness, in the sentence that follows its theorem, and omits part (e).

The carrier construction. Let $(U = [3h]; C_1, \dots, C_H)$ be an X3C instance in which every element lies in some set (otherwise output a fixed no-instance), and let $M \ge H$. Let $C' = (C'_1, \dots, C'_M)$ be the sets $C_1, \dots, C_H$ followed by $M - H$ copies of C_1. On the coordinates $U \cup \{*\}$ ($n = 3h + 1$) the instance $\mathcal{I}(C, M)$ has the $T = M + 2$ items

$$\chi(C'_j) \ \ (j \in [M]), \qquad w_{\mathrm{c}} := e_* - \mathbf{1}_U \ \ \text{(carrier)},$$
$$w_{\mathrm{b}} := \mathbf{1}_U - \textstyle\sum_{j \in [M]} \chi(C'_j) \ \ \text{(balancer)}.$$

Then $D = e_*$, so $\|D\|_p = 1$ for every p, and all entries are integers in $[-M, M]$. Identifying an item with its index, let

$$\mathcal{F} := \{[T]\} \cup \big\{ \{w_{\mathrm{c}}\} \cup \{\chi(C'_j)\}_{j \in J} : \{C'_j\}_{j \in J} \text{ is an exact cover of } U \big\}.$$

Lemma A.6 (Feasible family). *For every* $p \in [1, \infty]$ *and every* $c \in [0, 1)$*, the feasible deletion sets of* $\mathcal{I}(C, M)$ *are exactly the members of* $\mathcal{F}$*. Hence* $\mathrm{OPT}_c = h + 1$ *if* C *has an exact cover and* $\mathrm{OPT}_c = T$ *otherwise, for every* $c \in [0, 1)$ *simultaneously.*

Proof. Every residual is an integer vector, and a nonzero integer vector x has $\|x\|_p \ge \|x\|_\infty \ge 1 = \|D\|_p > c\|D\|_p$. So R is feasible iff $r(R) = 0$. The $*$-coordinate of $r(R)$ is $\mathbf{1}[w_{\mathrm{c}} \notin R]$, so $w_{\mathrm{c}} \in R$. On U the residual is the sum of the kept sets, plus w_{b} if the balancer is kept. If the balancer is kept, this equals $\mathbf{1}_U$ minus the sum of the deleted sets, which vanishes iff the deleted sets form an exact cover. If the balancer is deleted, it is a sum of nonzero 0/1 vectors, which vanishes iff no set is kept, that is, iff $R = [T]$. Conversely both kinds of sets have residual 0. An exact cover of C' never contains two copies of C_1, so exact covers of C' and of C have size h and exist simultaneously; and $T = M + 2 > h + 1$ whenever a cover exists. □

Proof of Theorem A.5, parts (a)–(c) and (e). We write ρ for $\rho(c, c')$ and ρ_c for $\rho(c, c)$.

(a) Use $\mathcal{I}(C, H)$ with the budget $h + 1$. By Lemma A.6 it is a yes-instance iff C has an exact cover. The entries are bounded by H, so the reduction is strong.

(b) Use $\mathcal{I}(C, M)$ with $M := H + \lceil (h+1)^{1/\varepsilon} \rceil$, polynomial for fixed ε. Then $T > (h+1)^{1/\varepsilon}$, so $h + 1 < T^\varepsilon$ and $T^{1-\varepsilon}(h+1) < T$. By Lemma A.6 the feasible family at c' is $\mathcal{F}$. An algorithm as in (b) returns some $R \in \mathcal{F}$ with $|R| \le T^{1-\varepsilon}\mathrm{OPT}_c$: on yes-instances $|R| < T$, on no-instances $R = [T]$. This decides X3C, and it shows that $\mathrm{OPT}_c \le T^\varepsilon$ versus $\mathrm{OPT}_{c'} = T$ is NP-hard to distinguish.

(c) *Stability.* Let $|\xi| < \rho$ (recall $\|D\|_p = 1$). Then $\|D'\|_p \in [1 - |\xi|, 1 + |\xi|]$ and $\|r'(R) - r(R)\|_p \le \sum_{i \notin R} \|\xi_i\|_p \le |\xi|$. If $R \in \mathcal{F}$, then $r(R) = 0$ and $\|r'(R)\|_p \le |\xi| \le c(1 - |\xi|) \le c\|D'\|_p$, because $|\xi| < c/(1 + c') \le c/(1 + c)$. If $R \notin \mathcal{F}$, then $\|r(R)\|_p \ge 1$, so $\|r'(R)\|_p \ge 1 - |\xi| > c'(1 + |\xi|) \ge c'\|D'\|_p$, because $|\xi| < (1 - c')/(1 + c')$. So at every tolerance in $[c, c']$ the feasible family of the perturbed instance is $\mathcal{F}$, and the reductions of (a) and (b) composed with any polynomial-time computable perturbation of mass less than ρ remain reductions. The set of such perturbed instances is a union of open balls in $\mathbb{R}^{T \times n}$.

Zero-sum-free instances. Let $Q := \lceil T(1 + c')/\min(c, 1 - c') \rceil$ and shift every item by $\xi_i := e_*/(2Q)$. Then $|\xi| = T/(2Q) \le \rho/2$, so the family is unchanged. For every nonempty kept set E the $*$-coordinate of the perturbed residual is $\mathbf{1}[w_{\mathrm{c}} \in E] + |E|/(2Q) > 0$, so the instance is zero-sum-free and $\mathrm{OPT}_0 = T$. With the common denominator $2Q$ all numerators are bounded by $2Q(M + 1)$, so the reduction stays strong.

Smoothed version. Let $\mathcal{D}$ be a distribution of perturbations that can be sampled in polynomial time and satisfies $\mathbb{P}[\,|\xi| < \rho_c] \ge 5/6$. Suppose a randomized polynomial-time algorithm returns, on every input and with probability at least 5/6 over $\xi \sim \mathcal{D}$ and its coins, a $T^{1-\varepsilon}$-approximate solution of the perturbed input at tolerance c. Given an X3C instance, build $\mathcal{I}(C, M)$ as in (b), sample ξ, run the algorithm on the perturbed instance, and answer "yes" iff it returns $|R| < T$. With probability at least 5/6 the perturbation lies in the stability ball, where the family is $\mathcal{F}$ and the answer is correct whenever the algorithm succeeds; so the answer is correct with probability at least 2/3, and NP $\subseteq$ BPP. In the smoothed model, scale the instance by $1/M_\infty$, where M_∞ is its largest absolute entry (polynomial), and add independent noise of mean square at most σ^2 to every entry. Feasibility is invariant under scaling, so this is noise of mean square at most $(\sigma M_\infty)^2$ on $\mathcal{I}(C, M)$, whose $\|D\|_2 = 1$. By Jensen's inequality $\mathbb{E}\|\xi_i\|_2 \le \sigma M_\infty \sqrt{n}$, so $\mathbb{E}\,|\xi| \le TM_\infty \sigma\sqrt{n} \le \rho_c/6$ for $\sigma \le \sigma_0 := \rho_c/(6M_\infty T\sqrt{n})$, and Markov's inequality gives $\mathbb{P}[\,|\xi| \ge \rho_c] \le 1/6$. An algorithm that always returns an optimal (or $T^{1-\varepsilon}$-approximate) solution and has expected running time $f(T, n, 1/\sigma)$ halts within $6f(T, n, 1/\sigma_0)$ steps with probability at least 5/6; stopped there, it is an algorithm as above, and it runs in polynomial time because $1/\sigma_0$ is polynomial. The noise must be samplable in polynomial time for the reduction to run; Gaussian noise rounded to polynomially many bits qualifies.

(e) Replace the X3C instance by $\lceil 4/(\theta^2 h) \rceil$ disjoint copies of itself, which has an exact cover iff the original has one, so that now $h \ge 4/\theta^2$. Let $\deg u$ be the number of sets through u, $\deg^* := \max_u \deg u$ and $\pi_0 \ge \deg^* + 1$. Pad with *doubled singletons* instead of copies of C_1: $\pi_u \in \{\pi_0, \pi_0 + 1\}$ items $2e_u$ for every $u \in U$ (their number $\sum_u \pi_u$ can be made as large as required in (b)), absorbed by the balancer $w_{\mathrm{b}} := \mathbf{1}_U - \sum_j \chi(C_j) - \sum_u 2\pi_u e_u$; the carrier is unchanged and $D = e_*$. *Family.* Residuals are integer vectors, so feasibility at any $c < 1$ means $r(R) = 0$, hence $w_{\mathrm{c}} \in R$. If the balancer is kept, the deleted sets and the deleted doubled singletons are nonnegative vectors summing to $\mathbf{1}_U$, so no doubled singleton is deleted and the deleted sets form an exact cover. If the balancer is deleted, the kept items are nonnegative and sum to 0, so $R = [T]$. So the feasible family is again $\mathcal{F}$ (with the new items), and (a)–(c) hold verbatim. *Sign structure.* Sets and doubled singletons are nonnegative, so every negative pair contains the carrier or the balancer, and $\nu \le 2$; $\langle w_{\mathrm{c}}, w_{\mathrm{b}} \rangle = \sum_u (\deg u - 1 + 2\pi_u) \ge 0$. *Angles.* $\cos(w_{\mathrm{c}}, \chi(C)) = -\sqrt{3/(3h+1)}$ and $\cos(w_{\mathrm{c}}, 2e_u) = -1/\sqrt{3h+1}$. Every entry of the balancer has absolute value between $2\pi_0$ and $\deg^* + 1 + 2\pi_0$, so $\|w_{\mathrm{b}}\|_2 \ge 2\pi_0\sqrt{3h}$, $\cos(w_{\mathrm{b}}, \chi(C)) \ge -(1 + \frac{\deg^*+1}{2\pi_0})/\sqrt{h}$ and $\cos(w_{\mathrm{b}}, 2e_u) \ge -(1 + \frac{\deg^*+1}{2\pi_0})/\sqrt{3h}$. With $h \ge 4/\theta^2$ and $\pi_0 \ge \deg^* + 1$ every cosine is at least $-\frac{3}{2} \cdot \frac{\theta}{2} > -\theta$. All entries stay polynomially bounded. □

Proof of Theorem A.5, part (d). Fix a rational $\kappa \in (c, 1)$, for instance $\kappa = (1 + c)/2$. Take orthonormal $e_0, e_1, f_1, \dots, f_L$, $\varphi_j :=$

$1 + j/(4L) \in (1, \frac{5}{4}]$, and the items

$$w_1 = \tfrac{1}{2}e_0 + 2e_1, \qquad w_2 = \tfrac{1}{2}e_0 - 2e_1,$$
$$\ell_j = \varphi_j e_0 + \kappa f_j, \qquad \ell'_j = -\ell_j \qquad (j \in [L]).$$

Then $D = e_0$, $T = 2L + 2$, and the squared tolerance is $c^2 < \kappa^2 < 1$.

Optimum. $R = \{w_1, w_2\}$ leaves the lure pairs ℓ_j, ℓ'_j, which sum to 0. No set of size at most 1 is feasible: the squared residuals are 1, $\frac{1}{4} + 4$, $(\varphi_j - 1)^2 + \kappa^2$ and $(1 + \varphi_j)^2 + \kappa^2$, all at least $\kappa^2 > c^2$. So $\mathrm{OPT}_c = 2$.

FO-greedy. Invariant at the start of a phase: some lure pairs are deleted, all other items are present, and the residual is e_0. The scores $\langle e_0, \cdot \rangle$ are $\frac{1}{2}$ for w_1, w_2, $\varphi_j > 1$ for ℓ_j and $-\varphi_j$ for ℓ'_j, so the unique maximiser is the remaining ℓ_j with the largest φ_j; call its value φ and its direction f. Afterwards the residual is $(1-\varphi)e_0 - \kappa f$, and the scores are $\varphi(\varphi-1)+\kappa^2$ for its partner, $\varphi_j(1-\varphi) < 0$ for every other ℓ_j, $\varphi_j(\varphi - 1) < \varphi(\varphi - 1)$ for every other ℓ'_j (all remaining φ_j are smaller than φ), and $(1 - \varphi)/2 < 0$ for w_1, w_2. So the partner is the unique maximiser, and its deletion restores the residual e_0. After L phases only w_1, w_2 remain, with equal scores; either deletion leaves the squared residual $\frac{17}{4}$, and then the other is deleted.

BI-greedy. Same invariant, with the lures removed in increasing order of φ_j. At the residual e_0 the squared residuals after one deletion are $(\varphi_j - 1)^2 + \kappa^2 \le \frac{1}{16} + \kappa^2 < \frac{17}{4}$ for ℓ_j (unique minimum at the smallest remaining φ_j), $(1 + \varphi_j)^2 + \kappa^2$ for ℓ'_j, and $\frac{17}{4}$ for w_1, w_2. After the deletion of ℓ_j with value φ, the candidates give: exactly 1 for the partner; $(\varphi + \varphi_j - 1)^2 + 2\kappa^2 > 1$ for every other ℓ_j; $(1 + \varphi_j - \varphi)^2 + 2\kappa^2 > 1$ for every other ℓ'_j (all remaining φ_j are larger than φ); and $(\varphi - \frac{1}{2})^2 + 4 + \kappa^2 > 1$ for w_1, w_2. So the partner is the unique minimiser and the residual is e_0 again; the final phase is as for FO-greedy.

Every residual visited before the last has squared norm at least $\min(1, \kappa^2) = \kappa^2 > c^2$, so neither rule stops early. The only tie is between w_1 and w_2, and it is harmless. □

In the basic family ν can equal 1: when the sets partition U and $M = H$, the balancer is 0. This is why part (e) says $\nu \le 2$. At $c = 0$ Theorem A.5 specialises to strong NP-completeness and $T^{1-\varepsilon}$-inapproximability of exact restoration. The hard instances cancel heavily ($\sum_i \|w_i\|_2$ grows linearly in T while $\|D\|_2 = 1$), and the perturbations that the theorem tolerates are of the size of $\|D\|$, not of the size of the individual contributions; Proposition A.8 shows that this cancellation is necessary for the bicriteria form. At $c = 0$, Min-Rollback is also the subclass of binary Min-RVLS$^=$ in which the system is $\sum_i x_i w_i = D$ ($x_i = 1$ for the deleted edits); the latter problem is NPO PB-complete [6, Prop. 10], but hardness of a superclass does not transfer to a subclass, so the theorem is needed even there. Sparse approximation with real coefficients is NP-hard [53], and bicriteria hardness of sparse regression is known under a complexity assumption [23]; here the coefficients are 0/1 and the target is the sum of all columns, so neither result implies Theorem A.5. The smoothed statement uses the perturbation model of smoothed analysis [62] but no theorem from it. In part (d), for κ close to 1 the first step of BI-greedy increases the norm of the residual; a variant that stops when no deletion strictly decreases it stops at once, at an infeasible point. By Lemma A.4 the contributions of part (d) are those of an edit log.

Remark A.7 (One output dimension: a pseudo-polynomial algorithm). Let the output dimension be $n = 1$ and let the contributions be integers of absolute value at most $w_{\max}$. Then Min-Rollback$_{p,c}$ is solved in time polynomial in T and $w_{\max}$, for every p and every rational $c \ge 0$. The residual of a kept set is an integer in $[-Tw_{\max}, Tw_{\max}]$. A table over these at most $2Tw_{\max} + 1$ sums holds, after the first i contributions, the largest size of a subfamily of them with each sum, and each further contribution updates it in one pass. The optimum is T minus the largest entry of the final table at a sum r with $|r| \le c\,|D|$. The same table over integer vectors gives a pseudo-polynomial algorithm for every *fixed* output dimension n: the residual of a kept set is one of at most $(2Tw_{\max} + 1)^n$ integer vectors, and the feasible ones are found by comparing $\|r\|_p$ with $c\,\|D\|_p$ (in exact arithmetic for $p = \infty$ and for every integer p). So, unless P = NP, minimum rollback is not strongly NP-hard for any fixed output dimension: the strong NP-hardness of Theorem A.5(a) needs an output dimension that grows with the instance: its reduction has one coordinate per element of the instance of X3C. For $n = 1$ and $c = 0$ the problem is NP-hard, weakly (Proposition B.69(d)). The nearest result on databases has the same shape: a smallest repair by tuple deletions of a constraint on a sum or avg aggregate is NP-hard for numbers in binary and is found in pseudo-polynomial time [2]. Minimum rollback is a selection problem on signed vectors; we state the relation, not a reduction between the two problems.

A.5 Where the hardness of rollback lives

Let $p = 2$ and assume that no contribution is zero. Let G be the Gram matrix of the contributions, with the entries $\langle w_i, w_j \rangle$, let $\Lambda := \mathrm{diag}(G)$, and let $\hat{G} := \Lambda^{-1/2} G \Lambda^{-1/2}$ have the extreme eigenvalues $\gamma \le \Gamma$. For $x \ge 0$ let $\mu(x)$ be the least number of contributions whose deletion leaves the others with $\sum_{i \notin R} \|w_i\|_2^2 \le x$; it is attained by deleting the contributions of largest norm first (exchange argument). LF deletes the contributions in the order of decreasing norm until the tolerance criterion holds; $R_{\mathrm{LF}}(c)$ is its deletion set.

Proposition A.8 (Conditioning sandwich). *For all contributions without a zero vector and every $c \ge 0$,*

$$\mathrm{OPT}_c \;\le\; |R_{\mathrm{LF}}(c)| \;\le\; \mu\big(c^2 \|D\|_2^2/\Gamma\big).$$

If $\hat{G}$ is nonsingular ($\gamma > 0$), then also $\mu(c^2\|D\|_2^2/\gamma) \le \mathrm{OPT}_c$, so LF is feasible at c and uses at most $\mathrm{OPT}_{c'}$ deletions for $c' = c\sqrt{\gamma/\Gamma}$, for every sign pattern.

Proof. For a kept set E with indicator vector $\chi(E)$, $\|r\|_2^2 = \chi(E)^\top G \chi(E) = x^\top \hat{G} x$ with $x = \Lambda^{1/2}\chi(E)$ and $\|x\|_2^2 = \sum_{i \in E} \|w_i\|_2^2$, so

$$\gamma \sum_{i \in E} \|w_i\|_2^2 \;\le\; \|r\|_2^2 \;\le\; \Gamma \sum_{i \in E} \|w_i\|_2^2.$$

The LF prefix of length $\mu(c^2\|D\|_2^2/\Gamma)$ leaves a kept set with $\|r\|_2^2 \le \Gamma \sum_{i \in E} \|w_i\|_2^2 \le c^2 \|D\|_2^2$, so LF stops no later, and $\mathrm{OPT}_c \le |R_{\mathrm{LF}}(c)|$ because the output of LF is feasible. If $\gamma > 0$, a feasible kept set has $\sum_{i \in E} \|w_i\|_2^2 \le c^2 \|D\|_2^2/\gamma$, which gives the lower bound; and with $c' := c\sqrt{\gamma/\Gamma}$, $\mathrm{OPT}_{c'} \ge \mu(c'^2\|D\|_2^2/\gamma) = \mu(c^2\|D\|_2^2/\Gamma) \ge |R_{\mathrm{LF}}(c)|$. □

The hypothesis $\gamma > 0$ cannot be dropped from the last claim. For the scalar contributions $(1, -1, 1)$, $\hat{G}$ has rank one, so $\gamma = 0$

and $c\sqrt{\gamma/\Gamma} = 0$; all norms are equal, so LF may delete in the order 2, 1, 3, and at $c = \frac{1}{2}$ it deletes all three items, while $\mathrm{OPT}_0 = 1$. Since $\operatorname{rank}\hat{G} \le n$, $\gamma = 0$ whenever $T > n$. Taking $E = [T]$ gives $\gamma \le \|D\|_2^2/\sum_i \|w_i\|_2^2$. The instances constructed for Theorem A.5(a), (b), (d) and (e) have $\|D\|_2 = 1$ and items of norm at least 1 (nonzero integer vectors; in (d), $\|w_1\|_2^2 = \|w_2\|_2^2 = \frac{17}{4}$ and $\|\ell_j\|_2^2 > 1$), so without a zero item they have $\gamma \le 1/T$; a perturbation as in (c), for $p = 2$, changes this bound by a constant factor only (its mass is below $\frac{1}{3}$, so $\|D'\|_2 \le \frac{4}{3}$ and $\sum_i \|w_i'\|_2^2 \ge T - \frac{2}{3}$). The bicriteria form of the hardness therefore cannot hold on logs with $\Gamma/\gamma \le \kappa$ once the relaxed tolerance reaches $c\sqrt{\kappa}$. The lower bound $\mu(c^2\|D\|_2^2/\gamma)$ can be certified in exact arithmetic by checking $\hat{G} - \gamma_0 I \succeq 0$ for a rational $\gamma_0 \le \gamma$ (for instance by an $LDL^\top$ factorisation) and using γ_0.

A.6 Compensation: closed form, non-locality, reported keys

Fix the protected keys K_{prot}, let P_{prot} be the orthogonal projector onto $\operatorname{col} K_{\mathrm{prot}}$ and $K_\perp := (I - P_{\mathrm{prot}})K_q$. A target is a matrix $V \in \mathbb{R}^{n\times A}$, where A is the number of complaint keys; for the minimum compensation of Def. 2.6, $V = -dWK_q$.

Lemma A.9 (Closed form; standard). *The system $XK_q = V$, $XK_{\mathrm{prot}} = 0$ is solvable iff $\operatorname{null} K_\perp \subseteq \operatorname{null} V$. If it is solvable:*

(a) $X^\star = VK_\perp^+$ is its unique solution of minimum Frobenius norm, and it is minimal in every unitarily invariant norm;

(b) for one complaint key $k_q \notin \operatorname{col} K_{\mathrm{prot}}$ with the target v_q, let $k_\perp := (I - P_{\mathrm{prot}})k_q$; then $X^\star = v_q k_\perp^\top/\|k_\perp\|_2^2$ and $\|X^\star\|_F = \|v_q\|_2/\|k_\perp\|_2$;

(c) for the target $V = -dWK_q$, the system is solvable iff dW vanishes on $\operatorname{col} K_q \cap \operatorname{col} K_{\mathrm{prot}}$.

The system is solvable for every target iff the rank of $[K_{\mathrm{prot}}, K_q]$ is $\operatorname{rank} K_{\mathrm{prot}} + A$.

Proof. $XK_{\mathrm{prot}} = 0$ iff $XP_{\mathrm{prot}} = 0$, and then $XK_q = XK_\perp$. *Solvability.* If X is a solution and $K_\perp \pi = 0$ for a vector $\pi \in \mathbb{R}^A$, then $K_q\pi \in \operatorname{col} K_{\mathrm{prot}}$, so $V\pi = XK_q\pi = 0$. Conversely, if $\operatorname{null} K_\perp \subseteq \operatorname{null} V$, then $VK_\perp^+ K_\perp = V$, because $K_\perp^+ K_\perp$ is the orthogonal projector onto the orthogonal complement of $\operatorname{null} K_\perp$; with $K_\perp^+ = (K_\perp^\top K_\perp)^+ K_\perp^\top$ and $K_\perp^\top K_{\mathrm{prot}} = 0$ we get $X^\star K_{\mathrm{prot}} = 0$ and $X^\star K_q = X^\star K_\perp = V$. *Minimality.* Let B be a basis of $\operatorname{col} K_{\mathrm{prot}}$. The set of solutions is $\{X : X[B \mid K_\perp] = [0 \mid V]\}$, whose element of minimum norm is $[0 \mid V][B \mid K_\perp]^+ = VK_\perp^+$ (as $B^\top K_\perp = 0$), unique by strict convexity. Every solution X satisfies $XK_\perp K_\perp^+ = X^\star$, and right multiplication by the orthogonal projector $K_\perp K_\perp^+$ does not increase singular values; in the Frobenius norm $\|X\|_F^2 = \|X^\star\|_F^2 + \|X - X^\star\|_F^2$. *One key.* $K_\perp = k_\perp$ is one column, and $k_\perp^+ = k_\perp^\top/\|k_\perp\|_2^2$; $X^\star$ has rank one. *Target.* $\{K_q\pi : K_\perp\pi = 0\} = \operatorname{col} K_q \cap \operatorname{col} K_{\mathrm{prot}}$. *Every target.* A target that does not vanish on $\operatorname{null} K_\perp$ exists unless $\operatorname{null} K_\perp = 0$; and $\operatorname{null} K_\perp = 0$ iff $\operatorname{rank} K_\perp = A$, where $\operatorname{col}[K_{\mathrm{prot}}, K_q]$ is the orthogonal direct sum of $\operatorname{col} K_{\mathrm{prot}}$ and $\operatorname{col} K_\perp$. □

The closed form is the minimum-norm null-space update of AlphaEdit [21] and its successors; we do not claim it. The next lemma states in which sense a compensation is not local and is fitted to the reported keys.

Lemma A.10 (Non-locality of a linear correction). *Let $X \in \mathbb{R}^{n\times d}$, $X \neq 0$.*

(a) $\operatorname{null} X$ is a subspace of dimension $d - \operatorname{rank} X \le d - 1$. Hence $Xk \neq 0$ for every key k outside a proper subspace, and $\mathbb{P}(Xk = 0) = 0$ if the keys have a density.

(b) If $\operatorname{rank} K_\perp = A$, then $X^\star k = V\pi(k)$ with $\pi(k) := (K_\perp^\top K_\perp)^{-1}K_\perp^\top k$ for every key k. For the target $V = -dWK_q$, a key k that was not reported is restored, $X^\star k = -dWk$, iff $dWk = dWK_q\pi(k)$: iff its damage is this combination of the reported damages.

(c) Let the keys be random with second moment $\Sigma = \mathbb{E}\,kk^\top \succ 0$, and let $\operatorname{rank} K_q = A$. Put $G_\Sigma := K_q^\top \Sigma^{-1} K_q$ and $X_\Sigma := VG_\Sigma^{-1}K_q^\top\Sigma^{-1}$. Every X with $XK_q = V$ satisfies

$$\mathbb{E}\|Xk\|_2^2 = C^\star + \operatorname{tr}\big((X - X_\Sigma)\Sigma(X - X_\Sigma)^\top\big) \;\ge\; C^\star := \operatorname{tr}\big(VG_\Sigma^{-1}V^\top\big),$$

with equality iff $X = X_\Sigma$; $C^\star > 0$ if $V \neq 0$. For one complaint key, $C^\star = \|v_q\|_2^2/(k_q^\top\Sigma^{-1}k_q)$.

(d) Let $k^{(1)}, \dots, k^{(N)}$ be a population of keys whose second-moment matrix $\Sigma = \frac{1}{N}\sum_l k^{(l)}k^{(l)\top}$ is positive definite. Let the complaint key be its member $k_q = k^{(l_0)}$, and put $h_q := k_q^\top\Sigma^{-1}k_q$. Every X with $Xk_q = v_q$ moves the other keys by

$$\sum_{l\neq l_0}\|Xk^{(l)}\|_2^2 \;\ge\; \|v_q\|_2^2\,(N/h_q - 1),$$

with equality for $X = X_\Sigma$. The mean of $k^{(l)\top}\Sigma^{-1}k^{(l)}$ over the population is d, so at least half of the keys have $h_q \le 2d$; for such a complaint key the bound is at least $\|v_q\|_2^2\,(N/(2d) - 1)$, which exceeds $\|v_q\|_2^2$ once $N > 4d$.

Proof. (a) Rank–nullity; a proper subspace is a Lebesgue null set. (b) $K_\perp^+ = (K_\perp^\top K_\perp)^{-1}K_\perp^\top$ when $K_\perp$ has full column rank. (c) $\mathbb{E}\|Xk\|_2^2 = \operatorname{tr}(X\Sigma X^\top)$. The matrix G_Σ is positive definite because $\operatorname{rank} K_q = A$, and $X_\Sigma K_q = V$. So $Y := X - X_\Sigma$ satisfies $YK_q = 0$, and $\Sigma X_\Sigma^\top = K_q G_\Sigma^{-1}V^\top$ gives $\operatorname{tr}(Y\Sigma X_\Sigma^\top) = \operatorname{tr}(YK_qG_\Sigma^{-1}V^\top) = 0$. Hence $\operatorname{tr}(X\Sigma X^\top) = \operatorname{tr}(X_\Sigma\Sigma X_\Sigma^\top) + \operatorname{tr}(Y\Sigma Y^\top)$, the first term equals $\operatorname{tr}(VG_\Sigma^{-1}V^\top)$, and the second is nonnegative and vanishes only for $Y = 0$. (d) $\sum_l\|Xk^{(l)}\|_2^2 = N\operatorname{tr}(X\Sigma X^\top) \ge N\|v_q\|_2^2/h_q$ by (c), and the term of the complaint key equals $\|v_q\|_2^2$ for every such X. The mean of $k^{(l)\top}\Sigma^{-1}k^{(l)}$ is $\operatorname{tr}(\Sigma^{-1}\frac{1}{N}\sum_l k^{(l)}k^{(l)\top}) = d$, and by Markov's inequality fewer than half of the keys exceed twice the mean. □

Part (c) is least squares with equality constraints, and for one complaint key its minimiser X_Σ has the form of the update of ROME [49]. By (b) and Lemma A.9(b), the change at a key relative to the repaired change is $\|X^\star k\|_2/\|v_q\|_2 = |\cos(k, k_\perp)|\,\|k\|_2/\|k_\perp\|_2$, which is not bounded by 1: it grows without bound as the complaint key approaches the protected span. Per key the bound of (c) is of order $\|v_q\|_2^2/d$, which is small; the contrast with rescoping is (a) and the total of (d).

A.7 The capacity law of exact protection

Compensation retains every recorded term. Its combinatorial part is the choice of the edits whose keys stay protected. Edit i has a *protection group*, a set of keys at which the correction must vanish (for instance the keys of the needed set of its request, Theorem A.1), stored as the columns of a matrix K_i with *protection span* $S_i := \operatorname{col} K_i$ of dimension r_i. For complaint keys K_q with A columns and a target

V of rank $a \geq 1$, a *sacrifice set* $R \subseteq [T]$ is *feasible* if some X satisfies $XK_i = 0$ for all $i \notin R$ and $XK_q = V$; the edits of R lose their protection, and no edit is deleted. Min-Sacrifice asks for a smallest feasible sacrifice set; OPT is its size. We write $U_R := \sum_{i \notin R} S_i$.

Let $Z \subseteq \bigcap_i S_i$ be a subspace of dimension z that all groups share (the keys of a common prompt prefix, Lemma A.3; $z = 0$ is allowed), and put $\mathrm{rk}_Z(R) := z + \sum_{i \notin R}(r_i - z)$ for $R \neq [T]$. The groups are in *general position modulo* Z (GP/Z) if, for every nonempty $J \subseteq [T]$,

$$\dim \sum_{i \in J} S_i = \min\bigl(d,\ z + \textstyle\sum_{i \in J}(r_i - z)\bigr).$$

Let $s_Z^\star$ be the least s such that $s = T$, or z plus the sum of the $T - s$ smallest numbers $r_i - z$ is at most $d - a$.

Lemma A.11 (Monotonicity). *If* $\operatorname{col} K_1 \subseteq \operatorname{col} K_2$*, every* X *with* $XK_2 = 0$ *satisfies* $XK_1 = 0$*. Hence the feasible sacrifice sets are closed under taking supersets, and the minimum norm of a correction does not decrease when protected keys are added.*

Proof. The columns of K_1 are linear combinations of those of K_2; and $U_{R'} \subseteq U_R$ for $R \subseteq R'$. □

Theorem A.12 (Capacity law of exact protection). *With this notation, the following hold.*

(a) *In general, one rank test decides whether* OPT $= 0$*, and every feasible sacrifice set* R *satisfies* $\dim U_R \leq d - a$.
(b) *Under GP/Z,* OPT $\geq s_Z^\star$ *for every complaint and every target of rank* a.
(c) *Let, in addition, the complaint keys be* $A = a$ *linearly independent keys with* $\dim(\sum_{i \in J} S_i + \operatorname{col} K_q) = \min\bigl(d,\ z + \sum_{i \in J}(r_i - z) + a\bigr)$ *for every nonempty* J*. Then* OPT $= s_Z^\star$*. In particular,* OPT $= 0$ *iff* $z + \sum_i (r_i - z) \leq d - a$.
(d) $s_Z^\star$ *is computed in* $O(T \log T)$ *time by sacrificing the largest groups first.*

Proof. For a sacrifice set R apply Lemma A.9 with the kept groups as protected keys, so that $\operatorname{col} K_{\mathrm{prot}} = U_R$ and $K_\perp = (I - P_{U_R})K_q$, where P_{U_R} is the orthogonal projector onto U_R.

(a) $R = \emptyset$ is feasible iff $\operatorname{null} K_\perp \subseteq \operatorname{null} V$, that is, iff stacking V under $K_\perp$ does not increase the rank. If R is feasible with solution X, then $V = XK_\perp$, so $a \leq \operatorname{rank} K_\perp$, and $\operatorname{col} K_\perp \subseteq U_R^\perp$; hence $a \leq d - \dim U_R$.

(b) Let $R \neq [T]$ be feasible. Under GP/Z the dimension of U_R is $\min(d, \mathrm{rk}_Z(R))$, and by (a) it is at most $d - a < d$, so $\mathrm{rk}_Z(R) \leq d - a$. Since $r_i - z \geq 0$, the kept groups contribute at least as much as the $T - |R|$ smallest ones, so $s = |R|$ satisfies the defining condition of $s_Z^\star$; hence $|R| \geq s_Z^\star$. For $R = [T]$ this is trivial.

(c) A target of rank A has trivial null space, so R is feasible iff $\operatorname{null} K_\perp = 0$, that is, iff $\dim(U_R + \operatorname{col} K_q) = \dim U_R + A$ (Lemma A.9). For $R \neq [T]$ the left side is $\min(d, \mathrm{rk}_Z(R) + A)$ and the right side is $\min(d, \mathrm{rk}_Z(R)) + A$ under the two assumptions; they agree iff $\mathrm{rk}_Z(R) \leq d - A$. $R = [T]$ is feasible because $\operatorname{rank} K_q = A$, so OPT $\leq T$. If $s_Z^\star < T$, sacrificing the $s_Z^\star$ groups of largest dimension leaves $\mathrm{rk}_Z(R) \leq d - A$ by the definition of $s_Z^\star$, a feasible set of size $s_Z^\star$. With (b), OPT $= s_Z^\star$ in both cases. The last claim is the case $R = \emptyset$.

(d) Among the sets of a given size, sacrificing the largest groups minimises rk_Z; sort the dimensions and scan the prefix sums. By Lemma A.11, feasibility is preserved when further groups are sacrificed. □

The theorem is a dimension count. That a linear map recalls stored items exactly only up to the dimension of its keys is classical [39]; what the theorem adds is the closed form of the minimum sacrifice in (b)–(d).

Genericity. For fixed sizes d, A, z and (r_i), let S_i be spanned by the columns of $[B_Z, B_i]$ and $\operatorname{col} K_q$ by those of B_q, where B_Z has z, B_i has $r_i - z$ and B_q has A columns. For Lebesgue-almost every choice of the entries, GP/Z and the hypothesis of (c) hold: each condition asks a maximal minor of $[B_Z, B_i\ (i \in J)]$, or of this matrix extended by B_q, to be nonzero; a choice of standard basis vectors makes that minor ± 1, so it is a nonzero polynomial in the entries, and the product of these polynomials over all J vanishes only on a null set. General position cannot be certified for a given log: a lower bound on OPT without it would require solving Min-Sacrifice, which is NP-hard (Prop. A.26).

Remark A.13 (A measured log). On the log analysed in the first round ($d = 11008$, $T = 100$), every prompt shares a template prefix of 15 tokens, so $z = 15$, and the needed sets have between 288 and 1058 keys (median 540). Protecting the last prompt key of every edit together with the full needed sets of the shortest edits stays feasible for every complaint up to 37 edits (per-complaint maximum 37), so OPT ≤ 63; this is a floating-point measurement (`results_real_keys.json`). It reaches rank 10903; under GP/Z the same protection set would have rank 10888, a gap consistent with near-duplicate copies of the prefix in low precision. Under GP/Z for the non-prefix keys, Theorem A.12 gives $s_Z^\star = 63$ (`check/c5_real_log_gp.json`), so this sacrifice is irreducible under that assumption. Plain general position ($z = 0$) would predict 65, which the measurement refutes; under plain general position, zero sacrifice needs a protected dimension per edit of at most $\lfloor (d-1)/T \rfloor = 110$.

A.8 The capacity of linear certification

This subsection proves Theorem 3.2 of the paper. It applies the certificate criterion of Def. 2.5(a) to *every* repair that is a weight, and counts. The paper says that a weight *restores* a complaint; here, as in the sources, a weight that restores every complaint of Q^- is said to *certify* Q^-. Part (a) of the paper's theorem is Theorem A.15(a)–(c) with Corollary A.16(i),(ii), for the required values (dep); its part (b) is Proposition A.18(a),(d) with Proposition A.19(c); what the paper says about the stored records is Sec. A.8.4.

Setting. $\mathcal{L}$ is an edit log (Def. 2.1); $U := [u_1, \ldots, u_T] \in \mathbb{R}^{n \times T}$, $\Psi := [\psi_1, \ldots, \psi_T] \in \mathbb{R}^{d \times T}$ and $K_E := [k_1, \ldots, k_T] \in \mathbb{R}^{d \times T}$ hold its write vectors, read-out vectors and own keys, so that $dW = U\Psi^\top$; for $J \subseteq [T]$, U_J, Ψ_J, K_J consist of the columns in J. A *linear repair* is a weight $W' = W_0 + D'$ with an arbitrary *update* $D' \in \mathbb{R}^{n \times d}$. The rollback of R is the linear repair with $D' = \sum_{i \notin R} \Delta_i$, a compensation has $D' = dW + X$, and re-editing (further edits appended to W_E) and a rebuild from W_0 by any editor that writes into the weight of the edited module are linear repairs as well. The *trace keys* of a query q are the keys k_t at *all* positions t of its pre-edit trace $x^0(q)$. For a finite set Q^- of wrong-answer complaints, $K_Q \in \mathbb{R}^{d \times A}$ holds the trace keys of all its queries as columns, $r_Q := \operatorname{rank} K_Q$, and P_Q is the orthogonal projector onto $\operatorname{col} K_Q$. The linear repair *certifies* Q^- if $D'K_Q = 0$ (Def. 2.5(a) for a weight). Let $H = [h_1, \ldots, h_T] \in \mathbb{R}^{n \times T}$

be a matrix of *required values*. The linear repair *preserves* edit i if $D'k_i = h_i$. We treat two choices of H:

(dep) $h_i = dWk_i$: the module output at the own key stays the deployed one, $W'k_i = W_E k_i$;

(tgt) $h_i = v_i^{\mathrm{rb}} = z_i^\star - W_0 k_i$: the module output at the own key is the target of the editor, $W'k_i = z_i^\star$ (Lemma B.65); $V^{\mathrm{rb}} := [v_1^{\mathrm{rb}}, \dots, v_T^{\mathrm{rb}}]$.

A set $J \subseteq [T]$ is *preservable* (given Q^- and H) if some linear repair certifies Q^- and preserves every edit of J; $\mathrm{cap}(Q^-)$ is the largest size of a preservable set. We write $M_J := (I - P_Q)K_J$; then $\mathrm{rank}\, M_J = \mathrm{rank}[K_Q, K_J] - r_Q$, because $\mathrm{col}[K_Q, K_J]$ is the orthogonal direct sum of $\mathrm{col}\, K_Q$ and $\mathrm{col}\, M_J$.

Lemma A.14 (The certificate of a linear repair). *Assume* (H1)–(H3) *in the form of Sec. A.2 (keys do not depend on the weight of the edited module) and exact arithmetic, and let $W' = W_0 + D'$ and q be a query.*

(a) If $D'k_t = 0$ at every position t of $x^0(q)$, then $\mathcal{M}[W']$ evaluates q on the trace $x^0(q)$ with the pre-edit module outputs and the pre-edit logits at every position, and $\hat{y}(q; W') = y_0(q)$.

(b) Conversely, the module outputs of $\mathcal{M}[W']$ at the positions of $x^0(q)$ are the pre-edit ones iff *$D'k_t = 0$ at every position of $x^0(q)$. The converse does not hold at the level of answers: $\hat{y}(q; W') = y_0(q)$ does not imply $D'k_t = 0$.*

Proof. (a) is Proposition A.2; we repeat the argument, which is that of Proposition B.1(a). By induction on the generation step, $\mathcal{M}[W']$ and $\mathcal{M}[W_0]$ hold the same prefix of $x^0(q)$. By (H1) the keys of its positions do not depend on the weight of the edited module, so they are the keys k_t of $x^0(q)$, and the module of $\mathcal{M}[W']$ returns $W'k_t = W_0k_t + D'k_t = W_0k_t$ there. By (H2) the logits are equal, and by (H3) so is the next token or the decision to stop. (b) At a position t of the fixed sequence $x^0(q)$ the key is k_t under both weights (H1), and $W'k_t = W_0k_t$ iff $D'k_t = 0$. For the last claim let the logits not depend on one coordinate of the module output and let the columns of $D' \neq 0$ be multiples of that coordinate vector. □

So for weights the certificate criterion is a statement about module outputs on the whole pre-edit trace, as for scoped logs: a certified complaint is resolved, with no approximation, and the certificate is equivalent to exact restoration of the module outputs along the trace. For the teacher-forced answer alone the needed set of $x^0(q)$ suffices (Proposition A.2), which is one key less per complaint; restoring fewer positions gives no guarantee, even at the restored positions.

Theorem A.15 (Capacity of linear certification). *Let Q^- be a finite set of wrong-answer complaints and H a matrix of required values.*

(a) Criterion. *J is preservable iff $H_J c = 0$ for every $c \in \mathbb{R}^J$ with $K_J c \in \mathrm{col}\, K_Q$. For* (dep) *this reads: dW vanishes on $\mathrm{col}\, K_Q \cap \mathrm{col}\, K_J$. The updates that certify Q^- and preserve J are then $D' = H_J M_J^+ + Y(I - P)$, $Y \in \mathbb{R}^{n \times d}$ arbitrary, with P the orthogonal projector onto $\mathrm{col}[K_Q, K_J]$; $H_J M_J^+$ is the one of least Frobenius norm.*

(b) Rank bound, for all values. *If J is preservable, then $\mathrm{rank}\, H_J \leq \mathrm{rank}\, M_J = \mathrm{rank}[K_Q, K_J] - r_Q \leq d - r_Q$: a certifying linear repair preserves at most $d - r_Q$ dimensions of required values. If M_J has full column rank $|J|$, then J is preservable, whatever H; if H_J has full column rank, the converse holds.*

(c) Capacity. *Let H have full column rank T* (ND). *Then J is preservable iff M_J has full column rank, that is, iff $\mathrm{rank}[K_Q, K_J] = r_Q + |J|$. Hence the preservable sets are the independent sets of a linear matroid on $[T]$, that of the vectors $(I - P_Q)k_i$, and*

$$\mathrm{cap}(Q^-) = \mathrm{rank}\big((I - P_Q)K_E\big) = \mathrm{rank}[K_Q, K_E] - r_Q,$$

which is $\mathrm{rank}\, K_E - \dim(\mathrm{col}\, K_Q \cap \mathrm{col}\, K_E)$: beyond dependent own keys, the edits that must be given up are counted by the dimension in which the complaint span meets the span of the own keys. It is at most $\min(T, d - r_Q)$, with equality iff $\mathrm{rank}[K_Q, K_E] = \min(d, r_Q + T)$, which holds for Lebesgue-almost every K_E. Every preservable set extends to one of size $\mathrm{cap}(Q^-)$, and for weights $\omega_i \geq 0$ on the edits the greedy rule, which scans the edits by decreasing weight and keeps an edit iff the kept set stays preservable, preserves a set of maximum weight.

(d) When (ND) holds. *For* (tgt), *$V^{\mathrm{rb}} = U(I + \Lambda)$ with a strictly triangular matrix Λ (in the order of the batches), so* (ND) *holds iff U has full column rank. For* (dep), *$H = U\,\Psi^\top K_E$, so* (ND) *holds iff U has full column rank and the $T \times T$ matrix $\Psi^\top K_E$ of the coordinates $\langle \psi_i, k_j \rangle$ is nonsingular; this forces linearly independent own keys. Both need $n \geq T$.*

(e) Without (ND). *The preservable sets are closed under taking subsets and $\mathrm{cap}(Q^-) \geq \mathrm{rank}((I - P_Q)K_E)$, but the inequality can be strict, the preservable sets need not form a matroid, and the greedy rule can fail; for* (dep) *and one complaint with one trace key, $T - \mathrm{cap}(Q^-)$ is the optimum of* Min-Sacrifice *for protection groups of one key, which is NP-hard to compute (Proposition A.26).*

Proof. (a) The requirement is a linear matrix equation,

$$D'\,[K_Q, K_J] = [0, H_J].$$

An equation $XA = B$ is solvable iff $\mathrm{null}\, A \subseteq \mathrm{null}\, B$ [56, Thm. 2]: necessity is clear, and if the inclusion holds, then $BA^+A = B$, because $I - A^+A$ is the orthogonal projector onto $\mathrm{null}\, A$. A vector (a, c) lies in $\mathrm{null}[K_Q, K_J]$ iff $K_J c = -K_Q a$, and $[0, H_J](a, c) = H_J c$; as a is arbitrary, the inclusion says that $H_J c = 0$ whenever $K_J c \in \mathrm{col}\, K_Q$. For (dep), $H_J c = dWK_J c$ and $\{K_J c : K_J c \in \mathrm{col}\, K_Q\} = \mathrm{col}\, K_Q \cap \mathrm{col}\, K_J$. The solution of least norm and the general solution are those of Lemma A.9(a), applied with the protected keys K_Q, the complaint keys K_J and the target H_J: there $K_\perp = M_J$, and two solutions differ by a matrix that vanishes on $\mathrm{col}[K_Q, K_J]$.

(b) Let D' certify Q^- and preserve J. Then $D'P_Q = 0$, so $H_J = D'K_J = D'(P_Q K_J + M_J) = D'M_J$ and $\mathrm{rank}\, H_J \leq \mathrm{rank}\, M_J$; and $\mathrm{col}\, M_J \subseteq (\mathrm{col}\, K_Q)^\perp$, a space of dimension $d - r_Q$. If M_J has full column rank and $K_J c \in \mathrm{col}\, K_Q$, then $M_J c = (I - P_Q)K_J c = 0$, so $c = 0$ and $H_J c = 0$; apply (a). If J is preservable and H_J has full column rank, then $|J| = \mathrm{rank}\, H_J \leq \mathrm{rank}\, M_J \leq |J|$.

(c) Under (ND) every H_J has full column rank, so by (b) J is preservable iff M_J has full column rank: iff the vectors $(I - P_Q)k_i$, $i \in J$, are linearly independent. These sets are the independent sets of a linear matroid [54], namely the contraction, by the columns of K_Q, of the matroid of the columns of $[K_Q, K_E]$, whose rank function is $J \mapsto \mathrm{rank}[K_Q, K_J] - r_Q$ [54]. Its rank is $\mathrm{rank}\, M_{[T]}$, all maximal independent sets have this size, and the greedy rule returns an independent set of maximum weight [54]. The second form of the capacity is the dimension formula $\mathrm{rank}[K_Q, K_E] = r_Q + \mathrm{rank}\, K_E - \dim(\mathrm{col}\, K_Q \cap \mathrm{col}\, K_E)$. Finally $\mathrm{rank}[K_Q, K_E] \leq \min(d, r_Q + T)$; for

fixed K_Q, equality asks a minor of $[B, K_E]$ of order $\min(d, r_Q + T)$ to be nonzero, where B is a basis of $\operatorname{col} K_Q$, and this minor is a nonzero polynomial in the entries of K_E (complete B by standard basis vectors).

(d) By Lemma B.65, $v_i^{\mathrm{rb}} = u_i + \sum_{j \prec i} \langle \psi_j, k_i \rangle u_j$, that is, $V^{\mathrm{rb}} = U(I + \Lambda)$ with $\Lambda_{ji} = \langle \psi_j, k_i \rangle$ for $j \prec i$ and $\Lambda_{ji} = 0$ otherwise. With the edits ordered by batch, Λ is strictly upper triangular, $I + \Lambda$ is invertible and $\operatorname{rank} V^{\mathrm{rb}} = \operatorname{rank} U$. For (dep), $H = dWK_E = U(\Psi^\top K_E)$ has rank T iff both factors have rank T, and then $\operatorname{rank} K_E = T$.

(e) Dropping requirements keeps a solution, and the inequality is the second claim of (b) for a largest J with M_J of full column rank. *Example.* Let $d = n = 2$, $T = 3$, one complaint key e_1, $k_1 = e_2$, $k_2 = -e_2, k_3 = e_1 + e_2, \psi_1 = e_2, \psi_2 = -e_2, \psi_3 = e_1$ (all own coordinates are 1), $u_1 = e_1$, $u_2 = e_2$, $u_3 = e_1 + e_2$, with targets as in Lemma A.4. Then $dWe_1 = u_3 \neq 0$ and $dWe_2 = u_1 - u_2$, and for (dep) the requirements are $D'e_1 = 0$, $D'e_2 = u_1 - u_2$ for edit 1 and for edit 2, and $D'e_2 = 2e_1$ for edit 3. The preservable sets are $\emptyset$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$: the maximal ones, $\{1, 2\}$ and $\{3\}$, differ in size, so they are not the bases of a matroid; greedy with the largest weight on edit 3 keeps one edit; and $\mathrm{cap} = 2 > 1 = \operatorname{rank}((I - P_Q)K_E)$. *Hardness.* Let the complaint have one trace key k_q and put $X := D' - dW$. Then D' certifies and preserves J for (dep) iff $Xk_i = 0$ for $i \in J$ and $Xk_q = -dWk_q$: the edits outside J are a feasible sacrifice set for the groups $\{k_i\}$ and the target $-dWk_q$ (Sec. A.7). Conversely, vectors $k_i \neq 0$ and k_q, not all k_i orthogonal to k_q, are the own keys and a complaint key of a log with $dWk_q \neq 0$: take $\psi_i := k_i$ and $n = 1$, $u_i := \langle k_i, k_q \rangle$, and targets as in Lemma A.4. □

Corollary A.16 (How many complaints, and three special cases). *Let $\mathcal{Q}^-$ be a finite set of wrong-answer complaints and H a matrix of required values.*

(i) All edits survive. *Under* (ND)*, some linear repair certifies $\mathcal{Q}^-$ and preserves all T edits iff* $\operatorname{rank}[K_Q, K_E] = r_Q + T$*; this needs $r_Q \leq d - T$. Let $\mathcal{Q}^-$ consist of N complaints with L trace keys each, and let the keys be in general position:*

$$\operatorname{rank} K_Q = \min(d, NL), \qquad \operatorname{rank}[K_Q, K_E] = \min(d, NL + T).$$

Then, under (ND)*,* $\mathrm{cap}(\mathcal{Q}^-) = \min\big(T, \max(0, d - NL)\big)$*. So all edits survive iff $N \leq \lfloor (d - T)/L \rfloor$, each further trace key costs one edit, and none survives once $NL \geq d$. Without* (ND)*,* $\operatorname{rank} H \leq d - r_Q$ *is still necessary.*

(ii) Full rank forces total rollback. *Every certifying update satisfies $D' = D'(I - P_Q)$ and $\operatorname{rank} D' \leq d - r_Q$. If $r_Q = d$, the only linear repair that certifies $\mathcal{Q}^-$ is W_0 itself, whatever the method: a rollback, a compensation, a re-edit or a rebuild from W_0.*

(iii) Rollback. *The rollback of R certifies $\mathcal{Q}^-$ iff $\sum_{i \notin R} u_i \psi_i^\top K_Q = 0$. If the write vectors of the kept edits are linearly independent, this holds iff $\psi_i^\top K_Q = 0$ for every $i \notin R$: a certifying rollback deletes every edit that has a nonzero coordinate at some trace key of some complaint, and it keeps at most $|\{i : P_Q \psi_i = 0\}|$ edits, against* $\mathrm{cap}(\mathcal{Q}^-)$ *for the best linear repair. If, in addition, U has full column rank, a kept edit i is preserved in the sense* (dep) *iff $\langle \psi_j, k_i \rangle = 0$ for every deleted j.*

Proof. (i) Theorem A.15(c) with $J = [T]$; $\operatorname{rank}[K_Q, K_E] \leq d$. Under general position $\mathrm{cap}(\mathcal{Q}^-) = \min(d, NL + T) - \min(d, NL)$, which is the stated value. The last claim is Theorem A.15(b). (ii) $D'K_Q = 0$ iff $D'P_Q = 0$; then the row space of D' lies in $(\operatorname{col} K_Q)^\perp$. (iii) The update of the rollback is $U_{[T]\setminus R} \Psi_{[T]\setminus R}^\top$, and a matrix $U_{[T]\setminus R}$ of full column rank can be cancelled on the left. Then $\psi_i^\top K_Q = 0$ iff $\psi_i \perp \operatorname{col} K_Q$ iff $P_Q \psi_i = 0$. The change at k_i against the deployed log is $-\sum_{j \in R} \langle \psi_j, k_i \rangle u_j$. □

Generically no read-out vector is orthogonal to the trace keys of a complaint, so a certifying rollback is the total one (Theorem 3.1(a) of the paper; Propositions B.42 and B.69), while the best linear repair keeps $\min(T, d - r_Q)$ edits: the gap between rollback and compensation under the certificate criterion. Both end at W_0 when $r_Q = d$.

Relation to Theorem A.12. Both statements are corollaries of one classical fact, the solvability of a linear matrix equation $XA = B$ [56, Thm. 2]. Its best-known case is the exact recall of stored pairs by a matrix when the keys are linearly independent [39], which gives the second claim of Theorem A.15(b). Lemma A.9 is the fact for $X[K_{\mathrm{prot}}, K_q] = [0, V]$, and Theorem A.15(a) is the same lemma with the roles exchanged: the complaint keys are the keys on which the update vanishes, and the own keys carry the values. Three things differ. *What is counted.* Theorem A.12(a) bounds the protected dimension by $d - a$, where $a = \operatorname{rank} V$ is the rank of the *damage* at the keys of one complaint; since $\operatorname{rank} dW \leq T$, this bound is never below $d - \min(r_Q, T)$. Theorem A.15 subtracts the rank r_Q of the *trace keys* of all complaints, which is the number that grows with the complaints, and it bounds the preserved edits, not the protected keys. *For which repairs.* The count holds for every weight $W_0 + D'$, not only for a correction of minimum norm appended to W_E. *What is hard.* With one value per edit and (ND) the preservable sets form a matroid and the optimum is a rank: the number of edits given up is $\dim(\operatorname{col} K_Q \cap \operatorname{col} K_E)$ for linearly independent own keys, at most one per complaint dimension. Without (ND), or with protection groups of several keys, the optimum is a minimum sacrifice, which is NP-hard in the worst case (Proposition A.26); its hard instances there have one complaint key and linearly dependent own keys; independent own keys and independent write vectors do not remove the hardness (Remark A.17). Where both theorems apply (as many complaint keys as the rank of the damage, general position), they give the same count, $T - \min(T, d - r_Q)$ edits given up. The price of the stronger count is the hypothesis (ND) on the values, which Theorem A.12 does not need. A protected span enters as follows: if a repair has to keep the deployed outputs on protected keys K_{prot} of rank p (the answer keys of the edits, say), then $\operatorname{rank}(dWK_{\mathrm{prot}}) \leq \operatorname{rank}\big((I - P_Q)K_{\mathrm{prot}}\big) \leq d - r_Q$ is necessary by the argument of (b), and $\operatorname{rank}[K_Q, K_{\mathrm{prot}}] = r_Q + p$ is sufficient; for U of full column rank, $\Psi^\top K_{\mathrm{prot}} \neq 0$ and Lebesgue-almost every K_Q with a given number A of columns (so that $r_Q = \min(d, A)$), the repair exists iff $r_Q \leq d - p$: if $r_Q + p > d$, the intersection $\operatorname{col} K_Q \cap \operatorname{col} K_{\mathrm{prot}}$ is almost surely a subspace of $\operatorname{col} K_{\mathrm{prot}}$ of positive dimension that is not contained in $\operatorname{null} \Psi^\top$. The rank conditions of general position alone do not give the "only if": for $d = 3$, $K_{\mathrm{prot}} = [e_1, e_2]$, $K_Q = [e_1, e_3]$ and one edit with $\psi_1 = k_1 = e_2$, $u_1 \neq 0$, they hold with $r_Q = 2 > d - p = 1$, and the repair exists, because $dWe_1 = 0$. So with N complaints of L trace keys the generic count is $N \leq \lfloor (d - p)/L \rfloor$.

Remark A.17 (Both halves of (ND) *are needed for deployed outputs).* Let $k_1, \ldots, k_T \in \mathbb{R}^m$ be nonzero and $k_q \neq 0$ in their span: an instance of Min-Sacrifice with groups of one key, whose feasible kept sets

are the $J \subseteq [T]$ with $k_q \notin \operatorname{span}\{k_i : i \in J\}$. There is an edit log with $d = n = T$, own keys $\pm e_1, \dots, \pm e_T$, write vectors $e_1, \dots, e_T$ and a set of complaint keys such that the sets that are preservable in the sense (dep) are exactly these J. So with linearly independent own keys and linearly independent write vectors, but a singular matrix of coordinates, the preservable sets need not form a matroid, and computing the capacity is NP-hard. *Proof.* Let $K = [k_1, \dots, k_T]$, $S := \{x \in \mathbb{R}^T : Kx \in \operatorname{span}\{k_q\}\}$, and let $g \in \mathbb{R}^T$ represent a linear functional with $Kx = \langle g, x\rangle\, k_q$ for $x \in S$ (it exists on S, as $k_q \neq 0$, and extends to $\mathbb{R}^T$). For every i some $\psi_i \in g + S^\perp$ has a nonzero i-th entry: otherwise $e_i \in S$ and $\langle g, e_i\rangle = 0$, that is, $k_i = 0$. Let the own key of edit i be e_i or $-e_i$, with the sign of that entry, so that $s_i > 0$, let $u_i := e_i$, with targets as in Lemma A.4, and let the complaint keys be a basis of S. Then $dW = \Psi^\top$ and $dWx = \langle g, x\rangle\, (1, \dots, 1)^\top$ for $x \in S$. By Theorem A.15(a), J is preservable iff dW vanishes on $S \cap \operatorname{span}\{e_i : i \in J\}$, that is, iff no x supported on J has $Kx = \varsigma k_q$ with $\varsigma \neq 0$: iff $k_q \notin \operatorname{span}\{k_i : i \in J\}$. The rest is Proposition A.26.

A.8.1 What a certifying linear repair leaves behind. The statements of this part are identities for an arbitrary matrix of keys in the place of K_Q. They are stated for the trace keys, for which they describe a certifying repair; for a matrix of selected keys (the answer positions of the complaints, say) they describe a repair that restores these keys and certifies nothing (Lemma A.14).

Proposition A.18 (The closest certifying repair). *Let K_Q hold the trace keys of a finite set $\mathcal{Q}^-$ of wrong-answer complaints.*

(a) No protected keys. *Every update D' with $D'K_Q = 0$ satisfies*

$$\|D' - dW\|_F^2 = \|D' - dW(I - P_Q)\|_F^2 + \|dWP_Q\|_F^2.$$

So the certifying update closest to dW is

$$dW(I - P_Q) = \textstyle\sum_{i\in[T]} u_i\big((I - P_Q)\psi_i\big)^\top,$$

the log with every read-out vector projected off the span of the complaint keys. It is the deployed weight plus the minimum compensation $X^\star = -dWP_Q$ for $K_{\text{prot}} = \emptyset$.

(b) Protected keys. *Let K_{prot} be protected keys, $K_\perp = (I - P_{\text{prot}})K_Q$, and $\Pi := K_Q K_\perp^+ \in \mathbb{R}^{d\times d}$. Updates with $D'K_Q = 0$ and $D'K_{\text{prot}} = dWK_{\text{prot}}$ exist iff dW vanishes on $\operatorname{col}K_Q \cap \operatorname{col}K_{\text{prot}}$, and the one closest to dW is then $dW + X^\star = dW(I - \Pi)$. The matrix Π is idempotent. Its kernel is $(\operatorname{col}K_\perp)^\perp$, which is $\operatorname{col}K_{\text{prot}} \oplus (\operatorname{col}[K_Q, K_{\text{prot}}])^\perp$, and its range lies in $\operatorname{col}K_Q$ and has the dimension $\operatorname{rank}K_\perp = r_Q - \dim(\operatorname{col}K_Q \cap \operatorname{col}K_{\text{prot}})$. If $\operatorname{col}K_Q \cap \operatorname{col}K_{\text{prot}} = \{0\}$, it is the oblique projector onto $\operatorname{col}K_Q$ along $\operatorname{col}K_{\text{prot}} \oplus (\operatorname{col}[K_Q, K_{\text{prot}}])^\perp$, it depends on K_Q only through $\operatorname{col}K_Q$, and $\|\Pi\|_2 = 1/\sin\vartheta$ for the smallest principal angle ϑ between $\operatorname{col}K_Q$ and $\operatorname{col}K_{\text{prot}}$; $\Pi = P_Q$ iff the two spans are orthogonal.*

(c) Three orthogonal pieces. *Let $P_\perp$ be the orthogonal projector onto $\operatorname{col}K_\perp$. In the situation of* (b),

$$dW(I - \Pi) = \underbrace{dWP_{\text{prot}}}_{\textit{kept}} \underbrace{- dWP_{\text{prot}}K_Q K_\perp^+}_{\textit{created}} + \underbrace{dW(I - P_{\text{prot}} - P_\perp)}_{\textit{untouched}},$$

and the rows of the three terms lie in the mutually orthogonal spaces $\operatorname{col}K_{\text{prot}}$, $\operatorname{col}K_\perp$ and $(\operatorname{col}[K_Q, K_{\text{prot}}])^\perp$; hence $\|dW(I - \Pi)\|_F^2 = \|dWP_{\text{prot}}\|_F^2 + \|dWP_{\text{prot}}K_Q K_\perp^+\|_F^2 + \|dW(I - P_{\text{prot}} - P_\perp)\|_F^2$. The sum $F := dWP_{\text{prot}}(I - \Pi)$ of the first two terms is forced*: every update with $D'K_Q = 0$ and $D'K_{\text{prot}} = dWK_{\text{prot}}$ equals F on $\operatorname{col}[K_Q, K_{\text{prot}}]$, and these updates are $F + Y(I - P_{\text{prot}} - P_\perp)$ with Y arbitrary; F is the one of least norm.*

(d) What remains at a key. *At a key k the repaired module of* (a) *differs from the pre-edit one by $dW(I - P_Q)k = \sum_i \langle (I - P_Q)\psi_i, k\rangle\, u_i = dWk - dWP_Q k$, and that of* (b) *by $dW(I - \Pi)k$. The factor $(I - P_Q)k$ vanishes exactly for $k \in \operatorname{col}K_Q$, and $(I - \Pi)k$ exactly on the range of Π. The remaining update $dW(I - P_Q)$ itself vanishes on the larger space $\operatorname{col}K_Q \oplus \big((\operatorname{col}K_Q)^\perp \cap \operatorname{null}dW\big)$, of dimension $d - \operatorname{rank}(dW(I - P_Q)) \geq d - T$, and nowhere else. Against the deployed model the repair changes the module output at k by $-dWP_Q k$ (respectively $-dW\Pi k$): if $X^\star \neq 0$, it moves every key outside a proper subspace, and unless dWP_Q vanishes on $\operatorname{null}dW$ these include keys with $dWk = 0$, which the log did not damage.*

Proof. (a) $D'K_Q = 0$ iff $D'P_Q = 0$ iff $D' = D'(I - P_Q)$. Then $D' - dW = (D' - dW(I - P_Q)) - dWP_Q$, the rows of the first matrix are orthogonal to $\operatorname{col}K_Q$ and those of the second lie in it. With $K_{\text{prot}} = \emptyset$, $K_\perp = K_Q$ and $X^\star = -dWK_Q K_Q^+ = -dWP_Q$ (Lemma A.9; the best approximate solution of a linear matrix equation is due to Penrose [57]).

(b) Put $X := D' - dW$; the conditions read $XK_{\text{prot}} = 0$ and $XK_Q = -dWK_Q$, and Lemma A.9(a),(c) gives the criterion and $X^\star = -dWK_Q K_\perp^+ = -dW\Pi$. Since $K_\perp^+ = (K_\perp^\top K_\perp)^+ K_\perp^\top$ and $K_\perp^\top P_{\text{prot}} = 0$, we have $K_\perp^+ P_{\text{prot}} = 0$ and $K_\perp^+ K_Q = K_\perp^+(P_{\text{prot}}K_Q + K_\perp) = K_\perp^+ K_\perp$, so $\Pi^2 = K_Q K_\perp^+ K_\perp K_\perp^+ = \Pi$. *Kernel.* $\operatorname{null}K_Q \subseteq \operatorname{null}K_\perp$, and the range of $K_\perp^+$ is $(\operatorname{null}K_\perp)^\perp$, which meets $\operatorname{null}K_Q$ only in 0; so $\Pi x = 0$ iff $K_\perp^+ x = 0$ iff $x \perp \operatorname{col}K_\perp$. *Range.* It is $K_Q(\operatorname{null}K_\perp)^\perp \subseteq \operatorname{col}K_Q$, of dimension $\operatorname{rank}\Pi = \operatorname{rank}K_\perp$, and $\operatorname{rank}K_\perp = r_Q - \dim(\operatorname{col}K_Q \cap \operatorname{col}K_{\text{prot}})$ because $K_Q c \in \operatorname{col}K_{\text{prot}}$ iff $K_\perp c = 0$. If the intersection is trivial, the range has dimension r_Q and is $\operatorname{col}K_Q$. Then let the columns of Q be an orthonormal basis of $\operatorname{col}K_Q$ and $K_Q = QB$ with B of full row rank; $(I - P_{\text{prot}})Q$ has full column rank, so $K_\perp^+ = B^+((I - P_{\text{prot}})Q)^+$ and $\Pi = Q((I - P_{\text{prot}})Q)^+$, which depends on $\operatorname{col}K_Q$ only (another basis QO, O orthogonal, gives the same matrix). Its norm is the inverse of the smallest singular value of $(I - P_{\text{prot}})Q$, and the singular values of $(I - P_{\text{prot}})Q$ are the sines of the principal angles between $\operatorname{col}K_Q$ and $\operatorname{col}K_{\text{prot}}$. An idempotent matrix is an orthogonal projector iff its kernel is orthogonal to its range, here iff $\operatorname{col}K_{\text{prot}} \perp \operatorname{col}K_Q$.

(c) Since $K_Q = P_{\text{prot}}K_Q + K_\perp$ and $K_\perp K_\perp^+ = P_\perp$, we have $dW\Pi = dWP_{\text{prot}}K_Q K_\perp^+ + dWP_\perp$, which gives the display. The rows of $K_\perp^+$ lie in $\operatorname{col}K_\perp$, and $\operatorname{col}[K_Q, K_{\text{prot}}]$ is the orthogonal direct sum of $\operatorname{col}K_{\text{prot}}$ and $\operatorname{col}K_\perp$. The matrix $F = dW(I - \Pi)(P_{\text{prot}} + P_\perp)$ is the restriction of $dW(I - \Pi)$ to $\operatorname{col}[K_Q, K_{\text{prot}}]$, on which all updates with $D'K_Q = 0$ and $D'K_{\text{prot}} = dWK_{\text{prot}}$ agree; its rows lie in this space, so it has least norm (Lemma A.23).

(d) The two formulas are (a) and (b). $dW(I - P_Q)k = 0$ iff $(I - P_Q)k \in \operatorname{null}dW$, that is, iff $k \in \operatorname{col}K_Q \oplus ((\operatorname{col}K_Q)^\perp \cap \operatorname{null}dW)$; the dimension is rank–nullity, and $\operatorname{rank}(dW(I - P_Q)) \leq \operatorname{rank}dW \leq T$. The change against W_E is $X^\star k$, and $X^\star \neq 0$ vanishes on a proper subspace only (Lemma A.10(a)). □

Part (d) corrects a tempting reading: the remaining update does not vanish "exactly on the complaint span". What vanishes exactly there is the component of the key that the repair removes; the edits act on the rest of every key. With U of full column rank, the repaired

weight of (a) is the log with the read-out vectors $(I-P_Q)\psi_i$, edit i acts at k iff $\langle (I-P_Q)\psi_i, k\rangle \neq 0$, and the next proposition counts these edits.

Proposition A.19 (Unreported keys). *Let k be a key that was not reported, and $x := r_Q/(d-p)$ with $p := \operatorname{rank} K_{\text{prot}}$ ($p = 0$ without protected keys).*

(a) Deterministic. *$dW(I-P_Q) = 0$ iff $\operatorname{col}(dW^\top) \subseteq \operatorname{col} K_Q$; for U of full column rank, iff every read-out vector lies in $\operatorname{col} K_Q$, which needs $r_Q \geq \operatorname{rank}\Psi$. The key k is restored, $dW(I-P_Q)k = 0$, iff $dWk = dWP_Q k$: iff its damage is the damage of its projection onto the reported span. For U of full column rank, the number of edits that act at k under the repaired weight, $|\{i : \langle (I-P_Q)\psi_i, k\rangle \neq 0\}|$, equals $|\{i : \psi_i \notin \operatorname{col} K_Q\}|$ for every k outside finitely many hyperplanes, each of which contains $\operatorname{col} K_Q$; it is T for Lebesgue-almost all Ψ if $r_Q < d$. Moreover*

$$\|dW(I-P_Q)k\|_2 \leq \|dW\|_2 \operatorname{dist}(k, \operatorname{col} K_Q),$$
$$\|dW(I-P_Q)k\|_2 \geq \varsigma_{\min}(U)\, \|\Psi^\top (I-P_Q)k\|_2,$$

with $\varsigma_{\min}(U)$ the least singular value of U.

(b) In expectation. *Let k be random with second moment $\Sigma = \mathbb{E}\, kk^\top$. For every update D', $\mathbb{E}\|D'k\|_2^2 = \operatorname{tr}(D'\Sigma D'^\top)$. For $\Sigma = \sigma^2 I$ the repair of Proposition A.18*(a) *leaves the fraction*

$$\frac{\mathbb{E}\|dW(I-P_Q)k\|_2^2}{\mathbb{E}\|dWk\|_2^2} = 1 - \frac{\|dWP_Q\|_F^2}{\|dW\|_F^2}$$

of the damage energy, which is between $\sum_{j>r_Q}\varsigma_j^2/\sum_j \varsigma_j^2$ and $1-\sum_{j>d-r_Q}\varsigma_j^2/\sum_j\varsigma_j^2$ for the singular values $\varsigma_1 \geq \cdots \geq \varsigma_d \geq 0$ of dW; both bounds are attained, so for $\operatorname{rank} dW \leq r_Q \leq d - \operatorname{rank} dW$ the fraction can be anything between 0 and 1: it depends on where the complaint span lies, not on its rank. If $\operatorname{col} K_Q$ is uniformly distributed among the subspaces of dimension r_Q (as for complaint keys with independent standard normal entries), the expected fraction is $1 - r_Q/d$, for every log. With protected keys, $\Sigma = \sigma^2 I$, $a := \|dWP_{\text{prot}}\|_F^2/\|dW\|_F^2$ and $\operatorname{col} K_Q$ uniformly distributed with $r_Q \leq d-p-2$, the repair of Proposition A.18(b) *leaves in expectation the fraction*

$$a\,\frac{d-p-1}{d-p-r_Q-1} + (1-a)\Big(1-\frac{r_Q}{d-p}\Big)$$
$$\geq\ \frac{a}{1-x} + (1-a)(1-x)\ \geq\ 2\sqrt{a(1-a)},$$

the first term being the share of the forced part F; the fraction is at least 1 for every r_Q if $a \geq \frac{1}{2}$, and it exceeds 1 as soon as $x > (1-2a)/(1-a)$.

(c) A floor that no complaint lowers. *Let $\Sigma \succ 0$, let the columns of B be an orthonormal basis of $\operatorname{col} K_{\text{prot}}$, and let D' be* any *update with $D'K_{\text{prot}} = dWK_{\text{prot}}$, certifying or not. Then*

$$\mathbb{E}\|D'k\|_2^2\ \geq\ C^\star_{\text{prot}} := \operatorname{tr}\big(dWB\,(B^\top\Sigma^{-1}B)^{-1}B^\top dW^\top\big),$$

with equality iff $D' = dWB(B^\top\Sigma^{-1}B)^{-1}B^\top\Sigma^{-1}$; for $\Sigma = \sigma^2 I$ the floor is $\sigma^2\|dWP_{\text{prot}}\|_F^2$, attained by $D' = dWP_{\text{prot}}$ alone. Requiring $D'K_Q = 0$ in addition can only raise the minimum; for $\Sigma = \sigma^2 I$ it becomes $\sigma^2\|F\|_F^2 = \sigma^2(\|dWP_{\text{prot}}\|_F^2 + \|dWP_{\text{prot}}K_Q K_\perp^+\|_F^2)$, attained by $D' = F$.

Proof. (a) $dW(I-P_Q) = 0$ iff the rows of dW lie in $\operatorname{col} K_Q$, and $\operatorname{col}(dW^\top) = \Psi\operatorname{col}(U^\top)$, which is $\operatorname{col}\Psi$ for U of full column rank. The second claim is Proposition A.18(d). For the count, let $\psi_i' := (I-P_Q)\psi_i$; it is zero iff $\psi_i \in \operatorname{col} K_Q$, and otherwise $\langle \psi_i', k\rangle = 0$ defines a hyperplane, which contains $\operatorname{col} K_Q$; a fixed proper subspace contains ψ_i for a null set of Ψ only. For the bounds, $dW(I-P_Q)k = U\,\Psi^\top(I-P_Q)k$ and $\|(I-P_Q)k\|_2 = \operatorname{dist}(k, \operatorname{col} K_Q)$.

(b) $\mathbb{E}\|D'k\|_2^2 = \mathbb{E}\operatorname{tr}(D'kk^\top D'^\top)$. For $\Sigma = \sigma^2 I$, $\|dW(I-P_Q)\|_F^2 = \|dW\|_F^2 - \|dWP_Q\|_F^2$, and $\|dWP_Q\|_F^2 = \operatorname{tr}(P_Q\, dW^\top dW)$ is between the sums of the r_Q smallest and of the r_Q largest eigenvalues of $dW^\top dW$ (Ky Fan). If the distribution of $\operatorname{col} K_Q$ is invariant under orthogonal maps, $\mathbb{E}P_Q$ commutes with all of them, so $\mathbb{E}P_Q = (r_Q/d)I$ by its trace, and $\mathbb{E}\|dWP_Q\|_F^2 = (r_Q/d)\|dW\|_F^2$. *Protected keys.* By Proposition A.18(b), Π depends on $\operatorname{col} K_Q$ only (the intersection with $\operatorname{col} K_{\text{prot}}$ is trivial almost surely, as $r_Q + p < d$), so we may take K_Q with independent standard normal entries. Let B and $B_\perp$ be orthonormal bases of $\operatorname{col} K_{\text{prot}}$ and of its orthogonal complement; $G_1 := B^\top K_Q$ and $G_2 := B_\perp^\top K_Q$ are independent Gaussian matrices, $K_\perp = B_\perp G_2$ and $K_\perp^+ = G_2^+ B_\perp^\top$. In the three pieces of Proposition A.18(c), the created one is $-dWBG_1G_2^+B_\perp^\top$, and $\mathbb{E}[G_1SG_1^\top \mid G_2] = \operatorname{tr}(S)\,I$ for $S := G_2^+(G_2^+)^\top = (G_2^\top G_2)^{-1}$, whose expected trace is $r_Q/(d-p-r_Q-1)$ (mean of an inverse Wishart matrix; this needs $d-p-r_Q-1 > 0$). So the created piece has expected energy $\|dWP_{\text{prot}}\|_F^2 r_Q/(d-p-r_Q-1)$. In the untouched piece $P_\perp$ is the projector onto a uniformly distributed subspace of dimension r_Q of the $(d-p)$-dimensional space $(\operatorname{col} K_{\text{prot}})^\perp$, so its expected energy is $\|dW(I-P_{\text{prot}})\|_F^2(1-r_Q/(d-p))$. Adding the kept piece gives the formula. The first inequality is $(N-1)/(N-r_Q-1) \geq N/(N-r_Q)$ for $N = d-p$, the second is the inequality of the arithmetic and the geometric mean, and $a/(1-x)+(1-a)(1-x) > 1$ iff $x > (1-2a)/(1-a)$, for $x \in (0,1)$.

(c) Lemma A.10(c) with the keys B and the target $V = dWB$ (there $XK_q = V$ is the only hypothesis on X). For $\Sigma = \sigma^2 I$, $C^\star = \sigma^2\|dWB\|_F^2$ and $X_\Sigma = dWBB^\top$. A minimum over a smaller set is not smaller; with $D'K_Q = 0$ the admissible updates are $F + Y(I - P_{\text{prot}} - P_\perp)$, whose least Frobenius norm is that of F (Proposition A.18(c)). □

Reading. Compensation is interpolation. The correction $X^\star = -dWK_Q K_\perp^+$ is, row by row, minus the least-norm interpolant of the damage map $k \mapsto dWk$ from the reported keys, fitted on the directions orthogonal to the protected span; what it leaves at an unreported key is the prediction error of that interpolant. For Gaussian keys the formula of (b) is the noise-free case of the risk of the least-norm interpolant with r_Q samples that uses $d-p$ of the d features [11, Thm. 1] (the cited theorem assumes label noise; the formula is proved above): a bias that falls like $1-x$ and a variance that grows like $1/(1-x)$ and diverges at the interpolation threshold $r_Q = d-p$, where the protected part of the damage plays the role of the noise. So under protection the damage on unreported keys is U-shaped in the number of reported keys: it has the floor $2\sqrt{a(1-a)}$, and (for $a < \frac{1}{2}$ and large $d-p$) beyond $x = 1-\sqrt{a/(1-a)}$ every further reported key *raises* it. *The count of the target statement.* Without protection, removing a fraction f of the damage energy on fresh keys takes $r_Q \approx fd$ dimensions (isotropic keys, generic span), and by Corollary A.16(i) all edits survive only if $r_Q \leq d-T$; so $f \leq 1 - T/d$. For $T \ll d$ this count does not bind, and rank is not what removes damage: with no complaint at all, the update dWP_{prot} keeps the protected outputs and leaves the share a on isotropic keys.

What binds is the forced part: a repair that keeps the edits keeps dWP_{prot}, and each certified dimension inflates it; in the generic model the forced part F, the best certifying repair for isotropic keys, has the expected share $a(d-p-1)/(d-p-r_Q-1)$, which grows with r_Q and is unbounded as r_Q approaches $d-p$. *Complaints are constraints, not information.* By (c), the linear repair that is best for unreported keys among those that keep the protected outputs does not depend on the complaints, and certifying them can only make it worse. Reported keys help unreported ones only as a sample of the key distribution Σ, which a tolerance uses and a certificate does not (Proposition A.20). *What these statements do not say.* They are module-level statements: about the update at a key, not about answers. The laws with a uniformly distributed span and isotropic keys are models, and real keys are neither (the measurements are confronted in Sec. A.8.4). And they concern the keys in K_Q: if K_Q holds all trace keys, the repair certifies the reported complaints; if it holds selected positions only, it uses less rank, certifies nothing, and leaves at every other position t of a reported complaint the update $dW(I-\Pi)k_t$, like at an unreported key.

A.8.2 Tolerance instead of the certificate.

Proposition A.20 (Approximate certification with a ridge). *Let $\mu > 0$, $G := K_Q K_Q^\top$ and $S_\mu := G(G+\mu I)^{-1} = K_Q(K_Q^\top K_Q + \mu I)^{-1}K_Q^\top$, and let $K_Q = \sum_{j\le r_Q} \varsigma_j\, g_j h_j^\top$ be a singular value decomposition, $\varsigma_1 \ge \cdots \ge \varsigma_{r_Q} > 0$.*

(a) Solution. *The unique minimiser of $\|X\|_F^2 + \frac{1}{\mu}\|(dW+X)K_Q\|_F^2$ over $X \in \mathbb{R}^{n\times d}$ is $X_\mu = -dWS_\mu$. The repaired update is $dW(I-S_\mu) = \sum_i u_i\big((I-S_\mu)\psi_i\big)^\top$ with $I-S_\mu = (I-P_Q) + \sum_j \frac{\mu}{\varsigma_j^2+\mu}\, g_j g_j^\top$: the log with its read-out vectors shrunk by the factor $\mu/(\varsigma_j^2+\mu)$ along g_j and unchanged orthogonally to $\operatorname{col} K_Q$. Among the corrections that leave at most its remaining damage on the complaint keys, X_μ has the least Frobenius norm.*

(b) Remaining damage on the complaints. *It is*

$$(dW+X_\mu)K_Q = \mu\, dWK_Q(K_Q^\top K_Q+\mu I)^{-1} = \sum_j \frac{\mu\,\varsigma_j}{\varsigma_j^2+\mu}\,(dWg_j)\,h_j^\top.$$

Let $dWK_Q \neq 0$. The relative remaining damage, *the ratio of $\|(dW+X_\mu)K_Q\|_F$ to $\|dWK_Q\|_F$, is non-decreasing in μ and lies between $\mu/(\varsigma_1^2+\mu)$ and $\mu/(\varsigma_{r_Q}^2+\mu)$. For $\mu\to 0$ it tends to 0 and $X_\mu \to -dWP_Q$, the certifying repair of Proposition A.18*(a)*; for $\mu\to\infty$ it tends to 1 and $X_\mu \to 0$. So a tolerance $c\in(0,1)$ on this ratio is met if $\mu \le \varsigma_{r_Q}^2\, c/(1-c)$ and only if $\mu \le \varsigma_1^2\, c/(1-c)$.*

(c) Edits and unreported keys. *At a key k the correction changes the module output by $-dWS_\mu k$, and $\|S_\mu k\|_2 \le \|P_Q k\|_2$ is non-increasing in μ. So an edit whose own key is orthogonal to $\operatorname{col} K_Q$ is preserved for every μ, and the output at any other own key moves by at most $\|dW\|_2\|S_\mu k_i\|_2$ (the change itself need not be monotone in μ). For keys with second moment $\sigma^2 I$ the remaining damage energy is $\sigma^2\|dW(I-S_\mu)\|_F^2 = \sigma^2\big(\|dW(I-P_Q)\|_F^2 + \sum_j(\frac{\mu}{\varsigma_j^2+\mu})^2\|dWg_j\|_2^2\big)$, never below that of the certifying repair.*

(d) Edits preserved exactly. *Let $J\subseteq[T]$, P_J the orthogonal projector onto $\operatorname{col} K_J$ and $K_\perp := (I-P_J)K_Q$. Among the corrections with $XK_J = 0$, the minimiser of the objective of* (a) *is $X_{J,\mu} = -dWK_Q(K_\perp^\top K_\perp+\mu I)^{-1}K_\perp^\top$, with the remaining damage $\mu\, dWK_Q(K_\perp^\top K_\perp+\mu I)^{-1}$, which tends to dWK_QP_0 as $\mu\to 0$, where P_0 is the orthogonal projector onto $\operatorname{null} K_\perp$. The least remaining damage of an update that preserves J in the sense* (dep) *is*

$$\delta_J := \min\{\|D'K_Q\|_F : D'K_J = dWK_J\} = \|dWK_QP_0\|_F;$$

$\delta_J = 0$ iff J is preservable, and $\delta_J \le \delta_{J'}$ for $J\subseteq J'$. If K_Q has A columns with independent standard normal entries and $\operatorname{rank} K_J = |J|$, then

$$\mathbb{E}\,\delta_J^2 = (A+|J|-d)_+\,\|dWP_J\|_F^2, \qquad \mathbb{E}\|dWK_Q\|_F^2 = A\,\|dW\|_F^2.$$

(e) No count survives a tolerance in the worst case. *For all $T\ge 1$ and $c\in(0,1)$ there are an edit log with $d=2T$, complaint keys that span $\mathbb{R}^d$, in any number and each with nonzero damage, and a linear repair that preserves all T edits in the sense* (dep) *and leaves at every complaint key at most c times its damage; the only linear repair that certifies these complaints is W_0, which preserves no edit.*

(f) Reported keys as a sample. *Let the columns of B be an orthonormal basis of $\operatorname{col} K_{\text{prot}}$. Among the updates with $D'K_{\text{prot}} = dWK_{\text{prot}}$, the minimiser of $\|D'K_Q\|_F^2 + \mu\|D'\|_F^2$ is*

$$dWB\big(B^\top(G+\mu I)^{-1}B\big)^{-1}B^\top(G+\mu I)^{-1},$$

the minimiser of Proposition A.19(c) *for the second moment $(G+\mu I)/A$.*

Proof. (a) The objective $\operatorname{tr}(XX^\top) + \frac{1}{\mu}\operatorname{tr}((dW+X)G(dW+X)^\top)$ is strictly convex, and its gradient $2X + \frac{2}{\mu}(dW+X)G$ vanishes iff $X(G+\mu I) = -dWG$. The two forms of S_μ agree because $K_Q(K_Q^\top K_Q+\mu I) = (K_QK_Q^\top+\mu I)K_Q$, and $S_\mu = \sum_j \frac{\varsigma_j^2}{\varsigma_j^2+\mu} g_jg_j^\top$. If X' left at most the remaining damage of X_μ and were shorter, its objective would be smaller. (b) $(I-S_\mu)K_Q = \sum_j \frac{\mu}{\varsigma_j^2+\mu}\varsigma_j g_j h_j^\top$, and the h_j are orthonormal, so $\|(dW+X_\mu)K_Q\|_F^2 = \sum_j(\frac{\mu}{\varsigma_j^2+\mu})^2\varsigma_j^2\|dWg_j\|_2^2$, against $\|dWK_Q\|_F^2 = \sum_j \varsigma_j^2\|dWg_j\|_2^2$. Every factor $\mu/(\varsigma_j^2+\mu)$ increases with μ from 0 to 1 and lies between those of ς_1 and ς_{r_Q}; and $\mu/(\varsigma^2+\mu)\le c$ iff $\mu\le\varsigma^2 c/(1-c)$. (c) $\|S_\mu k\|_2^2 = \sum_j(\frac{\varsigma_j^2}{\varsigma_j^2+\mu})^2\langle g_j,k\rangle^2 \le \sum_j\langle g_j,k\rangle^2 = \|P_Qk\|_2^2$, and every term decreases in μ. The energy is $\sigma^2\|dW(I-S_\mu)\|_F^2$, and the two parts of $I-S_\mu$ in (a) have orthogonal ranges. For the remark in brackets take $K_Q = [e_1, 3e_2]$, $dWe_1 = -dWe_2 \neq 0$ and $k = e_1+e_2$: $dWS_\mu k = (\frac{1}{1+\mu}-\frac{9}{9+\mu})dWe_1$ vanishes for $\mu\to 0$ and for $\mu\to\infty$ and not in between. (d) A correction with $XK_J = 0$ satisfies $X = X(I-P_J)$ and $XK_Q = XK_\perp$. The function $\|X\|_F^2 + \frac{1}{\mu}\|dWK_Q + XK_\perp\|_F^2$ of an unconstrained X is minimised, as in (a), by $X = -dWK_Q K_\perp^\top(K_\perp K_\perp^\top+\mu I)^{-1} = -dWK_Q(K_\perp^\top K_\perp+\mu I)^{-1}K_\perp^\top$, whose rows lie in $\operatorname{col} K_\perp \perp \operatorname{col} K_J$; so it is the constrained minimiser. Its remaining damage is $dWK_Q\big(I-(K_\perp^\top K_\perp+\mu I)^{-1}K_\perp^\top K_\perp\big) = \mu\, dWK_Q(K_\perp^\top K_\perp+\mu I)^{-1}$, and $\mu(K_\perp^\top K_\perp+\mu I)^{-1}\to P_0$. For every X with $XK_J = 0$, $\|dWK_Q + XK_\perp\|_F \ge \|(dWK_Q+XK_\perp)P_0\|_F = \|dWK_QP_0\|_F$, with equality for $X = -dWK_QK_\perp^+$; this is δ_J, with $D' = dW+X$. $\delta_J = 0$ iff $dWK_Qc = 0$ whenever $K_\perp c = 0$, that is, whenever $K_Qc \in \operatorname{col} K_J$: iff dW vanishes on $\operatorname{col} K_Q \cap \operatorname{col} K_J$ (Theorem A.15(a)). Monotonicity: fewer updates are admitted. *Gaussian keys.* Let B and $B_\perp$ be orthonormal bases of $\operatorname{col} K_J$ and of its orthogonal complement; $G_1 := B^\top K_Q$ and $G_2 := B_\perp^\top K_Q$ are independent Gaussian matrices and $\operatorname{null} K_\perp = \operatorname{null} G_2$, of dimension

$(A + |J| - d)_+$ almost surely. Then $dWK_Q P_0 = dWBG_1P_0$, and $\mathbb{E}[G_1P_0G_1^\top \mid G_2] = \mathrm{tr}(P_0)\, I$. (e) Let $t := 1/c - 1$, $n = T$, $u_i = e_i$, $k_i = e_i$ and $\psi_i = e_i + t\, e_{T+i}$ for $i \in [T]$, in $\mathbb{R}^{2T}$, with targets as in Lemma A.4; $\langle \psi_j, k_i \rangle = \mathbf{1}[i = j]$. The complaint keys are the vectors e_{T+j} and $e_j + e_{T+j}$, $j \in [T]$, and any number of nonzero multiples of them; their damages are multiples of $dWe_{T+j} = t\, u_j$ and of $dW(e_j + e_{T+j}) = (1 + t)\, u_j$. The update $D' := \sum_i u_i k_i^\top$ has $D'k_i = u_i = dWk_i$, $D'e_{T+j} = 0$ and $D'(e_j + e_{T+j}) = u_j = c\, dW(e_j + e_{T+j})$. The complaint keys span $\mathbb{R}^{2T}$, so W_0 is the only certifying linear repair (Corollary A.16(ii)), and $dWk_i \neq 0$. (f) The objective is $\mathrm{tr}(D'(G + \mu I)D'^\top)$; apply Lemma A.10(c) with $\Sigma := G + \mu I$, the keys B and the target dWB. □

Reading. The trade-off between the complaints and the edits is continuous under a tolerance and a threshold under the certificate. Along the ridge path the complaints are exact only in the limit $\mu \to 0$. With the edits of J kept exactly, generic complaints are exact up to the capacity $A \leq d - |J|$ of Theorem A.15, and each complaint key beyond it leaves, in mean square, the whole energy $\|dWP_J\|_F^2$ of the preserved part. With $a_J := \|dWP_J\|_F^2 / \|dW\|_F^2$, a root-mean-square tolerance c is therefore met by any number of generic complaint keys if $a_J \leq c^2$, and by $A \leq (d - |J|)/(1 - c^2/a_J)$ of them otherwise: a tolerance inflates the capacity by the factor $1/(1 - c^2/a_J)$. By (e) this is a property of generic keys, not a bound: the count of Theorem A.15 belongs to the certificate. Part (f) is the sense in which reported complaints help unreported ones under a tolerance: as a sample of the key distribution. If $(G + \mu I)/A$ equals the second moment Σ of the unreported keys, the update of (f) attains the floor of Proposition A.19(c), and no linear repair that keeps the protected outputs does better, with any number of complaints.

A.8.3 Rescoping, for contrast.

Corollary A.21 (Threshold certification has no dimension count). *Assume* (H1)–(H4). *Let $\tau^0 \in \Theta^T$ be the thresholds of a scoped log, Q^- any finite set of wrong-answer complaints and $\tau^\star$ its least repair (Theorem B.8).*

(a) $\tau^\star$ certifies Q^-, whatever the number of complaints and the rank r_Q of their trace keys, also for $r_Q = d$; every certified complaint is answered with $y_0(q)$.

(b) Edit i still fires on its own request under thresholds $\tau \geq \tau^0$ iff $\zeta_i(q_i) \geq \tau_i$. The edits that keep their own request under some raise-only repair that certifies Q^- are those of

$$J^\star := \{ i : \ \zeta_i(q_i) \geq \tau_i^0 \ \text{ and } \ \zeta_i(q_i) > \zeta_i(q) \text{ for every } q \in Q^- \},$$

and the least repair keeps all of them at once. Each edit is decided by its own coordinates: there is no interaction between the edits and no bound in terms of d.

(c) For all $d \geq 1$, $T \geq 1$ and $N \geq d$ there are an edit log and N wrong-answer complaints whose trace keys span $\mathbb{R}^d$ such that the least repair of the deployed scoping certifies all complaints and every edit keeps its own request, while the only linear repair that certifies them is W_0, which preserves no edit.

Proof. (a) Theorem B.8(a) and Proposition B.1(a); neither has a hypothesis on the keys. (b) The first claim is Def. 2.3. By Theorem B.8(a), thresholds $\tau \geq \tau^0$ certify Q^- iff $\tau \geq \tau^\star$, where $\tau_i^\star$ is the larger of τ_i^0 and $\max_{q \in Q^-} \zeta_i(q)^+$. So if edit i fires on q_i under such thresholds, then $\zeta_i(q_i) \geq \tau_i^\star$, which says $i \in J^\star$, because $\zeta \geq a^+$ means $\zeta > a$. Under $\tau^\star$ every edit of $J^\star$ fires on its own request. (c) Let $\psi_i = k_i = e_1$ and $u_i \neq 0$ for every i, all edits in one batch, with targets as in Lemma A.4, and let the request q_i have the single eligible position with the key k_i, so that $\zeta_i(q_i) = 1$. Choose the u_i with $\sum_j u_j \neq 0$. Let each complaint have one position; the keys are $\frac{1}{2}e_1$, $\frac{1}{2}e_1 + e_j$ for $j = 2, \ldots, d$ and, for $N > d$, further multiples of e_1 by factors in $(0, \frac{1}{2}]$. At such a key k the damage $dWk = \langle e_1, k \rangle \sum_j u_j$ is nonzero and every coordinate $\zeta_i(k) = \langle e_1, k \rangle$ lies in $(0, \frac{1}{2}]$, so from $\tau^0 \equiv -\infty$ the least repair is $\tau_i^\star = (\frac{1}{2})^+$, and $J^\star = [T]$. The keys span $\mathbb{R}^d$; by Corollary A.16(ii) only W_0 certifies, and $dWk_i = \sum_j u_j \neq 0$ and $v_i^{\mathrm{rb}} = u_i \neq 0$. □

Firing is coverage, not preservation. Applied to the deployed scoping, the least repair leaves edit i firing on its own request iff $\zeta_i(q_i) > \zeta_i(q)$ for every complaint q, and the module output at k_i is then the deployed one iff the contributions of the edits that are silent at k_i sum to zero; for linearly independent write vectors, iff every edit with a nonzero coordinate at k_i still fires there. Indeed the module returns $W_0k_i + \sum_j \langle \psi_j, k_i \rangle u_j$, the sum over the edits that fire at k_i. Applied to the retrofitted log of Theorem B.66, the least repair leaves the module output at k_i equal to the target $z_i^\star$ iff $\zeta_i(q) < 1$ for every complaint q (for $v_i^{\mathrm{rb}} \neq 0$): the thresholds satisfy $1 - \gamma_j^{\mathrm{sep}} < \tau_j \leq 1$, so at an eligible position with the key k_i no edit $j \neq i$ fires, before or after, as $\zeta_j(k_i) \leq 1 - \gamma_j^{\mathrm{sep}} < \tau_j$; and edit i, with $\zeta_i(k_i) = 1$, fires iff $(\max_q \zeta_i(q))^+ \leq 1$. If it fires, the module returns $W_0k_i + v_i^{\mathrm{rb}} = z_i^\star$, and otherwise W_0k_i. "Keeps its own request" is coverage in the sense of Def. 2.5: the edit still fires there. What the module then returns depends on the values: with the logged values it is the pre-edit output plus the contributions of the edits that still fire at the key, and with the rebased values and thresholds in the window of Theorem B.66 it is the target $z_i^\star$, for every number of edits. So under the certificate criterion the three semantics are ordered by what a complaint costs: a rollback certifies only by deleting every edit with a nonzero coordinate on the trace (generically all T, Corollary A.16(iii)); the best linear repair pays one edit per trace key beyond $d - T$ (Corollary A.16(i)); a threshold repair pays an edit only if a complaint coordinate reaches the own coordinate of that edit. The price of rescoping is elsewhere: it is not a weight, it needs the gate and W_0 at inference time, and its statement about unreported queries is statistical (Theorem 3.6), not a restoration.

A.8.4 Confrontation with the measurements.

No experiment was run for this part: the numbers are read from the stored records of the linear repair streams and of the checks on the two development logs, Q-A100-r3 and L-A100-r3 ($T = 100$, $d = 11{,}008$, output dimension 2,048 and 4,096).

What the joint compensation restores and protects. Comp-B computes $X = VK_\perp^+$ with $V = -dWK_q$. Its complaint keys K_q are the keys at the *answer-predicting* positions of the reported complaints, the last prompt position and the positions of the pre-edit answer except the last, about ten keys per complaint. Its protected keys are the keys at the answer-predicting positions of the T requests, of rank 400 and 483. One variant adds the answer keys of known-good rows to the protected keys (rank 2,254 and 1,355), and one restores *every* position of the reported complaints. So Comp-B is the

Table 2: Rank consumed by reported complaints and the counts that Corollary A.16(i) then gives ($d = 11{,}008$, $T = 100$; records of the linear streams, three seeds at ten complaints, half of them image complaints). "All positions" certifies; "answer positions" is the joint compensation of the paper, which restores and does not certify. Predicted columns extrapolate the rank per complaint measured at ten complaints and assume (ND) and general position; no record exists beyond ten complaints at all positions.

	all positions, 10 complaints			predicted number of complaints			answer positions: rank	
Log	keys	rank	rank per complaint	all T edits kept	with K_{ans} kept	only W_0 certifies from	10 complaints	all reported
Q-A100-r3	3,219–4,684	3,117–4,583	312–458	23–34	23–34	25–36	99	2,841
L-A100-r3	3,111–3,125	3,093–3,105	309–310	35	33–34	36	57	1,973

Table 3: The generic law of Proposition A.19(b) against the recorded module-level ratio $\|(W' - W_0)K\|_F / \|dWK\|_F$ at the answer keys K of held-out rows (median over the held-out complaints; 1: unrepaired, 0: pre-edit), seed 0. Share: norm share of a held-out key in the row space of the correction. D1: held-out rows whose pre-edit answer is not reproduced; the stored records use the held-out pool with the 25 queries that duplicate Repair queries (hence 54.7 and 61.5 unrepaired, against 54.2 and 62.3 on the disjoint pool of the paper).

				share		ratio		D1 (%)	
Log	Repair (restored keys; complaints)	protected rank	rank	generic	measured	generic law	measured	unrepaired	repaired
Q-A100-r3	answer positions; 10	400	99	0.09	0.35	1.00	0.94	54.7	51.4
Q-A100-r3	answer positions; 100	400	986	0.30	0.61	0.97	0.82	54.7	49.4
Q-A100-r3	answer positions; all	400	2,841	0.51	0.72	0.92	0.77	54.7	47.3
Q-A100-r3	all positions; 10	400	4,583	0.65	0.68	0.88	0.92	54.7	60.2
Q-A100-r3	answer positions, larger protection; 10	2,254	99	–	–	1.00–1.01	1.01	54.7	53.7
Q-A100-r3	answer positions, larger protection; 100	2,254	986	–	–	0.96–1.06	1.05	54.7	55.9
Q-A100-r3	answer positions, larger protection; all	2,254	2,841	–	–	0.90–1.22	1.23	54.7	66.9
L-A100-r3	answer positions; 10	483	57	0.07	0.20	1.00	0.98	61.5	61.5
L-A100-r3	answer positions; 100	483	611	0.24	0.44	0.98	0.95	61.5	59.1
L-A100-r3	answer positions; all	483	1,973	0.42	0.61	0.93	0.91	61.5	58.5
L-A100-r3	all positions; 10	483	3,099	0.53	0.57	0.90	0.96	61.5	63.4
L-A100-r3	answer positions, larger protection; 10	1,355	57	–	–	1.00–1.00	1.00	61.5	63.6
L-A100-r3	answer positions, larger protection; 100	1,355	611	–	–	0.98–1.03	1.07	61.5	66.4
L-A100-r3	answer positions, larger protection; all	1,355	1,973	–	–	0.93–1.12	1.08	61.5	66.8

repair of Proposition A.18(b) for selected keys. It does not certify (Lemma A.14): with the module output restored at every reported answer key it resolves 168/297 and 164/305 of the reported complaints, whereas the variant that restores all positions resolves 29/30 and 30/30 at ten complaints. This variant is a numerical fit of the restoration constraints: a pseudo-inverse with a cut of small singular values, whose correction is added to the module output in the precision of the model. Exact restoration at every position would resolve every complaint (Lemma A.14(a)); the complaint that stays unresolved is restored up to round-off at each of its positions, and the cause was not established in the record.

Rank. Table 2. At the answer positions a complaint consumes 9.6 and 6.5 dimensions on average, 2,841 and 1,973 for all reported complaints. At all positions a text complaint has 37–48 and 21–30 keys and an image complaint 299–1,054 and 593–602, most of them image tokens. The figure that the automated review passes of the project quote, rank 4,583 of 11,008 for ten complaints, is that of the all-position variant on Q-A100-r3 with the first seed: five text and five image complaints, 4,684 keys, of which 101 are dropped as dependent or inside the protected span.

What Theorem A.15 predicts. (ND) holds for target values on both logs, in floating point: the output dimension exceeds T and the write vectors are linearly independent (the least eigenvalue of the Gram matrix of the normalised contributions is 0.31 and 0.41). With one value per edit, Corollary A.16(i) allows $r_Q \le d - T$. At the rank per complaint measured at ten complaints, and under general position beyond them, all T edits survive up to 23–34 complaints on Q-A100-r3, depending on the seed, and 35 on L-A100-r3; with the answer keys of the edits kept, 23–34 and 33–34; and from 25–36 and 36 complaints on, $r_Q = d$ and the only certifying weight is W_0. The count is one of image complaints: text complaints alone, 270 and 411; image complaints alone, 13 and 18. If an edit counts as kept only when its whole request sequence is protected, no linear repair keeps all edits even for one complaint key (at most 37 and 18 of them, the measured capacity of exact protection). Beside *certified* complaints no generic count applies to whole request sequences: the trace of a complaint and a request sequence that begin with the same tokens have the same leading keys (Lemma A.3), a certifying update vanishes on them, and it therefore protects no request sequence that contains such a key k with $dWk \neq 0$ (Corollary A.22). (ND) for deployed outputs, the nonsingularity of the matrix of coordinates, was not checked on the logs. The thresholds themselves were not measured in the records: they stop at ten complaints at all positions, where the repair is feasible, as predicted. The prediction was written down before its measurement, which is experiment E5 (f) of the registered protocol; the paper reports the measured values next to it. The count does not bind a repair that restores answer positions only: about 1,108 and 1,627 complaints would fit. Under a tolerance the count is softer (Proposition A.20(d)): the span of the T own keys carries the share $a_J = 0.15$ and 0.12 of the energy of dW, so for generic complaint keys a root-mean-square tolerance of 0.1 raises

the capacity by the factor 1.07 and 1.09 only, and a tolerance of 0.39 and 0.35 is met by any number of them. Real keys are not generic, as the next paragraph shows.

Corollary A.22 (Shared keys under the certificate). *Let edit i have a protection group, a set of keys with matrix K_i (the keys of its request, say). If some linear repair certifies Q^- and keeps the deployed module output at every key of K_i, then dW vanishes on* $\operatorname{col} K_Q \cap \operatorname{col} K_i$. *In particular, if a key k of the group is a trace key of a complaint and $dWk \neq 0$, no certifying linear repair protects the group of edit i, whatever d.*

Proof. Theorem A.15(a) with the keys of K_i in the place of K_J and the values dWK_i; for a shared key, $D'k = 0$ and $D'k = dWk$. □

The plateau. With all reported complaints, the share of held-out rows whose pre-edit answer is not reproduced falls from 54.7 to 47.3 per cent and from 61.5 to 58.5 per cent, and the change of the module output at held-out answer keys stays at 0.77 and 0.91 of the unrepaired one (Table 3). *It is not explained by the missing certificate.* Restoring all positions of ten complaints certifies and resolves them, consumes rank 4,583 and 3,099, and leaves the ratio at 0.92 and 0.96 and the held-out rate *above* the unrepaired one (56.7 and 65.0 per cent on average over the seeds). *It is explained in kind by Propositions A.18 and A.19.* The correction is the interpolant of the damage from the reported keys. It acts on a held-out key through the component of the key in the reported span, complaint or not: on Q-A100-r3 the ratio is the same for held-out complaints and non-complaints (area under the curve 0.53; 0.60 on L-A100-r3), and the repair fixes 34 and 22 per cent of the held-out complaints while it breaks 29 and 30 per cent of the held-out non-complaints. Held-out rows with the keys of a reported complaint are restored to round-off. *In size the generic law is rough.* It gives the ratios 1.00, 0.98, 0.93 for 10, 100 and all complaints on L-A100-r3, against the measured 0.98, 0.95, 0.91; on Q-A100-r3 it gives 1.00, 0.97, 0.92 against 0.94, 0.82, 0.77. Real keys are not generic: a held-out answer key has the share 0.72 of its norm in the span of the reported answer keys, against 0.51 for a generic span of that rank, and the all-position span of ten complaints, of larger rank, covers it less and restores less. *The forced part shows in the direction of the effect.* With the larger protected span the ratio rises to 1.23 and 1.08 and the held-out rate to 66.9 and 66.8 per cent, above the unrepaired log: the reported keys keep only the share 0.44 and 0.69 of their norm outside the protected span, against 0.89 and 0.94 for generic keys, and the created part of Proposition A.18(c) grows as the complaint keys approach the protected span. The size is not predicted, because the energy share of the larger protected span is not in the records. The kept part dWP_{prot} alone leaves the ratio 1.02 and 0.87 at held-out complaints: held-out answer keys have the share 0.51 and 0.43 of their norm inside the protected span, against 0.19 and 0.21 for generic keys. Likewise the update that keeps the deployed output on the span of the own keys and nothing else, which leaves the ratio 0.39 and 0.35 on isotropic keys and is the best edit-preserving linear repair for them (Proposition A.19(c)), leaves 1.21 and 0.94 at the held-out complaints: the isotropic statements describe the Frobenius norm of the update, not the keys of real queries.

What the theory explains, and what it does not. It explains why reported complaints do not repair unreported ones under a linear repair, which keys the correction moves, why certifying more positions costs rank without helping other queries, and why protecting more makes unreported rows worse. It predicts the number of complaints at which a certifying linear repair must give up edits; this prediction is not tested by the records. It does not predict answers: every statement is about the module output. It does not give a bound that matches the plateau: for isotropic keys the floor of Proposition A.19(c) is the ratio 0.41 and 0.38, far below the measured 0.77 and 0.91; the floor for the real keys needs their second moment, which is not in the records (that of a few thousand held-out keys in $d = 11{,}008$ dimensions is singular and gives the floor 0; the key statistics of the editor would be a candidate), and the ratio 1.02 of the kept part is a property of one particular repair, not a lower bound.

A.9 Choosing what to sacrifice: price of protection and budgets

Budgeted-Sacrifice asks for a smallest sacrifice set R whose minimum compensation satisfies $\|X^\star(R)\|_F^2 \le \beta$, for a given cost budget β.

Membership. Min-Sacrifice and Budgeted-Sacrifice are in NP. The certificate is R. Feasibility is the consistency of a rational linear system, decided by a rank computation with polynomially bounded intermediate numbers [19]. For the budget, the pseudo-inverse of a rational matrix is computed from a full-rank factorisation, and the cost is a rational of polynomial size.

Lemma A.23 (Minimum-norm solutions). *A solution X of a solvable system $XM = B$ has minimum Frobenius norm iff its rows lie in* $\operatorname{col} M$.

Proof. The solution BM^+ has its rows in $\operatorname{col} M$, and two solutions differ by a matrix whose rows are orthogonal to $\operatorname{col} M$; so the solution with rows in $\operatorname{col} M$ is unique, and every other solution is longer by Pythagoras. □

Proposition A.24 (Price of one more protected key; standard). *Let X_1 be the minimum-norm solution of $XK_1 = 0$, $XK_q = V$, of cost $f_1^2 := \|X_1\|_F^2$. Add one protected key k and let $k_{\text{new}} := (I - P_{[K_1,K_q]})k$ be its component outside* $\operatorname{col}[K_1, K_q]$. *If $k_{\text{new}} \neq 0$, the enlarged system is solvable, its minimum-norm solution is $X_2 = X_1 - (X_1k)\, k_{\text{new}}^\top / \|k_{\text{new}}\|_2^2$, and*

$$f_2^2 = f_1^2 + \|X_1k\|_2^2 / \|k_{\text{new}}\|_2^2.$$

If $k_{\text{new}} = 0$, the enlarged system is solvable iff $X_1k = 0$, and then $X_2 = X_1$. Summed along any order of insertion, the increments give the cost under the whole protection set minus the cost without protection: the curve of cost against protection is computed in one pass.

Proof. Since $k_{\text{new}} \perp \operatorname{col}[K_1, K_q]$, $X_2K_1 = 0$ and $X_2K_q = V$; since $\langle k_{\text{new}}, k\rangle = \|k_{\text{new}}\|_2^2$, $X_2k = 0$. The rows of X_1 lie in $\operatorname{col}[K_1, K_q]$ (Lemma A.23), so those of X_2 lie in $\operatorname{col}[K_1, k, K_q]$ and X_2 has minimum norm; the cross term of $\|X_2\|_F^2$ vanishes because $X_1k_{\text{new}} = 0$. If $k_{\text{new}} = 0$, every solution of the first system agrees with X_1 on $\operatorname{col}[K_1, K_q] \ni k$. Telescoping is immediate. □

Proposition A.25 (Dictionary). *Assume one complaint key k_q, a target $v_q \neq 0$, and groups of one key each, $k_1, \dots, k_T$.*

(1) R is feasible iff some y has $\langle y, k_i\rangle = 0$ for all $i \notin R$ and $\langle y, k_q\rangle \neq 0$.
(2) For a system of linear equations $\langle g_i, x\rangle = \varrho_i$, $i \in [T]$, in $d-1$ unknowns, put $k_i = (g_i; -\varrho_i)$ and $k_q = e_d$. Then R is feasible iff the equations with $i \notin R$ are consistent. The map preserves every number and the optimum, and Min-Sacrifice *is the problem of minimising the number of unsatisfied equations subject to one mandatory equation, C Min ULR$^{=,=}$ of [6, Sec. 3.2].*

Proof. (1) By Lemma A.9, R is feasible iff k_q is not in the span of the kept keys, that is, iff some y is orthogonal to the kept keys and not to k_q. (2) Write $y = (x; \pi)$ with $\pi = \langle y, k_q\rangle \neq 0$ and scale to $\pi = 1$; then $\langle y, k_i\rangle = \langle g_i, x\rangle - \varrho_i$. In general, the mandatory equation is $\langle y, k_q\rangle = 1$ and the others are $\langle y, k_i\rangle = 0$. □

Proposition A.26 (Placing Min-Sacrifice; corollaries of known results). *With groups of one key and one complaint key: (a)* Min-Sacrifice *is strongly NP-complete with keys in $\{-1, 0, 1\}$; (b) unless* P = NP *it has no constant-factor approximation. Both persist for every fixed number of complaint keys and for prescribed group dimensions.*

Proof. (a) The reduction in the proof of [5, Thm. 1] maps an X3C instance $(U = [3h]; C_1, \ldots, C_H)$ to a system with the equations $\sum_{j:u\in C_j} x_j = 1$ for every u, $x_j = 1$ for every j, and $x_j = 0$ for every j, all coefficients in $\{-1, 0, 1\}$; $3h + H$ of them can be satisfied together iff an exact cover exists, and since $x_j = 1$ and $x_j = 0$ conflict in pairs, the minimum number of unsatisfied equations is at most H iff a cover exists. Proposition A.25(2) turns this system, number for number, into a Min-Sacrifice instance with keys in $\{-1, 0, 1\}^{H+1}$, $v_q = 1$ and budget H. (b) By [6, Sec. 3.1], minimising the number of unsatisfied equations cannot be approximated within any constant unless P = NP, a result of Arora, Babai, Stern and Sweedyk [8]. The map of Proposition A.25(2) preserves the optimum, and a feasible R maps back to an x that violates at most $|R|$ equations. *Extensions.* For A complaint keys, append $A - 1$ fresh coordinates and set $V = I_A$. For prescribed dimensions r_i, give group i $r_i - 1$ private fresh coordinates, which change no feasibility. □

Lemma A.27 (Cover gadget). *Given an* X3C *instance $(U = [3h]; C_1, \ldots, C_H)$, use the keys $E_j = e_j$ ($j \in [H]$) and $h + 1$ copies of $F_u = (\chi_u; -1) \in \mathbb{Q}^{H+1}$ for each $u \in U$, where $(\chi_u)_j = \mathbf{1}[u \in C_j]$, and $k_q = e_{H+1}$. The feasible R with $|R| \le h$ are exactly the sets $\{E_j : j \in J\}$ for exact covers $\{C_j\}_{j\in J}$, and for each such R, $U_R^\perp$ is spanned by $(\chi(J); 1)$.*

Proof. For an exact cover, $y = (\chi(J); 1)$ is orthogonal to every kept key and $\langle y, k_q\rangle = 1$. Conversely, let $y = (x; 1)$ witness feasibility (Prop. A.25(1)). Every F_u keeps a copy, so $\sum_{j\ni u} x_j = 1$ for every u. With $J = \{j : E_j \in R\}$, the kept E_j force $\operatorname{supp} x \subseteq J$ and $|J| \le h$, so at most h 3-sets cover $3h$ elements; they are disjoint and $|J| = h = |R|$. Every $y = (x; \pi)$ orthogonal to the kept keys has $x_j = 0$ off J and $x_j = \pi$ on J. □

Theorem A.28 (Choosing what to sacrifice under a cost budget). *Consider* Budgeted-Sacrifice *with one complaint key and groups of one key.*
(a) It is strongly NP-complete with $\beta = h + 1$ and keys in $\{-1, 0, 1\}$.
(b) It stays NP-hard when the protected keys and the complaint key are linearly independent, so that general position holds and the optimum without a budget is 0.
(c) It stays NP-hard for soft (ridge-penalised) protection.

Proof. (a) Use Lemma A.27 with the bound h on $|R|$ and $v_q = 1$. If a cover exists, Lemma A.9(b) gives $\|X^\star\|_F = 1/\operatorname{dist}(k_q, U_R)$, and the distance is $|\langle k_q, \hat{y}\rangle| = 1/\sqrt{h+1}$ for the unit vector $\hat{y}$ that spans $U_R^\perp$; so the cost is exactly $h + 1 = \beta$. If no cover exists, no R with $|R| \le h$ is feasible.

(b) Lift the keys to $\tilde{k}_i = (k_i; \kappa e_i)$ and $\tilde{k}_q = (k_q; 0)$ in dimension $H + 1 + T$, with $\kappa := 1/\big(3^{\lceil h/2\rceil}(3H+1)(h+1)\big)$. The lifted keys, including $\tilde{k}_q$, are linearly independent, and with K_{-R} the matrix of the kept keys,

$$\operatorname{dist}^2(\tilde{k}_q, \tilde{U}_R) = \min_\pi \|k_q - K_{-R}\pi\|_2^2 + \kappa^2\|\pi\|_2^2. \qquad (\star)$$

For a cover, $(\star) \ge \operatorname{dist}^2(k_q, U_R) = 1/(h+1)$, so the cost is at most β. Any other R with $|R| \le h$ is infeasible in the unlifted instance, so $k_q \in U_R$ and $(\star) \le \kappa^2\|\pi\|_2^2$ for every exact representation $K_{-R}\pi = k_q$. Let $J = \{j : E_j \in R\}$ and let M_J hold the incidence vectors of $\{C_j\}_{j\in J}$; the kept system is inconsistent iff $\mathbf{1} \notin \operatorname{col} M_J$. Let P be the orthogonal projector onto $(\operatorname{col} M_J)^\perp$ and $g := -P\mathbf{1}/\|P\mathbf{1}\|_2^2 \in \mathbb{R}^U$; place the coefficient g_u on one kept copy of each F_u and the coefficient $-\sum_{u\in C_j} g_u$ on each kept E_j. This represents e_{H+1}, because $g \perp \operatorname{col} M_J$ and $\langle g, \mathbf{1}\rangle = -1$. Moreover $\|g\|_2^2 = 1/\operatorname{dist}^2(\mathbf{1}, \operatorname{col} M_J) \le 3^h$ (the distance is a ratio of two Gram determinants of integer vectors; the numerator is a positive integer, and Hadamard's inequality bounds the denominator by 3^h), and the coefficients on the E_j have squared sum at most $3H\|g\|_2^2$. So $\|\pi\|_2^2 \le (1 + 3H)3^h$, and the cost is at least $1/\big(\kappa^2(1+3H)3^h\big) \ge (3H+1)(h+1)^2 > \beta$. The numbers have polynomial bit length but exponential values, so this is ordinary NP-hardness.

(c) Write a correction of the lifted instance as $\tilde{X} = [X \mid Y]$. The kept constraints force $Ye_i = -Xk_i/\kappa$, so $\min\|\tilde{X}\|_F^2 = \min_X\{\|X\|_F^2 + \kappa^{-2}\sum_{i\notin R}\|Xk_i\|_2^2 : Xk_q = v_q\}$, which is ridge-penalised protection; (b) is therefore the same statement for soft protection. □

So hardness moves from degenerate spans to conditioning once a budget is imposed.

A.10 Numerical verification

Every reduction and closed form of this appendix was checked by scripts independent of the proofs, in exact rational arithmetic wherever the claim is combinatorial, and the round-2 results were attacked again by a checker that shares no code with the provers. *Theorem A.5.* Feasible family equal to $\mathcal{F}$ in 468 and, independently, 1488 triples (instance, c, p), with c close to 0 and to 1; family unchanged under 2808 random and 18228 adversarial perturbations of mass just below $\rho(c, c')$, designed to break a witness or to make the cheapest non-family set feasible; zero-sum-free shift: 270 and 248; bicriteria pairs: 450; doubled-singleton padding: 5 families by exact search. Part (d): 42 logs with T up to 100 and c up to 99/100, and independently 72 logs, every tie-breaking branch, with $\mathrm{OPT}_c = 2$ certified in each. *Proposition A.8.* Sandwich in 1782 cases with arbitrary signs. *Lemma A.9.* Solvability criterion and closed form:

150/150 instances. *Theorem A.12.* Capacity law modulo a shared subspace: exact in 60/60 instances, while the bound for plain general position fails in 29 of them; lower bound without general position of the complaint keys: 0 violations in 80; 60 further random instances in the second check. *Section A.9.* The reduction of [5] under Proposition A.25(2): 60/60. Cover gadget: all 34 feasible small sets are covers (34), each of cost $h + 1$. Lifted instances of Theorem A.28(b): 12/12. Prescribed dimensions: 12/12.

B RESCOPING

This appendix proves the statements of the paper on rescoping. Scoped logs, firing, traces, complaints and the repair problems are those of Defs. 2.2–2.6, and (H1)–(H4) are the assumptions of Sec. 2. Every argument of this appendix was read by a referee pass (see the statement on the use of AI tools on the first page); the file `proof_gaps.txt` next to the sources lists the places that were closed last.

B.1 Setting and conventions

Thresholds. Thresholds range over the chain Θ of Def. 2.2. For a real a and $\tau_i \in \Theta$ we write $\tau_i > a$ for $\tau_i \geq a^+$: an edit with threshold τ_i does not fire at the coordinate a. Threshold vectors are compared coordinatewise. *Rolling back* or *silencing* an edit means setting its threshold to $+\infty$. This is a mask: the list of edits and every other computation of the module stay as they are. We put $(+\infty)^+ := +\infty$ and $(-\infty)^+ := -\infty$, so that a threshold of the form $(b + \eta\,\tau^0)^+$ is $+\infty$ for the scale $\eta = +\infty$.

What "equal" means. By (H4) the coordinate, the firing decision and the response of edit i at a position are computed from the parameters of edit i and the key and role of the position alone, not from the other edits, their number, or the batch in which the position is evaluated; and where no edit fires the module returns $W_0 k_t$ itself. Under (H4), and in one fixed *execution configuration* (precision, batch shape, number of edits, threads), equality of module outputs and logits below is equality of the computed numbers. Where (H4) fails, the statements hold in exact arithmetic (Remark B.5).

Coordinates of queries. All firing sets of queries are taken on the pre-edit trace (Def. 2.3): $\zeta_i(q)$ is the largest $\zeta_i(k_t)$ over the positions of $x^0(q)$ with a role in $\mathcal{R}_i$, and $i \in \mathrm{Fire}_\tau(q)$ iff $\zeta_i(q) \geq \tau_i$. A wrong-answer complaint q is certified iff $\tau_i > \zeta_i(q)$ for every i, and a pair (i, q) is covered iff $\zeta_i(q) \geq \tau_i$. At equal coordinates a certified complaint and a covered pair of one edit exclude each other. From Sec. B.4 on, an instance consists of the finite sets Q^-, Q^+ and $Q^!$ with their weights and these coordinates; Q_i^+ and $Q_i^!$ are the pairs of edit i, we assume $Q^+ \cap Q^! = \emptyset$, and a threshold vector is *feasible* if it covers every pair of $Q^!$. An instance that is derived from another one may give a complaint the coordinate $-\infty$ under an edit; this means that the edit fires on the complaint under no threshold, so that the edit does not constrain it.

Two firing sets. The paper calls $\mathrm{Fire}_{\mathcal{S}}(q) = \mathrm{Fire}_{\mathcal{S}}(x^0(q))$ the *pre-edit firing set* of q and reserves the word *lineage* for $\mathrm{Fire}_{\mathcal{S}}(x^{\mathcal{S}}(q))$, the set of edits that fire on the trace of $\mathcal{M}[\mathcal{S}]$ itself (Remark B.6). Sections B.5, B.6 and B.9 of this report were written before this distinction was drawn: there "lineage size" and $\mathrm{wd}(q)$ denote the size $|\mathrm{Fire}_\tau(q)|$ of the pre-edit firing set, which is the size of the lineage in the cases of Remark B.6(ii),(iii).

B.2 Locality certificate and frame rule

Proposition B.1 (Locality certificate and frame rule; Prop. 3.3 of the paper). *Assume* (H1)–(H4). *Let $\mathcal{S}$ be a scoped log and q a query.*

(a) Certificate. *If* $\mathrm{Fire}_{\mathcal{S}}(x^0(q)) = \emptyset$, *then* $x^{\mathcal{S}}(q) = x^0(q)$, *the logits of* $\mathcal{M}[\mathcal{S}]$ *on q are the pre-edit logits at every position, and* $\hat{y}(q; \mathcal{S}) = y_0(q)$. *The same holds if* $\mathrm{Fire}_{\mathcal{S}}(x^{\mathcal{S}}(q)) = \emptyset$; *the two conditions are equivalent, and the second one is observed at run time.*

(b) Frame rule. *Let $\mathcal{S}'$ differ from $\mathcal{S}$ only in the parameters (detector, role set, threshold, response, value) of the edits of a set $J \subseteq [T]$, and let $x = x^{\mathcal{S}}(q)$. If $J \cap \mathrm{Fire}_{\mathcal{S}}(x) = \emptyset$ and $J \cap \mathrm{Fire}_{\mathcal{S}'}(x) = \emptyset$, then q has the same trace, the same logits and the same answer under $\mathcal{S}$ and $\mathcal{S}'$.*

(c) Raise-only repairs. *If $\mathcal{S}'$ only raises thresholds or shrinks role sets, then* $\mathrm{Fire}_{\mathcal{S}'}(x) \subseteq \mathrm{Fire}_{\mathcal{S}}(x)$ *for every sequence x, so the first hypothesis of* (b) *suffices.*

(d) Lineage. *Silencing every edit outside* $\mathrm{Fire}_{\mathcal{S}}(x^{\mathcal{S}}(q))$ *leaves* $\hat{y}(q; \mathcal{S})$ *unchanged. Silencing the edits of* $\mathrm{Fire}_{\mathcal{S}}(x^0(q))$ *certifies q, so the answer becomes* $y_0(q)$; *and a scoped log that certifies q differs from $\mathcal{S}$ in the parameters of every edit of* $\mathrm{Fire}_{\mathcal{S}}(x^0(q))$.

Proof. (b) We show by induction on the generation step that both models generate the tokens of x. Suppose that they hold the same prefix, a prefix of x. By (H1) the key and the role of every position of the prefix are the same under both logs. An edit outside J has the same parameters in $\mathcal{S}$ and $\mathcal{S}'$, so by (H4) it fires at the same positions, with the same response and value. An edit of J fires at no position of x under $\mathcal{S}$ and at none under $\mathcal{S}'$, by the two hypotheses. So at every position the same edits fire with the same parameters, and the module outputs are equal (H4). Then the logits are equal (H2), and so is the next token, or the decision to stop (H3).

(a) Let $\mathcal{S}'$ be $\mathcal{S}$ with all thresholds $+\infty$. No edit fires under $\mathcal{S}'$, so by (H4), (H2) and (H3) $\mathcal{M}[\mathcal{S}']$ computes what $\mathcal{M}[W_0]$ computes: its trace on q is $x^0(q)$ and its answer is $y_0(q)$. Part (b) is symmetric in the two logs. Applied to the pair $(\mathcal{S}', \mathcal{S})$ with $J = [T]$ and $x = x^0(q)$, it gives the first claim; applied to $(\mathcal{S}, \mathcal{S}')$ with $x = x^{\mathcal{S}}(q)$, it gives the second. If one of the two conditions holds, the two traces coincide, hence so does the other.

(c) By (H1) the keys and roles of a fixed sequence do not depend on the log, and at a fixed key and role the firing test $\mathrm{role}(t) \in \mathcal{R}_i$, $\zeta_i(k_t) \geq \tau_i$ can only turn from true to false when τ_i grows or $\mathcal{R}_i$ shrinks.

(d) For the first claim let $x := x^{\mathcal{S}}(q)$ and apply (b) with J the complement of $\mathrm{Fire}_{\mathcal{S}}(x)$: these edits fire on x neither before nor after they are silenced. For the second claim, the silenced edits do not fire on $x^0(q)$, and by (c) no other edit starts to fire; apply (a). Finally, whether edit i fires on $x^0(q)$ depends on the parameters of edit i alone, by (H1) and (H4); an edit of $\mathrm{Fire}_{\mathcal{S}}(x^0(q))$ whose parameters are kept still fires. □

By (b), an answer depends only on the edits that fire on its trace, its *lineage*. The statement is true by construction of a gated module. It has analogues in data slicing for historical what-if queries [12,

Thm. 2], whose condition is the pair of hypotheses of (b), in provable repair of networks [24, Thm. 3], and in gated editors, which return the pre-trained computation outside the scope of every edit [22, 29]. The proposition states the implication for autoregressive generation explicitly; the following example separates the trace on which a complaint has to be repaired from the trace on which the log in service is observed.

Example B.2 (A complaint is repaired on its pre-edit firing set, not on its lineage). *One module with scalar keys and $W_0 = 1$; a prompt of one position, with role* last *and key* 0*; one answer position, with role* ans*. The first answer token is* a *if the module output at the prompt is below* 1*, and* b *otherwise. The key of the answer position is* -3 *after* a *and* 3 *after* b*, and the second token is* c *if the module output there is below* 2*, and* d *otherwise. Three edits with response* 1*: e_1 fires at the role* last *where $k \geq -1$, with value* 2*; e_2 fires at the role* ans *where $k \geq 2$, with value $\frac{1}{2}$; e_3 fires at the role* ans *where $-k \geq 2$, with value* 7*. The pre-edit answer is* ac*. The scoped log answers* bd*, and e_1, e_2 fire on its trace. On the pre-edit trace e_1, e_3 fire. Silencing $\{e_1, e_2\}$ gives* ad*: the prompt output returns to* 0*, the answer key to* -3*, and there e_3 fires. Silencing $\{e_1, e_3\}$ gives* ac*, and no edit fires. So silencing the lineage of the deployed trace does not repair the complaint, and this lineage contains an edit, e_2, that need not be changed; this is why Def. 2.3 places the firing set of a query on $x^0(q)$, and why a repair needs W_0. The pre-edit firing set of the query is $\{e_1, e_3\}$ and its lineage is $\{e_1, e_2\}$: e_1 changes a generated token, and e_2 fires on the continuation.*

Example B.3 (Two edited modules: firing is not antitone). *(H1) fails if a second module is edited, because its keys depend on the log. Let two scalar modules with $W_0 = 1$ be composed, the output h of the first being the key of the second. Edit e_1 of the first module fires where $k \geq 1$, with value* -5*; edit e_2 of the second fires where $h \geq 0$, with value* 10*; both have response* 1*. At the key $k = 2$ the pre-edit output is* 2*. The scoped log gives $h = -3$ and the output* -3*, and only e_1 fires. Raising the threshold of e_1 above* 2 *gives $h = 2$, e_2 fires, and the output is* 12*. So raising a threshold released an edit: with "fires" read as "is released in the pass of the log itself", (c) and (d) fail. With firing sets taken on the calls of the pre-edit run, the certificate and the least repair extend to several edited modules: no term is released in the pass of the scoped log iff none is open on the pre-edit keys of every module (Theorem B.82(b)). Sec. B.10 has the statements, and what does not extend.*

Example B.4 (Two edited modules: the frame rule fails on stored keys). *In the setting of Example B.3 let e_2 fire where $-h \geq 1$. At $k = 2$ the scoped log gives $h = -3$, e_2 fires, and the output is* 7*. Changing only the value of e_2 from* 10 *to* 20 *changes the output to* 17*. On the stored pre-edit keys ($k = 2$ for both modules) e_2 does not fire, so a lineage computed on stored keys is disjoint from $J = \{e_2\}$ although the answer changes. The frame rule needs the trace of the log itself, as in (b).*

Remark B.5 (Masks, branches and arithmetic). (i) If a rollback removed edits from the list instead of masking them, (H4) could fail: a gate that computes all coordinates of a key with one matrix–vector product need not return the same last bits for the remaining edits when a detector is removed, when the batch changes, or when an edit is appended. Then (b) and (d) hold in exact arithmetic only, and an edit of the lineage that sits exactly at its threshold can change its firing decision, and with it the answer. (ii) The module has to branch: where no edit fires it returns the product $W_0 k_t$ that the pre-edit model computes. Adding a correction that is zero, or subtracting the terms of the edits that do not fire from $W_E k_t$, does not return these bits in floating point. (iii) (H3) is used only for the choice of the next token. Under sampling the statements hold for each realised token sequence.

Remark B.6 (Pre-edit firing set and lineage). Let $\mathcal{S}$ be a scoped log and q a query, $\mathrm{Fire}_{\mathcal{S}}(x^0(q))$ its pre-edit firing set and $\mathrm{Fire}_{\mathcal{S}}(x^{\mathcal{S}}(q))$ its lineage. (i) Silencing any edits outside the lineage leaves the answer of q unchanged (Proposition B.1(d)): the lineage is a superset of the edits whose masking can change the answer. The pre-edit firing set is what a certificate and a least repair read (Proposition B.1(a), Theorem B.8). (ii) One of the two sets is empty iff the other is (Proposition B.1(a)): if an edit fires on the pre-edit trace, let t be the first position at which one does; the log's own run has the same tokens up to t, since nothing fired before, hence the same key at t (H1), and the edit fires there too; conversely, if nothing fires on the log's own run, that run is the pre-edit run. (The position at which an edit fires is not the token generated there: the argument compares keys, not answers.) (iii) If every eligible position of every edit lies in the prompt (role sets within {pre, last}, as for the migration of the paper), the two sets are equal: both traces begin with the prompt, the keys and roles of its positions do not depend on the log (H1), and no edit fires at an answer position. (iv) Otherwise neither set need contain the other: in Example B.2 they are $\{e_1, e_3\}$ and $\{e_1, e_2\}$. An edit can change a generated token, and another edit then fires on the continuation.

Remark B.7 (Top-1 selection). The migration of the paper does not sum over the firing edits (Sec. 4). Call the edits of $\mathrm{Fire}_{\mathcal{S}}(x, t)$ the *candidates* at a position, give every edit a further parameter $b_i \in \mathbb{R}$, and let the module return $W_0 k_t + g_i(\zeta_i(k_t))\, v_i$ for the *selected* candidate i, the candidate of largest margin $\zeta_i(k_t) - b_i$ and, among these, of least index; where there is no candidate it returns $W_0 k_t$ itself. (i) *What holds.* (H4) holds for this module: whether an edit is a candidate depends on its own parameters and on the position, and the output is determined by the key and the candidates with their parameters. So Proposition B.1(a)–(d) holds as stated, with "fires" read as "is a candidate": a query without a candidate on its pre-edit trace is answered as by the pre-edit model, bit for bit, and the frame rule holds if no edit of J is a candidate on the trace under either log (a change of b_i is a change of edit i). Theorem B.8, Theorem B.14 and the bounds of Sec. B.7 are statements about firing sets and coordinates and hold with candidates in the place of firing edits; a pair (i, q) is covered if edit i is a candidate on $x^0(q)$. Theorem B.66 holds as well: under its hypotheses a solver position has one candidate. (ii) *What does not.* The selected edit is not monotone in the thresholds: if two edits are candidates at a key, raising the threshold of the selected one above its coordinate selects the other, which was not selected before. A covered pair need not be selected, since another candidate can have the larger margin. Statements about module outputs that are proved for the sum (the thresholds τ^{r} of Proposition B.72(b), the tolerance criteria, output-preserved edits) are not claimed for top-1 selection.

B.3 Least repair

In this subsection detectors and role sets are fixed and a repair changes thresholds only. Coordinates and thresholds are the values that the deployed gate computes in one fixed execution configuration; they form a finite chain, in which a^+ is the successor of a. The *scope* of edit i under τ is the set of keys k with $\zeta_i(k) \geq \tau_i$.

Theorem B.8 (Least repair; Thm. 3.4(a)–(c) of the paper). *Let $\tau^0 \in \Theta^T$ be the current thresholds and Q^- a finite set of wrong-answer complaints, and put $\tau_i^\star := \max\big(\tau_i^0,\ \max_{q \in Q^-} \zeta_i(q)^+\big)$.*

(a) Existence and uniqueness. *A threshold vector $\tau \geq \tau^0$ certifies every $q \in Q^-$ iff $\tau \geq \tau^\star$. The set of these vectors is closed under coordinatewise minimum and maximum, and $\tau^\star$ is its least element: the unique least repair of Q^-.*

(b) Order independence. *The update U_q that repairs one complaint, $U_q(\tau)_i := \max(\tau_i, \zeta_i(q)^+)$, is inflationary and idempotent, and two updates commute. Hence every sequence of updates in which each complaint of Q^- occurs at least once turns τ^0 into $\tau^\star$. The update U_q changes exactly the thresholds of the edits that fire on q under the current thresholds, and none if q is certified. Certified queries stay certified.*

(c) Scope maximality. *For every $\tau \geq \tau^0$ that certifies Q^- and every edit, the scope under τ is contained in the scope under $\tau^\star$, which is contained in the scope under τ^0. Hence $\tau^\star$ at once minimises the set of changed edits, every threshold, the set of why-not pairs that lose coverage, and the set of queries on whose trace under τ^0 a changed edit fires (under τ^0); by Prop. B.1*(b),(c) *only these queries can change their answer.*

(d) Cost. *A reported position costs $O(Td)$ arithmetic operations, and the repair keeps $O(T)$ numbers.*

Proof. (a) By Prop. B.1(a) and the definition of the firing set, q is certified under τ iff $\tau_i > \zeta_i(q)$, that is, $\tau_i \geq \zeta_i(q)^+$, for every i: a conjunction of lower bounds, one per coordinate. Together with $\tau \geq \tau^0$ these bounds read $\tau \geq \tau^\star$. The set $\{\tau : \tau \geq \tau^\star\}$ is closed under coordinatewise minimum and maximum and has the least element $\tau^\star$.

(b) The maximum is associative, commutative and idempotent. So $U_q(\tau) \geq \tau$, $U_qU_q = U_q$ and $U_qU_{q'} = U_{q'}U_q$, and a sequence of updates that contains every complaint turns τ_i^0 into the maximum of τ_i^0 and all $\zeta_i(q)^+$, which is $\tau_i^\star$. U_q changes τ_i iff $\zeta_i(q)^+ > \tau_i$, that is, iff $\zeta_i(q) \geq \tau_i$, iff edit i fires on q. Raising thresholds shrinks firing sets (Prop. B.1(c)), so a certified query stays certified.

(c) The scope is antitone in the threshold, and $\tau \geq \tau^\star \geq \tau^0$ by (a). If $\tau_i^\star \neq \tau_i^0$ then $\tau_i \geq \tau_i^\star > \tau_i^0$, so every edit changed by $\tau^\star$ is changed by τ. A pair (i, q') that is covered under τ^0 and not under $\tau^\star$ has $\zeta_i(q') < \tau_i^\star \leq \tau_i$, so it is not covered under τ. The set of queries on whose trace under τ^0 a changed edit fires grows with the set of changed edits; traces and firing sets under τ^0 do not depend on the repair.

(d) The T coordinates of a key are T inner products in $\mathbb{R}^d$; the state is the vector τ. □

Remark B.9 (Domain, execution configuration and guard band). (i) With real thresholds and the gate "fires iff $\zeta \geq \tau$", the set of certifying vectors $\{\tau_i > \zeta_i(q)\}$ is open and has no least element. This is why thresholds live in a chain with a successor. (ii) The successor of a coordinate is defined only relative to the way in which the coordinate is computed. Theorem B.8 therefore holds in one fixed execution configuration. If thresholds are computed in one configuration and served in another (another batch shape or precision, or a log that has grown by an edit), a certified complaint can fire again and an edit whose threshold equals its own coordinate can be silenced. (iii) With an additive guard band, $\tau_i^\star := \max\big(\tau_i^0, \max_q \zeta_i(q) + \varepsilon_i\big)$ with ε_i larger than the largest discrepancy of the coordinate between the configurations in use, the repair certifies the complaints in each of these configurations and is least up to ε (a band that only equals the discrepancy can leave a complaint at its threshold, where the edit fires). A multiplicative band is wrong for coordinates that are not positive. The update of Sec. 4 of the paper, $\tau_i \leftarrow \max(\tau_i, (\zeta_i(q)+\xi)^+)$ with $\xi > 0$, is this repair with $\varepsilon_i = \xi$. It is inflationary, idempotent and independent of the order, as in Theorem B.8(b), and it certifies the complaints in every configuration whose coordinates differ by less than ξ from those of the repair; it is a conservative repair and *not* the least repair of the theorem: every threshold that it changes lies above that of $\tau^\star$, and it raises the threshold of an edit that does not fire on the complaint if the coordinate lies within the band below the threshold. The implementation uses a band that is relative to the coordinate, with an absolute floor. (iv) The theorem assumes one edited module, complaints evaluated on the pre-edit trace, which needs W_0, and "least" refers to the repairs that certify the complaints.

Remark B.10 (A least repair can change other answers). A query on which a raised edit fired can change its answer. If two edits fire on a query that is not a complaint, with contributions that cancel, silencing one of them for another complaint leaves the contribution of the other. The least repair minimises the set of queries of Theorem B.8(c), outside which no answer changes; it need not minimise the set of answers that do change. In the example, a certifying vector that also raises the threshold of the second edit above its coordinate on the query silences both edits on it: if no other edit fires on the query, it is then certified and keeps its answer, which it can lose under the least repair.

Remark B.11 (Relation to query repairs). Read the selection $\zeta_i(k) \geq \tau_i$ as the query of edit i, wrong-answer complaints as negative and why-not complaints as positive examples. Ten Cate, Kolaitis and Lutz [65] define, for a pre-order of proximity to a given query, a repair as a query that fits the labelled examples and is minimal in this pre-order; a *specialisation* is in addition contained in the given query. In their containment-of-difference order one query is closer than another if the examples on which it disagrees with the given query are among those on which the other disagrees (Def. 4.1; we use the numbering of the full version, arXiv:2501.11162v2). A threshold vector $\tau \geq \tau^0$ is a specialisation of τ^0, and edit i disagrees with τ^0 on the keys with $\tau_i^0 \leq \zeta_i(k) < \tau_i$, a set that grows with τ_i. So by Theorem B.8(a),(c) the least repair $\tau^\star$ is the containment-of-difference specialisation of τ^0 for the negative examples Q^-, and it is the least one, not only a minimal one. The notion is thus theirs, and the computation is the closure algorithm for intersection-closed classes [30]. The theorem is not a special case of a result of [65]: all their results are for conjunctive queries without comparison predicates, where such specialisations may not exist, may be two,

or infinitely many (Ex. 4.9, 4.10 and 4.15, Prop. 4.11), and a threshold selection compares a real-valued feature with a constant. For threshold selections on fixed detectors and negative examples the specialisation exists, is unique and does not depend on the order of the complaints. This relates to two directions that their Sec. 6 names as future work, repairs that differ only in the selection condition of a query, and repairing with two sets of examples in sequence against repairing with their union; it is the elementary case of nested scopes (Proposition B.12) and of negative examples, and we claim nothing beyond it. QFix [69] remains the closest task: it repairs constants of logged updates from complaints by a mixed-integer program under a distance, and the certificate criterion is what makes the repair a closed form. Editors adjust thresholds heuristically [29, 63]; a least repair driven by complaints is not among them.

The least repair is also the least fitting vector in the containment-of-difference order among *all* threshold vectors, lowering allowed. Indeed, let τ certify $\mathcal{Q}^-$, so that $\tau_i \geq \max_q \zeta_i(q)^+$. If $\tau_i \geq \tau_i^0$, then $\tau_i \geq \tau_i^\star$ and the keys on which edit i disagrees with τ^0 under $\tau^\star$ are among those under τ. If $\tau_i < \tau_i^0$, then $\max_q \zeta_i(q)^+ < \tau_i^0$, so $\tau_i^\star = \tau_i^0$ and edit i does not disagree with τ^0 under $\tau^\star$ at all.

Proposition B.12 (Least repairs and union-closed scope classes; the status of Thm. 3.4 of the paper). *Let $\mathcal{F}$ be a finite family of subsets of a set $\mathcal{K}$, the scopes, with $\emptyset \in \mathcal{F}$. Every set $B \subseteq \mathcal{K}$ has a greatest member of $\mathcal{F}$ disjoint from B iff $\mathcal{F}$ is closed under union.*

Proof. ($\Leftarrow$) The union of all members disjoint from B is a member, is disjoint from B, and contains each of them. ($\Rightarrow$) For $C_1, C_2 \in \mathcal{F}$ let $B := \mathcal{K} \setminus (C_1 \cup C_2)$. The greatest member disjoint from B contains C_1 and C_2 and is contained in $C_1 \cup C_2$. □

This is a standard fact about closure systems: a class is closed under intersection iff the class of the complements is closed under union [30]. With B the keys of the complaints and $\mathcal{F}$ the scopes that a repair may choose for an edit, a greatest member disjoint from B is a least repair. On the finite chain of this subsection the consequences are the following.

(i) The scopes of thresholds on one fixed detector are nested, hence closed under union. The scopes of one threshold per role are closed under union as well: the union takes the smaller threshold at each role. So these classes have least repairs (Theorem B.8).

(ii) A role set chosen together with one threshold does not. Let an edit have all roles and a threshold τ^0, and let its scope contain a complaint key at the role last with the coordinate $c \geq \tau^0$, a key k_a at another role with a coordinate in $[\tau^0, c]$, and a key k_b at the role last with a coordinate above c. Removing the role last keeps k_a and loses k_b; raising the threshold to c^+ keeps k_b and loses k_a: two repairs with incomparable scopes. A scope of the class that contains k_a and k_b has the role last and a threshold of at most the coordinate of k_a, hence of at most c, and so contains the complaint key: no repair has a scope that contains both, and there is no least repair. Half-spaces with a free detector have no least repair either: if two half-spaces avoid a complaint key that lies on the segment between a point of the one and a point of the other, every half-space that contains both half-spaces contains the two points, hence the segment and the key, by convexity.

(iii) The tolerance criterion of Def. 2.6(i) has no least repair either: for the scalar contributions $+1, -1, +1$ and $c = \frac{1}{2}$ the sets of silenced edits that meet the criterion are $\{1\}$, $\{3\}$ and $\{1, 2, 3\}$.

(iv) Why-not complaints are upper bounds, $\tau_i \leq \zeta_i(q')$ for a pair (i, q'). Let lo_i be the largest lower bound and hi_i the smallest upper bound on τ_i. If $\mathrm{lo}_i \leq \mathrm{hi}_i$ for all i, repairing each complaint by the least move, $\tau_i \leftarrow \max(\tau_i, \cdot)$ or $\tau_i \leftarrow \min(\tau_i, \cdot)$, ends for every order in the vector with the coordinates $\min(\max(\tau_i^0, \mathrm{lo}_i), \mathrm{hi}_i)$: a threshold inside $[\mathrm{lo}_i, \mathrm{hi}_i]$ is never moved, one below lo_i only rises, up to lo_i, and one above hi_i only falls, down to hi_i. If some $\mathrm{lo}_i > \mathrm{hi}_i$, the result depends on the order: repairing first all wrong-answer and then all why-not complaints ends at hi_i, and the reverse order ends at lo_i.

Proposition B.13 (Cost of a stream of complaints). *Let the complaints $q_1, \ldots, q_N$ arrive in this order and be repaired one by one, starting from τ^0.*

(a) The h-th complaint writes $\mathrm{wd}(q_h)$ thresholds, the lineage size of q_h under the thresholds in force when it arrives; the stream costs $\sum_{h \leq N} \mathrm{wd}(q_h)$ writes.

(b) If the coordinate vectors $(\zeta_i(q_h))_{i \in [T]}$, $h \leq N$, are exchangeable, the expected number of writes to the threshold of one edit is at most $\sum_{h \leq N} 1/h$, with equality if $\tau_i^0 = -\infty$ and the coordinates of the edit are almost surely distinct. The expected number of writes is at most $T \sum_{h \leq N} 1/h$.

Proof. (a) Theorem B.8(b). (b) The h-th complaint writes the threshold of edit i only if $\zeta_i(q_h) > \zeta_i(q_{h'})$ for all $h' < h$: it sets a strict record of the coordinate. Among the first h complaints at most one holds the strict maximum of the coordinate, and by exchangeability each of them is equally likely to do so; so the probability of a record at h is at most $1/h$, with equality if the coordinates are distinct. Without an initial threshold every record is a write. □

This is the mistake bound of the closure algorithm [30, Lemma 4] for a class that is closed under intersection, applied to lower orthants. The hypotheses are that every query of the stream is repaired, that the queries are exchangeable, and that the coordinate of a query is its maximum over the eligible positions. The same argument bounds the firing of a fresh query after N repairs (Theorem B.50).

B.4 Two-sided threshold repair: Scope-Cut

Scope-Cut (Def. 2.6(iii)) asks for feasible thresholds that maximise $\Phi_\lambda(\tau) = \omega(\mathrm{Clean}(\tau)) - \lambda\,\omega(\mathrm{Unc}(\tau))$, for a given $\lambda \geq 0$; $\Phi^*(\lambda)$ denotes the maximum. A *pair* is an element of $\mathcal{Q}^+ \cup \mathcal{Q}^!$. For a threshold vector τ let $U(\tau)$ be the set of all pairs (i, q') with $\tau_i > \zeta_i(q')$, so that $\mathrm{Unc}(\tau) = U(\tau) \cap \mathcal{Q}^+$ and τ is feasible iff $U(\tau) \cap \mathcal{Q}^! = \emptyset$, and let $C(\tau) := \mathrm{Clean}(\tau) \cup U(\tau)$. Let $\mathcal{G}$ be the digraph on $\mathcal{Q}^- \cup \mathcal{Q}^+ \cup \mathcal{Q}^!$ with an arc from $q \in \mathcal{Q}^-$ to a pair (i, q') whenever $\zeta_i(q') \leq \zeta_i(q)$. A set C of nodes is *closed* if no arc leaves it, and *admissible* if it is closed and disjoint from $\mathcal{Q}^!$; its weight is $\omega_\lambda(C) := \omega(C \cap \mathcal{Q}^-) - \lambda\,\omega(C \cap \mathcal{Q}^+)$. For a set C of nodes the *canonical*

thresholds are

$$\tau_i^C := \min\{\zeta_i(q') : (i,q') \in (Q_i^+ \cup Q_i^!) \setminus C\},$$

and $+\infty$ if no pair of edit i lies outside C. Let Q^-_{free} be the set of complaints without an out-arc and Q^-_∞ the set of complaints with an arc into $Q^!$.

Theorem B.14 (Scope-Cut is a maximum-weight closure; Thm. 3.7(b) of the paper). *Let $\lambda \geq 0$.*

(a) For every τ, the set $C(\tau)$ is closed, $\omega_\lambda(C(\tau)) = \Phi_\lambda(\tau)$, and τ is feasible iff $C(\tau)$ is admissible.

(b) For every closed C: $\text{Clean}(\tau^C) \supseteq C \cap Q^-$ and $U(\tau^C) \subseteq C$, so $\Phi_\lambda(\tau^C) \geq \omega_\lambda(C)$. If C is admissible, τ^C is feasible and finite.

(c) $\Phi^*(\lambda) = \max\{\omega_\lambda(C) : C \text{ admissible}\} = \omega(Q^-) - \text{mincut}(N_\lambda)$, where the network N_λ has an arc of capacity $\omega(q)$ from the source to each $q \in Q^-$, an arc to the sink of capacity $\lambda\,\omega(i,q')$ from each pair of Q^+ and of capacity ∞ from each pair of $Q^!$, and the arcs of $\mathcal{G}$ with capacity ∞. The source side C of a minimum cut gives optimal thresholds τ^C. If $\lambda > 0$, every optimal admissible C equals $C(\tau^C)$.

(d) $q \in Q^-_\infty$ iff q is certified under no feasible τ. The minimum cut is at most $\omega(Q^-)$. Deleting Q^-_∞ and $Q^!$ from the network gives the same optimum; the thresholds are then recovered with the pairs of $Q^!$ put back.

(e) Chained network. *Order the pairs of each edit i by nondecreasing coordinate, $\pi_{i,1}, \pi_{i,2}, \ldots$. Replace the arcs of $\mathcal{G}$ by the chain arcs $\pi_{i,l} \to \pi_{i,l-1}$ and, for every complaint q and edit i, one arc from q to $\pi_{i,l}$ with the largest l such that the coordinate of $\pi_{i,l}$ is at most $\zeta_i(q)$. The network has $|Q^-| + |Q^+| + |Q^!| + 2$ nodes and at most $|Q^-|(T+1) + 2(|Q^+| + |Q^!|) - T$ arcs, and it has the same optimum and the same recovery rule.*

Proof. (a) If q is certified and $\zeta_i(q') \leq \zeta_i(q)$, then $\tau_i > \zeta_i(q) \geq \zeta_i(q')$, so (i,q') is in $U(\tau)$; pairs have no out-arcs. The weight of $C(\tau)$ is $\omega(\text{Clean}(\tau)) - \lambda\,\omega(U(\tau) \cap Q^+)$, and $C(\tau) \cap Q^! = U(\tau) \cap Q^!$.

(b) Let $q \in C$. Closedness puts every pair (i,q') with $\zeta_i(q') \leq \zeta_i(q)$ into C, so every pair of edit i outside C has a coordinate above $\zeta_i(q)$, and $\tau_i^C > \zeta_i(q)$: q is certified. A pair in $U(\tau^C)$ has a coordinate below every coordinate of a pair of its edit outside C, so it is in C. As $\lambda \geq 0$ and the weights are positive, $\Phi_\lambda(\tau^C) \geq \omega_\lambda(C)$. If C is admissible, the pairs of $Q^!$ lie outside C, so τ_i^C is at most each of their coordinates; and τ_i^C is finite because the own request of edit i is in $Q^!$.

(c) By (a) and (b) the two maxima are equal. A cut with the source side C and the source has finite capacity iff no arc of $\mathcal{G}$ leaves C and C contains no pair of $Q^!$, that is, iff C is admissible; its capacity is then $\omega(Q^- \setminus C) + \lambda\,\omega(C \cap Q^+) = \omega(Q^-) - \omega_\lambda(C)$. For an optimal C, $\omega_\lambda(C) \leq \Phi_\lambda(\tau^C) \leq \Phi^*(\lambda) = \omega_\lambda(C)$; if $\lambda > 0$, positivity of the weights turns the two inclusions of (b) into equalities.

(d) A certified $q \in Q^-_\infty$ uncovers a pair of $Q^!$, by (a). For $q \notin Q^-_\infty$ the set of q and its out-neighbours is admissible, so q is certified under its canonical thresholds. The source alone is a cut of capacity $\omega(Q^-)$. A complaint of Q^-_∞ lies in no admissible set, and no other complaint has an arc into $Q^!$; so the admissible sets and their weights are those of the instance without Q^-_∞ and $Q^!$, in which every closed set is admissible. The canonical thresholds remain minima over the pairs of $Q^+ \cup Q^!$ outside C.

(e) A set is closed in the chained digraph iff it is closed in $\mathcal{G}$ and contains, with a pair $\pi_{i,l}$, the pairs $\pi_{i,1}, \ldots, \pi_{i,l-1}$: the pairs of edit i with a coordinate of at most $\zeta_i(q)$ are exactly the $\pi_{i,l}$ up to the largest such l. Every $C(\tau)$ has this form, and (b) applies to every set that is closed in $\mathcal{G}$. The network has $|Q^-|$ source arcs, $|Q^+| + |Q^!|$ sink arcs, $|Q^+| + |Q^!| - T$ chain arcs and at most $T|Q^-|$ arcs from complaints. □

Computation and attribution. The chained network has $N_{\text{nd}} = |Q^-| + |Q^+| + |Q^!| + 2$ nodes and $N_{\text{arc}} = O(T|Q^-| + |Q^+| + |Q^!|)$ arcs, and one maximum flow on it solves Scope-Cut, for instance in time $O\big(N_{\text{nd}} N_{\text{arc}} \log(N_{\text{nd}}^2/N_{\text{arc}})\big)$ [27]; an infinite capacity can be replaced by $\omega(Q^-) + 1$. Theorem B.14 is a corollary of known results, in two ways. *As a closure problem.* Scope-Cut is a monotone integer program with a separable objective: with x_i the number of coordinate levels of the pairs of edit i that are not covered and $z_q \in \{0,1\}$ the indicator of a certified complaint, the constraints are $x_i \geq l_i(q)\,z_q$, where $l_i(q)$ is the number of levels of edit i that are at most $\zeta_i(q)$. Parts (a)–(c) are Picard's reduction of a maximum-weight closure to a minimum cut [58], which generalises the selection problem of Rhys and Balinski [9, 59], as recounted in [31, Sec. 2.2], and the chained network of (e) is the chain graph that Hochbaum and Naor gave for such programs [32], as described in [31, Sec. 2.4]. *As a database-theory result.* Scope-Cut reduces to passive weighted monotone classification, which Tao and Wang solve by one maximum flow [64, Thm. 4]:

Proposition B.15 (Scope-Cut as monotone classification). *Let the weights be nonnegative and $\lambda \geq 0$. Map a complaint $q \in Q^-$ to the point $(\zeta_1(q), \ldots, \zeta_T(q)) \in \mathbb{R}^T$ with the label 0 and the weight $\omega(q)$, and a pair (i,q') of Q^+ or of $Q^!$ to the point with $\zeta_i(q')$ in dimension i and a value below every coordinate of the instance in the other dimensions, with the label 1 and the weight $\lambda\,\omega(i,q')$ for Q^+ and a weight above $\omega(Q^-)$ for $Q^!$ (for $T = 1$ add a second dimension, with the value 1 for complaints and 0 for pairs, so that the points of the two labels are distinct; points of one label that coincide are merged, with their weights added). A classifier $h\colon \mathbb{R}^T \to \{0,1\}$ is* monotone *if $h(x) \geq h(x')$ whenever $x \geq x'$ coordinatewise. Then the least weighted error of a monotone classifier on these points is $\omega(Q^-) - \max_\tau \Phi_\lambda(\tau)$, the maximum over the feasible τ, and the flow network of [64, Thm. 4] for these points (its vertices are the points that are* contending, *a point of label 0 that dominates a point of label 1 or a point of label 1 that is dominated by one of label 0; an arc of infinite capacity from a point of label 0 to each point of label 1 that it dominates) is the network N_λ of Theorem B.14 without the complaints that have no out-arc and the pairs that have no in-arc, which lie on no path from the source to the sink and do not change the minimum cut. Points of weight zero, the pairs of Q^+ for $\lambda = 0$, are left out.*

Proof. A threshold vector gives the monotone classifier "some coordinate reaches its threshold", which is 0 exactly on the certified complaints and 1 on the covered pairs; its weighted error is *at most* the weight of the complaints that are not certified plus that of the pairs that are not covered, with equality for thresholds among the coordinates of the instance and $+\infty$ (a threshold below the value that fills the other dimensions of the pair points classifies every pair point as 1, covered or not); thresholds of this kind suffice for the optimum, as the canonical thresholds of Theorem B.14 are of

this kind. Conversely let h be monotone, and let τ_i be the least coordinate $\zeta_i(q')$ of a pair of edit i with $h = 1$ ($+\infty$ if there is none). A pair with $h = 1$ is covered under τ. A complaint q with $h(q) = 0$ is certified: if $\zeta_i(q) \geq \tau_i = \zeta_i(q')$, the point of q dominates the point of the pair (i, q'), as it is larger in every other dimension, so that $h(q) \geq 1$. So the error of τ is at most that of h. A classifier of least error labels every point of $\mathcal{Q}^!$ with 1, since labelling all points with 1 costs $\omega(\mathcal{Q}^-)$. The point of q dominates the point of (i, q') iff $\zeta_i(q) \geq \zeta_i(q')$, the arcs of the digraph $\mathcal{G}$. □

What this report adds to these results, by elementary arguments, is the reading as a repair that only raises thresholds, the must-cover pairs, the bound on the number of breakpoints of the nested frontier (Theorem B.16), the chained network with $O(T|\mathcal{Q}^-| + |\mathcal{Q}^+| + |\mathcal{Q}^!|)$ arcs, and the bound for thresholds optimised edit by edit (Proposition B.18). The experiments of the paper use the least repair and the calibration rule, not the cut.

Theorem B.16 (Parametric frontier). *Let $\mathcal{A}$ be the family of admissible sets and $O(\lambda)$ the set of maximisers of ω_λ on $\mathcal{A}$.*

(a) $\mathcal{A}$ is closed under union and intersection, ω_λ is modular, and $O(\lambda)$ is closed under union and intersection.

(b) Strict nesting. *If $\lambda > \lambda' \geq 0$, $C \in O(\lambda)$ and $C' \in O(\lambda')$, then $C \subseteq C'$.*

(c) Φ^* is convex and piecewise linear with $h \leq \min\big(|\mathcal{Q}^- \setminus (\mathcal{Q}^-_{\text{free}} \cup \mathcal{Q}^-_\infty)|, |\mathcal{Q}^+|\big)$ breakpoints $\lambda_1 > \cdots > \lambda_h > 0$. Put $\lambda_0 := \infty$ and $\lambda_{h+1} := 0$. For $\lambda_{\ell+1} < \lambda < \lambda_\ell$ the optimum C_ℓ is unique, and $C_0 \subsetneq C_1 \subsetneq \cdots \subsetneq C_h$, with $C_0 = \mathcal{Q}^-_{\text{free}}$ and C_h the set of the complaints outside $\mathcal{Q}^-_\infty$ and their out-neighbours. Each $C_\ell \setminus C_{\ell-1}$ contains a complaint and a pair of $\mathcal{Q}^+$, and $O(\lambda_\ell)$ has the least element $C_{\ell-1}$ and the greatest element C_ℓ.

(d) Monotone thresholds and firing sets. *$\tau^{C_0} \leq \cdots \leq \tau^{C_h}$ coordinatewise, and for every query q, reported or not, $\text{Fire}_{\tau^{C_0}}(q) \supseteq \cdots \supseteq \text{Fire}_{\tau^{C_h}}(q)$. For an admissible C let $L_i(C)$ be the largest coordinate under edit i of a complaint or a pair of edit i in C ($-\infty$ if there is none), and $U_i(C) := \tau^C_i$. For $\lambda_{\ell+1} < \lambda < \lambda_\ell$ a threshold vector τ is optimal iff, for all i, $\tau_i \leq U_i(C_\ell)$ and ($\tau_i > L_i(C_\ell)$ or $L_i(C_\ell) = -\infty$). Both ends are nondecreasing in ℓ.*

(e) Budgets. *Let $B_\ell := \omega(C_\ell \cap \mathcal{Q}^+)$, $\Omega_\ell := \omega(C_\ell \cap \mathcal{Q}^-)$ and $\Omega(B) := \max\{\omega(\text{Clean}(\tau)) : \tau \text{ feasible}, \omega(\text{Unc}(\tau)) \leq B\}$. Then $0 = B_0 < \cdots < B_h$, $\lambda_\ell = (\Omega_\ell - \Omega_{\ell-1})/(B_\ell - B_{\ell-1})$, and the least concave majorant of Ω on $[0, \infty)$ is the piecewise linear interpolation of the points (B_ℓ, Ω_ℓ), constant after B_h. For a feasible τ, the pair $(\omega(\text{Unc}(\tau)), \omega(\text{Clean}(\tau)))$ is optimal for some $\lambda > 0$ only if it lies on the majorant, and for all λ of an interval of positive length iff it is a vertex. So $\Omega(B_\ell) = \Omega_\ell$, and $\Omega_{\ell-1} \leq \Omega(B) \leq \Omega_{\ell-1} + \lambda_\ell(B - B_{\ell-1})$ on $[B_{\ell-1}, B_\ell)$.*

(f) Computation. *All breakpoints and the chain $C_0, \ldots, C_h$ are computed in time $O\big(N_{\text{nd}} N_{\text{arc}} \log(N^2_{\text{nd}}/N_{\text{arc}})\big)$, the bound of one push–relabel maximum flow on the network with N_{nd} nodes and N_{arc} arcs.*

Proof. (a) Unions and intersections of closed sets are closed, and those of sets disjoint from $\mathcal{Q}^!$ are disjoint from $\mathcal{Q}^!$. ω_λ is a sum of node weights, so $\omega_\lambda(C) + \omega_\lambda(C') = \omega_\lambda(C \cap C') + \omega_\lambda(C \cup C')$; if C and C' are optimal, neither term on the right can exceed the optimum, so both attain it.

(b) Let $E := C \setminus C'$. Optimality of C against $C \cap C'$ gives $\omega(E \cap \mathcal{Q}^-) - \lambda\,\omega(E \cap \mathcal{Q}^+) \geq 0$, and optimality of C' against $C \cup C'$ gives $\omega(E \cap \mathcal{Q}^-) - \lambda'\omega(E \cap \mathcal{Q}^+) \leq 0$. Subtracting, $(\lambda - \lambda')\omega(E \cap \mathcal{Q}^+) \leq 0$, so E contains no pair of $\mathcal{Q}^+$; then $\omega(E \cap \mathcal{Q}^-) \leq 0$, so it contains no complaint; and it contains no pair of $\mathcal{Q}^!$.

(c) Φ^* is a maximum of finitely many functions that are affine in λ. Two distinct optima at some $\lambda > 0$ can be taken nested by (a); their difference E is nonempty and has $\omega(E \cap \mathcal{Q}^-) = \lambda\,\omega(E \cap \mathcal{Q}^+)$, so both sides are positive, as the weights and λ are. So the two optima have different slopes $-\omega(C \cap \mathcal{Q}^+)$, and λ is a breakpoint of Φ^*. Hence the optimum is unique off the breakpoints, the optima of two adjacent pieces are both optimal at the breakpoint between them and are nested by (b), and their difference contains a complaint and a pair of $\mathcal{Q}^+$. A complaint of $\mathcal{Q}^-_{\text{free}}$ is in every optimum, because adding it keeps a set admissible and raises the weight, and a complaint of $\mathcal{Q}^-_\infty$ is in none; this bounds h. For large λ an optimum contains no pair of $\mathcal{Q}^+$, which gives C_0, and for small $\lambda > 0$ it contains every complaint that can be certified, which gives C_h. By (b), every optimum at λ_ℓ contains $C_{\ell-1}$ and is contained in C_ℓ.

(d) $C \subseteq C'$ gives $\tau^C \leq \tau^{C'}$, and firing sets shrink when thresholds rise. Let $\lambda_{\ell+1} < \lambda < \lambda_\ell$ and $C := C_\ell$. If τ satisfies the two conditions, every pair of edit i in C has a coordinate of at most $L_i(C) < \tau_i$ and every pair outside C a coordinate of at least $U_i(C) \geq \tau_i$, so $U(\tau)$ is the set of pairs of C and τ is feasible; and every complaint q of C has $\zeta_i(q) \leq L_i(C) < \tau_i$ for all i, so $\text{Clean}(\tau) \supseteq C \cap \mathcal{Q}^-$. Hence $\Phi_\lambda(\tau) \geq \omega_\lambda(C) = \Phi^*(\lambda)$. Conversely, if τ is optimal, $C(\tau)$ is an optimal admissible set by Theorem B.14(a), so $C(\tau) = C$ by uniqueness: the pairs outside C are covered, which is $\tau_i \leq U_i(C)$, and the pairs and complaints of C are uncovered and certified, which is $\tau_i > L_i(C)$ wherever $L_i(C)$ is finite. L_i and U_i are nondecreasing in C.

(e) C_0 contains no pair, and each $C_\ell \setminus C_{\ell-1}$ contains a pair of $\mathcal{Q}^+$, of positive weight. Both $C_{\ell-1}$ and C_ℓ are optimal at λ_ℓ, which gives the formula for λ_ℓ. For a feasible τ with $\omega(\text{Unc}(\tau)) \leq B$ and every $\lambda \geq 0$, $\omega(\text{Clean}(\tau)) = \Phi_\lambda(\tau) + \lambda\,\omega(\text{Unc}(\tau)) \leq \Phi^*(\lambda) + \lambda B$. So $\Omega \leq \bar{\Omega}$ with $\bar{\Omega}(B) := \min_{\lambda \geq 0}(\Phi^*(\lambda) + \lambda B)$, a concave function. On the piece of $C_{\ell'}$, $\Phi^*(\lambda) + \lambda B = \Omega_{\ell'} + \lambda(B - B_{\ell'})$, so for $B \in [B_{\ell-1}, B_\ell]$ the minimum is attained at λ_ℓ and equals $\Omega_{\ell-1} + \lambda_\ell(B - B_{\ell-1})$, and for $B \geq B_h$ it is attained at $\lambda = 0$ and equals Ω_h: $\bar{\Omega}$ is the interpolation. The thresholds τ^{C_ℓ} show $\Omega(B_\ell) \geq \Omega_\ell = \bar{\Omega}(B_\ell)$, so $\bar{\Omega}$ agrees with Ω at the vertices; as Ω is nondecreasing, every concave majorant of Ω is at least $\bar{\Omega}$. If τ is optimal for $\lambda > 0$, then $\omega(\text{Clean}(\tau)) = \Phi^*(\lambda) + \lambda\,\omega(\text{Unc}(\tau)) \geq \bar{\Omega}(\omega(\text{Unc}(\tau)))$, so its pair lies on the majorant; by (c) and (d) it is a vertex if λ is not a breakpoint, and the thresholds τ^{C_ℓ} attain the vertices.

(f) In the parameter $-\lambda$ the capacities of the arcs from the source are constant, those of the arcs into the sink are nonincreasing, and the others are constant. These are the hypotheses of the parametric maximum-flow algorithm of Gallo, Grigoriadis and Tarjan [25], which computes all breakpoints and the nested minimum cuts within the time bound of one maximum flow. □

Nesting and the computation of the chain are those of parametric maximum flow [25], and (e) is Everett's principle for Lagrange multipliers [20]. The bound on h and the monotone firing sets are new and elementary.

Example B.17 (A Pareto-optimal repair that no multiplier finds). *One edit with its own request at the coordinate* 1*, two pairs of* Q^+ *with weight* 1 *at the coordinates* $a < a' < 1$*, and two complaints, of weight* 1 *at* a *and of weight* 3 *at* a'*. The feasible repairs realise the pairs* $(\omega(\mathrm{Unc}), \omega(\mathrm{Clean})) = (0,0), (1,1)$ *and* $(2,4)$*, and* $\Phi^*(\lambda) = \max(0, 4-2\lambda)$*. The repair with one uncovered pair is optimal for no* $\lambda > 0$*, although it is optimal for the budget* 1*. Budgets between the vertices of the frontier are the subject of Sec. B.5.2.*

Proposition B.18 (Independent thresholds). *For edit* i *and* $\vartheta \in \Theta$ *let* $\phi_i(\vartheta) := \omega\{q \in Q^- : \vartheta > \zeta_i(q)\} - \lambda\,\omega\{(i,q') \in Q^+ : \vartheta > \zeta_i(q')\}$*, and let* τ^I *be a vector of maximisers of the* ϕ_i *over the thresholds* ϑ *that cover the pairs of* $Q_i^!$*. Let* $\mathrm{cost}(\tau) := \omega(Q^- \setminus \mathrm{Clean}(\tau)) + \lambda\,\omega(\mathrm{Unc}(\tau)) = \omega(Q^-) - \Phi_\lambda(\tau)$*.*

(a) $\Phi_\lambda(\tau) = \sum_i \phi_i(\tau_i) - G(\tau)$ *with* $G(\tau) := \sum_{q \in Q^-} \omega(q)\big(y_q - \mathbf{1}[y_q = T]\big)$*, where* y_q *is the number of edits* i *with* $\tau_i > \zeta_i(q)$*. For* $T = 1$ *the rule is optimal.*

(b) Let $\lambda > 0$ *and let* $\tau^{\min}$ *be the canonical thresholds of the least optimal admissible set. Then* $\mathrm{Unc}(\tau^I) \supseteq \mathrm{Unc}(\tau^{\min})$ *and* $\mathrm{Clean}(\tau^I) \supseteq \mathrm{Clean}(\tau^{\min})$*; if every* τ_i^I *is the coordinate of a pair of edit* i*, then* $\tau^I \geq \tau^{\min}$*. Moreover* $\tau^C \leq \tau^{\mathrm{top}}$ *for every optimal admissible* C*, where* τ_i^{top} *is the largest maximiser of* ϕ_i*.*

(c) For every optimal τ^**,* $\Phi^*(\lambda) - \Phi_\lambda(\tau^I) \leq (T-1)\,\omega(Q^- \setminus \mathrm{Clean}(\tau^*))$*, and*

$$\mathrm{cost}(\tau^I) \;\leq\; T \cdot \min_\tau \mathrm{cost}(\tau).$$

(d) The factor T *is tight. For* $T \geq 2$ *and* $\varepsilon \in (0, \frac{1}{2})$ *let every edit have its own request at the coordinate* 1 *and one pair of* Q^+ *of weight* 1 *at a coordinate* $a < 1$*, let one complaint of weight* 1 *have the coordinate* a *under every edit, and let* $\lambda = 1 - \varepsilon$*. Then* $\max_\tau \Phi_\lambda = 0$ *and* $\min_\tau \mathrm{cost} = 1$*, while* $\Phi_\lambda(\tau^I) = 1 - T(1-\varepsilon) < 0$ *and* $\mathrm{cost}(\tau^I) = (1-\varepsilon)T$*.*

Proof. (a) $\sum_i \omega\{q : \tau_i > \zeta_i(q)\} = \sum_q \omega(q)\, y_q$, a complaint is certified iff $y_q = T$, and the second terms of the ϕ_i sum to $\lambda\,\omega(\mathrm{Unc}(\tau))$. For $T = 1$, $G = 0$.

(b) The admissible sets are finitely many and closed under union and intersection, and ω_λ is modular; so with two optimal admissible sets their intersection is optimal, and the intersection of all of them is the least optimal admissible set. By Theorem B.14(c) it equals $C(\tau^{\min})$, as $\lambda > 0$. Suppose $\tau_i^I < \tau_i^{\min}$ for some i. Let E_+ be the set of pairs of Q_i^+ and E_- the set of complaints with a coordinate under edit i in $[\tau_i^I, \tau_i^{\min})$. As $\tau_i^{\min}$ covers the pairs of $Q_i^!$, $\phi_i(\tau_i^I) \geq \phi_i(\tau_i^{\min})$, that is, $\lambda\,\omega(E_+) \geq \omega(E_-)$. Let τ' be $\tau^{\min}$ with the i-th threshold lowered to τ_i^I. It is feasible, it covers the pairs of E_+ again and certifies all complaints that $\tau^{\min}$ certifies except some of E_-, so $\Phi_\lambda(\tau') \geq \Phi_\lambda(\tau^{\min}) + \lambda\,\omega(E_+) - \omega(E_-) \geq \Phi^*(\lambda)$. So $C(\tau')$ is an optimal admissible set, and it is contained in $C(\tau^{\min})$; hence the two are equal. This says that no pair of edit i has a coordinate in $[\tau_i^I, \tau_i^{\min})$ and that every complaint certified by $\tau^{\min}$ has a coordinate below τ_i^I under edit i. For the edits with $\tau_i^I \geq \tau_i^{\min}$ the same two facts hold trivially, which proves the two inclusions. If τ_i^I is the coordinate of a pair of edit i, this pair has a coordinate in $[\tau_i^I, \tau_i^{\min})$, a contradiction; so $\tau_i^I \geq \tau_i^{\min}$. For the last claim suppose $\tau_i^C > \tau_i^{\mathrm{top}}$ for an optimal C. Then τ_i^C is not a maximiser of ϕ_i, the inequality above is strict for the interval $[\tau_i^{\mathrm{top}}, \tau_i^C)$, and lowering the i-th threshold of τ^C to τ_i^{top} raises Φ_λ above the optimum.

(c) $\phi_i(\tau_i^I) \geq \phi_i(\tau_i^*)$ for all i and (a) give $\Phi^*(\lambda) - \Phi_\lambda(\tau^I) \leq G(\tau^I) - G(\tau^*)$. A certified complaint contributes $(T-1)\omega(q)$ to G and every other one at most $(T-1)\omega(q)$ and at least 0, so $G(\tau^I) - G(\tau^*) \leq (T-1)\omega(Q^-) - (T-1)\omega(\mathrm{Clean}(\tau^*))$. Finally $\omega(Q^- \setminus \mathrm{Clean}(\tau^*)) \leq \mathrm{cost}(\tau^*)$.

(d) A threshold of at most a gives $\phi_i = 0$, and a threshold in $(a, 1]$ gives $\phi_i = 1 - (1-\varepsilon) = \varepsilon > 0$. So every edit raises its threshold above a: the complaint is certified and all T pairs are uncovered. Jointly, certifying the complaint uncovers T pairs and gives $1 - T(1-\varepsilon) < 0$; uncovering N pairs without certifying it gives $-N(1-\varepsilon)$; so the optimum is 0, with cost 1. □

B.5 Beyond Scope-Cut: the hardness frontier

Scope-Cut is polynomial because a complaint is certified by a conjunction of lower bounds, one threshold per edit, and a named pair is covered by one upper bound. This subsection lists what lies beyond: cardinality budgets (Sec. B.5.2), why-not complaints that do not name their edit (Sec. B.5.3), tolerance criteria (Sec. B.5.4) and richer predicate classes (Sec. B.5.5).

Table 4 collects the results, with these conventions. Every row holds over feasible thresholds. (a) *Budgets:* the pairs of Q^+ have unit weight. (b) *Missing provenance:* a why-not complaint names a query and at most κ candidate edits and is *satisfied* if one of them fires on the query ($\kappa = 1$ is Scope-Cut). (c) *Tolerance:* τ^0 is feasible, $c \in [0, 1)$, lineage sizes are taken under τ^0, and $\tau \geq \tau^0$ *repairs* a complaint if the damage at its key under τ, the sum of the contributions of the edits that fire, has p-norm at most c times that under τ^0 (for zero damage: a zero residual); the classification is a worst case over the values of the edits. In the first row of (c) there are no must-cover pairs: with them the identification with Min-Rollback$_{p,c}$ is exact only where un-firing an edit on the complaint uncovers no pair, and a repair need not exist; the hardness is not affected. (d) *Richer predicates:* a repair *satisfies* a complaint of Q^- if its edit is silent on it and a pair of Q^+ if it covers it. Except for the free detector, the reductions build logs whose key dimension is at least their number of detectors, and their hardness at a fixed key dimension is open. An automated review pass of the project relates the rows (b) to positive–negative partial set cover [51] and the boundary of the second row of (c) to the supermodularity of constraint languages on chains (Cohen, Cooper, Jeavons and Krokhin, 2005). We could open neither paper, so the tag "new" of these rows means: derived here, and not checked against these two sources.

Two detectors per edit. The status of the paper's complexity theorem (Theorem 3.7) points here for the rows (d) "Conjunction" and "Disjunction". With two detectors per edit: if an edit fires where both detectors meet their thresholds, the gain $\max_\tau \Phi_\lambda$ has no polynomial-time approximation scheme, and the cost $\omega(Q^-) - \Phi_\lambda$ has none either and is 3-approximable (Propositions B.35 and B.36); if it fires where one of them does, the gain is not approximable within $|Q^-|^{1-\varepsilon}$, and on the same instances the least cost is a minimum vertex cover (Theorem B.40). The method of the paper has one detector per edit; a why-not repair that adds a detector (Sec. B.11) assigns each granted pair to the detector that covers it, which is the polynomial case of the row "Disjunction".

B.5.1 Coordinate tables.

Table 4: The hardness frontier of rescoping beyond Scope-Cut. Gain: Φ_λ, to be maximised; cost: $\omega(Q^-) - \Phi_\lambda$, to be minimised. PTAS: polynomial-time approximation scheme.

Direction: problem	Result	Corollary of / new; statement
(a) Budget: maximise $\omega(\mathrm{Clean}(\tau))$ with at most B uncovered pairs	weighted densest-B-subhypergraph [15] on the vertices Q^+ with the hyperedges $\mathcal{G}(q) \subseteq Q^+$, value for value, and every weighted hypergraph on T vertices arises so; strongly NP-hard already for $\lvert\mathcal{G}(q)\rvert = 2$	known pattern [35, 41]; Prop. B.21
(a) Budgets that a multiplier λ reaches	the chain of Theorem B.16 solves the budgets $\lvert C_\ell \cap Q^+\rvert$, with no approximation ratio for the others. Deciding whether B is the number of uncovered pairs of some maximiser of Φ_λ, for some $\lambda \geq 0$, is NP-complete	new; Props. B.23, B.24
(b) Missing provenance: $\kappa = 2$ and a rational $\lambda > 0$	gain: maximising $\omega(\mathrm{Clean}(\tau))$ minus λ times the weight of the unsatisfied complaints is not approximable within $T^{1-\varepsilon}$, already with unit weights. Cost: on the same instances it is a minimum vertex cover; with at most κ candidates it is $(\kappa+1)$-approximable	new, from maximum clique [74]; Thms. B.25, B.28
(b) All complaints must be satisfied	minimising the weight of the uncertified complaints of Q^- is equivalent to red–blue set cover [13]; κ-approximable [42], optimally under the unique games conjecture [17, 38]	pattern of rule selection [40]; Thms. B.26, B.28
(c) Tolerance: one complaint q of nonzero damage, fewest silenced edits, no must-cover pairs	Min-Rollback$_{p,c}$ on the $\mathrm{wd}(q)$ contributions that fire on q, and every instance of Min-Rollback$_{p,c}$ arises so: Theorem A.5(a),(b) hold with $\mathrm{wd}(q)$ in place of T	corollary of Thm. A.5; Prop. B.29
(c) Any number of complaints: maximise the repaired weight minus λ times the newly uncovered weight	lineage size at most 2, nonzero damage: one minimum cut if $c < \frac{1}{2}$, NP-hard if $c \geq \frac{1}{2}$. Zero damage keeps the cut for $c < \frac{1}{2}$, and lineage size 3 is NP-hard for every c: polynomial for lineage size at most 1 and for lineage size 2 with $c < \frac{1}{2}$, NP-hard otherwise	new; Props. B.31, B.32
(d) Conjunction: an edit fires where each of its two detectors meets its threshold	gain: $\max_\tau \Phi_\lambda$ has no PTAS, already with unit weights and $\lambda = 1$. Cost: no PTAS either; 3-approximable	new, from maximum cut [4, 36]; Props. B.35, B.36
(d) Free detector: ψ_i is repaired as well, with $\langle \psi_i, k_i \rangle = s_i$ and the own request covered	NP-hard to distinguish complaint sets of which some repair satisfies a fraction $1-\varepsilon$ from those of which none satisfies $\frac{1}{2}+\varepsilon$, already for one edit. With linearly independent own and reported keys, every labelling is realisable, and the closest detector with prescribed coordinates is a closed-form projection	corollary of [28]; realisable case standard; Thm. B.38, Prop. B.37
(d) Disjunction: an edit fires where one of several detectors does	each pair assigned to one detector, which alone can cover it: Scope-Cut with one edit per detector. Any detector of its edit may cover a pair: the gain is not approximable within $\lvert Q^-\rvert^{1-\varepsilon}$, already with two detectors per edit, unit weights and any $\lambda \geq 1$; on the same instances the least cost is a minimum vertex cover, so that for the cost only its hardness follows	new, from independent set [74]; Prop. B.39, Thm. B.40

Lemma B.19 (Grid and realisability). *The following hold.*

(a) The sets $\mathrm{Clean}(\tau)$ *and* $U(\tau)$ *depend on* τ_i *only through the set of coordinates of complaints and pairs under edit* i *that lie strictly below* τ_i*. So it suffices to let* τ_i *range over these coordinates and* $+\infty$*.*

(b) Let a finite set of queries and reals $\tilde{\zeta}_i(q)$ *be given, for every edit* $i \in [T]$ *and query* q*, with* $\tilde{\zeta}_i(q_i) > 0$ *for one query* q_i *per edit. For every* $d \geq T$ *there is an edit log with these* T *edits, own requests* q_i *and one-position queries such that* $\zeta_i(q) = \tilde{\zeta}_i(q)/\tilde{\zeta}_i(q_i)$ *for all* i *and* q*.*

Proof. (a) Whether $\tau_i > \zeta_i(q)$ depends only on the side of τ_i on which the coordinate lies. (b) Let $\psi_i = e_i$, let every query have one position, of a role that all role sets contain, with the key $\sum_i (\tilde{\zeta}_i(q)/\tilde{\zeta}_i(q_i))\, e_i$, and let the own key k_i be the key of q_i. Then $s_i = \langle \psi_i, k_i \rangle = 1$ and $\zeta_i(q) = \tilde{\zeta}_i(q)/\tilde{\zeta}_i(q_i)$. Write vectors are arbitrary; the targets are defined from them as in Lemma A.4. □

The constructions below therefore give *coordinate tables*: numbers $\tilde{\zeta}_i(q)$ in which the own request of an edit need not have the coordinate 1. Dividing the coordinates and the thresholds of edit i by $\tilde{\zeta}_i(q_i)$ changes neither certified complaints nor covered pairs; where an edit has a value, the value is multiplied by $\tilde{\zeta}_i(q_i)$, so that contributions do not change. All hardness results of this subsection that are stated for coordinate tables thus hold for scoped logs whose key dimension is at least the number of detectors ($d \geq T$ with one detector per edit); whether they hold for a fixed key dimension is open. The agnostic problem of Theorem B.38 is not of this kind: its key dimension is that of the examples.

B.5.2 Cardinality budgets. In this part the pairs of Q^+ have unit weight, and the *budget problem* asks, for an integer $B \geq 0$, for

$$\Omega(B) := \max\{\omega(\mathrm{Clean}(\tau)) :\ \tau \text{ feasible},\ |\mathrm{Unc}(\tau)| \leq B\},$$

the function of Theorem B.16(e). The *demand set* $\mathcal{G}(q)$ of $q \in Q^-$ is the set of its out-neighbours in $\mathcal{G}$, the pairs (i, q') with $\zeta_i(q') \leq \zeta_i(q)$; its *width* is the number of edits with a pair in $\mathcal{G}(q)$. For $U \subseteq Q^+$ let $\mathrm{Cl}(U) := \{q \in Q^- \setminus Q^-_\infty : \mathcal{G}(q) \subseteq U\}$.

Lemma B.20 (Hypergraph form). (a) *For every feasible* τ*, the set* $\mathrm{Clean}(\tau)$ *is contained in* $\mathrm{Cl}(\mathrm{Unc}(\tau))$. (b) *For every* $U \subseteq Q^+$ *there is a feasible* τ *with* $\mathrm{Unc}(\tau) \subseteq U$ *and* $\mathrm{Clean}(\tau) = \mathrm{Cl}(U)$*. Hence*

$$\Omega(B) = \max\{\omega(\mathrm{Cl}(U)) :\ U \subseteq Q^+,\ |U| \leq B\}.$$

Proof. (a) If q is certified, then $\tau_i > \zeta_i(q) \geq \zeta_i(q')$ for every pair (i, q') of $\mathcal{G}(q)$, so $\mathcal{G}(q) \subseteq U(\tau)$; as τ is feasible, $\mathcal{G}(q)$ contains no pair of $Q^!$, so $q \notin Q^-_\infty$ and $\mathcal{G}(q) \subseteq \mathrm{Unc}(\tau)$. (b) Let $U' \subseteq U$ be the union of the sets $\mathcal{G}(q)$, $q \in \mathrm{Cl}(U)$. Among the pairs of one edit, each $\mathcal{G}(q)$ contains, with a pair, all pairs of the same or of a smaller coordinate; so does U'. Let τ_i be the smallest coordinate of a pair of edit i outside U'; it exists, because the pairs of $Q^!$ are outside U'. A pair of edit i with a coordinate below τ_i lies in U', and one with a coordinate of at least τ_i lies outside. So $U(\tau) = U'$, and τ is feasible. For $q \in \mathrm{Cl}(U)$, every pair of edit i outside U' has a coordinate above $\zeta_i(q)$, because the others are in $\mathcal{G}(q) \subseteq U'$; so $\tau_i > \zeta_i(q)$ and q is certified. With (a), $\mathrm{Clean}(\tau) \subseteq \mathrm{Cl}(U') \subseteq \mathrm{Cl}(U) \subseteq \mathrm{Clean}(\tau)$. □

Proposition B.21 (Budgets and densest subhypergraphs). *Let the pairs of* Q^+ *have unit weight.*

(a) *$\Omega(B)$ is the optimum of the weighted densest-B-subhypergraph problem on the hypergraph with the vertex set Q^+ and the hyperedges $\mathcal{G}(q)$, $q \in Q^- \setminus Q^-_\infty$, of weight $\omega(q)$: choose at most B vertices so as to maximise the weight of the hyperedges that they contain.*

(b) *Conversely, every weighted hypergraph with T vertices is the hypergraph of a coordinate table: one edit per vertex v, with its own request at the coordinate 1 and one pair of Q^+ at the coordinate 0; one complaint q_e per hyperedge e, of the weight of e, with $\tilde{\zeta}_v(q_e) = 0$ for $v \in e$ and -1 otherwise. Then $\mathcal{G}(q_e)$ is the set of the pairs of the vertices of e.*

(c) *Hence the budget problem is strongly NP-hard already when every demand set has width 2, with unit weights and one pair of Q^+ per edit: a graph has a clique of k vertices iff $\Omega(k) = k(k-1)/2$ on its table. An algorithm with a ratio for the budget problem has the same ratio for densest-k-subgraph and densest-k-subhypergraph, and conversely.*

Proof. (a) is Lemma B.20. (b) In the table every complaint is outside Q^-_∞, and $\tilde{\zeta}_v(q') \le \tilde{\zeta}_v(q_e)$ holds for the pair of v at the coordinate 0 iff $v \in e$. (c) follows from (a), (b) and the NP-completeness of Clique [37]; the numbers of the table are -1, 0 and 1. □

This is an instance of a known pattern, not a new reduction. A budget on a closure problem is the partially ordered knapsack problem, which generalises densest-k-subgraph [41], and a budgeted repair problem of databases is tied to minimum k-union in [35]. No better guarantee than for densest-k-subhypergraph [15] is available for the budget problem.

Proposition B.22 (Demand sets of width one). *If every demand set of a complaint outside Q^-_∞ has width at most 1, the budget problem is solved in $O((|Q^+| + T)\,|Q^+|)$ operations once the demand sets are computed and sorted.*

Proof. For edit i let $x_1 < \cdots < x_L$ be the distinct coordinates of its pairs of Q^+ that lie below every pair of $Q^\dagger_i$. A feasible τ uncovers, in edit i, the pairs with a coordinate of at most x_l for some level $l \in \{0, \dots, L\}$, none for $l = 0$. Let $N_i(l)$ be their number and $\omega_i(l)$ the weight of the complaints outside Q^-_∞ whose demand set is nonempty, lies in edit i and within the level l. The complaints outside Q^-_∞ with an empty demand set are in $\mathrm{Cl}(U)$ for every U; let F be their weight. By Lemma B.20 the best weight with the levels (l_i) is $F + \sum_i \omega_i(l_i)$, and it uncovers $\sum_i N_i(l_i)$ pairs. This is a multiple-choice knapsack with integer sizes of at most $|Q^+|$, solved by the recursion $\mathrm{best}_i[B] = \max_l \big(\mathrm{best}_{i-1}[B - N_i(l)] + \omega_i(l)\big)$. □

With Proposition B.21 this is a dichotomy in the width of the demand sets: width one is polynomial, width two is as hard as densest-k-subgraph. The frontier of Theorem B.16 relates to budgets as follows. Call a budget B *supported* if some maximiser τ of Φ_λ, for some $\lambda \ge 0$, has $|\mathrm{Unc}(\tau)| = B$; the budgets $B_0, \dots, B_h$ of the vertices are supported, and further budgets can be. Let $\bar{\Omega}(B) = \min_{\lambda \ge 0}(\Phi^*(\lambda) + \lambda B)$ be the least concave majorant of Ω, and (B_ℓ, Ω_ℓ), $\ell \le h$, its vertices.

Proposition B.23 (The Lagrangian frontier and budgets). *Let the pairs of Q^+ have unit weight.*

(a) *If τ maximises Φ_λ, then τ is optimal for the budget $|\mathrm{Unc}(\tau)|$.*

(b) *$\Omega(B) \le \bar{\Omega}(B)$ for every B, and for $B < B_h$: $\Omega(B) = \bar{\Omega}(B)$ iff B is supported.*

(c) *For $B_{\ell-1} < B < B_\ell$, the repair of the vertex $\ell - 1$ is within the budget and loses at most $\lambda_\ell(B - B_{\ell-1}) < \Omega_\ell - \Omega_{\ell-1}$ against $\Omega(B)$; the repair of the vertex ℓ has a weight of at least $\Omega(B)$ but uncovers $B_\ell > B$ pairs.*

(d) *No approximation ratio follows. On the table of Proposition B.21(b) for a connected graph (V, E) in which every vertex has degree ϱ, the only maximisers of Φ_λ uncover no pair ($\lambda \ge \varrho/2$) or all $|V|$ pairs ($\lambda \le \varrho/2$): the supported budgets are 0 and $|V|$. For the complete bipartite graph with ϱ vertices on each side, $\Omega(2) = 1$, the frontier offers the value 0 within the budget or uncovers 2ϱ pairs, and $\bar{\Omega}(2) = \varrho$.*

(e) *If a polynomial-time algorithm returns, for every graph table and every k, thresholds with at most $\kappa_2 k$ uncovered pairs and at least $\Omega(k)/\kappa_1$ certified complaints, then densest-k-subgraph has a polynomial-time $2\kappa_1\kappa_2^2$-approximation.*

Proof. (a) If $|\mathrm{Unc}(\tau')| \le |\mathrm{Unc}(\tau)|$, then

$$\begin{aligned}\omega(\mathrm{Clean}(\tau')) &\le \Phi_\lambda(\tau') + \lambda|\mathrm{Unc}(\tau)| \\ &\le \Phi_\lambda(\tau) + \lambda|\mathrm{Unc}(\tau)| = \omega(\mathrm{Clean}(\tau)).\end{aligned}$$

(b) The bound is Theorem B.16(e). If B is supported by τ and λ, then $\Omega(B) \ge \omega(\mathrm{Clean}(\tau)) = \Phi^*(\lambda) + \lambda B \ge \bar{\Omega}(B)$. Conversely let $\Omega(B) = \bar{\Omega}(B) = \Phi^*(\lambda) + \lambda B$ for a minimising λ, and let τ^* be optimal for the budget B. Then $\Phi^*(\lambda) \ge \Phi_\lambda(\tau^*) \ge \Omega(B) - \lambda B = \Phi^*(\lambda)$, so τ^* maximises Φ_λ and, if $\lambda > 0$, $|\mathrm{Unc}(\tau^*)| = B$. If $\lambda = 0$, then $\Omega(B) = \Phi^*(0) = \Omega_h$, which requires $B \ge B_h$ because a repair that certifies every complaint outside Q^-_∞ uncovers all B_h pairs of their demand sets.

(c) By (b) and Theorem B.16(e), $\Omega(B) \le \bar{\Omega}(B) = \Omega_{\ell-1} + \lambda_\ell(B - B_{\ell-1})$ and $\Omega_\ell = \bar{\Omega}(B_\ell) \ge \bar{\Omega}(B) \ge \Omega(B)$.

(d) By Lemma B.20 the outcomes are the sets $U \subseteq V$ of vertices whose pair is uncovered, with the edges inside U certified: $\Phi_\lambda = e(U) - \lambda|U| = (\varrho/2 - \lambda)|U| - |\partial U|/2$, where $e(U)$ counts the edges inside U and ∂U is the set of edges that leave U; and $\partial U = \emptyset$ only for $U \in \{\emptyset, V\}$. In the bipartite example, two uncovered pairs certify at most one edge, and $\bar{\Omega}(2) = \Phi^*(\varrho/2) + \varrho = \varrho$.

(e) Let U be the set of uncovered pairs and $N := |U| \le \kappa_2 k$. By Lemma B.20(a) the certified complaints are edges inside U, so $e(U) \ge \Omega(k)/\kappa_1$. If $N > k$, delete vertices of minimum degree one by one: deleting such a vertex from a graph with $N' \ge 3$ vertices and e' edges leaves at least $e'(N' - 2)/N'$ edges, so k vertices keep at least $e(U)\,k(k-1)/(N(N-1))$ edges. With $N \le \kappa_2 k$ and $k \ge 2$, $k(k-1)/(N(N-1)) \ge (k-1)/(\kappa_2^2 k) \ge 1/(2\kappa_2^2)$. □

The parametric cut of Theorem B.16 thus solves the budgets $B_0, \dots, B_h$ of its vertices and no others. Part (b) says when the Lagrangian bound $\bar{\Omega}$ is tight, not that the frontier attains it: a budget strictly between two vertices is supported only if one of the tied optima at a breakpoint has exactly this size.

Proposition B.24 (Deciding whether a budget is supported). *It is NP-complete to decide whether a given budget is supported; by Proposition B.23(b) this is, below B_h, the question whether the Lagrangian bound is tight. Whether a budget is one of the vertex budgets $B_0, \dots, B_h$ is decided in polynomial time (Theorem B.16(f)).*

Proof. *Membership.* A certificate is a feasible threshold vector τ with $|\operatorname{Unc}(\tau)| = B$. The function $\lambda \mapsto \Phi^*(\lambda) - \Phi_\lambda(\tau)$ is nonnegative, convex and piecewise linear on $[0, \infty)$, with kinks only at the breakpoints of Φ^*; so if it vanishes somewhere, it vanishes at 0 or at a breakpoint. The breakpoints and the values of Φ^* are computed by minimum cuts (Theorem B.16). *Hardness.* Let (V, E) be a graph and $M := |V| + 1$. The table has one edit per vertex v, with its own request at 1, one pair of Q^+ at 0 and a complaint q_v of weight 1 with $\tilde{\zeta}_v(q_v) = 0$ and -1 under every other edit; and one edit per edge e, with its own request at 1, M pairs of Q^+ at 0 and a complaint q_e of weight M with the coordinate 0 under e and under the two edits of the end points of e, and -1 elsewhere. A threshold of at most -1 leaves no complaint certified; for $\lambda > 0$ such a vector maximises Φ_λ only if it uncovers no pair, and for $\lambda = 0$ it is not a maximiser. So let all thresholds exceed -1. By Lemma B.19(a) a threshold vector is then described by the set $I \subseteq V$ of vertex edits and the set $F \subseteq E$ of edge edits whose thresholds exceed 0. It uncovers $|I| + M|F|$ pairs and certifies q_v for $v \in I$ and q_e for $e \in F$ with both end points in I. So

$$\Phi_\lambda = (1-\lambda)\,|I| + (1-\lambda)M\,|F \cap E(I)| - \lambda M\,|F \setminus E(I)|,$$

where $E(I)$ is the set of edges inside I. For $\lambda > 1$ the only maximiser is $I = F = \emptyset$; for $\lambda < 1$ it is $I = V$, $F = E$; for $\lambda = 1$ the maximisers are the pairs with $F \subseteq E(I)$. The supported budgets are therefore 0, $|V| + M|E|$ and the numbers $|I| + M|F|$ with $F \subseteq E(I)$. As $|I| < M$, the budget $k + M\,k(k-1)/2$ with $2 \le k \le |V|$ is of this form iff some I with $|I| = k$ has $k(k-1)/2$ edges inside, that is, iff the graph has a clique of k vertices. The complaints q_v of the vertices are needed: without them only $I = F = \emptyset$ would be optimal at $\lambda = 1$. □

On these tables the frontier has two vertices only, with the budgets 0 and $|V| + M|E|$.

B.5.3 Complaints without provenance. An *anonymous* why-not complaint a names a query q_a and a set $\operatorname{cand}(a) \subseteq [T]$ of candidate edits, and has a weight $\omega(a) > 0$. It is *satisfied* by τ if some candidate fires on the query, $\tau_i \le \zeta_i(q_a)$ for some $i \in \operatorname{cand}(a)$. A named pair is the case of one candidate. Let Q^* be the set of anonymous complaints, $\kappa := \max_a |\operatorname{cand}(a)|$ the *ambiguity*, and $\operatorname{Unsat}(\tau)$ the set of anonymous complaints that τ does not satisfy. Two problems: maximise

$$\Psi_\lambda(\tau) := \omega(\operatorname{Clean}(\tau)) - \lambda\,\omega(\operatorname{Unc}(\tau)) - \lambda\,\omega(\operatorname{Unsat}(\tau))$$

over the feasible τ; and, with *hard* anonymous complaints, minimise the weight $\omega(Q^- \setminus \operatorname{Clean}(\tau))$ of the complaints that are not certified over the feasible τ that satisfy every anonymous complaint. The thresholds $-\infty$ satisfy everything, so the first optimum is at least 0 and the second problem is feasible. Let hi_i be the smallest coordinate of a pair of $Q_i^!$, the largest feasible threshold of edit i.

Theorem B.25 (Two candidates suffice for hardness). *Fix a rational $\lambda > 0$. Maximising Ψ_λ is NP-hard, and for every $\varepsilon > 0$ no polynomial-time algorithm returns thresholds of value at least $\max \Psi_\lambda / T^{1-\varepsilon}$, unless* P = NP. *This holds with unit weights, $Q^+ = \emptyset$, exactly two candidates per anonymous complaint, and every complaint of Q^- at or above the coordinate of an anonymous complaint under exactly one edit.*

Proof. Let (V, E) be a graph and $M := \lceil 1/\lambda \rceil$. The table has one edit per vertex v, with its own request at 1; one complaint $q_v \in Q^-$ with $\tilde{\zeta}_v(q_v) = 0$ and $\tilde{\zeta}_u(q_v) = -1$ for $u \neq v$; and for every edge $\{u, v\}$, M anonymous complaints with the candidates u, v and both coordinates 0. We claim that $\max \Psi_\lambda$ is the independence number $\operatorname{ind}(V, E)$ of the graph, and that every feasible τ yields in linear time an independent set of at least $\Psi_\lambda(\tau)$ vertices. Feasibility is $\tau_v \le 1$. Let $S := \{v : \tau_v > 0\}$. The complaint q_v is certified only if $v \in S$, and the complaints of an edge are unsatisfied iff both end points are in S. So $\Psi_\lambda(\tau) \le |S| - \lambda M\, e(S) \le |S| - e(S)$, where $e(S)$ counts the edges inside S, and deleting one end point of every edge inside S leaves an independent set of at least $|S| - e(S)$ vertices. Conversely, for an independent set S the thresholds $\tau_v = 1$ for $v \in S$ and $\tau_v = 0$ otherwise certify the complaints q_v, $v \in S$, and satisfy every anonymous complaint, so $\Psi_\lambda = |S|$. The table has $T = |V|$ edits, $|V|$ complaints and $M|E|$ anonymous complaints; for λ given in binary, use one anonymous complaint of weight M per edge. Hence thresholds of value at least $\max \Psi_\lambda / T^{1-\varepsilon}$ yield an independent set of at least $\operatorname{ind}(V, E)/|V|^{1-\varepsilon}$ vertices, and the independence number is NP-hard to approximate within $|V|^{1-\varepsilon}$ [74]. □

The complementary objective, $\omega(Q^- \setminus \operatorname{Clean}) + \lambda\,\omega(\operatorname{Unsat})$ to be minimised, equals the vertex cover number on this table, so that form is NP-hard as well. The hardness does not depend on small or different candidate sets: let every anonymous complaint have *all* edits as candidates, the complaints of the edge $\{u, v\}$ having the coordinate 0 under u and v and -2 under every other edit. An edit outside $\{u, v\}$ fires on them only if its threshold is at most -2, which leaves no complaint of Q^- certified and gives a value of at most 0; and thresholds above -2 have the same value as before. So $\max \Psi_\lambda = \operatorname{ind}(V, E)$ again. The disjunction that makes the problem hard is created by the coordinates of the complaint keys, not by the declared candidates.

Theorem B.26 (Hard anonymous complaints are red–blue set cover). *Red–blue set cover asks, for a family of sets of red and blue elements with weights on the red ones, for a subfamily that covers all blue elements and minimises the weight of the red elements that it covers [13].*

- *(a) Every instance of red–blue set cover is, with the same optimum, the problem with hard anonymous complaints of a table: one edit per set S_j, with its own request at 1; one complaint of Q^- per red element, with the coordinate 0 under j if the element is in S_j and -1 otherwise; one anonymous complaint per blue element, with the candidates $\{j :$ the element is in $S_j\}$ and the coordinates 0.*
- *(b) Conversely, let W_0 be the weight of the complaints $q \in Q^-$ with $\zeta_i(q) \ge \mathrm{hi}_i$ for some i; they are certified by no feasible τ. Drop them, and drop the anonymous complaints a with $\zeta_i(q_a) \ge \mathrm{hi}_i$ for some candidate i, which every feasible τ satisfies. For every edit i and every coordinate $x = \zeta_i(q_a) < \mathrm{hi}_i$ of a remaining anonymous complaint with the candidate i, let $S_{i,x}$ consist of the remaining anonymous complaints a' with the candidate i and $\zeta_i(q_{a'}) \ge x$ (blue) and of the remaining complaints $q \in Q^-$ with $\zeta_i(q) \ge x$ (red): what edit i fires on when its threshold is x. Then the optimum with hard anonymous complaints is W_0 plus the optimum of this red–blue set cover instance, and solutions are translated in both directions without loss.*

Proof. (a) A threshold $\tau_j \le 0$ selects S_j. If all thresholds exceed -1, the complaints that are not certified are the red elements of the selected sets, and the anonymous complaints are satisfied iff the selected sets cover the blue elements. A threshold of at most -1 leaves no complaint certified, which no cover beats.

(b) From a subfamily that covers the blue elements take τ_i as the least x with $S_{i,x}$ selected, and hi_i if there is none. It is feasible; a blue a covered by a selected $S_{i,x}$ has $\zeta_i(q_a) \ge x \ge \tau_i$, so it is satisfied; and a remaining complaint that is not certified has $\zeta_i(q) \ge \tau_i$ for some i with $\tau_i < \mathrm{hi}_i$, so it is a red element of the selected set S_{i,τ_i}. Conversely, from a feasible τ that satisfies all anonymous complaints pick for every remaining a a candidate i_a with $\tau_{i_a} \le \zeta_{i_a}(q_a)$ and select $S_{i_a,x}$ with $x = \zeta_{i_a}(q_a)$. All blue elements are covered, and a red element q of such a set has $\zeta_{i_a}(q) \ge x \ge \tau_{i_a}$, so it is not certified. □

So every hardness result for red–blue set cover holds for hard anonymous complaints, already with one threshold level per edit, and every approximation algorithm for it applies to the avoidable weight. The reduction follows the pattern of rule selection [40]. With two candidates per complaint and complaints of $\boldsymbol{Q}^-$ that one edit can reach, the table of Theorem B.25 with hard complaints is minimum vertex cover.

Proposition B.27 (Provenance is the boundary: designations). *A* designation ς *picks one candidate* $\varsigma(a) \in \mathrm{cand}(a)$ *for every anonymous complaint, which turns* a *into the named pair* $(\varsigma(a), q_a)$ *of weight* $\omega(a)$ *and the instance into a* Scope-Cut *instance with optimum* $\Phi^*_\varsigma(\lambda)$. *Then*

$$\max_\tau \Psi_\lambda(\tau) = \max_\varsigma \Phi^*_\varsigma(\lambda),$$

so the problem is solved by $\prod_a |\mathrm{cand}(a)|$ *minimum cuts. For* $\kappa = 1$ *it is* Scope-Cut*, one minimum cut; for* $\kappa = 2$ *it is NP-hard and not approximable within* $T^{1-\varepsilon}$ *(Theorem B.25).*

Proof. Fix τ and ς. The value of τ in the designated instance is $\omega(\mathrm{Clean}(\tau)) - \lambda\,\omega(\mathrm{Unc}(\tau)) - \lambda\,\omega(N_\varsigma(\tau))$, where $N_\varsigma(\tau)$ is the set of anonymous complaints whose designated edit does not fire on them. An unsatisfied complaint lies in $N_\varsigma(\tau)$ for every ς, so this value is at most $\Psi_\lambda(\tau)$, with equality for a designation that sends every satisfied complaint to an edit that fires on it. Maximise over τ and ς. □

Alternating the two steps, designate every satisfied complaint to an edit that fires on it and solve the cut of the designated instance, never decreases Ψ_λ. It is a heuristic: Theorem B.25 excludes a polynomial-time guarantee better than $T^{1-\varepsilon}$.

Theorem B.28 (The approximation ratio is the ambiguity). *Consider hard anonymous complaints with at most* κ *candidates each, and call* $\omega(\boldsymbol{Q}^- \setminus \mathrm{Clean}(\tau)) - W_0$ *the* avoidable *weight of* τ*, with* W_0 *as in Theorem B.26*(b).

- *(a) There is a polynomial-time* κ*-approximation of the avoidable weight, hence of* $\omega(\boldsymbol{Q}^- \setminus \mathrm{Clean})$. *For* $\kappa = 1$ *it is exact.*
- *(b) With penalties instead of hard complaints, minimising* $\omega(\boldsymbol{Q}^- \setminus \mathrm{Clean}(\tau)) + \lambda\,\omega(\mathrm{Unsat}(\tau))$*, where pairs of* $\boldsymbol{Q}^+$ *count as anonymous complaints with one candidate, has a polynomial-time* $(\kappa+1)$*-approximation.*
- *(c) With exactly* κ *candidates per complaint, unit weights and complaints of* $\boldsymbol{Q}^-$ *on which only one edit can fire, the problem contains minimum vertex cover in* κ*-uniform hypergraphs. So it is NP-hard to approximate within* 1.3606 *for* $\kappa = 2$ *[18] and within* $\kappa - 1 - \varepsilon$ *for* $\kappa \ge 3$ *[17]; under the unique games conjecture it is hard within* $\kappa - \varepsilon$ *for every* $\kappa \ge 2$ *[38], so that the ratio of* (a) *is then optimal.*

Proof. (a) *Algorithm.* Let the *residual weight* ω' be ω on the complaints of $\boldsymbol{Q}^-$ that are certified under $\tau = (\mathrm{hi}_i)_i$ and 0 on the others, and start from these thresholds. While some anonymous complaint a is not satisfied: for each candidate i let $N_i := \{q \in \boldsymbol{Q}^- : \zeta_i(q) \ge \zeta_i(q_a)\}$, the complaints on which edit i fires if it is to satisfy a; let $\mu := \min_i \omega'(N_i)$, attained by i^*; for each candidate i pick a function $\omega^{(i)} \le \omega'$ that vanishes outside N_i and has the total μ, with $\omega^{(i^*)} = \omega'$ on N_{i^*}; replace ω' by $\omega' - \min(\omega', \sum_i \omega^{(i)})$, and lower τ_{i^*} to $\zeta_{i^*}(q_a)$.

Analysis. Let $\varphi^k := \min(\omega', \sum_i \omega^{(i)})$ be the function subtracted in round k, μ_k its μ and a_k its complaint. (1) $\varphi^k \ge \omega^{(i)}$ for each candidate i, so $\varphi^k(N_i) \ge \mu_k$. Thresholds τ' that satisfy a_k have $\tau'_i \le \zeta_i(q_{a_k})$ for some candidate i, so the complaints that they do not certify include N_i and have φ^k-weight at least μ_k. (2) The total of φ^k is at most $\kappa\mu_k$. (3) The complaints that lose their certificate in round k lie in N_{i^*}, whose residual weight is 0 after the subtraction; residual weights never increase, and the complaints that are not certified at the start have residual weight 0. So the complaints that the output τ does not certify have final residual weight 0, and with ω written, on the complaints that can be certified, as $\sum_k \varphi^k$ plus the final ω', the avoidable weight of the output is at most $\sum_k \varphi^k$ of all complaints, which is at most $\kappa \sum_k \mu_k$, which is at most κ times the avoidable weight of an optimal solution by (1). Thresholds only decrease, so every round satisfies one more complaint for good, and the output is feasible because $\tau \le (\mathrm{hi}_i)_i$. For $\kappa = 1$ the factor is 1.

(b) Give every complaint one more option, to pay its penalty $\lambda\,\omega(a)$: let μ be the minimum of the $\omega'(N_i)$ and the remaining penalty, and subtract μ from the penalty as well. If the penalty attains the minimum, the complaint is bought out and not looked at again; otherwise τ_{i^*} is lowered. Every solution pays at least μ_k in round k, at most $(\kappa+1)\mu_k$ is subtracted, and the output has residual cost 0.

(c) Take one edit per vertex, with its own request at 1, a complaint q_v with $\tilde{\zeta}_v(q_v) = 0$ and -1 under the other edits, and one hard anonymous complaint per hyperedge, with its vertices as candidates and the coordinates 0. A feasible τ has $\tau_v \le 1$. A threshold of at most -1 leaves no complaint certified, which is never better than a vertex cover. If all thresholds exceed -1, let $X := \{v : \tau_v \le 0\}$: q_v is not certified iff $v \in X$, and the complaint of a hyperedge is satisfied iff the hyperedge meets X. So the optimum is the size of a minimum vertex cover. □

Parts (a) and (b) are instances of the greedy algorithm for covering with a submodular cost [42]: the weight of the complaints that are not certified is a nondecreasing submodular function of the lowered thresholds, and every anonymous complaint constrains at most κ of them. The maximisation form of Theorem B.25 and the minimisation form of Theorem B.28 differ as independent set and vertex cover do.

B.5.4 Tolerance criteria. Let every edit have a value $v_i \in \mathbb{Q}^n$ and the identity response, and let complaints have one position. Under thresholds τ the *damage* of q at the edited module is $\mathrm{dmg}_\tau(q) := \sum_{i \in \mathrm{Fire}_\tau(q)} \zeta_i(q)\, v_i$. Let τ^0 be the current thresholds; for the deployed log they are $-\infty$. Fix $p \in [1, \infty]$ and a rational $c \in [0, 1)$. Thresholds $\tau \geq \tau^0$ *c-repair* q if $\|\mathrm{dmg}_\tau(q)\|_p \leq c\, \|\mathrm{dmg}_{\tau^0}(q)\|_p$, the tolerance criterion of Def. 2.6(i) relative to the damage under τ^0. The certificate criterion, no edit fires, is stronger. The lineage size of q is $\mathrm{wd}(q) = |\,\mathrm{Fire}_{\tau^0}(q)|$.

Proposition B.29 (One complaint is Min-Rollback on its lineage). *Let q be a complaint with $\mathrm{dmg}_{\tau^0}(q) \neq 0$ and $F := \mathrm{Fire}_{\tau^0}(q)$.*

(a) For $\tau \geq \tau^0$ the damage of q depends on τ only through the set $S := \{i \in F : \tau_i > \zeta_i(q)\}$ of edits that are un-fired on q, and equals $\sum_{i \in F \setminus S} \zeta_i(q) v_i$. Minimising $|S|$ subject to the c-repair of q is the instance $(\zeta_i(q) v_i)_{i \in F}$ of Min-Rollback$_{p,c}$: the same feasible sets and the same objective.

(b) Every instance $w_1, \ldots, w_T$ of Min-Rollback$_{p,c}$ arises in this way, from the coordinate table with T edits of values w_i, one complaint with the coordinate 1 under every edit, own requests at the coordinate 2, and $\tau_i^0 = \frac{1}{2}$ (or any value of at most 1, also $-\infty$). There $F = [T]$, the contributions are the w_i, and a threshold in $(1, 2]$ un-fires an edit on the complaint and on nothing else.

(c) Hence parts (a) *and* (b) *of Theorem A.5 hold for the repair of one complaint with $\mathrm{wd}(q)$ in place of T; and the problem is solved by enumerating the $2^{\mathrm{wd}(q)}$ subsets of F, whatever the length of the log.*

(d) The sets S that repair q are not closed under taking supersets: for the scalar contributions $2, 1, -1$ and $c = \frac{1}{4}$, $S = \{1\}$ repairs and $S = \{1, 2\}$ does not.

Proof. (a) Thresholds only rise, so an edit outside F stays outside, and the firing set of q under τ is $F \setminus S$. (b) Every $S \subseteq [T]$ is realised by the thresholds 2 on S and $\frac{1}{2}$ elsewhere, which cover the own requests. (c) follows from (a), (b) and Lemma B.19. (d) The damage is 2; the residual is 0 for $S = \{1\}$ and -1 for $S = \{1, 2\}$. □

Remark B.30 (Traces with several positions). Proposition B.29 is stated for a complaint of one position. If the pre-edit trace of q has several positions and the tolerance criterion is applied at one reported position t_q, with the key k_q (Def. 2.5(b)), let $F_q := \mathrm{Fire}_{\tau^0}(x^0(q), t_q)$ be the set of edits that fire *at* t_q, and for $\tau \geq \tau^0$ let $S := \{i \in F_q : \tau_i > \zeta_i(k_q)\}$ be the set of edits that are silenced at t_q. The module output at t_q under τ is $W_0 k_q + \sum_{i \in F_q \setminus S} \zeta_i(k_q)\, v_i$ (identity response, as in Proposition B.29), so parts (a)–(d) hold with F_q in the place of F and $\zeta_i(k_q)$ in the place of $\zeta_i(q)$, by the same proofs. The cost has to be counted at t_q: an edit of S can still fire at another position of the trace, at which its coordinate is larger, and it then remains in $\mathrm{Fire}_\tau(q)$, so that the number of edits of $\mathrm{Fire}_{\tau^0}(q)$ that no longer fire on q does not count it (one edit with the coordinates 0.2 at the reported position and 0.8 at another one, and a threshold raised to 0.3). For the deployed scoping with thresholds τ^0 the contributions $\zeta_i(k_q) v_i^{\log} = \Delta_i k_q$ are those of Def. 2.6(i). Theorem 3.1(e) of the paper is stated in this form.

So in the deployed log, where every edit fires on every key, the hard instance of a complaint has the size T, and in a scoped log it has the size $\mathrm{wd}(q)$: scoping does not remove the hardness of the tolerance criterion, it confines it to the lineage of the complaint. By (d), the closure structure of Scope-Cut is lost for several complaints that share edits. For these, let τ^0 be feasible, and let the *tolerance form* of Scope-Cut ask for feasible $\tau \geq \tau^0$ that maximise the weight of the c-repaired complaints minus λ times the weight of the newly uncovered pairs, $\omega(\mathrm{Unc}(\tau) \setminus \mathrm{Unc}(\tau^0))$. A complaint with zero damage under τ^0 counts as repaired iff its residual is zero, which is the criterion read literally.

Proposition B.31 (Several complaints of lineage size two). *Consider the tolerance form on instances in which every complaint has a nonzero damage or an empty firing set under τ^0.*

(a) For every p and every $c \in [\frac{1}{2}, 1)$ it is NP-hard, already when every complaint has lineage size 2, with scalar values, unit weights, one pair of Q^+ per edit and $\lambda = 1$.

(b) For every p and every $c \in [0, \frac{1}{2})$, on complaints of lineage size at most 2 it is one Scope-Cut instance, hence one minimum cut.

Proof. (a) Let (V, E) be a graph. The table has one edit per vertex u, with the scalar value 1, its own request at the coordinate 2, one pair of Q^+ at the coordinate 1, and $\tau_u^0 = 0$; and one complaint q_e per edge $e = \{x, y\}$, with the coordinate 1 under x and y and -1 under the others. Then $\mathrm{Fire}_{\tau^0}(q_e) = \{x, y\}$, the damage is 2, and q_e is c-repaired iff at least one of x, y is un-fired, because the residual 1 is at most $2c$ and the residual 2 is not. With $U := \{u \in V : \tau_u > 1\}$ the value is the number of edges that meet U minus $|U|$, that is, $|E| - |V| + (|W| - e(W))$ for $W := V \setminus U$, where $e(W)$ counts the edges inside W. The maximum of $|W| - e(W)$ is the independence number: an independent set attains it, and deleting one end point of every edge inside W leaves an independent set of at least $|W| - e(W)$ vertices. So the optimum is $|E| - |V|$ plus the independence number, which is NP-hard to compute: it is the clique number of the complement graph [37].

(b) Let a complaint of lineage size 2 have the contributions x, y with $x + y \neq 0$. Un-firing no edit never repairs it, as $c < 1$, and un-firing both always does. If un-firing either one alone repaired it, then $\|y\|_p \leq c\|x + y\|_p$ and $\|x\|_p \leq c\|x + y\|_p$, so $\|x\|_p + \|y\|_p \leq 2c\|x + y\|_p < \|x + y\|_p$, a contradiction. So there is a nonempty set $\mathrm{Req}(q) \subseteq \mathrm{Fire}_{\tau^0}(q)$ such that q is repaired iff every edit of $\mathrm{Req}(q)$ is un-fired; for lineage size 1 it is the firing set, and for lineage size 0 it is empty and q is always repaired. Let the instance I' have the pairs that τ^0 covers and, for every complaint, the coordinate $\zeta_i(q)$ under the edits of $\mathrm{Req}(q)$ and $-\infty$ under the others. For $\tau \geq \tau^0$ the Scope-Cut value of τ in I' is its value in the tolerance form; and replacing any τ by its coordinatewise maximum with τ^0 changes no uncovered set of I' and loses no certified complaint. □

So at lineage size 2 the tolerance criterion is polynomial below $c = \frac{1}{2}$ and NP-hard from $c = \frac{1}{2}$ on: the disjunction "un-fire one edit or the other" is what breaks the cut, as for anonymous complaints (Theorem B.25).

Proposition B.32 (Completion of the classification). *For the tolerance form with an unbounded number of complaints the following hold.*

(a) For every p and every $c \in [0, 1)$ it is NP-hard on complaints of lineage size 3 with nonzero damage.

(b) *For $c < \frac{1}{2}$ and $\lambda > 0$, on complaints of lineage size at most 2 it is a* Scope-Cut *instance plus a constant, also when complaints of lineage size 2 have zero damage.*
(c) *Hence, unless* P = NP*, it is polynomial exactly for lineage size at most* 1*, and for lineage size* 2 *with $c < \frac{1}{2}$.*

Proof. (a) Add to the table of Proposition B.31(a) one edit z with the value -1 and $\tau_z^0 = 0$, whose own request has the coordinate 1 and under which every complaint has the coordinate 1. Then $\mathrm{Fire}_{\tau^0}(q_e) = \{x, y, z\}$ and the damage is $1+1-1 = 1$. Un-firing z on a complaint would uncover its own request, so z fires in every feasible repair. The residual is 1 if neither x nor y is un-fired, 0 if exactly one is, and -1 if both are: q_e is c-repaired iff exactly one end point of e is un-fired, for every $c \in [0, 1)$. With $U := \{u \in V : \tau_u > 1\}$ and $\lambda := 1/(|V| + 1)$ the value is the number of edges between U and $V \setminus U$ minus $\lambda|U|$, and $\lambda|U| < 1$. So the optimum, rounded up, is the size of a maximum cut, which is NP-hard to compute [4].

(b) A complaint q of lineage size 2 with the contributions x and $-x \neq 0$ has the damage 0 and is repaired iff the residual is 0, that is, iff both or neither of its two edits i, j are un-fired; if both contributions are 0, it is always repaired. With $z_i := \mathbf{1}[\tau_i > \zeta_i(q)]$, $\mathbf{1}[z_i = z_j] = 1 - z_i - z_j + 2z_iz_j$. So the term $\omega(q)\mathbf{1}[z_i = z_j]$ of the objective is the constant $\omega(q)$, plus the weight $2\omega(q)$ of a complaint that has the coordinates of q under i and j and $-\infty$ elsewhere, which is certified iff both edits are un-fired, minus λ times the weight of two pairs of $\boldsymbol{Q}^+$, one of each edit, at the coordinates $\zeta_i(q)$ and $\zeta_j(q)$ and of weight $\omega(q)/\lambda$, which are uncovered iff their edit is un-fired on q. Adding these complaints and pairs to the instance I' of Proposition B.31(b) gives a Scope-Cut instance with the same values up to the constant. In its network the two pairs have sink arcs of capacity $\omega(q)$, so the same minimum cut also serves $\lambda = 0$.

(c) Lineage size at most 1 is the case $\mathrm{Req}(q) = \mathrm{Fire}_{\tau^0}(q)$ of the proof of Proposition B.31(b), for every c; a complaint of lineage size 1 with zero damage is always repaired. Lineage size 2 with $c < \frac{1}{2}$ is (b). The hard cases are Proposition B.31(a), for lineage size 2 and $c \geq \frac{1}{2}$, and (a), for lineage size 3. □

The classification is a worst case over the values of the edits. If a complaint has nonzero damage and every nonempty proper subfamily of its contributions sums to a vector of norm above c times the norm of the damage, then only un-firing its whole lineage repairs it, and it is an ordinary complaint of Scope-Cut at any lineage size. So at $c = 0$ the hardness at lineage size 3 needs contributions that cancel exactly, as in the gadget of (a).

B.5.5 *Richer predicate classes.*

Conjunctions of two thresholds. Let every edit i have two fixed coordinates ζ_i, ζ_i' and two thresholds, and fire at a key k iff $\zeta_i(k) \geq \tau_i$ and $\zeta_i'(k) \geq \tau_i'$. A complaint of $\boldsymbol{Q}^-$ is certified iff no edit fires on it, a pair (i, q') is covered iff edit i fires on q', and Φ_λ is as before. Let P_i and P_i' be the sets of the first and of the second coordinates of the pairs of edit i. Edit i *touches* a complaint q if $\zeta_i(q) \geq \min P_i$ and $\zeta_i'(q) \geq \min P_i'$.

Lemma B.33 (Normal form). *For all feasible thresholds there are feasible thresholds with $\tau_i \in P_i$ and $\tau_i' \in P_i'$ for every i and an objective that is at least as large.*

Proof. Feasibility gives $\tau_i \leq \zeta_i(q_i) \in P_i$, so the smallest element of P_i that is at least τ_i exists; raise τ_i to it, and likewise τ_i'. Thresholds only rise, so certified complaints stay certified. A covered pair has a first coordinate in P_i of at least the old threshold, hence of at least the new one, and likewise for the second; so it stays covered. □

Proposition B.34 (Conjunctions: polynomial cases). *For conjunctions of two thresholds the following hold.*
(a) *If every pair must be covered ($\boldsymbol{Q}^+ = \emptyset$), the thresholds $\tau_i = \min P_i$, $\tau_i' = \min P_i'$ are optimal.*
(b) *If every complaint must be certified, the problem splits into T problems with one edit each; it is feasible iff for no edit i a complaint has both coordinates at least the componentwise minimum of the pairs of $\boldsymbol{Q}_i^!$.*
(c) *If every complaint is touched by at most one edit, then on the grid of Lemma B.33 the objective is a sum of one term per edit plus the weight of the complaints that no edit touches, and the problem is solved by T searches over the grids $P_i \times P_i'$.*

Proof. (a) The feasible thresholds are those with $\tau_i \leq \min P_i$ and $\tau_i' \leq \min P_i'$, no pair can be given up, and raising thresholds keeps complaints certified. (b) Whether edit i fires on a complaint involves the thresholds of edit i only, and the weight of the uncovered pairs is a sum over the edits. Among the thresholds that cover the pairs of $\boldsymbol{Q}_i^!$, the componentwise minimum of these pairs fires on the fewest keys. (c) On the grid, $\tau_i \geq \min P_i$ and $\tau_i' \geq \min P_i'$, so an edit fires only on complaints that it touches. A complaint that only edit i touches is certified iff edit i does not fire on it. □

Proposition B.35 (Conjunctions: two edits per complaint are hard; the status of Thm. 3.7 of the paper). *Let (V, E) be a graph with maximum degree* $\deg^*$*, let $\lambda \geq \deg^* + 1$ be an integer and $M := \lambda + 1$. Consider the table with one edit per vertex u, with its own request at the coordinates $(2, 2)$ and two pairs of $\boldsymbol{Q}^+$ of weight* 1*, π_u at $(0, 2)$ and π_u' at $(2, 0)$; a complaint q_u^H of weight M with the coordinates $(1, 1)$ under u; and for every edge $\{u, v\}$ two complaints of weight* 1*, q_{uv} with $(1, 2)$ under u and $(2, 1)$ under v, and q_{vu} with $(2, 1)$ under u and $(1, 2)$ under v. Under every edit that is not mentioned a complaint has the coordinates $(-1, -1)$.*
(a) *$\max \Phi_\lambda = |V| + \mathrm{maxcut}(V, E)$, and from all feasible thresholds one computes in polynomial time a cut with at least $\Phi_\lambda - |V|$ edges.*
(b) *Replacing q_u^H by M and each of π_u, π_u' by λ copies of unit weight gives the same objective with unit weights and the multiplier* 1*. Every complaint is touched by at most two edits, and all coordinates are in $\{-1, 0, 1, 2\}$.*
(c) *Hence, for conjunctions of two thresholds, deciding whether $\max \Phi_\lambda$ is at least a given integer is NP-complete, and the gain $\max \Phi_\lambda$ has no polynomial-time approximation scheme unless* P = NP*: thresholds with $\Phi_\lambda \geq \varrho \max \Phi_\lambda$ yield a cut with at least $(1 - 5(1 - \varrho))$ maxcut edges on graphs without isolated vertices.*

Proof. (a) By Lemma B.33 let all thresholds be in $\{0, 2\}$, and put $A_u := \mathbf{1}[\tau_u = 2]$, $B_u := \mathbf{1}[\tau_u' = 2]$. Then π_u is uncovered iff A_u, and π_u' iff B_u; q_u^H is certified iff $A_u \vee B_u$; q_{uv} is certified iff $A_u \wedge B_v$, and q_{vu} iff $B_u \wedge A_v$. Vertex u contributes $M\,\mathbf{1}[A_u \vee B_u] - \lambda(A_u + B_u)$, which is 0 for $(A_u, B_u) = (0, 0)$, 1 for $(1, 0)$ and $(0, 1)$, and $1 - \lambda$ for $(1, 1)$. *Lower bound.* For $S \subseteq V$ let $(A_u, B_u) = (1, 0)$ on S and $(0, 1)$

off S. Exactly one complaint of each edge between S and $V \setminus S$ is certified and none of another edge, so $\Phi_\lambda = |V| + \mathrm{cut}(S)$. *Upper bound.* Changing a vertex from $(1,1)$ to $(1,0)$ gains λ and loses at most the complaints of its edges that need B_u, one per incident edge, at most $\deg^* < \lambda$. Changing $(0,0)$ to $(1,0)$ gains 1 and loses nothing. So the objective does not decrease on the way to thresholds in which every vertex is in the state $(1,0)$ or $(0,1)$, and these have the value $|V| + \mathrm{cut}(\{u : A_u = 1\})$. All steps are polynomial.

(b) A pair of weight λ and multiplier 1 counts like a pair of weight 1 and multiplier λ. A complaint is touched by an edit iff both coordinates are at least 0.

(c) Membership in NP follows from Lemma B.33. For hardness take $\lambda := |E| + 1$ in (b). On graphs without isolated vertices $|V| \le 2|E| \le 4\,\mathrm{maxcut}$, so $\max \Phi_\lambda \le 5\,\mathrm{maxcut}$, and by (a) the cut obtained from thresholds with $\Phi_\lambda \ge \varrho \max \Phi_\lambda$ has at least $\mathrm{maxcut} - (1-\varrho)\max\Phi_\lambda$ edges. The maximum cut problem has no polynomial-time approximation scheme unless P = NP [36], even on cubic graphs [4]. □

With Proposition B.34(c) this is a sharp boundary: complaints touched by one edit are polynomial, complaints touched by two edits are hard. It also shows why Scope-Cut is polynomial: with one threshold per edit, "the edit is silent on the complaint" is a chain condition, and a second threshold turns it into a disjunction. The statements above concern the *gain* Φ_λ, which is a difference and can be negative. For the *cost* $\omega(\mathcal{Q}^-) - \Phi_\lambda$, which is the weight of the complaints that are not certified plus λ times the weight of the uncovered pairs, the following holds.

Proposition B.36 (Conjunctions: both objective forms; the status of Thm. 3.7 of the paper). *For conjunctions of two thresholds the following hold.*

(a) On cubic graphs take the constant parameters $\lambda = 4$ and $M = 5$ in Proposition B.35. Then the largest gain and the least cost are

$$|V| + \mathrm{maxcut} \le \tfrac{7}{3}\,\mathrm{maxcut} \qquad \text{and} \qquad 7|V| - \mathrm{maxcut} \le \tfrac{25}{3}\,\mathrm{maxcut},$$

and the cut that Proposition B.35(a) computes from feasible thresholds misses at most $\max\Phi_\lambda - \Phi_\lambda$ edges of a maximum cut, which is their cost minus the least cost. So neither the gain nor the cost has a polynomial-time approximation scheme unless P = NP, already when, in the version with unit weights, every edit has 9 pairs and touches 11 complaints and every complaint is touched by at most two edits.

(b) The cost is 3-approximable in polynomial time.

Proof. (a) A cubic graph has $|E| = \frac{3}{2}|V|$ and a cut with at least $|E|/2$ edges, so $|V| \le \frac{4}{3}\,\mathrm{maxcut}$. The complaints have the total weight $M|V| + 2|E| = 8|V|$, and $\max\Phi_\lambda = |V| + \mathrm{maxcut}$ by Proposition B.35(a), which also gives the statement on the cut. A cut with at least $(1-\varepsilon')\,\mathrm{maxcut}$ edges for every $\varepsilon' > 0$ would be a polynomial-time approximation scheme for maximum cut on cubic graphs, which does not exist unless P = NP [4]. In the version with unit weights of Proposition B.35(b) an edit has its own request and four copies of each of its two pairs, and it touches the five copies of its heavy complaint and the two complaints of each of its three edges.

(b) Take the linear program with variables $A_{i,x}$ and $B_{i,x}$ in $[0,1]$ for the events $\tau_i > x$ and $\tau'_i > x$, nonincreasing in x along the coordinates of edit i and fixed to 0 where a pair of $\mathcal{Q}^!_i$ requires it; a variable φ_q for every complaint q, with $\varphi_q \ge 1 - A_{i,\zeta_i(q)} - B_{i,\zeta'_i(q)}$ for every edit i; a variable $\varphi_{i,q'}$ for every pair $(i,q') \in \mathcal{Q}^+$, with $\varphi_{i,q'} \ge A_{i,\zeta_i(q')}$ and $\varphi_{i,q'} \ge B_{i,\zeta'_i(q')}$; and the objective

$$\sum_{q\in\mathcal{Q}^-} \omega(q)\,\varphi_q + \lambda \sum_{(i,q')\in\mathcal{Q}^+} \omega(i,q')\,\varphi_{i,q'}.$$

Thresholds give a point of the program whose objective is their cost. Round every A and B to 1 iff it is at least $\frac{1}{3}$. The rounded variables are monotone and describe feasible thresholds. A complaint that is not certified has an edit with both rounded variables 0, so $\varphi_q > \frac{1}{3}$; an uncovered pair has a rounded variable 1, so $\varphi_{i,q'} \ge \frac{1}{3}$. So the cost of the rounded thresholds is at most three times the optimum of the program, which is at most the least cost. □

Half-spaces with a free detector. Let the detector of an edit be repaired as well: a repair of edit i is a pair (ψ', τ') of a vector and a real threshold, the edit fires at k iff $\langle\psi', k\rangle \ge \tau'$, and the own key keeps its coordinate and fires, $\langle\psi', k_i\rangle = s_i$ and $\tau' \le s_i$. Here coordinates are not normalised.

Proposition B.37 (Free detector: the realisable case; standard). *Let each edit have reported keys, some of which must fire and some of which must not, with a margin $\mu > 0$: $\langle\psi', z\rangle \ge \tau' + \mu$ for the former, $\langle\psi', x\rangle \le \tau' - \mu$ for the latter, and $\tau' + \mu \le s_i$.*

(a) Every constraint involves one edit, so minimising $\sum_i \|\psi'_i - \psi_i\|_2^2$ splits into T convex quadratic programs.

(b) Let $Y = [y_1, \dots, y_L]$ hold the own key $y_1 = k_i$ and reported keys, and let $\pi \in \mathbb{R}^L$ prescribe their coordinates, with $\pi_1 = s_i$. The system $Y^\top\psi' = \pi$ is solvable iff π is orthogonal to $\mathrm{null}\,Y$, that is, iff the prescribed coordinates satisfy every linear dependency among the keys. The solution closest to ψ_i is then $\psi_i + Y(Y^\top Y)^+(\pi - Y^\top\psi_i)$, at the squared distance $(\pi - Y^\top\psi_i)^\top(Y^\top Y)^+(\pi - Y^\top\psi_i)$.

(c) Hence if the own key and the reported keys of an edit are linearly independent, which requires at most $d-1$ reported keys, every labelling of the reported keys as firing or not firing is realisable, for every margin and every $\tau' \le s_i - \mu$.

Proof. (a) is clear. (b) The system is solvable iff π is in the range of $Y^\top$, which is the orthogonal complement of the null space of Y. The solutions form an affine subspace with the direction space $\mathrm{null}\,Y^\top$. Let $\psi^* := \psi_i + Y(Y^\top Y)^+(\pi - Y^\top\psi_i)$. The vector $\pi - Y^\top\psi_i$ is in the range of $Y^\top$, which is that of $Y^\top Y$, so $Y^\top\psi^* = \pi$. Then ψ^* is a solution and $\psi^* - \psi_i$, a vector in the range of Y, is orthogonal to $\mathrm{null}\,Y^\top$, so $\|\psi' - \psi_i\|_2^2 = \|\psi' - \psi^*\|_2^2 + \|\psi^* - \psi_i\|_2^2$ for every solution ψ'; and $\|\psi^* - \psi_i\|_2^2$ is the stated quadratic form because $(Y^\top Y)^+ Y^\top Y (Y^\top Y)^+ = (Y^\top Y)^+$. (c) In (b) let the entries of π after the first be $\tau' + \mu$ or $\tau' - \mu$; $\mathrm{null}\,Y = 0$. □

The case $\pi = (s_i, 0, \dots, 0)$ of (b) is the detector repair of the paper: the detector closest to ψ_i that keeps the own coordinate and has the coordinate 0 on the reported keys of wrong-answer complaints. It exists iff no linear dependency among these keys and the own key contradicts the prescribed coordinates, in particular if they are linearly independent, and it is then the closed form of (b), a projection of ψ_i. This is the polynomial side of the next theorem.

Theorem B.38 (Free detector: the agnostic case). *Consider one edit with its own key k_i and $s_i > 0$, why-not keys, which are*

satisfied *by a repair* (ψ', τ') *iff the edit fires on them,* $\langle \psi', z \rangle \geq \tau'$*, and keys of wrong-answer complaints, which are satisfied iff it does not,* $\langle \psi', x \rangle < \tau'$*. For all* $\varepsilon, \varepsilon' > 0$ *it is NP-hard to distinguish the sets of reported keys of which some repair satisfies a fraction of at least* $1 - \varepsilon$ *from those of which no repair satisfies a fraction of* $\frac{1}{2} + \varepsilon'$*. This holds with the threshold free and with the threshold fixed at any* $\tau^* \leq s_i$*. Consequently the least number of reported keys that are not satisfied has no polynomial-time constant-factor approximation unless* P = NP*.*

Proof. An instance of half-space maximum agreement consists of two multisets $S^+, S^- \subseteq \{-1, 1\}^N$ of examples; the agreement of a vector $\phi \in \mathbb{R}^N$ and a real ϑ is the number of $\xi \in S^+$ with $\langle \phi, \xi \rangle \geq \vartheta$ plus the number of $\xi \in S^-$ with $\langle \phi, \xi \rangle < \vartheta$. For all $\varepsilon, \varepsilon' > 0$ it is NP-hard to distinguish instances in which some (ϕ, ϑ) agrees with a fraction $1 - \varepsilon$ of the examples from instances in which none agrees with a fraction $\frac{1}{2} + \varepsilon'$ [28] (Defs. 2.2 and 2.3 and Theorem 8.1 in the numbering of the authors' full version). Let the key dimension be $N + 2$, map an example ξ to the key $(\xi, 1, 0)$, a why-not key if $\xi \in S^+$ and the key of a wrong-answer complaint if $\xi \in S^-$, and let the own key be e_{N+2}. For (ϕ, ϑ) and a threshold τ^* let $\psi' := (\phi, \tau^* - \vartheta, s_i)$. Then $\langle \psi', k_i \rangle = s_i$ and $\langle \psi', (\xi, 1, 0) \rangle = \langle \phi, \xi \rangle + \tau^* - \vartheta$, which is at least τ^* iff $\langle \phi, \xi \rangle \geq \vartheta$. Conversely, for a repair (ψ', τ') let ϕ be the first N coordinates of ψ' and $\vartheta := \tau' - \psi'_{N+1}$; then $\langle \psi', (\xi, 1, 0) \rangle \geq \tau'$ iff $\langle \phi, \xi \rangle \geq \vartheta$. So the number of satisfied keys equals the agreement, in both directions, with a free and with a fixed threshold. □

The read-out equation only fixes one further coordinate. By Proposition B.37(c) the hardness needs at least d reported keys for the edit, or dependent keys.

Disjunctions of detectors. Let an edit carry several detectors, each a threshold on its own coordinate, and fire where one of them fires. A complaint is certified iff no detector of any edit fires on it.

Proposition B.39 (Disjunctions with designated coverage). *Assign to every pair one detector of its edit, and call the pair covered iff this detector fires on its query. Then the problem is* Scope-Cut *with one edit per detector, and Theorems B.14 and B.16 hold for it, a detector without a pair of* $Q^!$ *being allowed the threshold* $+\infty$*. For all thresholds the objective with designated coverage is at most the objective with true coverage, in which any detector of its edit may cover a pair; the two optima coincide if no pair can be covered by a detector other than its own.*

Proof. The conditions are those of Scope-Cut with detectors in the place of edits: a complaint is certified iff every detector has its threshold above the coordinate of the complaint, and a pair is covered iff the threshold of its detector is at most its coordinate. A pair that is covered by its own detector is covered. □

Theorem B.40 (Disjunctions with true coverage; the status of Thm. 3.7 of the paper). *Let a pair be covered iff some detector of its edit fires on its query.*

(a) Maximising the gain Φ_λ *is NP-hard and has no polynomial-time approximation within* $|Q^-|^{1-\varepsilon}$ *unless* P = NP*, already with two detectors per edit, one pair besides the own request per edit and unit weights, if this pair is in* $Q^!$ *or if it is in* Q^+ *and* $\lambda \geq 1$*.*

(b) On the same tables the least cost $\omega(Q^-) - \max \Phi_\lambda$ *is the vertex cover number of the graph. So the bound of* (a) *is a statement about the gain; for the cost only the hardness of vertex cover follows.*

(c) The optimum is the maximum over the designations of the optimum with designated coverage. So choosing the designation is NP-hard, and the problem is solved by one minimum cut per designation of the pairs that more than one detector can cover.

Proof. (a) Let (V, E) be a graph with ordered vertices. The complaints are the vertices, of weight 1. Each edge $e = \{u, v\}$, $u < v$, gets one edit with two detectors and a pair π_e of weight 1. Under the first detector the own request has the coordinate 3, π_e has 1, the complaint u has 2 and the other complaints 0. Under the second detector π_e has 1, the complaint v has 2, and the own request and the other complaints have 0. The own request is covered by the first detector with a threshold in $(2, 3]$, which fires on no complaint. Covering π_e needs a threshold of at most 1 on one of the two detectors, which then fires on u or on v. So if every π_e is covered, the certified complaints form an independent set. Conversely an independent set I stays certified if each π_e is covered through an end point outside I: through u, with a threshold in $(0, 1]$ on the first detector and a threshold above 2 on the second, or through v, with a threshold in $(0, 1]$ on the second detector and a threshold in $(2, 3]$ on the first. So $\max \Phi_\lambda$ is the independence number if the pairs π_e are in $Q^!$. If they are in Q^+ and $\lambda \geq 1$, covering an uncovered π_e through the second detector changes the objective by at least $\lambda - 1 \geq 0$, so the optimum is the same. The independence number is NP-hard to approximate within $|V|^{1-\varepsilon}$ [74].

(b) The complaints that are not certified form a vertex cover, and every vertex cover arises.

(c) As in Proposition B.27: for fixed thresholds, the designated objective is at most the true one, with equality for a designation that sends every covered pair to a detector that fires on it. □

For $\lambda < 1$ the optimum of the table can exceed the independence number, so the condition of (a) is needed.

Remark B.41 (Disjunctions: the cost form). With at most L detectors per edit the cost $\omega(Q^-) - \Phi_\lambda$ under true coverage is $(L + 1)$-approximable. Write $\zeta_{i,e}$ and $\tau_{i,e}$ for the coordinate and the threshold of detector e of edit i. Take the linear program with variables $\chi_{i,e,x} \in [0, 1]$ for the events $\tau_{i,e} \leq x$, nondecreasing in x along the coordinates under the detector; a variable $\varphi_q \in [0, 1]$ for every complaint q, with $\varphi_q \geq \chi_{i,e,\zeta_{i,e}(q)}$ for every detector; a variable $\varphi_{i,q'} \in [0, 1]$ for every pair (i, q'), with $\varphi_{i,q'} + \sum_e \chi_{i,e,\zeta_{i,e}(q')} \geq 1$, the sum being over the detectors of edit i, and $\varphi_{i,q'} = 0$ for the pairs of $Q^!$; and the objective

$$\sum_{q \in Q^-} \omega(q)\, \varphi_q + \lambda \sum_{(i,q') \in Q^+} \omega(i, q')\, \varphi_{i,q'}.$$

Thresholds give a point of the program whose objective is their cost. Round every χ to 1 iff it is at least $1/(L + 1)$; the rounded variables are monotone and describe thresholds. A complaint that is not certified then has $\varphi_q \geq 1/(L+1)$, and a pair that is not covered has all its at most L variables below $1/(L + 1)$, so $\varphi_{i,q'} > 1/(L + 1)$; in particular the pairs of $Q^!$ stay covered (for them $\varphi_{i,q'} = 0$, so one of the at most L variables is at least $1/L$). So the cost of the rounded

thresholds is at most $L+1$ times the optimum of the program, which is at most the least cost.

B.6 Lineage size

Let $\mathcal{S}$ be a scoped log with arbitrary responses and values, q a wrong-answer complaint, $F := \mathrm{Fire}_{\mathcal{S}}(q)$ and $\mathrm{wd}(q) = |F|$ its lineage size. The *contribution* of edit i at a position t of $x^0(q)$ is $w_{i,t} := g_i(\zeta_i(k_t))\, v_i$ if edit i fires at t, and 0 otherwise; $w_i(q)$ is the vector of the contributions of edit i at all positions of $x^0(q)$. For the deployed log $\mathcal{S}^E$ every edit fires at every position, $w_{i,t} = \Delta_i k_t$, and $\mathrm{wd}(q) = T$.

Proposition B.42 (Lineage size). *Let $\mathcal{S}_R$ be $\mathcal{S}$ with the edits of $R \subseteq [T]$ silenced.*

(a) At a position t of $x^0(q)$ the module of $\mathcal{M}[\mathcal{S}_R]$ returns $W_0 k_t + \sum_{i\in F\setminus R} w_{i,t}$: on the pre-edit trace of q a rollback acts through $R \cap F$ only.

(b) Exact restoration. *If $R \supseteq F$, then q is certified. If no nonempty subfamily of the vectors $w_i(q)$, $i \in F$, sums to zero, the sets R that restore the pre-edit module output at every position of $x^0(q)$ are exactly the sets $R \supseteq F$, and the smallest has $\mathrm{wd}(q)$ elements. For the deployed log, under the same condition on all T vectors, only $R = [T]$ restores.*

(c) Tolerance. *Whether the residual $\sum_{i\in F\setminus R} w_i(q)$ meets a tolerance criterion depends on $R \cap F$, so a smallest R is a subset of F and has at most $\mathrm{wd}(q)$ elements. For complaints of one position this is* Min-Rollback$_{p,c}$ *on $\mathrm{wd}(q)$ contributions, and every instance of* Min-Rollback$_{p,c}$ *arises so (Proposition B.29): for unbounded lineage size the problem is strongly NP-hard and not $\mathrm{wd}(q)^{1-\varepsilon}$-approximable at any $c < 1$.*

Proof. (a) Keys and roles of the fixed sequence $x^0(q)$ do not depend on the log (H1). An edit outside F fires at no position of $x^0(q)$, and a silenced edit fires nowhere; the rest is the module map of Def. 2.2. (b) If $R \supseteq F$, no edit fires on $x^0(q)$. By (a), the output is restored at every position iff $\sum_{i\in F\setminus R} w_i(q) = 0$, which under the condition forces $F \setminus R = \emptyset$. (c) follows from (a) and Proposition B.29. □

So on a log that has scopes the parameter in the place of the length T is the lineage size: the hard instances of rollback live on the edits that fire on the complaint. This compares two logs, not two repairs: on one scoped log a rollback and the least repair change the same edits (Theorem B.76(a)). The condition of (b) is needed: two edits that fire with the contributions $+1$ and -1 have lineage size 2 and need no rollback. The paper reports the widths that we measure.

B.7 Calibration from benign queries

Complaints repair what was reported; calibration bounds what was not. Let $q_1, \dots, q_m$ be benign calibration queries and q_{m+1} a fresh benign query. For a query q let $\zeta(q) := (\zeta_i(q))_{i\in[T]} \in \mathbb{R}^T$ be its coordinate vector; it is taken on the pre-edit trace and does not depend on the thresholds.

(E) The coordinate vectors $\zeta(q_1), \dots, \zeta(q_{m+1})$ are *exchangeable*: their joint law is invariant under permutations of the queries.

Independent and identically distributed queries are exchangeable, and so are draws without replacement from a finite pool. (E) holds if the queries are exchangeable and the detectors, the role sets and W_0 are fixed without reference to them. It is the only assumption on the queries; nothing is assumed about the dependence among the T coordinates of one query. The calibration queries are a sample of benign queries, not the reported complaints. By Proposition B.1(a),

$$\mathrm{Fire}_\tau(q) = \emptyset \implies \hat{y}(q;\tau) = y_0(q). \tag{LC}$$

Let $\tau^0 \in (0,\infty)^T$ be reference thresholds that are fixed without the calibration queries. The *score* of a query q is the largest of the ratios $\zeta_i(q)/\tau_i^0$,

$$\mathrm{sc}(q) := \max_{i\in[T]} \zeta_i(q)/\tau_i^0.$$

A *shared scale* is a number $\eta \in [-\infty,\infty]$, read as the threshold vector with $\tau_i = (\eta\, \tau_i^0)^+$: edit i fires at a key k iff $\zeta_i(k)/\tau_i^0 > \eta$. Hence

$$\mathrm{Fire}_{(\eta\tau^0)^+}(q) \neq \emptyset \iff \mathrm{sc}(q) > \eta. \tag{1}$$

B.7.1 One shared scale.

Definition B.43 (Conformal scale). *Let $\alpha \in (0,1)$, $\kappa := \lceil (1-\alpha)(m+1) \rceil \in \{1,\dots,m+1\}$ and $l := \lfloor \alpha(m+1) \rfloor = m+1-\kappa$. The* conformal scale $\hat{\eta}$ *is the κ-th smallest of $\mathrm{sc}(q_1), \dots, \mathrm{sc}(q_m)$ if $\kappa \le m$, and $+\infty$ if $\kappa = m+1$. The* conformal thresholds *are $\hat{\tau}_i := (\hat{\eta}\, \tau_i^0)^+$.*

So $\hat{\eta} < \infty$, and some edit can fire, iff $l \ge 1$, iff $m \ge 1/\alpha - 1$.

Lemma B.44 (Rank lemma). *Let $\xi_1, \dots, \xi_N$ be exchangeable real random variables, $N = m+1 \ge 2$ and $\kappa \in \{1,\dots,m\}$. Let $\xi_{(\kappa:m)}$ be the κ-th smallest of $\xi_1, \dots, \xi_m$ and $\xi_{(1:N)} \le \cdots \le \xi_{(N:N)}$ the order statistics of all N variables. Then*

$$\{\xi_N > \xi_{(\kappa:m)}\} = \{\xi_N > \xi_{(\kappa:N)}\}, \qquad \mathbb{P}(\xi_N > \xi_{(\kappa:m)}) \le \frac{N-\kappa}{N}, \tag{2}$$

$$\{\xi_N < \xi_{(\kappa:m)}\} = \{\xi_N < \xi_{(\kappa+1:N)}\}, \quad \mathbb{P}(\xi_N < \xi_{(\kappa:m)}) \le \frac{\kappa}{N}, \tag{3}$$

and both inequalities are equalities if the ξ_h are almost surely pairwise distinct.

Proof. (2). If $\xi_N > \xi_{(\kappa:m)}$, the κ smallest of $\xi_1, \dots, \xi_m$ are below ξ_N, so at least κ of the N variables are below ξ_N and $\xi_{(\kappa:N)} < \xi_N$. Conversely, if $\xi_{(\kappa:N)} < \xi_N$, then κ of the N variables are below ξ_N; none of them is ξ_N, so κ of $\xi_1, \dots, \xi_m$ are below ξ_N and $\xi_{(\kappa:m)} < \xi_N$. As $\xi_{(\kappa:N)}$ is a symmetric function of $(\xi_1, \dots, \xi_N)$, exchangeability gives $\mathbb{P}(\xi_h > \xi_{(\kappa:N)}) = \mathbb{P}(\xi_N > \xi_{(\kappa:N)})$ for every h, so $\mathbb{P}(\xi_N > \xi_{(\kappa:N)}) = \mathbb{E}\,\#\{h \le N : \xi_h > \xi_{(\kappa:N)}\}/N$. At least κ of the variables are at most $\xi_{(\kappa:N)}$, so the count is at most $N-\kappa$, and it equals $N-\kappa$ when the values are distinct.

(3). If $\xi_N < \xi_{(\kappa:m)}$, then $m-\kappa+1$ of $\xi_1, \dots, \xi_m$ exceed ξ_N, so the $(m-\kappa+1)$-th largest of all N variables, which is $\xi_{(\kappa+1:N)}$, exceeds ξ_N. Conversely, if $\xi_N < \xi_{(\kappa+1:N)}$, then $N-\kappa = m-\kappa+1$ of the N variables exceed ξ_N, none of them is ξ_N, so the $(m-\kappa+1)$-th largest of $\xi_1, \dots, \xi_m$, which is $\xi_{(\kappa:m)}$, exceeds ξ_N. The rest is as before: at most κ of the variables are below $\xi_{(\kappa+1:N)}$, with equality for distinct values. □

Theorem B.45 (Shared scale: bound on false firing; the status of Thm. 3.6 of the paper). *Assume* (E) *and use the conformal thresholds of Definition B.43.*

(a) Upper bound, ties allowed.

$$\mathbb{P}\big(\mathrm{Fire}_{\hat{\tau}}(q_{m+1}) \neq \emptyset\big) = \mathbb{P}\big(\mathrm{sc}(q_{m+1}) > \hat{\eta}\big) \le \frac{l}{m+1} \le \alpha.$$

(b) Exactness. *If the scores are almost surely pairwise distinct, the probability equals* $l/(m+1)$*, and* $\alpha - \frac{1}{m+1} < \frac{l}{m+1} \le \alpha$.

(c) The conventions cannot be relaxed. *With the gate "edit i fires iff* $\zeta_i/\tau_i^0 \ge \hat{\eta}$*" at the same scale,* (b) *still holds but* (a) *fails: if all scores are equal, the fresh query fires with probability* 1 *whenever* $l \ge 1$*. Without distinctness the lower bound of* (b) *fails: if all scores are equal, the strict gate fires with probability* 0.

(d) Monotonicity. *The bound of* (a) *holds for every threshold vector* $\tau \ge \hat{\tau}$*, whatever else it depends on, in particular for the coordinatewise maximum of* $\hat{\tau}$ *and any other vector. It can fail for* $\tau \le \hat{\tau}$.

(e) Answers. $\mathbb{P}\big(\hat{y}(q_{m+1};\hat{\tau}) \ne y_0(q_{m+1})\big) \le \mathbb{P}\big(\mathrm{sc}(q_{m+1}) > \hat{\eta}\big) \le \alpha$.

All probabilities are joint over the calibration queries and the fresh query.

Proof. (a) By (1), some edit fires on q_{m+1} iff $\mathrm{sc}(q_{m+1}) > \hat{\eta}$. If $\kappa = m+1$ the event is empty and $l = 0$. Otherwise apply (2) to the scores, which are exchangeable by (E): the bound is $(N-\kappa)/N = l/(m+1)$, and $l \le \alpha(m+1)$. (b) The last sentence of Lemma B.44, which is trivial for $l = 0$; and $l > \alpha(m+1) - 1$. (c) For distinct scores the events that the fresh score is at least, and that it exceeds, the κ-th smallest calibration score differ by a null event. If all scores equal one number and $\kappa \le m$, then $\hat{\eta}$ is this number; the fresh score is always at least $\hat{\eta}$ and never exceeds it, while $\alpha - 1/(m+1) > 0$ whenever $\alpha(m+1) > 1$. (d) If $\tau \ge \hat{\tau}$, then $\mathrm{Fire}_\tau(q) \subseteq \mathrm{Fire}_{\hat{\tau}}(q)$ for every q, so the event of (a) shrinks. For the failure take $T = 1$, $\tau^0 = 1$, distinct scores and $\kappa \ge 2$, and lower the threshold to the next smaller calibration score: by (2) the fresh query then fires with probability $(l+1)/(m+1)$, which exceeds α because $l + 1 > \alpha(m+1)$. (e) By (LC) the event that the answer differs from $y_0(q_{m+1})$ is contained in the event that some edit fires; apply (a). □

Theorem B.46 (Shared scale: calibration-conditional form). *Let the scores* $\mathrm{sc}(q_1), \dots, \mathrm{sc}(q_{m+1})$ *be independent with the distribution function* F*, and let*

$$\mathrm{FF}_m := \mathbb{P}\big(\mathrm{sc}(q_{m+1}) > \hat{\eta} \,\big|\, q_1, \dots, q_m\big) = 1 - F(\hat{\eta}) \qquad (F(+\infty) := 1)$$

be the false-firing rate of the calibrated log on fresh benign queries.

(a) *For every* $\varepsilon \in (0,1)$, $\mathbb{P}(\mathrm{FF}_m > \varepsilon) \le \mathbb{P}(\mathrm{Bin}(m,\varepsilon) \le l-1)$*, where* $\mathrm{Bin}(m,\varepsilon)$ *is a binomial variable with* m *trials of success probability* ε.

(b) *If* F *is continuous,* (a) *is an equality for every* ε*:* $\mathrm{FF}_m = 0$ *if* $l = 0$*, and otherwise* FF_m *has the law of the* l*-th smallest of* m *independent uniform variables, a beta law with mean* $l/(m+1)$.

(c) *For every* $\delta \in (0,1)$, $\mathbb{P}\big(\mathrm{FF}_m > \alpha + \sqrt{\ln(1/\delta)/(2m)}\big) \le \delta$.

(d) *Independence cannot be weakened to exchangeability: if the scores are a uniformly random ordering of* $m+1$ *distinct numbers, then* $\mathrm{FF}_m \in \{0,1\}$ *and* $\mathbb{P}(\mathrm{FF}_m > \varepsilon) = l/(m+1)$ *for every* $\varepsilon \in (0,1)$.

Proof. If $l = 0$, then $\hat{\eta} = +\infty$, $\mathrm{FF}_m = 0$, and the right side of (a) is the probability of an empty event. Let $l \ge 1$, so that $\kappa = m+1-l \le m$ and, by independence, $\mathrm{FF}_m = 1 - F(\hat{\eta})$ with $\hat{\eta}$ the κ-th smallest calibration score.

(a) Fix ε and let $x^* := \inf\{x : F(x) \ge 1-\varepsilon\}$. By right-continuity $F(x^*) \ge 1-\varepsilon$, and $F(x) < 1-\varepsilon$ for $x < x^*$. Hence $\mathrm{FF}_m > \varepsilon$ iff $F(\hat{\eta}) < 1-\varepsilon$, iff $\hat{\eta} < x^*$, iff at least κ calibration scores are below x^*. Their number is binomial with m trials and success probability $\mathbb{P}(\mathrm{sc}(q_1) < x^*) \le 1-\varepsilon$. The probability that a binomial variable is at least κ is nondecreasing in its success probability, so $\mathbb{P}(\mathrm{FF}_m > \varepsilon) \le \mathbb{P}(\mathrm{Bin}(m, 1-\varepsilon) \ge \kappa) = \mathbb{P}(\mathrm{Bin}(m,\varepsilon) \le m-\kappa)$, and $m - \kappa = l - 1$.

(b) If F is continuous, $\mathbb{P}(\mathrm{sc}(q_1) < x^*) = 1-\varepsilon$ and the inequality is an equality. The l-th smallest of m independent uniform variables is at most ε iff at least l of them are, which has the probability $\mathbb{P}(\mathrm{Bin}(m,\varepsilon) \ge l) = \mathbb{P}(\mathrm{FF}_m \le \varepsilon)$.

(c) Let $\varepsilon := \alpha + x$ with $x := \sqrt{\ln(1/\delta)/(2m)}$; if $\varepsilon \ge 1$ there is nothing to prove. Since $l - 1 \le \alpha(m+1) - 1 \le \alpha m$, Hoeffding's inequality for the binomial law gives $\mathbb{P}(\mathrm{Bin}(m,\varepsilon) \le l-1) \le \mathbb{P}(\mathrm{Bin}(m,\varepsilon) - m\varepsilon \le -mx) \le e^{-2mx^2} = \delta$.

(d) Given the calibration scores, the fresh score is the remaining number, so FF_m is the indicator of the event that it exceeds $\hat{\eta}$, which has the probability $l/(m+1)$ by Theorem B.45(b). For $l \ge 1$ and ε close to 1 this exceeds the bound of (a), which tends to 0. □

Status. None of this is new. Theorem B.45(a)–(c) is the split-conformal test for outliers: the conformal p-value of [10] rejects exactly when the score is strictly above the κ-th smallest calibration score and is valid without a continuity assumption; see also [7, Sec. 4.4] and, for the matching lower bound under continuity, [44, Thm. 2.2]. Theorem B.46(a)–(c) is [67, Thm. 1(a) and Cor. 1(a)]. Part (e) is a corollary of Proposition B.1. The treatment of ties and of the gate is exposition, which the definition of a scoped log makes necessary.

What the statements do not say. The guarantee is for queries that are exchangeable with the calibration queries. It is marginal: it holds neither conditionally on the query nor under a shift of the query distribution. The reference τ^0 must not be fitted on the calibration queries. The guarantee is for the log as calibrated: after an edit is appended, the calibration queries have to be scored again. Nothing is said about why-not pairs; $\hat{\eta}$ can exceed the scores of the own requests (Sec. B.7.3).

Proposition B.47 (A threshold that can be stored). *Let coordinates, references and scales be floating-point numbers, let* $\mathrm{fl}(a/\tau_i^0)$ *be the correctly rounded quotient, let the score be computed as the largest of these quotients, and let the gate of the calibrated log be "edit i fires at the coordinate a iff* $\mathrm{fl}(a/\tau_i^0) > \hat{\eta}$*". Let* $a_i^\star$ *be the largest floating-point number* a *with* $\mathrm{fl}(a/\tau_i^0) \le \hat{\eta}$*. Then for every floating-point number* a*,*

$$\mathrm{fl}(a/\tau_i^0) > \hat{\eta} \iff a > a_i^\star \iff a \ge \mathrm{succ}(a_i^\star),$$

where succ *is the next larger floating-point number. Hence, with the stored thresholds* $\hat{\tau}_i := \mathrm{succ}(a_i^\star)$*, the comparison* $\zeta_i \ge \hat{\tau}_i$ *of Def. 2.2 is the calibrated comparison of ratios, and* (1) *and Theorem B.45*(a), (d), (e) *hold for the computed numbers.*

Proof. Correctly rounded division by a positive number is nondecreasing in the numerator. So the set of the floating-point numbers a with $\mathrm{fl}(a/\tau_i^0) > \hat{\eta}$ is closed under taking larger numbers, and its complement has the largest element $a_i^\star$. □

The successor of the rounded product $\mathrm{fl}(\hat{\eta}\,\tau_i^0)$ is not this threshold in general. In double precision, for the coordinate 0.03 and the reference 0.41, the product of the rounded quotient with 0.41 rounds to a number below 0.03. If all queries have this coordinate ($T = 1$), every score equals $\hat{\eta}$: the comparison of ratios never fires, and the

comparison with the rounded product fires on the fresh query with probability 1, for every α. The number $a_i^\star$ is found from $\mathrm{fl}(\hat\eta\tau_i^0)$ by stepping to neighbouring floating-point numbers. The rank κ has to be computed exactly as well: evaluated in floating point, $\lceil (m+1)(1-\alpha)\rceil$ can be one rank too high, which is conservative but invalidates (b) and the condition $m \ge 1/\alpha - 1$.

Remark B.48 (The stored threshold of Algorithm 1). Algorithm 1 (the rule of Proposition B.58 with unit references) compares differences: the score of a query is the largest of the computed margins $\mathrm{fl}(\zeta_i(q) - b_i)$, written $\zeta_i(q) \ominus b_i$ in the paper, and edit i is to fire at a coordinate a iff $\mathrm{fl}(a - b_i) > \hat\eta$. (i) Coordinates, baselines and scores are numbers of one floating-point format (double precision in the implementation), which contains $\pm\infty$; a coordinate that is not a number is an error. A coordinate is finite, or $-\infty$ for a query without an eligible position of the edit, which is never compared; a baseline is finite when some query of the first block has an eligible position for the edit, which is assumed (Def. 2.3 assumes one for every query). With these, no subtraction of infinities occurs, and a stored threshold $+\infty$ means that the edit never fires at a finite coordinate. (ii) Correctly rounded subtraction of a constant is non-decreasing in its first argument, so Proposition B.47 holds with $\mathrm{fl}(a - b_i)$ in the place of $\mathrm{fl}(a/\tau_i^0)$, by the same proof: the stored threshold is the successor, in the format, of the largest number $a_i^\star$ with $\mathrm{fl}(a_i^\star - b_i) \le \hat\eta$. If $\hat\eta = +\infty$, the stored thresholds are $+\infty$ and no edit can fire; if every finite number satisfies the inequality, the successor is $+\infty$ as well; if none does (a baseline $-\infty$), the stored threshold is the least finite number, and edit i fires at every finite coordinate. (iii) The successor of the rounded sum $\mathrm{fl}(b_i + \hat\eta)$ is not this threshold in general: under it the query that defines $\hat\eta$ can fire. (iv) The bounds of Proposition B.58(a) and of Sec. B.7.5 are applied to the computed scores. These are a fixed function of the coordinate vectors, hence exchangeable, or independent with one law, whenever the vectors are; the rank is computed in integer arithmetic; and ties, which rounding can create, only make the bounds conservative (Lemma B.44, Theorem B.61(a)).

B.7.2 The least repair of benign queries.

Definition B.49 (Running maxima). $\bar\zeta_i := \max_{h\le m}\zeta_i(q_h)$ *and* $\hat\tau_i := \bar\zeta_i^+$*: edit i fires on q iff $\zeta_i(q) > \bar\zeta_i$.*

By Theorem B.8, $\hat\tau$ is the least repair of the queries $q_1,\dots,q_m$ from the thresholds $-\infty$, and its coordinatewise maximum with τ^0 is their least repair from initial thresholds τ^0. For the $m+1$ queries let $H_i \subseteq [m+1]$ be the set of the maximisers of coordinate i. If $|H_i| = 1$, its element is the *holder* of coordinate i. Let $H \subseteq [m+1]$ be the set of all holders: the queries, among the $m+1$, that alone attain the maximum of some coordinate. Clearly $|H| \le \min(T, m+1)$.

Theorem B.50 (Least repair of benign queries; the status of Thm. 3.6 of the paper). *Assume* (E) *and use the thresholds of Definition B.49.*

(a) One edit. $\mathbb{P}\big(i \in \mathrm{Fire}_{\hat\tau}(q_{m+1})\big) = \mathbb{P}(|H_i| = 1)/(m+1) \le 1/(m+1)$.

(b) Number of firing edits.

$$\mathbb{E}\,|\,\mathrm{Fire}_{\hat\tau}(q_{m+1})| = \frac{\sum_{i\le T}\mathbb{P}(|H_i|=1)}{m+1} \le \frac{T}{m+1}.$$

(c) Some edit fires.

$$\mathbb{P}\big(\mathrm{Fire}_{\hat\tau}(q_{m+1}) \ne \emptyset\big) = \frac{\mathbb{E}\,|H|}{m+1} \le \min\Big(1, \frac{T}{m+1}\Big).$$

(d) No ties. *Suppose that almost surely no coordinate takes the same value on two of the $m+1$ queries. Then* (a) *and* (b) *are equalities, and the probability of* (c) *is at least $1/(m+1)$. Moreover:*

(i) if the T sequences $(\zeta_i(q_h))_{h\le m+1}$, $i \in [T]$, are mutually independent, then $\mathbb{P}\big(\mathrm{Fire}_{\hat\tau}(q_{m+1}) \ne \emptyset\big) = 1 - \big(\frac{m}{m+1}\big)^T$, and $(1-\frac{1}{e})\min(1, \frac{T}{m+1}) \le 1 - (\frac{m}{m+1})^T \le \min(1, \frac{T}{m+1})$;

(ii) if $\zeta_i(q_h) = f_i(\xi_h)$ with strictly increasing functions f_i and exchangeable, almost surely distinct reals $\xi_1,\dots,\xi_{m+1}$, then the fresh query fires all T edits with probability $1/(m+1)$ and no edit otherwise;

(iii) for all $m, T \ge 1$ there is an exchangeable law without ties with $\mathbb{P}\big(\mathrm{Fire}_{\hat\tau}(q_{m+1}) \ne \emptyset\big) = \min(1, \frac{T}{m+1})$.

(e) Monotonicity. *The upper bounds of* (a)–(c) *hold for every threshold vector $\tau \ge \hat\tau$, whatever else it depends on.*

(f) Answers. $\mathbb{P}\big(\hat y(q_{m+1};\hat\tau) \ne y_0(q_{m+1})\big) \le \min\big(1, T/(m+1)\big)$.

All probabilities are joint over $q_1,\dots,q_{m+1}$.

Proof. (a) Edit i fires on q_{m+1} iff $\zeta_i(q_{m+1}) > \bar\zeta_i$, that is, iff $H_i = \{m+1\}$. For $h \le m+1$ the event $H_i = \{h\}$ is the image of the event $H_i = \{m+1\}$ under the transposition of the queries h and $m+1$, so by (E) these events have the same probability; they are disjoint, and their union is the event $|H_i| = 1$. (b) Sum (a) over i. (c) Some edit fires on q_{m+1} iff $m+1 \in H$. A permutation of the queries permutes H in the same way, so by (E) $\mathbb{P}(h \in H)$ does not depend on h, and $(m+1)\mathbb{P}(m+1 \in H) = \sum_{h\le m+1}\mathbb{P}(h \in H) = \mathbb{E}\,|H|$. Finally $|H| \le \min(T, m+1)$. (d) Without ties $|H_i| = 1$ almost surely, which gives the equalities, and $|H| \ge 1$. (i) The event $\zeta_i(q_{m+1}) > \bar\zeta_i$ is a function of the i-th sequence, so the T events are independent, each of probability $1/(m+1)$ by (a). For the inequalities put $y := T/(m+1)$: $(1-\frac{1}{m+1})^T \le e^{-y}$, and $1-e^{-y}$ is concave and increasing in y and vanishes at 0, so it is at least $y(1-\frac{1}{e})$ on $[0,1]$ and at least $1-\frac{1}{e}$ on $[1,\infty)$; the upper bound is (c). (ii) $\zeta_i(q_{m+1}) > \bar\zeta_i$ iff $\xi_{m+1} > \max_{h\le m}\xi_h$, for every i at once; apply (a) to one coordinate. (iii) Let the vectors $\pi^h \in \mathbb{R}^T$, $h = 1,\dots,m+1$, have the coordinates $\pi_i^h := \mathbf{1}[h=i] - h/(m+2)$. In a coordinate $i \le \min(T, m+1)$ the unique maximum is at $h = i$, in the other coordinates it is at $h = 1$, and all values of a coordinate are distinct; so $|H| = \min(T, m+1)$. Let the coordinate vectors of the queries be a uniformly random ordering of $\pi^1,\dots,\pi^{m+1}$, which are coordinate vectors of queries by Lemma B.19(b). This law is exchangeable, and (c) gives the value $\min(T, m+1)/(m+1)$. (e) If $\tau \ge \hat\tau$, then $\mathrm{Fire}_\tau(q) \subseteq \mathrm{Fire}_{\hat\tau}(q)$ for every q. (f) follows from (LC) and (c). □

Corollary B.51 (Sample size for a false-firing level). *Let $\alpha \in (0,1)$ and $\Lambda := \ln\frac{1}{1-\alpha} \in \big(\alpha, \frac{\alpha}{1-\alpha}\big)$.*

(a) The least repair has $\mathbb{P}(\mathrm{Fire}_{\hat\tau}(q_{m+1}) \ne \emptyset) \le \alpha$ for every law that satisfies (E) *iff $m \ge T/\alpha - 1$.*

(b) For independent coordinates without ties it has this property iff $m \ge m_{\mathrm{ind}}(T,\alpha) := \big\lceil 1/\big((1-\alpha)^{-1/T} - 1\big)\big\rceil$, and $T/\Lambda - 1 < m_{\mathrm{ind}}(T,\alpha) \le \lceil T/\Lambda\rceil$.

Proof. (a) Sufficiency is Theorem B.50(c). Necessity is (d)(iii): $\min(1, \frac{T}{m+1}) \le \alpha < 1$ forces $T/(m+1) \le \alpha$. (b) By (d)(i) the condition is $(1+\frac{1}{m})^T \le (1-\alpha)^{-1}$, that is, $m \ge 1/(e^x - 1)$ with

$x := \Lambda/T$. For $x \in (0,1)$, $x < e^x - 1 < x/(1-x)$, hence $\frac{1}{x} - 1 < \frac{1}{e^x - 1} < \frac{1}{x}$; for $x \geq 1$ the lower bound is trivial and the upper bound still holds. Finally $\alpha < \Lambda < \alpha/(1-\alpha)$. □

Theorem B.50 is the record argument for exchangeable sequences. Part (c) is the mistake bound of the closure algorithm [30, Lemma 4] for a class that is closed under intersection, applied to lower orthants, whose closure is the vector of the coordinatewise maxima. Equivalently, it is the Bonferroni rule for conformal p-values of the single edits at the level $1/(m+1)$ [48]. The identity with $|H|$ and the attainment (d)(iii) are elementary sharpenings. Without ties, a fresh query fires $T/(m+1)$ edits in expectation under any dependence between the edits; the dependence only decides how these firings are distributed over the queries, one query that fires all edits in (d)(ii) or T different queries that fire one edit each in (d)(iii). The law of (d)(iii) is a random ordering of a finite population. As in Theorem B.45, the statements are marginal and require exchangeability, and by (e) they cover a threshold vector with further upper caps, for instance to keep pairs covered, only if it still dominates $\hat{\tau}$.

B.7.3 Why-not pairs and a high-probability form. Let (i_h, q'_h), $h = 1, \ldots, m'+1$, be why-not pairs, the last one fresh, with the *scores* $\mathrm{sc}(i_h, q'_h) := \zeta_{i_h}(q'_h)/\tau^0_{i_h}$. Under a shared scale η a pair is covered iff its score exceeds η.

(E′) The scores of the $m'+1$ pairs are exchangeable.

Nothing is assumed about the dependence between the benign queries and the pairs.

Definition B.52 (Two-sided rule). *Let $\alpha, \alpha' \in (0,1)$, $\kappa := \lceil (1-\alpha)(m+1) \rceil$ and $\kappa' := \lfloor \alpha'(m'+1) \rfloor$. Let η_- be the κ-th smallest benign calibration score, or $+\infty$ if $\kappa = m+1$, and η_+ the κ'-th smallest calibration score of the pairs, or $-\infty$ if $\kappa' = 0$. The rule outputs the scale $\hat{\eta} := \eta_-$ and* raises a flag *if $\eta_- \geq \eta_+$.*

Theorem B.53 (Two-sided calibration of a shared scale). *Assume* (E) *and* (E′).

(a) For every scale $\tilde{\eta}$, possibly depending on all calibration data, with $\tilde{\eta} < \eta_+$ almost surely, the fresh pair is not covered with a probability of at most $\kappa'/(m'+1) \leq \alpha'$.

(b) Joint statement. *The fresh benign query fires with a probability of at most α, and the event that the rule raises no flag and the fresh pair is not covered under $\hat{\eta}$ has a probability of at most α'.*

(c) The scales η that meet both calibrated requirements, $\eta \geq \eta_-$ and $\eta < \eta_+$, form the interval $[\eta_-, \eta_+)$, which is nonempty iff the rule raises no flag; each of its points satisfies (b).

(d) Without the flag, $\hat{\eta}$ carries no guarantee for pairs: if every score of a pair lies below every benign score, the fresh pair is never covered.

Proof. (a) If $\kappa' = 0$, no such $\tilde{\eta}$ exists. Let $\kappa' \geq 1$. The fresh pair is not covered iff its score is at most $\tilde{\eta}$, hence only if it is below η_+, which has a probability of at most $\kappa'/(m'+1)$ by (3), applied to the scores of the pairs with κ' in the place of κ; and $\kappa' \leq \alpha'(m'+1)$. (b) The first bound is Theorem B.45(a). If no flag is raised, $\hat{\eta} = \eta_- < \eta_+$, so the event is contained in the event that the score of the fresh pair is below η_+; apply (a). (c) The interval is nonempty iff $\eta_- < \eta_+$. For a scale $\eta \geq \eta_-$ the firing event shrinks, and for $\eta < \eta_+$ the argument of (b) applies. (d) Then the score of the fresh pair is below the smallest benign calibration score, which is at most $\hat{\eta}$. □

The bound of (b) is joint; the probability that the fresh pair is missed *given* that no flag is raised is not controlled. Example: let the benign score be the constant 0, the score of a pair 0 with probability $\frac{9}{10}$ and 1 with probability $\frac{1}{10}$, $m = m' = 1$ and $\alpha = \alpha' = \frac{1}{2}$. Then $\eta_- = 0$ and η_+ is the calibration score of the pair. No flag is raised with probability $\frac{1}{10}$; the event of (b) has the probability $\frac{9}{100} \leq \alpha'$, but given that no flag is raised the fresh pair is missed with probability $\frac{9}{10}$.

Proposition B.54 (Least repair: high-probability form). *Let the coordinate vectors of benign queries be independent and identically distributed. For a threshold vector τ let $\mathrm{FF}(\tau) := \mathbb{P}(\mathrm{Fire}_\tau(q) \neq \emptyset)$ for a fresh benign query q, and let $\mathrm{FF}_m(\tau)$ be the share of the calibration queries on which some edit fires. Then $\mathrm{FF}_m(\tau) = 0$ iff $\tau \geq \hat{\tau}$, with $\hat{\tau}$ the least repair of Definition B.49; the largest value of* FF *on these vectors is $\mathrm{FF}(\hat{\tau})$; $\mathbb{E}\,\mathrm{FF}(\hat{\tau}) \leq \min(1, \frac{T}{m+1})$; and for $\varepsilon \in (0,1)$*

$$\mathbb{P}\big(\mathrm{FF}(\hat{\tau}) > \varepsilon\big) \;\leq\; T\Big(1 - \frac{\varepsilon}{T}\Big)^m \;\leq\; T\, e^{-m\varepsilon/T}.$$

So if $m \geq \frac{T}{\varepsilon} \ln \frac{T}{\delta}$, then with probability at least $1-\delta$ every threshold vector on which no calibration query fires has a false-firing rate of at most ε.

Proof. No calibration query fires iff $\zeta_i(q_h) < \tau_i$ for all i and h, iff $\tau_i > \bar{\zeta}_i$ for all i, iff $\tau \geq \hat{\tau}$; and FF is nonincreasing in every threshold. The bound on the expectation is Theorem B.50(c), since $\mathbb{E}\,\mathrm{FF}(\hat{\tau})$ is the probability that a fresh query fires. For the tail bound let $\zeta = (\zeta_i)_{i \in [T]}$ be the coordinate vector of a fresh query and, for each i, $a_i := \inf\{a : \mathbb{P}(\zeta_i > a) \leq \varepsilon/T\}$, a real number. By right-continuity $\mathbb{P}(\zeta_i > a_i) \leq \varepsilon/T$, and $\mathbb{P}(\zeta_i \geq a_i) \geq \varepsilon/T$. On the event that $\bar{\zeta}_i \geq a_i$ for all i, $\mathrm{FF}(\hat{\tau}) = \mathbb{P}(\exists i : \zeta_i > \bar{\zeta}_i \mid q_1, \ldots, q_m) \leq \sum_i \mathbb{P}(\zeta_i > a_i) \leq \varepsilon$. Hence the event $\mathrm{FF}(\hat{\tau}) > \varepsilon$ is contained in the union over i of the events $\bar{\zeta}_i < a_i$, and $\mathbb{P}(\bar{\zeta}_i < a_i) = \mathbb{P}(\zeta_i < a_i)^m \leq (1-\varepsilon/T)^m$. □

Neither statement of this part is new. Theorem B.53 is the rank lemma applied to the lower tail of the scores of the pairs, a lower conformal quantile. Proposition B.54 is the standard argument for learning an orthant from positive examples: the least repair is the hypothesis of the closure algorithm [30] on lower orthants, the smallest orthant that contains the calibration vectors.

B.7.4 One shared scale against per-edit thresholds. A *calibration rule* is a measurable map $\hat{\tau}(\cdot)$ that sends the coordinate vectors of m benign calibration queries to a threshold vector in Θ^T; it is *symmetric* if it does not depend on the order of the queries. Edit i is *alive* on a sample if its threshold is below $+\infty$. For a class $\mathcal{P}$ of laws of $(\zeta(q_1), \ldots, \zeta(q_{m+1}))$ the rule is *α-valid over $\mathcal{P}$* if $\mathbb{P}(\mathrm{Fire}_{\hat{\tau}}(q_{m+1}) \neq \emptyset) \leq \alpha$ for every law in $\mathcal{P}$. We use $\mathcal{P}_{\mathrm{ex}}$, all exchangeable laws; $\mathcal{P}_{\mathrm{prod}}$, independent identically distributed vectors with independent coordinates; and $\mathcal{P}_{\mathrm{ind}} \subset \mathcal{P}_{\mathrm{prod}}$, in addition with continuous coordinates. Three types of rules:

- a *shared scale* relative to a fixed reference τ^0: $\hat{\tau}_i = (\hat{\eta}\,\tau^0_i)^+$ with $\hat{\eta}$ a function of the calibration scores; examples are the conformal scale and the *fit-clean scale* $\hat{\eta} = \max_{h \leq m} \mathrm{sc}(q_h)$, the least repair within the family of shared scales;
- an *interpolating* rule: $\hat{\tau}_i \leq \bar{\zeta}^+_i$ for all i on every sample, so that no threshold is placed above the largest calibration coordinate of its own edit; the least repair of Definition B.49 is the largest interpolating rule;

- a *coordinatewise* rule: $\hat{\tau}_i$ is a function of the calibration coordinates of edit i alone, that is, of $\zeta_i(q_1), \dots, \zeta_i(q_m)$.

Theorem B.55 (Sample sizes of the two families). *Let $\alpha \in (0,1)$, $T \geq 1$, and Λ and $m_{\mathrm{ind}}(T,\alpha)$ as in Corollary B.51.*

(a) One shared scale needs $m \geq 1/\alpha - 1$, and no rule needs less.
  (i) The conformal scale is α-valid over $\mathcal{P}_{\mathrm{ex}}$ for every m, and it keeps the edits alive iff $m \geq 1/\alpha - 1$.
  (ii) Let $m < 1/\alpha - 1$ and let a calibration rule be α-valid over $\mathcal{P}_{\mathrm{ex}}$. Then for every sample and every edit i, edit i is alive for at most a fraction $\alpha(m+1) < 1$ of the orderings of the sample. If the rule is symmetric, every edit is silenced on every sample.
  (iii) If a rule is α-valid over all laws of independent identically distributed vectors, then under every such law and for every i, $\mathbb{P}(\hat{\tau}_i < +\infty) \leq \alpha(m+1)(1+\frac{1}{m})^m < e\,\alpha(m+1)$.

(b) The least repair, interpolating rules and coordinatewise rules need m of the order T/α.
  (i) The least repair is α-valid over $\mathcal{P}_{\mathrm{ex}}$ iff $m \geq T/\alpha - 1$.
  (ii) Under an interpolating rule every query fires at least the edits that it fires under the least repair. Hence under every exchangeable law the fresh query fires with a probability of at least $\mathbb{E}\,|H|/(m+1)$, and no interpolating rule is α-valid over $\mathcal{P}_{\mathrm{ind}}$ unless $m \geq m_{\mathrm{ind}}(T,\alpha) > T/\Lambda - 1$.
  (iii) Let a coordinatewise rule keep every edit alive on every sample. If it is symmetric and α-valid over $\mathcal{P}_{\mathrm{ex}}$, then $m \geq T/\alpha - 1$. If it is α-valid over $\mathcal{P}_{\mathrm{prod}}$, then $m + 1 > T/(e\Lambda)$.

(c) Same sample. *For the fit-clean shared scale and the least repair computed from the same calibration queries, every query fires under the shared scale only edits that it fires under the least repair. Over $\mathcal{P}_{\mathrm{ex}}$ their false-firing probabilities are at most $\frac{1}{m+1}$ and $\min(1, \frac{T}{m+1})$, and both bounds are attained. Over $\mathcal{P}_{\mathrm{ind}}$ they equal $\frac{1}{m+1}$ and $1 - (\frac{m}{m+1})^T$.*

Proof. (a)(i) Theorem B.45(a); $\hat{\eta} < \infty$ iff $\lfloor \alpha(m+1) \rfloor \geq 1$.

(a)(ii) Fix vectors $\pi^1, \dots, \pi^m \in \mathbb{R}^T$, an edit i and a real B, and let $\pi^B \in \mathbb{R}^T$ have the i-th coordinate B. Let the coordinate vectors of the $m+1$ queries be a uniformly random ordering of $\pi^1, \dots, \pi^m, \pi^B$; this law is exchangeable. With probability $\frac{1}{m+1}$ the fresh vector is π^B; then the calibration vectors are a uniformly random ordering of the sample, and edit i fires on the fresh query iff $B \geq \hat{\tau}_i$. Validity gives that the threshold of edit i is at most B for at most a fraction $\alpha(m+1)$ of the orderings, for every B; letting $B \to \infty$, it is below $+\infty$ for at most this fraction. For a symmetric rule the fraction is 0 or 1.

(a)(iii) Fix a law P_0 on $\mathbb{R}^T$, $\varepsilon \in (0,1)$ and B, and consider independent vectors, each drawn from P_0 with probability $1-\varepsilon$ and equal to π^B with probability ε. With probability $(1-\varepsilon)^m\varepsilon$ the calibration vectors are drawn from P_0 and the fresh one equals π^B, so $\varepsilon(1-\varepsilon)^m P_0^m(\hat{\tau}_i \leq B) \leq \alpha$. Let $B \to \infty$ and take $\varepsilon = \frac{1}{m+1}$, for which $\varepsilon(1-\varepsilon)^m = \frac{1}{m+1}(1+\frac{1}{m})^{-m}$.

(b)(i) Corollary B.51(a). (b)(ii) The gate is antitone in each threshold, and the thresholds of an interpolating rule are at most those of the least repair. The rest is Theorem B.50(c), (d)(i) and Corollary B.51(b).

(b)(iii) *Exchangeable laws.* Let $N := m+1$ and $T' := \min(T, N)$. Choose reals π_i^h for $h \leq N$, $i \leq T$, $h \neq i$, arbitrarily. For $i \leq T'$ let a_i be the threshold that the rule gives to edit i on the calibration coordinates $(\pi_i^h)_{h \neq i}$, which is below $+\infty$ by assumption, and choose a real $\pi_i^i \geq a_i$. Let the coordinate vectors be a uniformly random ordering of $\pi^1, \dots, \pi^N$. If the fresh vector is π^i with $i \leq T'$, the calibration values of coordinate i are the π_i^h, $h \neq i$, in some order, so by symmetry the threshold of edit i is a_i and edit i fires. Hence $\alpha \geq T'/N$; as $\alpha < 1$, this forces $T < N$ and $T/(m+1) \leq \alpha$. *Product laws.* Let $a_i < +\infty$ be the threshold of edit i on the calibration coordinates $(0, \dots, 0)$, choose reals $B_i \geq a_i$, $B_i \neq 0$, and let the coordinates be independent, coordinate i being independent over the queries with the value B_i with probability ε and 0 otherwise. For each i, with probability $\varepsilon(1-\varepsilon)^m$ all calibration values of coordinate i are 0 and the fresh value is B_i, in which case edit i fires; these events are independent over i. So $1 - (1 - \varepsilon(1-\varepsilon)^m)^T \leq \alpha$, hence $T\varepsilon(1-\varepsilon)^m \leq \Lambda$. With $\varepsilon = \frac{1}{m+1}$, $\varepsilon(1-\varepsilon)^m > \frac{1}{e(m+1)}$.

(c) If edit i fires on q under the fit-clean scale, then $\zeta_i(q)/\tau_i^0 > \max_h \mathrm{sc}(q_h) \geq \bar{\zeta}_i/\tau_i^0$, so $\zeta_i(q) > \bar{\zeta}_i$. The bounds are (2) with $\kappa = m$ and Theorem B.50(c); they are attained by the last sentence of Lemma B.44 and by Theorem B.50(d)(iii). Over $\mathcal{P}_{\mathrm{ind}}$ the scores are almost surely distinct, and the second value is Theorem B.50(d)(i). □

In (a)(ii), (a)(iii) and (b)(iii) the laws live on $\mathbb{R}^T$: the statements concern validity for every law of the coordinate vectors. Every finite family of vectors in $\mathbb{R}^T$ is the family of the coordinate vectors of the queries of some log with $d \geq T$ (Lemma B.19(b)).

Definition B.56 (Heterogeneous logs). *Fix $T \geq 2$, a reference $\tau^0 \in (0,\infty)^T$, numbers $0 < x_{\mathrm{lo}} \leq x_{\mathrm{hi}} < 1$ and levels $\varphi_2, \dots, \varphi_T \in (0,1]$ with*

$$\varphi_i/\tau_i^0 < x_{\mathrm{lo}}/\tau_1^0 \qquad (i = 2, \dots, T). \tag{4}$$

The log has the key space $\mathbb{R}^T$ with the orthonormal basis $e_1, \dots, e_T$, arbitrary W_0 and write vectors $u_i \neq 0$, the detectors $\psi_i = e_i$ and the own keys e_i, so that $s_i = 1$ and $\zeta_i(k) = \langle e_i, k \rangle$. Queries have one eligible position. Benign queries *have the key $x\,e_1$ with $x \in [x_{\mathrm{lo}}, x_{\mathrm{hi}}]$ drawn from a law π. The* why-not pairs *are $(1, q')$ with the key e_1 and, for $i \geq 2$, (i, q') with the key $\varphi_i e_i$, each with probability $1/T$.*

Edit 1 is loud: unrelated keys reach the fraction $x \geq x_{\mathrm{lo}}$ of its own coordinate. The edits $2, \dots, T$ are quiet, but their rephrasings reach only the fraction φ_i of the own coordinate. For every reference τ^0 there are such numbers; for a constant reference (4) reads $\varphi_i < x_{\mathrm{lo}}$. For a threshold vector let FF be the probability that a benign query fires and COV the probability that a why-not pair is covered.

Proposition B.57 (Adapting to each edit can be necessary). *For every log of Definition B.56:*

(a) The per-edit thresholds $(x_{\mathrm{hi}}^+, \varphi_2, \dots, \varphi_T)$ have FF $= 0$ *and* COV $= 1$*: no edit fires on a benign query, which is therefore answered as before the edits* (LC)*, and every pair is covered.*

(b) The least repair of $m \geq 1$ benign queries that are exchangeable with the fresh one is $((\max_h x_h)^+, 0^+, \dots, 0^+)$, where $x_h e_1$ is the key of q_h. It covers every pair on every sample, and the fresh benign query fires with a probability of at most $\frac{1}{m+1}$.

(c) For every shared scale $\eta \in [-\infty, \infty]$: FF $= 1$ *or* COV $\leq \frac{1}{T}$*. Hence every shared scale with* FF < 1 *covers at most a fraction $1/T$ of the pairs; the conformal scale has* COV $\leq 1/T$ *on every sample; and every scale $\hat{\eta}$ that depends on the data has $\mathbb{E}\,\mathrm{COV}(\hat{\eta}) \leq \frac{1}{T} + \mathbb{E}\,\mathrm{FF}(\hat{\eta})$.*

Proof. A benign key $x\,e_1$ has the coordinates $(x, 0, \dots, 0)$; the pair of edit 1 has the coordinate 1 and the pair of edit $i \geq 2$ has the coordinate φ_i under its edit. (a) $x \leq x_{\text{hi}}$ does not reach x_{hi}^+, and $0 < \varphi_i$; $1 > x_{\text{hi}}$ and $\varphi_i \geq \varphi_i$. (b) The calibration coordinates are $(x_h, 0, \dots, 0)$. $1 > x_{\text{hi}} \geq \max_h x_h$ and $\varphi_i > 0$ give the coverage. A fresh benign query fires iff its x exceeds $\max_h x_h$; apply Theorem B.50(a). (c) The score of a benign query is x/τ_1^0, and the scores of the pairs are $1/\tau_1^0$ and φ_i/τ_i^0. If FF < 1 under η, then $x/\tau_1^0 \leq \eta$ with positive probability, so $\eta \geq x_{\text{lo}}/\tau_1^0 > \varphi_i/\tau_i^0$ for all $i \geq 2$ by (4): the $T-1$ pairs of the quiet edits are not covered. The conformal scale is a calibration score or $+\infty$, hence at least x_{lo}/τ_1^0. Finally COV $\leq \frac{1}{T} + \mathbf{1}[\text{FF} = 1] \leq \frac{1}{T} + \text{FF}$ for every scale; take expectations. □

Proposition B.58 (Two-block rule; Thm. 3.6(a), (c) of the paper). *Let the calibration queries be split, without looking at them, into a block $\mathcal{B}_1$ of $m_1 \geq 1$ queries and a block $\mathcal{B}_2$ of m_2 queries, $m = m_1 + m_2$. Let $\tau^0 \in (0,\infty)^T$ be fixed, let $b_i := \max_{q\in\mathcal{B}_1} \zeta_i(q)$ be the* baseline *of edit i, and let*

$$\text{sc}_2(q) := \max_{i\in[T]} (\zeta_i(q) - b_i)/\tau_i^0.$$

Let $\kappa := \lceil (1-\alpha)(m_2+1) \rceil$ and $l_2 := \lfloor \alpha(m_2+1) \rfloor$, let $\hat\eta$ be the κ-th smallest of the scores $\text{sc}_2(q)$, $q \in \mathcal{B}_2$, or $+\infty$ if $\kappa = m_2 + 1$, and let $\hat\tau_i := (b_i + \hat\eta\,\tau_i^0)^+$.

(a) Under (E), $\mathbb{P}\big(\text{Fire}_{\hat\tau}(q_{m+1}) \neq \emptyset\big) \leq l_2/(m_2+1) \leq \alpha$, *and every edit is alive iff $m_2 \geq 1/\alpha - 1$: the sample size of one shared scale, whatever T.*

(b) The rule gives per-edit thresholds, and it is neither interpolating nor coordinatewise.

(c) On every log of Definition B.56 with exchangeable benign queries, the rule covers every pair with a probability of at least $1 - \binom{m_2}{l_2}/\binom{m}{l_2}$, which is at least $1 - 2^{-l_2}$ for $m_1 = m_2$ and does not depend on T.

Proof. (a) Edit i fires on a query q iff $\zeta_i(q) > b_i + \hat\eta\tau_i^0$, so some edit fires iff $\text{sc}_2(q) > \hat\eta$. A permutation of the queries of $\mathcal{B}_2$ and the fresh query leaves the joint law of all coordinate vectors invariant, and the function sc_2 depends on the first block only; so the scores sc_2 of the queries of $\mathcal{B}_2$ and of the fresh query are exchangeable. Apply (2) as in Theorem B.45(a), with m_2 in the place of m. The scale is finite iff $l_2 \geq 1$. (b) The threshold of edit i depends on the coordinates of all edits through $\hat\eta$, and it exceeds the largest calibration coordinate of edit i whenever $\hat\eta$ is large enough. (c) On these logs $b = (b_1, 0, \dots, 0)$ with b_1 the largest x of the first block, and $\text{sc}_2(q) = \max\big((x - b_1)/\tau_1^0, 0\big)$ for a benign query with the key $x\,e_1$. So $\hat\eta \geq 0$, and $\hat\eta = 0$ unless at least l_2 queries of the second block have an x above b_1. If $\hat\eta = 0$, the thresholds are $(b_1^+, 0^+, \dots, 0^+)$ and cover every pair, as in Proposition B.57(b). If at least l_2 queries of the second block exceed every query of the first, the l_2 largest of the m values, ties broken at random, lie in the second block; by exchangeability their positions are a uniformly random set of l_2 of the m queries, so this event has the probability $\binom{m_2}{l_2}/\binom{m}{l_2}$, which for $m_1 = m_2$ is $\prod_{h<l_2} \frac{m_2-h}{2m_2-h} \leq 2^{-l_2}$. □

The split into two blocks is essential: with the baselines b_i taken over both blocks the scores of the second block and of the fresh query are no longer exchangeable, and the bound of (a) fails. The bound of (c) is void for $l_2 = 0$, when the scale is $+\infty$ and no edit can fire.

Remark B.59 (Reading of the sample sizes). The bound $T/(m+1)$ of Theorem B.50 and the sample sizes of Theorem B.55(b) are properties of the least repair, of interpolating rules and of coordinatewise rules. They are not the price of per-edit thresholds as such: the two-block rule adapts the threshold of each edit, through the baseline b_i, and is valid with the sample size of one shared scale. It is split conformal calibration with a score that is fitted on a separate block, the analogue for edits of the clustered rule between pooled and classwise calibration [16], and therefore standard. Proposition B.57 shows what one shared scale relative to a fixed reference cannot do, and Corollary B.51 what the least repair cannot do. As for one shared scale, the comparison of $(\zeta_i - b_i)/\tau_i^0$ with $\hat\eta$ has to be stored so that the stored comparison is the calibrated one (Proposition B.47, Remark B.48).

Remark B.60 (Guards and capped edits). Let a guard lower the thresholds of a set C of *capped* edits below the calibrated ones (Algorithm 1, line 7): the guarded vector τ' has $\tau_i' = \hat\tau_i$ for $i \notin C$ and $\tau_i' < \hat\tau_i$ for $i \in C$. Let τ'' equal $\hat\tau$ outside C and $+\infty$ on C. Then $\tau'' \geq \hat\tau$, and an uncapped edit fires on a query under τ' iff some edit fires on it under τ'' (coordinates are finite, so an edit with threshold $+\infty$ never fires). So the bound of Proposition B.58(a), and those of Theorems B.61(a) and B.62(a), which hold for every threshold vector above the calibrated one whatever else it depends on, bound the probability that an *uncapped* edit fires on the fresh benign query under the guarded thresholds. Nothing is bounded for the event that some edit fires, nor for the event that the answer differs from the pre-edit answer: a capped edit can fire on every benign query, and one capped edit suffices. The bounds are on firing: an edit that fires need not change the answer. Before the guard the bounds are those of the proposition; after it, capped edits have to be reported with the firing that they cause.

B.7.5 *The calibration-conditional form of the two-block rule.*

The bound of Proposition B.58(a), Theorem 3.6(a) of the paper, is marginal: it averages over the calibration sample. This part states what holds for the sample that was actually drawn. We use the two-block rule of Proposition B.58 with a free *rank parameter* $l \in \{0, \dots, m_2\}$: $\hat\eta_l$ is the $(m_2 + 1 - l)$-th smallest of the scores $\text{sc}_2(q)$, $q \in \mathcal{B}_2$, that is, their l-th largest, and $\hat\eta_0 := +\infty$; the thresholds are $\hat\tau_i^{(l)} := (b_i + \hat\eta_l\,\tau_i^0)^+$. The rule of the paper is the *default rank* $l = \lfloor \alpha(m_2+1) \rfloor$. We assume

(I) the coordinate vectors $\zeta(q_1), \dots, \zeta(q_m)$ and those of the fresh benign queries are independent with one common law, and the split into $\mathcal{B}_1$ and $\mathcal{B}_2$ does not depend on them.

For a fixed vector b of baselines let F_b be the distribution function of the score $\text{sc}_2(q) = \max_i(\zeta_i(q) - b_i)/\tau_i^0$ of a fresh benign query q. The *realised false-firing rate* of a threshold vector τ that was computed from the calibration queries is

$$\text{FF}(\tau) := \mathbb{P}\big(\text{Fire}_\tau(q_{m+1}) \neq \emptyset \;\big|\; q_1, \dots, q_m\big),$$
$$\text{FF}_l := \text{FF}(\hat\tau^{(l)}) = 1 - F_b(\hat\eta_l) \qquad (F_b(+\infty) := 1),$$

a random variable through the calibration sample; the second identity holds because some edit fires on q under $\hat\tau^{(l)}$ iff $\text{sc}_2(q) > \hat\eta_l$.

$\mathrm{Bin}(m_2, \varepsilon)$ is a binomial variable with m_2 trials of success probability ε.

Theorem B.61 (Law of the realised false-firing rate). *Assume* (I) *and let* $l \in \{1, \dots, m_2\}$*; for* $l = 0$, $\mathrm{FF}_0 = 0$.

(a) Ties allowed. *For* $\varepsilon \in (0,1)$ *let* $x_\varepsilon := \min\{x : F_b(x) \geq 1-\varepsilon\}$ *and* $p_\varepsilon := \mathbb{P}(\mathrm{sc}_2(q) \geq x_\varepsilon)$ *for the given baselines. Then, almost surely,*

$$\mathbb{P}\big(\mathrm{FF}_l > \varepsilon \,\big|\, \mathcal{B}_1\big) = \mathbb{P}\big(\mathrm{Bin}(m_2, p_\varepsilon) \leq l-1\big) \;\leq\; \mathbb{P}\big(\mathrm{Bin}(m_2, \varepsilon) \leq l-1\big),$$

and $p_\varepsilon \geq \varepsilon$. *The bound holds for every threshold vector* $\tau \geq \hat{\tau}^{(l)}$ *in the place of* $\hat{\tau}^{(l)}$*, whatever else it depends on.*

(b) Continuous scores. *If no coordinate* $\zeta_i(q)$ *of a benign query has an atom, then* (a) *is an equality for every* ε*: conditionally on* $\mathcal{B}_1$*, and hence unconditionally,* FF_l *has the beta law with the parameters* $(l, m_2 + 1 - l)$*, the law of the* l*-th smallest of* m_2 *independent uniform variables.* FF_l *is independent of* $\mathcal{B}_1$*, with*

$$\mathbb{E}\,\mathrm{FF}_l = \frac{l}{m_2+1}, \qquad \mathrm{Var}\,\mathrm{FF}_l = \frac{l\,(m_2+1-l)}{(m_2+1)^2(m_2+2)}.$$

(c) A finite test set; exchangeability suffices. *Instead of* (I) *let the coordinate vectors of the* m *calibration queries and of* N *test queries be exchangeable, and let* $\mathrm{FF}_{l,N}$ *be the share of the test queries on which some edit fires under* $\hat{\tau}^{(l)}$*. If the scores* sc_2 *of the queries of* $\mathcal{B}_2$ *and of the test queries are almost surely pairwise distinct, then for* $x = 0, \dots, N$

$$\mathbb{P}\big(N\,\mathrm{FF}_{l,N} = x \,\big|\, \mathcal{B}_1\big) = \binom{x+l-1}{l-1}\binom{N-x+m_2-l}{m_2-l}\Big/\binom{m_2+N}{N},$$

the beta-binomial law with the parameters $(N;\, l,\, m_2+1-l)$*, with mean* $N\,l/(m_2+1)$*. With ties,* $N\,\mathrm{FF}_{l,N}$ *is stochastically at most a variable with this law.*

(d) Answers. $\mathbb{P}\big(\hat{y}(q_{m+1}; \hat{\tau}^{(l)}) \neq y_0(q_{m+1}) \mid q_1, \dots, q_m\big) \leq \mathrm{FF}_l$*, and the share of the test queries of* (c) *whose answer differs from the pre-edit answer is at most* $\mathrm{FF}_{l,N}$*.*

Proof. Condition on $\mathcal{B}_1$. This fixes the baselines b and the function sc_2. By (I) the scores sc_2 of the m_2 queries of $\mathcal{B}_2$ and of the fresh queries are independent with the distribution function F_b, and $\mathrm{FF}_l = 1 - F_b(\hat{\eta}_l)$, where $\hat{\eta}_l$ is the κ-th smallest score of $\mathcal{B}_2$, $\kappa := m_2 + 1 - l \in \{1, \dots, m_2\}$.

(a) F_b is nondecreasing and right-continuous with the limits 0 and 1, and $0 < 1-\varepsilon < 1$. So the set $\{x : F_b(x) \geq 1-\varepsilon\}$ is a closed half-line $[x_\varepsilon, \infty)$ with a real x_ε, and

$$\begin{aligned} \mathrm{FF}_l > \varepsilon &\iff F_b(\hat{\eta}_l) < 1-\varepsilon \iff \hat{\eta}_l < x_\varepsilon \\ &\iff \#\{q \in \mathcal{B}_2 : \mathrm{sc}_2(q) < x_\varepsilon\} \geq \kappa \\ &\iff \#\{q \in \mathcal{B}_2 : \mathrm{sc}_2(q) \geq x_\varepsilon\} \leq l-1. \end{aligned}$$

The last number is binomial with m_2 trials and success probability p_ε. As $F_b(x) < 1-\varepsilon$ for $x < x_\varepsilon$, the left limit of F_b at x_ε is at most $1-\varepsilon$, so $p_\varepsilon \geq \varepsilon$; and the probability that a binomial variable is at most $l-1$ is nonincreasing in its success probability. For $\tau \geq \hat{\tau}^{(l)}$ the firing sets of all queries shrink, so $\mathrm{FF}(\tau) \leq \mathrm{FF}_l$.

(b) For a fixed b the score has no atom, since $\mathbb{P}(\mathrm{sc}_2(q) = x) \leq \sum_i \mathbb{P}(\zeta_i(q) = b_i + x\,\tau_i^0)$, which is 0. So F_b is continuous, $F_b(x_\varepsilon) = 1-\varepsilon$ and $p_\varepsilon = 1 - F_b(x_\varepsilon) = \varepsilon$. Hence $\mathbb{P}(\mathrm{FF}_l > \varepsilon \mid \mathcal{B}_1) = \mathbb{P}(\mathrm{Bin}(m_2, \varepsilon) \leq l-1)$, which is the probability that fewer than l of m_2 independent uniform variables are at most ε, that is, that their l-th smallest exceeds ε. This conditional law does not depend on $\mathcal{B}_1$, which gives the independence and the unconditional law; the moments are those of the beta law.

(c) Given $\mathcal{B}_1$, the coordinate vectors of the queries of $\mathcal{B}_2$ and of the test queries are exchangeable, and so are their scores sc_2, a fixed function of these vectors. As the scores are distinct, their order is a uniformly random ordering of the $m_2 + N$ queries; so the set of ranks, counted from the largest score, that the queries of $\mathcal{B}_2$ occupy is a uniformly random set of m_2 of the ranks $1, \dots, m_2+N$. Now $\hat{\eta}_l$ is the l-th largest score of $\mathcal{B}_2$, and exactly x test scores exceed it iff the rank $x+l$ belongs to $\mathcal{B}_2$ and exactly $l-1$ of the ranks $1, \dots, x+l-1$ do; the other m_2-l ranks of $\mathcal{B}_2$ are then among the last $m_2+N-x-l$. This gives the count $\binom{x+l-1}{l-1}\binom{m_2+N-x-l}{m_2-l}$ out of $\binom{m_2+N}{m_2}$ sets. Each test query fires with probability $l/(m_2+1)$ by Lemma B.44, which gives the mean. With ties, give the queries independent uniform labels and order the pairs (score, label) lexicographically: this order is exchangeable and has no ties, the l-th largest element of $\mathcal{B}_2$ in it has the score $\hat{\eta}_l$, and every test query whose score exceeds $\hat{\eta}_l$ lies above it. So the number of test queries that fire is at most the number that lie above this element, which has the stated law.

(d) By Proposition B.1(a), and by Theorem B.82 for several edited modules, the answer of a query on which no edit fires is the pre-edit answer. □

Theorem B.62 (The rank for a level α with confidence $1-\delta$). *Let* $\alpha, \delta \in (0,1)$*,*

$$l_\delta := \max\big\{l \in \{0, \dots, m_2\} :\; \mathbb{P}\big(\mathrm{Bin}(m_2, \alpha) \leq l-1\big) \leq \delta\big\},$$

and use the two-block rule with the rank parameter l_δ*, that is, with the scale* $\hat{\eta}_{l_\delta}$*, the* l_δ*-th largest score of* $\mathcal{B}_2$*.*

(a) Guarantee. *Under* (I)*, for every law of the coordinate vectors, ties allowed,*

$$\mathbb{P}\big(\mathrm{FF}_{l_\delta} > \alpha\big) \leq \mathbb{P}\big(\mathrm{Bin}(m_2, \alpha) \leq l_\delta - 1\big) \leq \delta,$$

also conditionally on $\mathcal{B}_1$ *and for every threshold vector* $\tau \geq \hat{\tau}^{(l_\delta)}$*: with probability at least* $1-\delta$ *over the calibration sample, at most a fraction* α *of the fresh benign queries fire an edit, and at most this fraction is answered differently from the pre-edit model.*

(b) No lower rank. *If no coordinate has an atom, then* $\mathbb{P}(\mathrm{FF}_l > \alpha) = \mathbb{P}(\mathrm{Bin}(m_2, \alpha) \leq l-1) > \delta$ *for every* $l > l_\delta$*: among the scales that are order statistics of the scores of* $\mathcal{B}_2$*,* $\hat{\eta}_{l_\delta}$ *is the smallest one with the guarantee of* (a) *for all laws.*

(c) Sample size. $l_\delta \geq 1$*, that is, the rule assigns finite thresholds, iff* $(1-\alpha)^{m_2} \leq \delta$*, that is, iff* $m_2 \geq \ln(1/\delta)\,\big/\,\ln\big(1/(1-\alpha)\big)$*.*

(d) Price. *The rule is the default rule at the smaller level* $\alpha' := l_\delta/(m_2+1)$*, so its marginal bound is* $l_\delta/(m_2+1)$*. If* $l_\delta \leq l := \lfloor \alpha(m_2+1) \rfloor$*, each threshold rises from* $(b_i + \hat{\eta}_l \tau_i^0)^+$ *to* $(b_i + \hat{\eta}_{l_\delta}\tau_i^0)^+$*: the scale moves up by* $l - l_\delta$ *order statistics of the scores of* $\mathcal{B}_2$ *(for large* δ *the selected rank can exceed the default one, and the thresholds then fall:* $m_2 = 1$*,* $\alpha = 0.1$*,* $\delta = 0.95$*). If no coordinate has an atom,* $F_b(\hat{\eta}_{l_\delta}) - F_b(\hat{\eta}_l)$ *has the beta law with the parameters* $(l - l_\delta,\, m_2 + 1 - l + l_\delta)$ *and the mean* $(l - l_\delta)/(m_2+1)$ *if* $l_\delta < l$*; and if* $1 \leq l_\delta < l$ *and, given* $\mathcal{B}_1$*, the score has an exponential upper tail above a level* y*, that is,* $\mathbb{P}(\mathrm{sc}_2(q) > y + x) = \mathbb{P}(\mathrm{sc}_2(q) > y)\, e^{-x/\beta}$ *for all* $x \geq 0$*, then on the event* $\hat{\eta}_l > y$*, and on the event* $\hat{\eta}_l \geq y$ *if the score has no*

atom at y,

$$\mathbb{E}\big(\hat\eta_{l_\delta} - \hat\eta_l \,\big|\, \mathcal{B}_1,\ \hat\eta_l\big) = \beta \sum_{x=l_\delta}^{l-1} \frac{1}{x}.$$

Proof. The set over which the maximum is taken contains $l = 0$, for which the probability is that of an empty event.

(a) Theorem B.61(a) with $\varepsilon = \alpha$ and $l = l_\delta$ if $l_\delta \geq 1$; if $l_\delta = 0$, $\mathrm{FF}_0 = 0$. The statement on answers is Theorem B.61(d).

(b) Theorem B.61(b), and the definition of l_δ together with the fact that $\mathbb{P}(\mathrm{Bin}(m_2, \alpha) \leq l - 1)$ is nondecreasing in l.

(c) $l_\delta \geq 1$ iff $l = 1$ is admissible, iff $\mathbb{P}(\mathrm{Bin}(m_2, \alpha) = 0) = (1 - \alpha)^{m_2} \leq \delta$.

(d) $\lfloor \alpha'(m_2 + 1) \rfloor = l_\delta$. Let F_b be continuous and let $\varphi_{(1)} \leq \cdots \leq \varphi_{(m_2)}$ be the ordered values $F_b(\mathrm{sc}_2(q))$, $q \in \mathcal{B}_2$, the order statistics of m_2 independent uniform variables. Then $F_b(\hat\eta_{l_\delta}) - F_b(\hat\eta_l) = \varphi_{(m_2+1-l_\delta)} - \varphi_{(m_2+1-l)}$, and the difference of two uniform order statistics whose ranks differ by $l - l_\delta$ has the law of $\varphi_{(l-l_\delta)}$. For the last claim, given that $\hat\eta_l \geq y$, the $l - 1$ scores of $\mathcal{B}_2$ above $\hat\eta_l$ exceed it by independent exponential variables with mean β (lack of memory), and the gap between the x-th and the $(x+1)$-th largest of them, and between the smallest of them ($x = l - 1$) and $\hat\eta_l$, is the minimum of x independent exponential variables, with mean β/x; sum over $x = l_\delta, \ldots, l - 1$. □

Theorem B.63 (Sample size of the calibration-conditional guarantee). *Let a calibration rule, possibly randomised, send the coordinate vectors of m benign calibration queries to a threshold vector $\hat\tau \in \Theta^T$, and let $\alpha, \delta \in (0, 1)$. Suppose that $\mathbb{P}(\mathrm{FF}(\hat\tau) > \alpha) \leq \delta$ under every law of independent identically distributed coordinate vectors, unbounded ones included (with a known bound on the coordinates the statement does not apply). Then under every such law and for every edit i*

$$\mathbb{P}\big(\hat\tau_i < +\infty\big) \ \leq\ \delta\,(1 - \alpha)^{-m}.$$

(a) If $m < \ln(1/\delta) \,/\, \ln\big(1/(1 - \alpha)\big)$, every such rule silences every edit with a probability of at least $1 - \delta(1 - \alpha)^{-m} > 0$, under every law; no such rule keeps an edit alive on every sample.

(b) The bound is attained. If $(1 - \alpha)^m \leq \delta$, the shared scale at the largest calibration score (the fit-clean scale of Sec. B.7.4) satisfies the hypothesis and keeps every edit alive on every sample. Otherwise the rule that uses this scale with probability $\delta(1 - \alpha)^{-m} < 1$ and silences all edits otherwise satisfies the hypothesis, and each edit is alive with exactly this probability.

(c) If the hypothesis is required only for the laws under which no coordinate has an atom, the bound holds under every such law.

(d) The two-block rule with the rank l_δ of Theorem B.62 keeps the edits alive iff $m_2 \geq \ln(1/\delta) \,/\, \ln\big(1/(1 - \alpha)\big)$: its second block has the least size that any rule needs for the whole sample (the two-block procedure itself uses $m_1 + m_2$ queries).

Proof. Fix a law P_0 on $\mathbb{R}^T$, an edit i, a real B and $\varepsilon \in (\alpha, 1)$, and let $\pi^B \in \mathbb{R}^T$ have the i-th coordinate B. Let the coordinate vectors be independent with the law $(1 - \varepsilon)P_0 + \varepsilon\,\delta_{\pi^B}$, realised as follows: the vector of q_h is π^B if $\iota_h = 1$ and a draw from P_0 if $\iota_h = 0$, the ι_h being independent indicators with $\mathbb{P}(\iota_h = 1) = \varepsilon$, independent of the draws and of the randomness of the rule. On the event that $\hat\tau_i \leq B$ in the chain Θ, edit i fires on every query whose vector is π^B, so $\mathrm{FF}(\hat\tau) \geq \varepsilon > \alpha$. On the event $\iota_1 = \cdots = \iota_m = 0$, which has the probability $(1 - \varepsilon)^m$, the calibration vectors are m independent draws from P_0. Hence

$$(1 - \varepsilon)^m\, P_0^m\big(\hat\tau_i \leq B\big) \ \leq\ \mathbb{P}\big(\mathrm{FF}(\hat\tau) > \alpha\big) \ \leq\ \delta.$$

Let $\varepsilon \downarrow \alpha$ and then $B \to \infty$: the events $\{\hat\tau_i \leq B\}$ increase to $\{\hat\tau_i < +\infty\}$.

(a) Then $\delta(1 - \alpha)^{-m} < 1$. (b) The largest of the scores of Sec. B.7 is the conformal scale of Definition B.43 at the level $1/(m + 1)$, for which $l = 1$. By Theorem B.46(a), applied at this level with $\varepsilon = \alpha$, its realised rate exceeds α with a probability of at most $\mathbb{P}(\mathrm{Bin}(m, \alpha) \leq 0) = (1 - \alpha)^m$; the randomised rule multiplies this probability by $\delta(1 - \alpha)^{-m}$. (c) Replace δ_{π^B} by a law without atoms under which the i-th coordinate is at least B; if P_0 has no atoms, neither has the mixture, and the argument is unchanged. (d) Theorem B.62(c). □

Table 5: The realised false-firing rate of the two-block rule for continuous scores (Theorems B.61 and B.62); values computed from the exact formulas. l: default rank parameter; l_δ: rank parameter for the confidence $1 - \delta$; the scale is the l-th largest score of $\mathcal{B}_2$.

α	m_2	l	$\mathbb{P}(\mathrm{FF}_l > \alpha)$	$l_{0.1}$	$\mathbb{P}(\mathrm{FF}_{l_{0.1}} > \alpha)$	$l_{0.05}$
0.05	20	1	0.358	0	0	0
0.05	45	2	0.335	1	0.099	0
0.05	59	3	0.429	1	0.048	1
0.05	100	5	0.436	2	0.037	2
0.05	200	10	0.455	6	0.062	5
0.05	500	25	0.471	19	0.086	17
0.05	1000	50	0.480	41	0.081	39
0.10	100	10	0.451	6	0.058	5
0.01	500	5	0.440	2	0.040	2

Numbers for $m_2 = 100$ and $\alpha = 0.05$. The default rank parameter is $l = 5$: the scale is the fifth largest of the 100 scores of $\mathcal{B}_2$ (rank 96). For continuous scores the realised rate has the beta law with the parameters $(5, 96)$: its mean is $5/101 = 4.95\,\%$, its median $4.66\,\%$, its 95th percentile $8.92\,\%$ and its 99th percentile $11.2\,\%$, and it exceeds α with the probability $\mathbb{P}(\mathrm{Bin}(100, 0.05) \leq 4) = 0.436$. For $\delta = 0.1$, and also for $\delta = 0.05$, the rank parameter is $l_\delta = 2$: the scale is the second largest score (rank 99). The realised rate then exceeds α with the probability 0.037 (rank 98 would give 0.118), its mean is $2/101 = 1.98\,\%$ and its 95th percentile $4.66\,\%$; this is the default rule at any level in $[2/101, 3/101)$. For $\delta = 0.01$, $l_\delta = 1$ (the largest score; probability 0.0059). The least sizes of Theorem B.63 for $\alpha = 0.05$ are 45, 59 and 90 queries for $\delta = 0.1$, 0.05 and 0.01, against 19 for the marginal bound. On a finite test set the measured rate spreads further (Theorem B.61(c)): on 200 test queries, with the default rank, its standard deviation is 2.6 percentage points and it exceeds $5\,\%$ with the probability 0.40; with $l_\delta = 2$, with the probability 0.051. The price in height (Theorem B.62(d)): the scale moves from the fifth to the second largest score, which gives up a further $3/101$ of the benign scores on average, and $\beta(\frac{1}{2} + \frac{1}{3} + \frac{1}{4}) = 1.08\,\beta$ of height if the upper tail of the scores is exponential with the scale β. The closed form of Theorem B.46(c) is void at this size: its slack $\sqrt{\ln(1/\delta)/(2m_2)}$ is 0.107 for $\delta = 0.1$ and falls below $\alpha = 0.05$ only for $m_2 \geq 461$. Table 5 gives other sizes: for a fixed δ the loss of rank $l - l_\delta$ grows like $\sqrt{m_2\,\alpha(1 - \alpha)}$, so that $l_\delta/l \to 1$ only for $m_2 \gg 1/\alpha$.

Status. None of this is new mathematics. With $\mathcal{B}_1$ as the proper training set and $\mathcal{B}_2$ as the calibration set, the two-block rule is an inductive conformal predictor, and Theorem B.61(a),(b) is [67, Thm. 1(a)], which is stated for every fixed proper training set and with "if and only if" for continuous scores; in the language of outlier tests it is [10, Prop. 1], who call the property *calibration-conditional validity*. The exact form with ties (p_ϵ) is an elementary sharpening. Part (c) is the beta-binomial law of the empirical coverage on a finite validation set [7, App. C], obtained there for independent data as a beta mixture of binomial laws and here by counting, which shows that exchangeability suffices. Theorem B.62 is a one-sided distribution-free tolerance limit from order statistics, applied to the scores of the second block as in [67, Sec. 3 and App. A], where it is traced to Wilks (1941) for continuous laws and to Scheffé and Tukey (1945) without continuity (we did not open these two papers); part (d) is the spacing law of uniform order statistics and the Rényi representation of exponential order statistics. The sufficient size of Theorem B.63 is the sample size of such tolerance limits; the necessity for arbitrary rules is the contamination argument of Theorem B.55(a)(iii), carried over to the conditional criterion. We did not find this necessity stated in this form and regard it as elementary.

Remark B.64 (Reading). (i) *What is conditioned on.* FF_l is a property of the deployed thresholds: under (I), the long-run share of benign queries that fire. The marginal bound says that its mean over calibration samples is at most α; with the default rank it exceeds α on almost half of the samples (Table 5, fourth column, which tends to $\frac{1}{2}$). (ii) *Independence is needed for the rate, not for a finite test set.* For a finite population the realised rate on "fresh" queries is not defined and the reading by the beta law is not available (Theorem B.46(d)); part (c) is the statement for the queries that an experiment holds out, and it is what a measured distribution of rates over calibration draws has to be compared with. (iii) *Several edited modules.* All statements hold with the terms (i, μ) of Definition B.78 in the place of the edits and "not certified" in the place of "fires" (Corollary B.90). (iv) *Capped edits.* As for the marginal bound, the statements cover a threshold vector with upper caps (guards of own requests) only if it still dominates $\hat{\tau}^{(l)}$. (v) *The baselines may be any finite quantities computed from the first block.* By part (b) the law of FF_l does not depend on $\mathcal{B}_1$: the size of the first block and the quality of the baselines affect coverage, not validity.

B.7.6 Thresholds from the log alone, guards and refusals. This part says what the implementation of Sec. 4 of the paper computes where the paper is brief. It contains no theorem.

What the log provides. The stored factors and solver keys give the detectors, the rebased values (Lemma B.65) and the first rule below; for the closed-form editors of Sec. 2 the read-out vector is a similarity to the own key, whitened by the editor's own key statistics [50]. The second rule, the guards and the near-scope variant of Sec. B.7.7 need the keys of the logged requests *as served*, that is, the requests and one pass of the model for each; no answer of another query, no further data and no encoder is needed.

Two rules without benign queries. The migration (role last). All thresholds are equal to $\max_{i \neq j} \zeta_i(k_j)$, the largest coordinate of an edit on the stored solver key of another edit. No margin is added, so the pair that attains the maximum fires (the gate is $\zeta \geq \tau$), and the value can be negative. If it exceeds 1, that is, if $\gamma_i^{\mathrm{sep}} < 0$ for some edit, the guard below lowers every threshold to just below 1. For $T = 1$ the rule is undefined. *The repair* (all positions). Let h_i be the largest coordinate of edit i at the positions of the logged request sequences other than its own solver position: all positions of the sequences of the other requests and the other positions of its own (on the public text logs the sequences are the request prompts). The threshold is $\tau_i = h_i + \kappa\,|h_i|$ with one constant $\kappa > 0$ of the registered setup. So $\tau_i > h_i$ iff $h_i \neq 0$; for $h_i = 0$ the position that attains h_i still fires; and for $h_i < 0$ the threshold is negative, so that edit i fires on a half-space of keys that contains the origin. For $T = 1$ the rule is defined if the request has a second position. Both rules are then passed through the guard.

Guards. Every edit has a finite set of *guard keys*: its stored solver key, at which its coordinate is 1, and the key of its own request as served, at the solver position (one key per request, also when all positions are eligible; a variant adds the keys of that position in the passes of a decode of the request). Line 7 of Algorithm 1 lowers a threshold that lies above the coordinate of a guard key of its edit to that coordinate minus a band, relative to the coordinate, and marks the edit as capped; afterwards every edit fires at each of its guard keys. The served solver position is an eligible position of the pre-edit trace of the own request, so a guarded edit covers the pair (i, q_i) in the sense of Def. 2.5; the guard asks more than this coverage, which needs one eligible position only, and less than its name suggests: at a guard key the edit need not be selected under top-1 selection (Remark B.7), and the request need not be answered with y_i. The must-cover pairs $\boldsymbol{Q}^!$ of Theorem B.14 and of Algorithm 1 are pairs of an edit and a query; the implementation holds each of them by one guard key (the served key of an own request, the key of a granted why-not complaint). On the public text logs a served key differs from the stored one by the round-off of a padded batch, and the band is widened until every guard key that fires is clear of the decision band of the gate. Where a benign key has a larger coordinate than a guard key, the guard and the bound exclude each other: a capped edit fires on every such key (Remark B.60). Without the guard (the literal rule, which the experiments report as "guards off") an edit whose guard key lies below its threshold is silent on its own request and is counted as lost.

Refusal. For a complaint q the least repair with the band ξ proposes $\tau_i' = \max(\tau_i, (\zeta_i(q) + \xi)^+)$, where $\zeta_i(q)$ is the largest coordinate at the eligible keys that the complaint presents. There is a *conflict* if a guard key k of an edit i fires under τ and not under τ', that is, if $\tau_i \leq \zeta_i(k) \leq \zeta_i(q) + \xi$. In the default form the complaint is then refused: no threshold changes, and the complaint is counted. In the second form that the paper reports the update is applied; the edit is then silent at that guard key and is counted as lost. (A third form of the library raises only the thresholds of the edits without a conflict, which does not certify the complaint.) With $\xi > 0$ a complaint is refused already when $\zeta_i(q)$ lies within the band below $\zeta_i(k)$.

Recalibration and stored complaints. Algorithm 1 does not start from the current thresholds: it replaces all of them. The certificate

of a repaired complaint therefore persists only while the log and the detectors are fixed and thresholds are raised (Theorem B.8(b), Proposition B.1(c)). After a recalibration, and after a change of detectors, which changes the coordinates of the complaints, the stored complaints are repaired again by the least repair of their recomputed coordinates, in any order; the library keeps the keys of the accepted complaints for this. Appending an edit is not an operation of the library: the longer log is a new log, on which thresholds are calibrated anew and the stored complaints have to be repaired again, since the new edit can fire on a complaint that was certified.

B.7.7 Registered variants of the calibration. The paper names three registered variants of Algorithm 1. They are defined here.

Null repair on benign keys. Each read-out vector is replaced by the closest vector that has the coordinate zero on the keys of a sample of benign queries and keeps the own coordinate: the case $\pi = (s_i, 0, \dots, 0)$ of Proposition B.37(b), a projection in closed form. A detector whose own key lies in the span of the sample keys cannot keep its own coordinate; it is left unchanged and is reported. Thresholds, values and guard keys are unchanged, and Algorithm 1 then runs as it is. The variant changes detectors: it is not a thresholds-only repair, and the paper registers it as a variant of its migration. The sample is disjoint, by query, from the calibration queries.

Log-derived near-scope calibration. For every edit, recombinations of its logged request are formed, without labels and without complaints: its image with an unrelated question, its question with an unrelated image, and its image with the question of another logged request. The baseline b_i of line 3 becomes the larger of its value and the largest coordinate of edit i on its own recombinations; the second block, the rank and the guard are unchanged.

Calibration-conditional rank. Line 5 uses the scale $\hat{\eta}_{l_\delta}$ of Theorem B.62 in the place of the default rank. If $l_\delta = 0$, that is, if $(1-\alpha)^{m_2} > \delta$, the scale is $+\infty$ and all thresholds are $+\infty$ before the guard.

Why the bounds are kept. Proposition B.58(a) needs the scores sc_2 of the queries of $\mathcal{B}_2$ and of the fresh query to be exchangeable given everything that defines the function sc_2. In the first variant this function depends, beyond $\mathcal{B}_1$, on the null-repair sample, and in the second on the recombinations. Both are fixed without looking at the calibration queries, or drawn so that, conditionally on them and on $\mathcal{B}_1$, the coordinate vectors of $\mathcal{B}_2$ and of the fresh benign query are still exchangeable; then the proof of the proposition applies with the raised baselines, which are quantities of the first block (Remark B.64(v)). That the auxiliary sample shares no query with the calibration queries is necessary for this and is not sufficient in general: a sample that is selected by looking at the calibration queries breaks the exchangeability without sharing a query with them, which is why the detector repair from complaints of Sec. B.11 is recalibrated on a fresh sample. The third variant is Theorem B.62(a); it needs assumption (I), independent and identically distributed queries, and under exchangeability alone it has the marginal bound of its rank only (Theorem B.62(d)).

What the bounds do not say. They bound the firing of uncapped edits (Remark B.60) on benign queries that are exchangeable with the sample. A recombination is not such a query: raising the baselines on recombinations is a heuristic against near-scope firing and gives no bound on it.

B.8 Rebased values and retrofit

In an edit log (Def. 2.1) the write vector is the residual $u_i = z_i^\star - W_{\prec i} k_i$ of the target $z_i^\star$ of the editor for the module output at the *solver key* k_i, against the weight $W_{\prec i} = W_0 + \sum_{j \prec i} \Delta_j$ that was in force when the batch of edit i was solved.

Lemma B.65 (Rebase; Lemma 4.1 of the paper). *For every edit i of an edit log, with arbitrary read-out vectors,*

$$v_i^{\mathrm{rb}} := z_i^\star - W_0 k_i = u_i + \sum_{j \prec i} \langle \psi_j, k_i \rangle \, u_j,$$

$$W_E k_i - z_i^\star = (s_i - 1)\, u_i + \sum_{j \not\prec i,\ j \neq i} \langle \psi_j, k_i \rangle \, u_j.$$

Proof. $z_i^\star = u_i + W_{\prec i} k_i = u_i + W_0 k_i + \sum_{j \prec i} \langle \psi_j, k_i \rangle u_j$, which is the first identity. With $W_E k_i = W_0 k_i + \sum_j \langle \psi_j, k_i \rangle u_j$ and $\langle \psi_i, k_i \rangle = s_i$, subtracting gives the second. □

So the rebased value is computed from the log alone: write vectors, read-out vectors, solver keys and the order of the batches. The sums run over the edits of earlier *batches*: the edits of one batch were solved against the same weight, and the first identity fails if the sum runs over all earlier edits. The second identity says what the deployed log does at its own keys: it misses the target by the shortfall $(1 - s_i)u_i$ of the attainment and by the leaks of the edits of the same and of later batches. A log that was written in low precision differs from $W_0 + \sum_i \Delta_i$ by a round-off term, which is not part of Def. 2.1 and has to be replayed separately. The lemma is a corollary of the recursion of sequential locate-and-edit solvers: for batches of one edit and $s_i = 1$ both identities are in [34], and NeuralDB stores the values v_i^{rb} when it is built from the edited facts [22]. The lemma adds batches, arbitrary read-out vectors and $s_i \neq 1$.

Let $\gamma_i^{\mathrm{sep}} := 1 - \max_{j \neq i} \zeta_i(k_j)$ be the *self-separation margin* of edit i on the solver keys ($\gamma_1^{\mathrm{sep}} := +\infty$ for $T = 1$, the maximum over the empty set being $-\infty$): the detector of edit i separates its own solver key from the solver keys of the other edits iff $\gamma_i^{\mathrm{sep}} > 0$.

Theorem B.66 (Retrofit; Prop. 4.2 of the paper). *Let $0 \le \varepsilon_i < \gamma_i^{\mathrm{sep}}$ for every edit i, and let $\mathcal{S}$ be the scoped log with the detectors ψ_i of the log, role sets that contain the role of the solver positions, the response $g_i \equiv 1$, the rebased values $v_i = v_i^{\mathrm{rb}}$, and thresholds with $1 - \gamma_i^{\mathrm{sep}} < \tau_i \le 1 - \varepsilon_i$.*

(a) At its solver position, exactly edit i fires, and the module returns $W_0 k_i + v_i^{\mathrm{rb}} = z_i^\star$: every edit attains its module-output target, for every number T of edits.

(b) At every position at which no edit fires, the module returns $W_0 k_t$, and Proposition B.1 applies.

(c) Edit i fires at every position with a role in $\mathcal{R}_i$ whose key has $\zeta_i(k_t) \ge 1 - \varepsilon_i$.

Proof. (a) $\zeta_i(k_i) = 1 \ge \tau_i$, so edit i fires. For $j \neq i$, $\zeta_j(k_i) \le \max_{l \neq j} \zeta_j(k_l) = 1 - \gamma_j^{\mathrm{sep}} < \tau_j$, so edit j does not. By the module map of Def. 2.2, the module returns $W_0 k_i + g_i(1)\, v_i^{\mathrm{rb}}$, which is $z_i^\star$ by

Lemma B.65. (b) is the module map again, and (c) is the firing rule of Def. 2.2 with $\tau_i \le 1 - \varepsilon_i$. □

The theorem is a corollary of Lemma B.65. The module, $W_0 k$ plus the stored residual of the matched edit and nothing otherwise, is that of gated key–value editors [22, 29]. Specific to the setting of the paper: the values come from a deployed log without solving again, and the read-out vectors of the log serve as detectors. Three cautions. *The margin is needed.* Self-separation alone allows $\tau_i = 1$, a threshold at the own solver coordinate. The solver keys come from the batched forward pass of the editor; at serving time the key of the same request can have a lower own coordinate, and thresholds at 1 then silence own requests. The margin ε_i has to cover this drop, which is measured on the keys of the requests as served. *Attainment is not behaviour.* The target $z_i^\star$ is a module output at one key. Whether q_i is answered with y_i depends on how the editor computed the target (on the logs of the paper, once, on the pre-edit model) and on the key of the request as served, so (a) does not imply that q_i is answered with y_i. *Thresholding alone is not enough.* With the logged values $v_i^{\log} = s_i u_i$ and the identity response, an edit that fires alone at its solver key leaves the error $W_0 k_i + s_i u_i - z_i^\star = -(1-s_i)\, u_i - \sum_{j<i} \langle \psi_j, k_i \rangle\, u_j$, by Lemma B.65. The retrofit preserves the targets, not the deployed weight: it changes the module output wherever a coordinate below a threshold is not zero, and it needs W_0. *The margin is a diagnostic, and the hypotheses are not established by calibration.* The margin γ_i^{sep} is computed from the log alone and compares the own solver key with the solver keys of the other edits only. If $\gamma_i^{\text{sep}} \le 0$, no threshold satisfies the hypothesis for edit i; if $\gamma_i^{\text{sep}} > 0$, nothing follows about benign keys, near-scope keys or keys of rephrased requests. The thresholds of Algorithm 1 are fitted to benign queries and to guard keys and need not lie in the interval of the theorem: the scoped log of the paper attains the target of an edit at a position with its solver key if the edit is selected there, which the theorem guarantees under its hypotheses only.

Proposition B.67 (Cross-firing). *Let the thresholds, responses and role sets of a scoped log be arbitrary, let $\Xi_{ji} := g_j(\zeta_j(k_i))$ if edit j fires at the solver position of edit i and $\Xi_{ji} := 0$ otherwise, and let V' and V^{rb} be the matrices with the columns v_j and v_i^{rb}.*

(a) Every edit attains its target at its solver position iff $V'\Xi = V^{\text{rb}}$. If Ξ is invertible, the values $V' = V^{\text{rb}}\Xi^{-1}$ attain every family of targets. If Ξ is singular, values exist iff $V^{\text{rb}}\pi = 0$ for every π with $\Xi\pi = 0$, which fails for almost every family of targets.

(b) For the thresholds $-\infty$ and the identity response, $\Xi_{ji} = \langle \psi_j, k_i \rangle / s_j$ has rank at most d: for $T > d$ a linear log attains almost no family of targets, whatever its values.

(c) A thresholded Ξ can have rank $T > d$. In key dimension 1, let the solver keys be reals $0 < k_1 < \cdots < k_T$, all detectors equal to 1, all thresholds equal to 1 and all responses equal to 1. Then edit j fires at k_i iff $i \ge j$, Ξ is triangular with unit diagonal, and the values $v_1 = v_1^{\text{rb}}$, $v_i = v_i^{\text{rb}} - v_{i-1}^{\text{rb}}$ attain all T targets.

(d) With the response 1, two solver positions at which the same edits fire give equal columns of Ξ, which is then singular. With the identity response it need not be: for two edits that fire at both solver keys with $\zeta_1(k_2) = \zeta_2(k_1) = \frac{1}{2}$, Ξ is invertible.

Proof. (a) By Def. 2.2 the module returns $W_0 k_i + \sum_j \Xi_{ji} v_j$ at the solver position of edit i, and the target is $W_0 k_i + v_i^{\text{rb}}$. A linear system $V'\Xi = V^{\text{rb}}$ is solvable iff the rows of V^{rb} lie in the row space of Ξ, that is, iff V^{rb} vanishes on the null space of Ξ; for a fixed $\pi \ne 0$ the matrices with $V^{\text{rb}}\pi = 0$ form a proper subspace. (b) Ξ is the product of a diagonal matrix, the $T \times d$ matrix with the rows $\psi_j^\top$ and the $d \times T$ matrix with the columns k_i. (c) $\zeta_j(k_i) = k_i/k_j \ge 1$ iff $i \ge j$, and column i of $V'\Xi$ is $\sum_{j \le i} v_j$. (d) Column i of Ξ is the indicator vector of the edits that fire at the solver position of edit i. In the example Ξ has the rows $(1, \frac{1}{2})$ and $(\frac{1}{2}, 1)$. □

Under top-1 selection, where only the firing edit of largest coordinate enters the sum and every solver key selects its own edit, Ξ is the identity. The construction of (c), thresholded units on ordered projections with a triangular matrix, is that of [72]. In contrast to Theorem A.12, where exact protection by a linear correction consumes key dimension, the number of targets that a scoped log attains is not bounded by d; its limit is the separation of the solver keys by the detectors and, after a repair of complaints, their separation from the complaints (Proposition B.72(e)).

B.9 The three repairs under one criterion

This subsection proves Theorem 3.1 of the paper and holds the table that its Sec. 3 cites (paragraph "One criterion"): it judges each of the three repairs by each of the two criteria. The parts of the paper's theorem are: (a) Proposition B.69; (b) Proposition B.74(a),(b),(e); (c) Proposition B.72(a),(c),(d),(e) and Theorem B.76(a),(c); (d) Proposition B.75(d); (e) Theorem B.76(d) and Proposition B.77(c)(v),(vi). Table 6 says, for each repair and each criterion, what is lost and at which cost, with the statement that proves each cell. One module is edited and (H1)–(H4) hold; statements about weights are in exact arithmetic, as in Sec. A.2. The module of a scoped log is the sum of Def. 2.2; statements about firing sets (certified, least repairs, lost edits, widths) hold under top-1 selection as well, statements about module outputs (restored, tolerances, output-preserved edits) are for the sum.

Keys of a complaint set. Q^- is a finite set of wrong-answer complaints. In this subsection K_q is the matrix whose columns are the keys at *all* positions of the pre-edit trace $x^0(q)$, K_Q is the matrix of the columns of all K_q, $q \in Q^-$, with A columns, the *trace keys* of the complaints, and P_Q is the orthogonal projector onto $\operatorname{col} K_Q$. The own keys of the log are the columns of $K_E := [k_1, \ldots, k_T]$.

Linear repairs. A *linear repair* is a weight $W' \in \mathbb{R}^{n \times d}$, with the model $\mathcal{M}[W']$. Rollbacks $W_R = W_0 + \sum_{i \notin R} \Delta_i$ (R is the set of *deleted* edits, Def. 2.6), compensations $W_E + X$, re-editing and a rebuild from W_0 are linear repairs.

Three criteria. Let $\mathcal{M}'$ be a repaired model, given by a weight W' or by a scoped log $\mathcal{S}'$, with the module map $F_{\mathcal{M}'}$: $F_{W'}(x,t) = W' k_t$ for a weight, and $F_{\mathcal{S}'}$ is that of Def. 2.2. A query q is

- *certified* (Def. 2.5(a)) if $\text{Fire}_{\mathcal{S}'}(q) = \emptyset$ for a scoped log, and if $(W' - W_0)K_q = 0$ for a weight;
- *restored* if $F_{\mathcal{M}'}(x^0(q), t) = W_0 k_t$ at every position t of $x^0(q)$: the module form of the certificate, which is defined for both kinds of models (an equality of exact arithmetic wherever an edit fires);

Table 6: The three repairs under each criterion, with the statements that prove each cell. A complaint is *restored* if the module returns the pre-edit output at every position of its pre-edit trace (Lemma B.68); *certified* (no edit fires) implies it. Tolerance: at one reported key per complaint with nonzero damage, $p \in [1, \infty]$, $c \in [0, 1)$. An edit is lost by a rollback if it is deleted, by a weight if the module output at its own key is no longer that of the deployed log (variant: no longer the target), by thresholds if it no longer fires at its own key. $\mathrm{Ctr}(Q^-)$: edits with a nonzero contribution at a trace key; μ_i: largest coordinate of edit i on the eligible trace keys of the complaints; $\varrho = \mathrm{rank}((I - P_Q)K_E)$.

	Certificate (restored on the whole pre-edit trace)	Tolerance (one key per complaint)
Rollback	*Feasible:* always; $R = [T]$ returns W_0. *Edits lost:* if the contributions of $\mathrm{Ctr}(Q^-)$ are zero-sum-free (independent write vectors, or independent coordinate vectors), the certifying deletion sets are the supersets of $\mathrm{Ctr}(Q^-)$, which is the unique least one. For a trace key outside a null set a rollback certifies iff $W_R = W_0$: all T edits, unless edits cancel as matrices. *Complexity:* one pass, $O(T(Ad + n))$. Without zero-sum-freeness: the tolerance criterion at $c = 0$, strongly NP-hard and not $T^{1-\varepsilon}$-approximable; for $n = 1$ weakly NP-hard (Subset Sum), and no approximation of the number of kept edits. *Proved by:* Proposition B.69; Theorem A.5 at $c = 0$; Proposition B.42(b).	*Feasible:* always ($R = [T]$). *Edits lost:* OPT_c deletions, between 1 and T; a kept edit i is output-preserved only if the deleted contributions at k_t sum to zero. In the family of Proposition B.75(g): $T - \max(0, \lfloor c^2(T+3) \rfloor - 3)$ edits. *Complexity:* strongly NP-hard; no polynomial-time algorithm within $T^{1-\varepsilon}$, even for a weaker tolerance c'; for $c > 0$ robust to perturbations; greedy can delete all T where 2 suffice. *Proved by:* Theorem A.5; Proposition B.77(a).
Any weight: compensation, re-editing, rebuild	*Feasible:* always (W_0); if $\mathrm{rank}\, K_Q = d$, W_0 is the only certifying weight. *Edits lost:* the preserved values span at most $\varrho \le \min(T, d - \mathrm{rank}\, K_Q)$ dimensions. If every $\varrho + 1$ of the values dWk_i (variant: v_i^{rb}) are linearly independent, at most ϱ edits are preserved, and ϱ can be; without this hypothesis more can be. *Complexity:* Gaussian elimination under the hypothesis (a matroid; greedy for weights); in general the choice of the preserved edits is Min-Sacrifice: strongly NP-hard, no constant-factor approximation. *Proved by:* Theorem A.15, Corollary A.16 (proof of record); restated as Proposition B.71.	*Feasible, with no edit lost:* if $\mathrm{rank}[K_{\mathrm{prot}}, K_q] = \mathrm{rank}\, K_{\mathrm{prot}} + A$; the minimum compensation is then a closed form and exact. At one complaint key: iff $k_q \notin \mathrm{col}\, K_{\mathrm{prot}}$, for every $c < 1$; a tolerance does not help. *Edits lost beyond:* the least sacrifice of Theorem A.12; in general position $\max(0, T + A - d)$ for exact restoration of A keys. At several keys a tolerance can remove the count: all but $\lceil 1/c \rceil$ edits, or all, can survive complaint keys that span $\mathbb{R}^d$; or none. *Complexity:* closed form for a fixed protection set (a convex feasibility problem for $c > 0$); the least sacrifice is strongly NP-hard, without constant-factor approximation. *Proved by:* Lemma A.9; Theorem A.12; Proposition A.26; Proposition B.77(b); Proposition B.75(f),(g); Proposition A.20(e).
Rescoping: thresholds of a scoped log	*Feasible:* always, for every log, key dimension, number of complaints and current thresholds, also when $\mathrm{rank}\, K_Q = d$, in the strong form that no edit fires on a complaint. *Edits lost:* by the least repair exactly $\{i : \mu_i \ge \pi_i\}$ ($\pi_i = 1$ for the own key), and at least these by every certifying threshold vector: a matter of separation, not of dimension. In a retrofit with rebased values (Proposition B.72(e)) the edits with $\mu_i < 1$ attain their targets, and at the own key of the others no edit fires; with logged values a released edit is output-preserved iff the contributions silenced at its own key sum to zero. 0 against T for both rows above in the families of Proposition B.75(d),(f). *Complexity:* one pass, $O(Td)$ per reported position; unique least repair, every arrival order; touches exactly the $\mathrm{wd}(q)$ edits that fire on the complaint, as a rollback of the same log does. *Proved by:* Proposition B.1; Theorem B.8; Proposition B.72; Theorem B.76.	*Feasible:* always (the certificate repair). *Edits lost:* never more than for rollback, for every cost of the silenced pairs (containment); deciding whether no edit need be lost is strongly NP-hard, so the number of lost edits has no approximation ratio; under the certificate the same question is one pass. *Complexity:* one complaint, fewest un-fired edits: Min-Rollback$_{p,c}$ on the $\mathrm{wd}(q)$ firing contributions, the same feasible sets as for a rollback of that log; exact in $O(2^{\mathrm{wd}(q)})$. Several complaints with newly uncovered pairs penalised: polynomial exactly for width ≤ 1 and for width 2 with $c < \frac{1}{2}$ (one minimum cut), NP-hard otherwise. No least repair. *Proved by:* Propositions B.29, B.31, B.32; Proposition B.74; Proposition B.77(c).

- *c-repaired* at a position t of $x^0(q)$, for a norm index p and $c \ge 0$, if $\|F_{\mathcal{M}'}(x^0(q), t) - W_0 k_t\|_p \le c \, \|F_{\mathcal{M}}(x^0(q), t) - W_0 k_t\|_p$, where $\mathcal{M}$ is the model that is repaired. For the deployed log the right side is $c \, \|dWk_t\|_p$, and for a rollback this is the tolerance criterion of Def. 2.6(i) at the complaint key k_t.

Lemma B.68 (The three criteria). *Let q be a query and $\mathcal{M}'$ a weight or a scoped log.*

(a) If q is restored, then $\mathcal{M}[\mathcal{M}']$ evaluates q on the pre-edit trace with the pre-edit logits, and $\hat{y}(q; \mathcal{M}') = y_0(q)$. The converse fails: the answer can be $y_0(q)$ although q is not restored.

(b) For a weight, certified and restored are the same condition. For a scoped log, certified implies restored, and q is restored iff, at every position t of $x^0(q)$, the contributions $g_i(\zeta_i(k_t))\, v_i$ of the edits that fire at t sum to zero; so restored implies certified if no nonempty family of contributions of edits that fire at one position of $x^0(q)$ sums to zero.

(c) Restored is the same as c-repaired with $c = 0$ at every position of $x^0(q)$, for every p; so certified implies restored, and restored implies c-repaired at every position, for every $c \ge 0$.

(d) A rollback of the scoped log $\mathcal{S}^E$ by a mask (thresholds $+\infty$ on R and $-\infty$ elsewhere) is certified as a scoped log iff $R = [T]$, and is restored iff the weight W_R is certified.

Proof. (a) The sequence $x^0(q)$ has the same keys and roles under every model (H1). If the module outputs of $\mathcal{M}'$ on it are those of $\mathcal{M}[W_0]$, so are the logits (H2), and by induction on the generation step both models choose the same next token and stop at the same step (H3); for a weight this is Proposition A.2. For the converse let $n = d = 1$, $W_0 = 1$, a prompt of one position with the key 1, and let the answer be the token a if the module output at the prompt is positive and b otherwise, followed by the end of the answer. The weight $W' = 2$ answers a, as W_0 does, and its module output at the prompt is $2 \ne 1$. (b) For a weight, $F_{W'}(x^0(q), t) - W_0 k_t = (W' - W_0)k_t$. For a scoped log, $F_{\mathcal{S}'}(x^0(q), t) - W_0 k_t$ is the sum of the contributions of the edits that fire at t (Def. 2.2), which is empty if no edit fires on $x^0(q)$; under the last hypothesis the sum vanishes only if it is empty. (c) A norm vanishes only at zero; the rest is (b). (d) Under the thresholds $-\infty$ an edit fires at every position, and under $+\infty$ at none, so the firing set of every query is $[T] \setminus R$. With

the logged values and the identity response the edits of $[T] \setminus R$ contribute $\zeta_i(k_t)\, v_i^{\log} = \langle \psi_i, k_t \rangle u_i = \Delta_i k_t$, so the module map is $k_t \mapsto W_R k_t$ (Proposition B.74). □

So *restored* is one criterion for all three repairs, and it is the one that Def. 2.5(a) of the paper names for both kinds of models; the paper says *certifies* of scoped logs only, and in this subsection "certified", said of a weight, means that it restores. For linear repairs it is the certificate; for scoped logs the certificate of the paper is sufficient for it, bit for bit by (H4), and equivalent to it in the absence of exact cancellations. Below, a statement "under the certificate" is proved for restored and for certified, and says so where the two differ. A weight that is certified in exact arithmetic returns the pre-edit module output up to round-off only (Remark B.5(ii)).

Preserved edits. A repair is charged with the edits that it loses. At the solver position of edit i, with the own key k_i, an edit is

- *output-preserved* if the module returns $W_E k_i$, what the deployed log returns: $(W' - W_E)k_i = 0$ for a weight;
- *target-preserved* if the module returns the target $z_i^\star$: $(W' - W_0)k_i = v_i^{\mathrm{rb}}$ for a weight, with the rebased value $v_i^{\mathrm{rb}} = z_i^\star - W_0 k_i$ of Lemma B.65;
- *released* (scoped logs only) if edit i fires there, that is, $\tau_i \le \zeta_i(k_i) = 1$ if the role of the solver position is in $\mathcal{R}_i$.

More generally, each edit may have a finite nonempty set M_i of *must-fire keys*, keys at positions with a role in $\mathcal{R}_i$ at which it has to keep firing, with the *protected coordinate* $\pi_i := \min_{k \in M_i} \zeta_i(k)$: the edit keeps firing on M_i under τ iff $\tau_i \le \pi_i$. Three choices occur: the solver key ($\pi_i = 1$, released); the own request as a pair of $Q^!$, which is covered iff the edit fires at *some* eligible position of $x^0(q_i)$ ($\pi_i = \zeta_i(q_i)$; this is one must-fire key, the eligible key of the largest coordinate); and all eligible keys of the own request (π_i their smallest coordinate). An edit is *lost* by thresholds τ if $\tau_i > \pi_i$.

B.9.1 Rollback under the certificate. For the log $\mathcal{L}$ and the complaints Q^- let $\mathrm{Ctr}(Q^-) := \{i \in [T] : \Delta_i K_Q \neq 0\} = \{i : u_i \neq 0 \text{ and } \psi_i^\top K_Q \neq 0\}$ be the set of edits that *contribute* on the complaints: an edit outside it adds nothing at any trace key. The family $(\Delta_i K_Q)_{i \in J}$ is *zero-sum-free* if no nonempty subfamily sums to the zero matrix (Sec. A.1, for the contributions stacked over the trace keys).

Proposition B.69 (Rollback under the certificate). *Let $\mathcal{L}$ be an edit log and Q^- a finite set of wrong-answer complaints.*

(a) Characterisation. *Let $J := \mathrm{Ctr}(Q^-) \setminus R$. The rollback of $R \subseteq [T]$ certifies Q^- iff the contributions of the kept edits that contribute sum to zero, $\sum_{i \in J} \Delta_i K_Q = 0$. Every $R \supseteq \mathrm{Ctr}(Q^-)$ certifies. If the write vectors of the kept edits that contribute, u_i with $i \in \mathrm{Ctr}(Q^-) \setminus R$, are linearly independent, then R certifies iff $R \supseteq \mathrm{Ctr}(Q^-)$; in particular, if the write vectors of all kept edits are linearly independent, R certifies iff $\psi_i^\top K_Q = 0$ for every kept edit i.*

(b) Least deletion set. *The certifying deletion sets are exactly the supersets of $\mathrm{Ctr}(Q^-)$ iff the family $(\Delta_i K_Q)_{i \in \mathrm{Ctr}(Q^-)}$ is zero-sum-free. This holds if the write vectors u_i, $i \in \mathrm{Ctr}(Q^-)$, are linearly independent, and if the coordinate vectors $K_Q^\top \psi_i$, $i \in \mathrm{Ctr}(Q^-)$, are. Then $\mathrm{Ctr}(Q^-)$ is the least certifying deletion set, for inclusion and hence for the number of deleted edits and for every cost that is monotone in R; it is the unique least certifying set under inclusion and is computed in one pass over the log, with $O(T(Ad + n))$ operations. In particular, if the write vectors are linearly independent and every edit has a nonzero coordinate at some trace key of some complaint, only $R = [T]$ certifies.*

(c) Generic complaints. *Let $\mathcal{E}(\mathcal{L}) \subset \mathbb{R}^d$ be the union of the null spaces of the nonzero matrices among $\sum_{i \in J} \Delta_i$, $J \subseteq [T]$: a union of at most $2^T - 1$ proper subspaces, closed and of Lebesgue measure zero. If some trace key of some complaint lies outside $\mathcal{E}(\mathcal{L})$, a rollback certifies Q^- iff $W_R = W_0$. If, in addition, no nonempty subfamily of $\Delta_1, \ldots, \Delta_T$ sums to zero, then only $R = [T]$ certifies: every certifying rollback deletes all T edits. This holds when the write vectors are linearly independent, when the read-out vectors are linearly independent and no write vector is zero, and for Lebesgue-almost every choice of the write and read-out vectors, whatever T, n and d.*

(d) Without these hypotheses. *At one complaint key k_q, the rollback of R certifies iff $r(R) = 0$: the certificate is the tolerance criterion of Def. 2.6(i) with $c = 0$, for every p, and for several trace keys it is this criterion for the contributions stacked over the keys. Hence, by Theorem A.5(a),(b) with $c = c' = 0$, finding a smallest certifying deletion set is strongly NP-hard, and unless* P = NP *no polynomial-time algorithm returns a certifying set of at most $T^{1-\varepsilon}$ times the smallest size; this holds already for one complaint with one position, on logs whose own keys and complaint key are linearly independent. For output dimension $n = 1$ and one complaint key, deciding whether some certifying rollback keeps at least one edit is NP-complete, weakly, so that the largest number of kept edits, which is zero or positive, is NP-hard to approximate within any factor; for $n = 1$ and integer contributions it is computed in time polynomial in T and the largest absolute contribution.*

Proof. (a) $(W_R - W_0)K_Q = \sum_{i \notin R} \Delta_i K_Q = \sum_{i \notin R} u_i\, (\psi_i^\top K_Q)$, and the terms of the edits outside $\mathrm{Ctr}(Q^-)$ vanish; for $R \supseteq \mathrm{Ctr}(Q^-)$ the sum is empty. Let the u_i, $i \in J := \mathrm{Ctr}(Q^-) \setminus R$, be linearly independent and let the sum vanish. Its column at a trace key k is $\sum_{i \in J} \langle \psi_i, k \rangle\, u_i = 0$, so $\langle \psi_i, k \rangle = 0$ for every $i \in J$ and every trace key: $\psi_i^\top K_Q = 0$ for $i \in J$, which contradicts $i \in \mathrm{Ctr}(Q^-)$ unless $J = \emptyset$. If the write vectors of all kept edits are linearly independent, none of them is zero, so a kept edit lies in $\mathrm{Ctr}(Q^-)$ iff $\psi_i^\top K_Q \neq 0$.

(b) If the family is zero-sum-free and R certifies, the subfamily of $\mathrm{Ctr}(Q^-) \setminus R$ sums to zero by (a), so it is empty. If a nonempty $J \subseteq \mathrm{Ctr}(Q^-)$ has $\sum_{i \in J} \Delta_i K_Q = 0$, then $R := [T] \setminus J$ certifies by (a) and does not contain $\mathrm{Ctr}(Q^-)$. For the first sufficient condition see (a). For the second, let $\sum_{i \in J} u_i\, (\psi_i^\top K_Q) = 0$ with $J \subseteq \mathrm{Ctr}(Q^-)$ and the vectors $K_Q^\top \psi_i$, $i \in J$, linearly independent: every row of the sum is a vanishing combination of them, whose coefficients are one coordinate of the u_i; so $u_i = 0$ for $i \in J$, and $J = \emptyset$. The set $\mathrm{Ctr}(Q^-)$ is found by computing the A coordinates $\langle \psi_i, k \rangle$ of every edit, each an inner product in $\mathbb{R}^d$.

(c) Each null space is a proper subspace because its matrix is not zero. Let the trace key k lie outside $\mathcal{E}(\mathcal{L})$, let R certify, and $J := [T] \setminus R$. Then $\sum_{i \in J} \Delta_i k = 0$; if the matrix $\sum_{i \in J} \Delta_i$ were not zero, k would lie in its null space, hence in $\mathcal{E}(\mathcal{L})$. So $W_R - W_0 = \sum_{i \in J} \Delta_i = 0$; conversely $W_R = W_0$ certifies every query. Under the additional hypothesis J is empty. If $\sum_{i \in J} u_i \psi_i^\top = 0$ and the u_i are

linearly independent, every ψ_i, $i \in J$, is zero, which $s_i > 0$ excludes; if the ψ_i are linearly independent, every u_i, $i \in J$, is zero. For a fixed nonempty J, the entries of $\sum_{i\in J} u_i\psi_i^\top$ are polynomials in the entries of the vectors, not all zero (for $u_i = e_1$, $\psi_i = e_1$ the sum is $|J|\, e_1e_1^\top$), so they vanish together only on a null set; there are finitely many J.

(d) At one key, $(W_R - W_0)k_q = \sum_{i\notin R} w_i = r(R)$, and $\|r(R)\|_p \le 0 \cdot \|D\|_p$ iff $r(R) = 0$. Theorem A.5 allows $c = c' = 0$ (on its instances the feasible family is the same for every tolerance in $[0, 1)$, Lemma A.6), and by Lemma A.4 its instances are the contributions of logs with linearly independent own keys and complaint key; the complaint is a query with one position (further positions may be added whose keys are orthogonal to every read-out vector). For $n = 1$: membership in NP holds because $r(R)$ is computed exactly. Let positive integers x_0 and $x_1, \dots, x_N$ be an instance of Subset Sum (the problem Knapsack of [37]): is some subfamily of $x_1, \dots, x_N$ of sum x_0? Take the log of Lemma A.4 with the $T = N + 1$ scalar contributions $x_1, \dots, x_N, -x_0$. A certifying rollback that keeps a nonempty set J exists iff a nonempty subfamily sums to zero; as $x_1, \dots, x_N$ are positive, such a subfamily contains $-x_0$, and its other members sum to x_0. An algorithm with any approximation factor for the largest number of kept edits would decide whether this number is at least 1. For integer contributions of absolute value at most $x_{\max}$, the largest zero-sum subfamily is found by dynamic programming over the at most $2Tx_{\max} + 1$ attainable sums. □

Example B.70 (Coordinates that do not vanish are not enough). *Let $n = d = 1$, $T = 3$, $\psi_i = k_i = 1$ and $u = (1, -1, 1)$, with one complaint key* 1*. No coordinate vanishes,* $\mathrm{Ctr}(\mathcal{Q}^-) = [3]$ *and the deployed damage is* 1*. The certifying deletion sets are* $\{1\}$*,* $\{3\}$ *and* $[3]$*: there are two incomparable minimal sets, no least one, and the smallest keeps two edits. So "the least deletion set is $\{i : \psi_i^\top K_Q \neq 0\}$" needs the hypothesis of* (b)*; the write vectors* $1, -1, 1$ *are linearly dependent.*

Status. Elementary. Part (b), for the deployed log and under zero-sum-freeness of all T contributions, is Proposition B.42(b); added here are the set $\mathrm{Ctr}(\mathcal{Q}^-)$ when coordinates vanish, the equivalence, the two sufficient conditions and the generic form (c). Part (d) is a corollary of Theorem A.5 and of the NP-completeness of Subset Sum; nothing in it is new. Four remarks. (i) Under the certificate, rollback is hard only through exact cancellations among the contributions, as in the instances of (d): for a complaint key outside a null set it returns the pre-edit weight, and without cancellations the least deletion set is read off in one pass. For $c > 0$ this is different: the hardness of Theorem A.5 survives perturbations and holds on instances on which exact restoration is trivial (its part (c)). (ii) Deciding whether the one-pass set $\mathrm{Ctr}(\mathcal{Q}^-)$ is the least certifying set is coNP-complete already for $n = 1$, by (b) and the reduction of (d). (iii) With thresholds $+\infty$ in the place of deletions, the analogue of (a) and (b) for the edits that fire on a complaint in any scoped log is Proposition B.42. (iv) If a trace key has a law that gives probability zero to every proper subspace, for instance a density, then with probability one every certifying rollback returns W_0. Genericity is a statement about measure, not about editors: an editor that projects its update onto the null space of a set of preserved keys [21] writes read-out vectors that are orthogonal to these keys, so a complaint whose trace keys lie in their span has no contributing edit; it is restored by every rollback, and it cannot be a wrong-answer complaint.

B.9.2 Linear repairs under the certificate.

Proposition B.71 (Capacity of certifying linear repairs, restated). *Let $W' = W_0 + D'$ be a linear repair that certifies $\mathcal{Q}^-$, that is, $D'K_Q = 0$, and let J be the set of the edits that it output-preserves, with $\varphi_i := dWk_i$, or the set of those that it target-preserves, with $\varphi_i := v_i^{\mathrm{rb}}$. Let K_J be the matrix of the own keys of J and $\varrho := \mathrm{rank}((I - P_Q)K_E)$.*

(a) $\dim \mathrm{span}\{\varphi_i : i \in J\} \le \mathrm{rank}\big((I - P_Q)K_J\big) \le \varrho \le \min(T,\ d - \mathrm{rank}\, K_Q)$.

(b) If every $\varrho + 1$ of the vectors $\varphi_1, \dots, \varphi_T$ are linearly independent, as when all T are (the condition is void for $\varrho = T$), a certifying linear repair preserves at most ϱ edits, and a set J is preserved by some certifying linear repair iff $(I - P_Q)K_J$ has full column rank.

(c) For every log, every set J for which $(I - P_Q)K_J$ has full column rank is preserved by some certifying linear repair; so ϱ edits can be preserved, in either sense.

(d) If $\mathrm{rank}\, K_Q = d$, the only certifying linear repair is W_0; it preserves exactly the edits with $\varphi_i = 0$.

(e) Choosing the edits. *A set J is output-preserved by some certifying linear repair iff $[T] \setminus J$ is a feasible sacrifice set of Sec. A.7 for the protection groups $\{k_i\}$, the complaint keys K_Q and the target $-dWK_Q$. Under the hypothesis of* (b) *a largest such set has ϱ elements and is found by Gaussian elimination. In general, finding one is* Min-Sacrifice *with groups of one key: strongly NP-hard, and without a constant-factor approximation of the number of edits that are not preserved unless* P = NP*, already for one complaint of one position (Proposition A.26).*

Proof. (a), (c), (d) and (e) are Theorem A.15(b),(e) and Corollary A.16(ii), and (b) is part (c) of that theorem under a weaker hypothesis. We repeat the argument as far as the hypothesis of (b) is concerned. $D'K_Q = 0$ means $D' = D'(I - P_Q)$. (a) $\varphi_i = D'k_i = D'(I - P_Q)k_i$ for $i \in J$, so the φ_i are the images of the columns of $(I - P_Q)K_J$ under one linear map; these columns lie in the orthogonal complement of $\mathrm{col}\, K_Q$, of dimension $d - \mathrm{rank}\, K_Q$. (c) If $(I-P_Q)K_J$ has full column rank, the columns of K_J are linearly independent and their span meets $\mathrm{col}\, K_Q$ in 0, so the system $D'K_Q = 0$, $D'k_i = \varphi_i$ $(i \in J)$ prescribes a linear map on a basis of a direct sum. (b) If $\varrho = T$, the matrix $(I - P_Q)K_E$ has full column rank, so has $(I - P_Q)K_J$ for every J, and both claims hold by (c). Let $\varrho < T$ and let J be preserved. If $|J| > \varrho$, some $J' \subseteq J$ has $\varrho + 1$ elements, and by (a) the vectors φ_i, $i \in J'$, span at most ϱ dimensions, against the hypothesis. So $|J| \le \varrho$; then J lies in a set of $\varrho+1$ edits, the φ_i, $i \in J$, are linearly independent, and by (a) $|J| \le \mathrm{rank}((I - P_Q)K_J) \le |J|$. The converse is (c). (d) $P_Q = I$. (e) Write $D' = dW + X$: then $D'K_Q = 0$ iff $XK_Q = -dWK_Q$, and edit i is output-preserved iff $Xk_i = 0$. Under the hypothesis of (b), the indices of ϱ linearly independent columns of $(I - P_Q)K_E$ form a largest set, by (b) and (c). For the hardness, an instance of Proposition A.26 has one complaint key k_q and nonzero keys k_i, and a sacrifice set is feasible iff k_q is not in the span of the kept keys, whatever the nonzero target (Proposition A.25(1)). Take the log with the own keys k_i, $\psi_i := k_i$, $n = 1$ and write vectors $u_i \in \{1, 2\}$ such that $dWk_q = \sum_i u_i\langle k_i, k_q\rangle \neq 0$, and the complaint of

one position with the key k_q. Such write vectors exist unless every key is orthogonal to k_q, in which case no key need be sacrificed. □

Without the hypothesis of (b) the count fails: for $T = 3$, $d = 2$, own keys and read-out vectors $e_1, e_2, e_1 + e_2$ and no complaint (or, in dimension 3, one complaint key e_3), W_E certifies, preserves three edits, and the rank is 2. *Status.* The statement and its proof of record are those of Sec. A.8, which traces the criterion to the solvability of a linear matrix equation and relates it to Theorem A.12. The hypothesis of (b) is weaker than the hypothesis (ND) there, that all T values are linearly independent, which needs $n \geq T$: it needs $n \geq \varrho + 1$ only, and it gives the same conclusion. As for rollback, the hard instances of (e) are degenerate, and under the hypothesis of (b) the repair is linear algebra.

B.9.3 Rescoping under the certificate.

Proposition B.72 (Rescoping under the certificate). *Let $\mathcal{S}$ be a scoped log with thresholds τ^0, $\mathcal{Q}^-$ a finite set of wrong-answer complaints, and $\mu_i := \max_{q \in \mathcal{Q}^-} \zeta_i(q)$ the largest coordinate of edit i on the eligible keys of the complaints ($\mu_i := -\infty$ for $\mathcal{Q}^- = \emptyset$, so that the least repair is then τ^0). Detectors and role sets are fixed.*

(a) Always feasible, in one pass. *The thresholds $\tau_i^\star = \max(\tau_i^0, \mu_i^+)$ certify $\mathcal{Q}^-$, for every log, every complaint set, every key dimension and all current thresholds τ^0, calibrated or not. $\tau^\star$ is the unique least raise-only repair that certifies $\mathcal{Q}^-$, every arrival order of the complaints ends in it, it changes exactly the thresholds of the edits that fire on some complaint under τ^0, and it costs $O(Td)$ operations per reported position (Theorem B.8).*

(b) Module form. *$\tau^\star$ restores $\mathcal{Q}^-$. Call a pair (i, t) of an edit and a position of the trace of a complaint* active *if edit i fires at t under τ^0 with a nonzero contribution, and let τ_i^{r} be the largest of τ_i^0 and the $\zeta_i(k_t)^+$ over the active pairs of edit i. Then $\tau^{\mathrm{r}} \leq \tau^\star$, τ^{r} restores $\mathcal{Q}^-$, and it changes exactly the thresholds of the edits that have an active pair. If at no position of a complaint a nonempty family of nonzero contributions of edits that fire there under τ^0 sums to zero, the raise-only repairs that restore $\mathcal{Q}^-$ are exactly the $\tau \geq \tau^{\mathrm{r}}$, and a mask restores $\mathcal{Q}^-$ iff it silences every edit that has an active pair. If moreover every pair that fires is active, $\tau^{\mathrm{r}} = \tau^\star$: restored and certified are then the same criterion for raise-only repairs.*

(c) Lost edits. *Let every edit have a protected coordinate π_i with $\tau_i^0 \leq \pi_i$. The least repair loses exactly the edits of $\{i : \mu_i \geq \pi_i\}$, those whose must-fire keys the detector does not* separate *from the complaints. Every threshold vector that certifies $\mathcal{Q}^-$, raise-only or not, loses these edits. So the least repair loses the fewest edits among all certifying threshold vectors, for inclusion, and it loses none iff $\mu_i < \pi_i$ for every i.*

(d) Deployed scoping. *For $\mathcal{S} = \mathcal{S}^E$, μ_i is the largest $\zeta_i(k)$ over all trace keys k of the complaints, $\tau_i^\star = \mu_i^+$, every threshold changes, and the repaired module returns $W_0 k + \sum_{i:\, \zeta_i(k) > \mu_i} \Delta_i k$ at a key k. Edit i is released iff $\mu_i < 1$. At its solver key the module returns $W_E k_i - \sum_{j:\, \zeta_j(k_i) \leq \mu_j} \Delta_j k_i$: the edit is output-preserved iff the contributions that the repair silences at k_i sum to zero, as when $\mu_i < 1$ and every other edit j has $\zeta_j(k_i) = 0$ or $\zeta_j(k_i) > \mu_j$. A released edit need not be output-preserved (Example B.73).*

(e) Rebased values. *Let $\mathcal{S}$ be a retrofit (Theorem B.66): response 1, values v_i^{rb}, margins $\gamma_i^{\mathrm{sep}} > 0$ and thresholds $1 - \gamma_i^{\mathrm{sep}} < \tau_i^0 \leq 1$. Its least repair target-preserves every edit with $\mu_i < 1$, and at the solver key of an edit with $\mu_i \geq 1$ no edit fires. So if $\mu_i < 1$ and $\gamma_i^{\mathrm{sep}} > 0$ for every edit, the thresholds τ_i with $\max(1 - \gamma_i^{\mathrm{sep}}, \mu_i) < \tau_i \leq 1$ certify $\mathcal{Q}^-$ and every edit attains its target, for every number of edits and every key dimension.*

(f) No dimension count. *The set of lost edits depends on the complaints only through the T numbers μ_i. For all $T, d \geq 1$, every $J \subseteq [T]$, every number $N \geq 1$ of complaints and every $1 \leq \varrho \leq \min(d, N)$ there are a log with T edits in key dimension d and N complaints with trace keys of rank ϱ whose least repair in $\mathcal{S}^E$ loses exactly the edits of J (for the solver keys as must-fire keys). In particular there are instances with $T > d$ whose complaint keys span $\mathbb{R}^d$ on which no edit is lost.*

Proof. (a) is Theorem B.8(a),(b),(d) with Proposition B.1(a). If τ^0 are calibrated thresholds, the bounds on false firing of Theorems B.45(a) and B.50(c) and of Proposition B.58(a) hold for $\tau^\star$ as well: $\tau^\star \geq \tau^0$, and raising thresholds shrinks every firing set (Proposition B.1(c)).

(b) Certified implies restored (Lemma B.68(b)). For $\tau \geq \tau^0$ the edits that fire at a position t under τ are among those that fire there under τ^0 (Proposition B.1(c)), and the module returns $W_0 k_t$ plus their contributions, which do not depend on the thresholds. An active pair (i, t) fires under τ iff $\zeta_i(k_t) \geq \tau_i$. Under τ^{r} no active pair fires, so every contribution that is left is zero; and τ_i^{r} exceeds τ_i^0 iff edit i has an active pair. As $\zeta_i(k_t)^+ \leq \zeta_i(q)^+$ for a position of $x^0(q)$, $\tau^{\mathrm{r}} \leq \tau^\star$. Under the hypothesis the contributions at a position sum to zero iff no active pair of the position fires, that is, iff $\tau_i > \zeta_i(k_t)$ for every active pair: a conjunction of lower bounds, which reads $\tau \geq \tau^{\mathrm{r}}$; for a mask it reads that the edits with an active pair are silenced. If every pair that fires is active and edit i fires on q, the eligible position of the largest coordinate is active, so $\tau_i^{\mathrm{r}} \geq \zeta_i(q)^+$; if edit i does not fire on q, $\zeta_i(q)^+ \leq \tau_i^0$.

(c) For reals x and y, $x^+ > y$ in the chain Θ iff $x \geq y$. So $\tau_i^\star > \pi_i$ iff $\tau_i^0 > \pi_i$ or $\mu_i \geq \pi_i$, and the first is excluded. If τ certifies $\mathcal{Q}^-$, then $\tau_i > \mu_i$, that is, $\tau_i \geq \mu_i^+$, for every i (Sec. B.1); if moreover $\mu_i \geq \pi_i$, then $\tau_i \geq \mu_i^+ > \pi_i$.

(d) In $\mathcal{S}^E$ all positions are eligible and $\tau^0 = -\infty$, so every edit fires on every complaint and the least repair changes every threshold. Under $\tau^\star$ edit i fires at k iff $\zeta_i(k) > \mu_i$, and it contributes $\zeta_i(k)\, v_i^{\log} = \Delta_i k$; at the solver key $\zeta_i(k_i) = 1$. Subtracting from $W_E k_i = W_0 k_i + \sum_j \Delta_j k_i$ the contributions of the edits that do not fire at k_i gives the output; an edit j with $\zeta_j(k_i) = 0$ contributes 0.

(e) $\tau_j^\star \geq \tau_j^0 > 1 - \gamma_j^{\mathrm{sep}} \geq \zeta_j(k_i)$ for $j \neq i$, so no other edit fires at k_i. If $\mu_i < 1$, then $\tau_i^\star = \max(\tau_i^0, \mu_i^+) \leq 1 = \zeta_i(k_i)$: edit i fires alone, and the module returns $W_0 k_i + v_i^{\mathrm{rb}} = z_i^\star$ (Lemma B.65). If $\mu_i \geq 1$, then $\tau_i^\star > 1$. The last claim is Theorem B.66(a) for thresholds that also exceed the μ_i.

(f) The first sentence is (c). Let $\psi_i = e_1$ for every edit, $k_i = e_1$ for $i \in J$ and $k_i = 3e_1$ for $i \notin J$, with arbitrary write vectors, and let the complaints have one position each, with the keys $2e_1$, $2e_1 + e_2, \ldots, 2e_1 + e_\varrho$ and $N - \varrho$ further copies of $2e_1$. The keys have rank ϱ, and $\zeta_i(k) = 2/s_i$ at each of them, which is $2 \geq 1$ for $i \in J$ and $\frac{2}{3} < 1$ for $i \notin J$. For the last claim take $J = \emptyset$ and $\varrho = d < T$. □

Example B.73 (Released but not output-preserved). *Let $d = 2$, $T = 2$, $\psi_1 = k_1 = e_1$, $\psi_2 = e_2$, $k_2 = e_2 + \frac{1}{2} e_1$, $u_1 \neq 0$, and one*

complaint with the key $\frac{3}{5}e_1$. *The least repair of* $\mathcal{S}^E$ *is* $\tau^\star = ((\frac{3}{5})^+, 0^+)$, *and both edits are released. At* k_2 *the first edit has the coordinate* $\frac{1}{2} \le \frac{3}{5}$ *and is silent, so the module returns* $W_0k_2 + u_2$ *in the place of* $W_Ek_2 = W_0k_2 + u_2 + \frac{1}{2}u_1$. *If the first edit is of an earlier batch, the target* $z_2^\star = W_0k_2 + \frac{1}{2}u_1 + u_2$ *is now missed by* $\frac{1}{2}u_1$*; if it is not, the target* $W_0k_2 + u_2$ *is attained after the repair and was missed before it.*

Status. (a) restates Theorem B.8; (b) is elementary, and for one complaint it refines Proposition B.42(b); (c) is a corollary of Theorem B.8, where the claim about vectors that are not raise-only is one line more than its part (c); (d) and (e) are corollaries of Lemma B.65 and Theorem B.66; (f) is a construction. Nothing is new mathematics. What the proposition adds to the paper is the accounting: under the certificate, rescoping is always feasible, and its loss is a matter of separation, not of dimension; and "no edit is lost" has three meanings. *Released* is what thresholds alone guarantee. *Output-preserved*, the notion that Proposition B.71 uses for linear repairs, holds for a released edit of the deployed scoping iff the contributions that are silenced at its solver key sum to zero (it suffices that no silenced edit leaks onto this key), which fails on logs with cross-talk. *Target-preserved* is what the rebased values add, and it needs the self-separation of the solver keys in addition.

B.9.4 Rescoping contains rollback. By (H4) the module of a model is a map from a key and a role to an output. Two models are *the same* if these maps agree at every key in $\mathbb{R}^d$ and every role; by (H1)–(H3) they then have the same trace, logits and answer on every query. A model is *reachable* by a class of repairs if it is the same as the model of some repair of the class. For a scoped log $\mathcal{S}$ with thresholds τ^0, the *mask* of $R \subseteq [T]$ is the threshold repair τ^R with $\tau_i^R = +\infty$ for $i \in R$ and $\tau_i^R = \tau_i^0$ otherwise.

A *cost* assigns a number to every threshold vector, and a rollback is charged the cost of its mask. Let $\mathcal{K}$ be a set of *accounting keys*, each the key of a position with its role (the positions of the own requests, of why-not pairs, of any population of queries). The *silenced set* $\mathrm{Sil}(\tau)$ is the set of the pairs $(i,k) \in [T] \times \mathcal{K}$ such that edit i fires at k under τ^0 and not under τ; the mask of R silences the pairs of the edits of R. A cost is *monotone* if it is a function of the silenced set that does not decrease when this set grows. The number of lost edits (for must-fire keys in $\mathcal{K}$), the weight $\omega(\mathrm{Unc}(\tau) \setminus \mathrm{Unc}(\tau^0))$ of the newly uncovered pairs and the number of edits with a silenced pair are monotone costs; the number of changed thresholds is a cost.

Proposition B.74 (Containment). *Let* $\mathcal{L}$ *be an edit log with the deployed scoping* $\mathcal{S}^E$.

(a) Identity. *In the deployed scoping* $\mathcal{S}^E$ *the mask of* R *has the module map* $k \mapsto W_Rk$ *at every key and role: it is the same model as the rollback* $\mathcal{M}[W_R]$. *More generally, under any thresholds* τ *the module of* $\mathcal{S}^E$ *returns* $W_{R_\tau(k)}\ k$ *at a key* k, *where* $R_\tau(k)$ *is the set of the edits that do not fire at* k*: a threshold repair is a rollback whose deletion set depends on the key.*

(b) Containment. *Every model that is reachable by a rollback is reachable by a threshold repair of* $\mathcal{S}^E$. *Hence, for every criterion that is a property of the repaired model (restored, c-repaired at any set of positions, resolved) or of the repaired scoped log, a rollback being read as its mask (certified), for conjunctions of these over complaints and protected queries, and for every cost, the least cost of a threshold repair that meets the criterion is at most the least cost of a rollback that meets it (an inequality of infima). The same holds for the masks of any scoped log* $\mathcal{S}$.

(c) Under the certificate. *For raise-only repairs of a scoped log,* $\mathrm{Sil}(\tau^\star) \subseteq \mathrm{Sil}(\tau)$ *for the least repair* $\tau^\star$ *of* Q^- *and every* $\tau \ge \tau^0$ *that certifies* Q^-. *So* $\tau^\star$ *minimises every monotone cost. The certifying masks are those of the sets* $R \supseteq F := \bigcup_{q \in Q^-} \mathrm{Fire}_{\tau^0}(q)$, *the mask of* F *is the cheapest of them for every monotone cost, and* $\mathrm{Sil}(\tau^\star) \subseteq \mathrm{Sil}(\tau^F)$.

(d) Strictness. *If some* $u_i \neq 0$, *some threshold repair of* $\mathcal{S}^E$ *is reachable by no linear repair: its module map is not additive in the key. On a set of keys on which no coordinate crosses its threshold, a threshold repair of* $\mathcal{S}^E$ *agrees with one rollback.*

(e) One criterion for both. *The inequality of* (b) *can fail if rollback is judged by restored and rescoping by certified: let* $n = d = 1$, $T = 3$, $\psi_i = k_i = 1$ *and* $u = (1, -1, 1)$, *with one complaint of one position with the key* 1, *at which the deployed damage is* 1. *The rollback of the third edit restores the complaint and silences one pair of an edit and this key, whereas every threshold repair that certifies the complaint silences all three.*

Proof. (a) $\mathcal{S}^E$ has all roles, the identity response and the logged values, so an edit that fires at a key k contributes $\zeta_i(k)\, v_i^{\log} = (\langle \psi_i, k\rangle / s_i)\, s_iu_i = \Delta_i k$, and the module returns $W_0k + \sum_{i \notin R_\tau(k)} \Delta_i k = W_{R_\tau(k)}k$. Under the threshold $-\infty$ an edit fires at every key and under $+\infty$ at none, so $R_{\tau^R}(k) = R$ for every k. (b) The mask of R is a threshold repair, by (a) it meets a criterion of the kind named iff the rollback of R does, and it has the cost of the rollback; a minimum over a larger set is not larger. For another scoped log, masks are threshold repairs by definition. (c) Theorem B.8(c): the scope of every edit under τ is contained in its scope under $\tau^\star$. A mask is raise-only, it certifies Q^- iff it silences every edit that fires on a complaint (Proposition B.1(c),(d)), and the silenced set of a mask grows with R. (d) Let $u_i \neq 0$, $\tau_i = 0$ and $\tau_j = +\infty$ for $j \neq i$. The repaired module returns $W_0k + \mathbf{1}[\langle \psi_i, k\rangle \ge 0]\, \langle \psi_i, k\rangle\, u_i$. Its outputs at k_i and at $-k_i$ sum to $s_iu_i \neq 0$, whereas those of a weight sum to zero. If the set of edits that fire is the same set J at every key of a set, the module returns $W_{[T]\setminus J}\, k$ there, by (a). (e) The kept contributions are 1 and -1. In $\mathcal{S}^E$ every edit fires at every position, so a certifying repair raises all three thresholds above the coordinate 1 of the complaint. □

Status. True by construction; the paper states (a) after Def. 2.2 ("the deployed log and its rollbacks are the extreme threshold settings of one family"). Four remarks on what "contains" means. (i) The identity (a) is special to the logged values $v_i^{\log} = s_iu_i$ with the identity response. In a scoped log with other values or responses a mask still removes the terms of its edits from the module map, but the result is in general the model of no weight: one edit with the response 1 and a value $v \neq 0$ returns $W_0k + v$ on its scope. (ii) (a) is an identity of exact arithmetic. Computed as the product W_Rk_t, a rollback agrees with its mask up to round-off only (in the check, on each of 2,000 random keys in double precision the two differ in the last bits); under the convention of the paper that a rollback *is* a mask (Sec. B.1), (b) holds bit for bit. (iii) The converse inclusion fails by (d), and linear repairs other than rollbacks are in general not threshold repairs of $\mathcal{S}^E$: a threshold repair returns W_0k plus a

vector of span$\{u_i\}$. (iv) By (e), a comparison of the two semantics is at one criterion only if both are judged by the same form of the certificate. The two forms agree on masks and on raise-only repairs when no nonempty family of contributions of edits that fire at one position of a complaint sums to zero (Lemma B.68(b)).

B.9.5 The granularity gap.

Proposition B.75 (Granularity gap under the certificate). *Let $\mathcal{L}$ be an edit log and Q^- a finite set of wrong-answer complaints; in* (a)–(c) *the trace keys of Q^- span $\mathbb{R}^d$, and* (d)–(h) *name their complaints.*

(a) A rollback certifies Q^- iff $W_R = W_0$. If no nonempty subfamily of $\Delta_1, \dots, \Delta_T$ sums to zero, every certifying rollback deletes all T edits.

(b) The only linear repair that certifies Q^- is W_0. It output-preserves exactly the edits with $dWk_i = 0$ and target-preserves exactly those with $v_i^{\mathrm{rb}} = 0$.

(c) The least threshold repair of $\mathcal{S}^E$ certifies Q^- and loses exactly the edits with $\mu_i \geq 1$, where μ_i is the largest coordinate of edit i on the trace keys of the complaints.

(d) A family. *For $T \geq 1$, $d \geq T$ and $0 < \xi < \frac{1}{2}$ let $\psi_i = k_i = e_i$ with arbitrary write vectors $u_i \neq 0$, and let Q^- consist of d queries of one position with the keys $\xi(\mathbf{1} + e_j)$, $j \in [d]$, where $\mathbf{1}$ is the all-ones vector (the queries of the family are wrong-answer complaints only if their damage $\xi\,(\sum_i u_i + \mathbf{1}[j \leq T]\,u_j)$ is not zero, as for linearly independent write vectors; the three conclusions below do not need this). These keys span $\mathbb{R}^d$, and every edit has the coordinate ξ or 2ξ on each of them. On this log every certifying rollback deletes all T edits; every certifying linear repair is W_0 and preserves no edit, in either sense; and the least threshold repair $\tau_i^\star = (2\xi)^+$ of $\mathcal{S}^E$ certifies all complaints while every edit is released, output-preserved and target-preserved, each edit firing alone at its solver key.*

(e) Every key dimension. *For $d < T$ let $\psi_i = k_i = e_{\ell(i)}$ with $\ell(i) := 1 + ((i-1) \bmod d)$, the complaints of* (d)*, with the same proviso, and write vectors such that no nonempty subfamily of those of one direction sums to zero. Then every certifying rollback deletes all T edits, the only certifying linear repair W_0 output-preserves no edit, and under the least threshold repair every edit is released and output-preserved.*

(f) Scaled own keys: both criteria. *In any log, let ξk_i with $\xi \neq 0$ be a trace key of a complaint. A linear repair W' c-repairs the complaint at this key iff $\|(W' - W_0)k_i\|_p \leq c\,\|dWk_i\|_p$; so if $dWk_i \neq 0$, no linear repair that c-repairs it for some $c < 1$, and none that certifies it, output-preserves edit i. For the log of* (d) *and the T complaints of one position with the keys ξk_i, $0 < \xi < 1$: for every p and every $c \in [0, 1)$, the only rollback that c-repairs all complaints deletes all T edits, a linear repair that c-repairs all complaints preserves no edit, in either sense, and the least threshold repair $\tau_i^\star = \xi^+$ certifies all complaints while every edit is released, output-preserved and target-preserved.*

(g) The family of (d) under a tolerance. *In* (d) *let $d = T$, $p = 2$, $c \in (0, 1)$ and let the write vectors be orthonormal. The largest number of edits that a rollback keeps while it c-repairs all complaints is $\max(0, \lfloor c^2(T+3) \rfloor - 3)$. If $1 \leq c^2(T+3)$ and $1 \leq \kappa \leq T$ satisfies $(T-\kappa)/\kappa^2 \leq c^2(T+3)$, some compensation c-repairs all complaints and output-preserves $T - \kappa$ edits; $\kappa = \lceil 1/c \rceil$ qualifies once $T \geq 1/c^2$. The least threshold repair loses no edit.*

(h) Fewer complaints. *In* (d) *let the write vectors be linearly independent and keep only the complaints $j \in [N]$, $1 \leq N \leq T$. Under the certificate, every certifying rollback still deletes all T edits; the largest number of edits that a certifying linear repair output-preserves is $T + 1 - N$ if $T < d$ and $T - N$ if $T = d$; and the least threshold repair loses no edit. So one complaint costs rollback the whole log, each further complaint costs a linear repair one edit, and no number of these complaints costs rescoping an edit.*

Proof. (a) K_Q has full row rank, so $(W_R - W_0)K_Q = 0$ iff $W_R = W_0$, that is, iff the kept Δ_i sum to zero. (b) $D'K_Q = 0$ forces $D' = 0$; then $W'k_i = W_0k_i$, which is W_Ek_i iff $dWk_i = 0$ and $z_i^\star$ iff $v_i^{\mathrm{rb}} = 0$. (c) Proposition B.72(c),(d) with the solver keys as must-fire keys.

(d) $K_Q = \xi(\mathbf{1}\mathbf{1}^\top + I)$ has the eigenvalues $\xi(d+1)$ and ξ, so it is invertible, and $\langle \psi_i, \xi(\mathbf{1} + e_j)\rangle = \xi(1 + \mathbf{1}[i = j])$. If $\sum_{i\in J} u_i e_i^\top = 0$, its column i is $u_i = 0$ for $i \in J$; so $J = \emptyset$ and (a) applies. $\Delta_j k_i = \mathbf{1}[i = j]u_i$, hence $dWk_i = u_i = v_i^{\mathrm{rb}} \neq 0$ for every order of the batches (Lemma B.65), and (b) applies. In (c), $\mu_i = 2\xi < 1$, attained at the complaint $j = i$. At k_i edit i has the coordinate $1 > 2\xi$ and every other edit the coordinate 0, so edit i fires alone and the module returns $W_0k_i + u_i = W_Ek_i = z_i^\star$.

(e) The complaint keys and the coordinates are as in (d), with $\mu_i = 2\xi$. If $\sum_{i\in J} u_i e_{\ell(i)}^\top = 0$, its column of a direction is the sum of the u_i, $i \in J$, of that direction, so $J = \emptyset$. At k_i the edits of the direction $\ell(i)$ have the coordinate 1 and fire, the others have the coordinate 0: the module returns $W_0k_i + \sum_{j:\ell(j)=\ell(i)} u_j = W_Ek_i$, and $dWk_i \neq 0$ by the hypothesis on the write vectors.

(f) A weight is homogeneous: $(W' - W_0)(\xi k_i) = \xi\,(W' - W_0)k_i$ and $dW(\xi k_i) = \xi\,dWk_i$; divide both sides of the criterion by $|\xi|$. If edit i were output-preserved, $(W' - W_0)k_i = dWk_i$, and the criterion would read $\|dWk_i\|_p \leq c\,\|dWk_i\|_p$. In the log of (d), $dWk_i = u_i = v_i^{\mathrm{rb}} \neq 0$, so no edit is output-preserved or target-preserved, and a rollback that keeps edit i has $(W_R - W_0)k_i = u_i$. The complaint ξk_i has the coordinate ξ under edit i and 0 under the others, so $\mu_i = \xi < 1$, and the rest is as in (d).

(g) Divide the keys by ξ. For a rollback that keeps J, the residual at the complaint j is $\sum_{i\in J}(1 + \mathbf{1}[i = j])u_i$, of squared norm $|J| + 3\,\mathbf{1}[j \in J]$, and the squared norm of the damage is $T+3$. If $J \neq \emptyset$, a complaint $j \in J$ requires $|J| + 3 \leq c^2(T+3)$, which implies the condition $|J| \leq c^2(T+3)$ of the complaints $j \notin J$. For the compensation let J be a set of $T - \kappa$ edits, $u_J := \sum_{i\in J} u_i$, and let D' be the matrix with $D'e_i = u_i$ for $i \in J$ and $D'e_i = -u_J/\kappa$ otherwise; $X := D' - dW$ vanishes on the own keys of J. Then $D'\mathbf{1} = u_J - \kappa\,u_J/\kappa = 0$, so the module output at the complaint j moves by $D'(\mathbf{1} + e_j) = D'e_j$, of norm 1 for $j \in J$ and $\sqrt{T - \kappa}/\kappa$ otherwise; both are at most $c\sqrt{T+3}$ under the two conditions. For $\kappa = \lceil 1/c \rceil$ and $T \geq 1/c^2$, $(T - \kappa)/\kappa^2 \leq c^2 T$ and $\kappa \leq T$.

(h) Every edit has a nonzero coordinate on every complaint key and the write vectors are linearly independent, so the least certifying deletion set is $[T]$ (Proposition B.69(b)). The vectors $dWk_i = u_i$ are linearly independent, so by Proposition B.71(b),(c) the largest number of output-preserved edits is $\operatorname{rank}[K_Q, K_E] - \operatorname{rank} K_Q$. The N complaint keys are linearly independent, and $\operatorname{col}[K_Q, K_E] = \operatorname{span}\{e_1, \dots, e_T, \mathbf{1}\}$, of dimension $T + 1$ if $T < d$ and T if $T = d$. For

rescoping, μ_i is 2ξ for $i \le N$ and ξ for $i > N$, and the argument of (d) applies. □

Status. New as a statement, elementary; (a)–(c) are corollaries of Propositions B.69–B.72. In the family the three repairs are judged by one criterion, and the gap is in one quantity: the edits lost, 0 for rescoping against T for rollback and for every linear repair. There is no gap in the number of modified statements, T thresholds against T deletions, nor in the cost of computing the repair, one pass in each case. The complaints of the family are queries that every edit touches slightly: once such keys span the key space, a weight cannot leave them alone and keep any edit, whereas a threshold separates them from the own keys, on which the coordinate is 1. The family is constructed and its own keys are orthonormal; the general statement for a given log is (c): the loss of rescoping is the set of edits with $\mu_i \ge 1$, which is measured, not assumed. The comparison counts lost edits and leaves one thing out: the weight W_0 restores every query, reported or not, whereas the threshold repair certifies exactly the queries all of whose coordinates are at most the μ_i; about the other unreported queries rescoping has the calibration bounds of Theorem 3.6 only.

The gap of (d) against *compensation* depends on the criterion, and (g) says by how much: under a tolerance the complaints of (d), which are nearly parallel, are repaired by giving up about $1/c$ edits, not T, with a correction of rank $\lceil 1/c \rceil$ (the rollback of these edits plus one term of rank one), while rollback still gives up a fraction $1 - c^2$ of the log. The gap that does not depend on the criterion is (f): a weight is homogeneous in the key and cannot treat ξk_i and k_i differently, a threshold can. Its complaints are scaled copies of own keys, which a benign query need not resemble; (f) is the extreme case of a complaint key with a large component along an own key.

B.9.6 What rescoping gains: granularity. In a scoped log $\mathcal{S}$ a rollback of R is the mask τ^R (Sec. B.9.4); for $\mathcal{S}^E$ this is the rollback W_R of the paper. So both semantics act on one object and are judged by one criterion.

Theorem B.76 (What rescoping gains under one criterion). *Let $\mathcal{S}$ be a scoped log with thresholds τ^0, $\mathcal{Q}^-$ a finite set of wrong-answer complaints, $F := \bigcup_{q\in\mathcal{Q}^-} \mathrm{Fire}_{\tau^0}(q)$ the set of edits that fire on a complaint, and $\mu_i := \max_{q\in\mathcal{Q}^-} \zeta_i(q)$. For one complaint, $|F|$ is the lineage size $\mathrm{wd}_{\mathcal{S}}(q)$.*

(a) The same edits are touched. *Every scoped log that certifies $\mathcal{Q}^-$ and differs from $\mathcal{S}$ in parameters of edits differs from it in every edit of F. The least threshold repair changes exactly the thresholds of the edits of F, and a mask certifies $\mathcal{Q}^-$ iff it silences F. For restored, under the hypothesis of Proposition B.72(b), the last two claims hold with the edits that have an active pair in the place of F, and every raise-only threshold repair that restores $\mathcal{Q}^-$ changes the threshold of each of these edits. The first claim does not carry over: a scoped log can restore $\mathcal{Q}^-$ and leave an edit with an active pair unchanged, by changing a value, or by lowering the threshold of another edit, so that contributions cancel. (Two edits with $\psi_i = k_i = 1$, the identity response and the complaint key 1: with the values $1, 1$ and the thresholds $-\infty$, change the second value to -1; with the values $1, -1$ and the thresholds $-\infty, 2$, lower the second threshold to $-\infty$.) The lineage size is a property of the log: $\mathrm{wd}_{\mathcal{S}^E}(q) = T$ for every query, and $\mathrm{wd}_{\tau'}(q) \le \mathrm{wd}_{\tau}(q)$ for $\tau' \ge \tau$.*

(b) Containment. *Masks are threshold repairs. Under each criterion, applied to both, the least cost of a threshold repair is at most that of a rollback, for every cost that is a function of the silenced set (Proposition B.74).*

(c) Granularity. *The least repair silences an edit $i \in F$ on the eligible keys k with $\tau_i^0 \le \zeta_i(k) \le \mu_i$ and on no other key; the mask of F silences it on its whole scope. If no edit is lost under τ^0, the least repair loses the edits of $\{i \in F : \mu_i \ge \pi_i\}$ and the mask loses all of F. The two sets coincide if no edit of F is separated from the complaints, and they are $\emptyset$ and $[T]$ on the logs of Proposition B.75(d).*

(d) No lower complexity. *For certified, both least repairs, $\tau^\star$ and the mask of F, are computed in one pass, with $O(Td)$ operations per reported position. For restored (the case $c = 0$) and under the tolerance criterion with $c \in (0, 1)$ (values v_i and the identity response, as in Sec. B.5.4), take one complaint q of one position with nonzero damage and, as the cost, the number of edits that fire on q under τ^0 and are un-fired on it (for a reported position of a longer trace: Remark B.30). Then the threshold repairs $\tau \ge \tau^0$ and the masks have the same feasible sets of un-fired edits and the same optimum, that of* Min-Rollback$_{p,c}$ *on the $\mathrm{wd}_{\mathcal{S}}(q)$ contributions that fire on q, and every instance of* Min-Rollback$_{p,c}$ *arises so, also for $\mathcal{S}^E$. So both problems are strongly NP-hard and not $\mathrm{wd}_{\mathcal{S}}(q)^{1-\varepsilon}$-approximable, both are solved by enumerating $2^{\mathrm{wd}_{\mathcal{S}}(q)}$ sets, and for restored both are solved by F if the contributions that fire on q are zero-sum-free.*

(e) Streams. *Let complaints $q_1, \dots, q_N$ arrive in turn and be repaired from $\mathcal{S}$, once by least repairs and once by the masks of their firing sets. After every complaint the thresholds of the masked log are at least those of the other log; so its firing sets are contained in those of the other log, and the set of edits that it has lost contains the set that the other log has lost. The masks silence at most T edits in total. The least repairs write $\sum_{h\le N} \mathrm{wd}(q_h)$ thresholds, the lineage sizes being taken when the complaints arrive; this number can be NT, and it is at most $T(1 + \frac{1}{2} + \cdots + \frac{1}{N})$ in expectation for exchangeable complaints (Proposition B.13).*

Hence, under one criterion, rescoping neither touches fewer edits than rollback nor lies in a lower complexity class: what is polynomial is the criterion certified, for both, and what is hard is a criterion on module outputs when contributions cancel, for both. What rescoping gains is (b) *and* (c)*: a touched edit is not deleted, it is restricted to the keys above the complaints, and no edit is lost whose must-fire keys its detector separates from the complaints.*

Proof. (a) The first claim is Proposition B.1(d), applied to every complaint. The second is Theorem B.8(b), and the third is Proposition B.74(c). For restored, the claims about raise-only threshold repairs and masks are Proposition B.72(b); in the first example the contributions become 1 and -1, and in the second the edits that fire contribute 1 and -1. In $\mathcal{S}^E$ every threshold is $-\infty$ and every role is eligible, so every edit fires at every position; raising thresholds shrinks firing sets (Proposition B.1(c)). (b) Proposition B.74(b). (c) Edit i fires at an eligible key k under τ^0 iff $\zeta_i(k) \ge \tau_i^0$, and under $\tau_i^\star = \max(\tau_i^0, \mu_i^+)$ iff in addition $\zeta_i(k) > \mu_i$. The lost sets are those of Proposition B.72(c), and a mask loses the edits that it silences.

If $\mu_i \geq \pi_i$ for every $i \in F$, the two sets are F. (d) Theorem B.8(d) for the least repair; the least mask is the union of the firing sets, which costs the same coordinates. For the other criteria, Proposition B.29(a), which allows $c = 0$, shows that the damage of q under $\tau \geq \tau^0$ depends only on the set of edits of $\mathrm{Fire}_{\tau^0}(q)$ that are un-fired on q, and a mask un-fires the edits of $R \cap \mathrm{Fire}_{\tau^0}(q)$; every subset is realised by a mask, hence by thresholds. The rest is Proposition B.29(b),(c), and Lemma A.4 for $\mathcal{S}^E$. If the firing contributions are zero-sum-free, a zero residual forces every firing edit to be un-fired (Proposition B.42(b)). (e) Let τ and τ' be the thresholds of the least-repaired and of the masked log, with $\tau' \geq \tau$ before a complaint q arrives. The mask raises τ'_i to $+\infty$ for the edits that fire on q under τ'. For the others, $\tau'_i > \zeta_i(q)$, that is, $\tau'_i \geq \zeta_i(q)^+$, hence $\tau'_i \geq \max(\tau_i, \zeta_i(q)^+)$, the new threshold of the other log. So $\tau' \geq \tau$ after the complaint, and firing sets and lost sets are monotone in the thresholds. A silenced edit fires on no later complaint, so every edit is silenced at most once. The number of writes is Theorem B.8(b); for $\mathcal{S}^E$ it is NT if every coordinate of q_{h+1} exceeds that of q_h. □

Status. New as a statement; every part is a corollary of results of this report (the parts name them), and (b), (c) contain Propositions B.74 and B.75. Two readings. (i) The sentence "the lineage size takes the place of T" would compare two logs, not two semantics: on the deployed log a complaint touches all T edits under either semantics, and on a log that already has scopes it touches $\mathrm{wd}_{\mathcal{S}}(q)$ edits under either. A log acquires scopes only through rescoping, by calibration or by earlier repairs; after a rollback of $\mathcal{S}^E$ the kept edits still fire at every position. (ii) For the weight form of the certificate on the deployed log, the least deletion set is the set $\mathrm{Ctr}(Q^-)$ of the edits that contribute on the complaints, under the zero-sum-freeness of Proposition B.69(b); it is $[T]$ unless a write vector vanishes or a read-out vector is orthogonal to every trace key of the complaints.

B.9.7 The tolerance column. In this part every complaint has one reported position, with the *complaint key* k_q and nonzero damage; $p \in [1, \infty]$ and a rational $c \in [0, 1)$ are fixed, and a complaint is c-repaired if it is so at its complaint key. A tolerance at a complaint key implies nothing about the answer (Sec. A.2); what implies the pre-edit answer is the certificate (Lemma B.68(a)).

Proposition B.77 (The three repairs under the tolerance criterion). *For complaints of one reported position with nonzero damage, $p \in [1, \infty]$ and $c \in [0, 1)$, the following hold.*

(a) Rollback *(Theorem A.5(a),(b), quoted). The set $R = [T]$ always c-repairs. Let $0 \leq c \leq c' < 1$ be rationals; then, for one complaint key, "*Min-Rollback$_{p,c}$ *is strongly NP-hard. For every $\varepsilon > 0$, unless* P = NP*, no polynomial-time algorithm returns R with $\|r(R)\|_p \leq c'\|D\|_p$ and $|R| \leq T^{1-\varepsilon}\mathrm{OPT}_c$. It is NP-hard to distinguish $\mathrm{OPT}_c \leq T^{\varepsilon}$ from $\mathrm{OPT}_{c'} = T$." At $c = 0$ this is rollback under the certificate at one key (Proposition B.69(d)).*

(b) Compensation. *Let K_{prot} be the protected keys and K_q the matrix of the A complaint keys.*

(i) If $\mathrm{rank}[K_{\mathrm{prot}}, K_q] = \mathrm{rank}\, K_{\mathrm{prot}} + A$, the minimum compensation $X^\star = VK_\perp^+$ restores every complaint key and vanishes on every protected key (Lemma A.9): every tolerance is met, in closed form, and no protection is given up. With the own keys as protected keys this holds iff the A complaint keys are linearly independent modulo $\mathrm{col}\, K_E$, which needs $\mathrm{rank}\, K_E + A \leq d$; all T edits are then output-preserved.

(ii) For one complaint key, a correction X with $XK_{\mathrm{prot}} = 0$ that c-repairs the complaint exists iff $k_q \notin \mathrm{col}\, K_{\mathrm{prot}}$, that is, iff an admissible X exists: at one key a tolerance $c < 1$ protects nothing that exact restoration does not. Hence the smallest set of protection groups that must be given up is that of Min-Sacrifice*, for every $c \in [0, 1)$: strongly NP-complete and without a constant-factor approximation for groups of one key (Proposition A.26), and equal to $s_Z^\star$ under the general position of Theorem A.12(c), which for T groups of one key is $\max(0, T + 1 - d)$. For the exact restoration of A complaint keys the count of Theorem A.12(c) is $\max(0, T + A - d)$.*

(iii) For several complaint keys a tolerance can protect more than exact restoration does, and the number of protected edits is then not bounded by a dimension count: in Proposition B.75(g) the complaint keys span $\mathbb{R}^d$, exact restoration protects no edit, and for $T \geq 1/c^2$ a c-repairing compensation protects all but $\lceil 1/c \rceil$ edits; Proposition A.20(e) *has logs on which all T edits are protected. It need not: in Proposition B.75(f) no edit can be protected under either criterion. For a fixed set of protected keys, c-repairing all complaints is a convex feasibility problem in X, a linear program for $p \in \{1, \infty\}$.*

(c) Rescoping. *Let $\mathcal{S}$ be a scoped log with thresholds τ^0, values v_i and the identity response.*

(i) The least repair of Theorem B.8 c-repairs every complaint, for every $c \geq 0$.

(ii) (Proposition B.29, quoted.) For one complaint q with $F := \mathrm{Fire}_{\tau^0}(q)$ and the set $S := \{i \in F : \tau_i > \zeta_i(q)\}$ of the edits that $\tau \geq \tau^0$ un-fires on q: "Minimising $|S|$ subject to the c-repair of q is the instance $(\zeta_i(q)v_i)_{i \in F}$ of Min-Rollback$_{p,c}$*: the same feasible sets and the same objective." Every instance of* Min-Rollback$_{p,c}$ *arises in this way. "Hence parts* (a) *and* (b) *of Theorem A.5 hold for the repair of one complaint with $\mathrm{wd}(q)$ in place of T; and the problem is solved by enumerating the $2^{\mathrm{wd}(q)}$ subsets of F, whatever the length of the log."*

(iii) (Propositions B.31 and B.32, quoted.) Let τ^0 be feasible, and let the tolerance form *of* Scope-Cut *ask for feasible $\tau \geq \tau^0$ that maximise the weight of the c-repaired complaints minus λ times the weight of the newly uncovered pairs, for any number of complaints; a complaint with zero damage under τ^0 counts as repaired iff its residual is zero. Then: "For every p and every $c \in [\frac{1}{2}, 1)$ it is NP-hard, already when every complaint has lineage size 2, with scalar values, unit weights, one pair of Q^+ per edit and $\lambda = 1$. For every p and every $c \in [0, 1)$ it is NP-hard on complaints of lineage size 3 with nonzero damage. For $c < \frac{1}{2}$ and $\lambda > 0$, on complaints of lineage size at most 2 it is a* Scope-Cut *instance plus a constant, also when complaints of width 2 have zero damage. Hence, unless* P = NP*, it is polynomial exactly for lineage size at most 1, and for width 2 with*

$c < \frac{1}{2}$." *The classification is a worst case over the values of the edits.*

(iv) Against rollback. *For every cost that is a function of the silenced set, the least cost of a c-repairing threshold repair is at most that of a c-repairing mask (Proposition B.74(b)). This is an inequality of optima and not a reduction: it transfers neither hardness nor an approximation ratio in either direction. For one complaint and the number of un-fired edits as the cost the two optima are equal (Theorem B.76(d)).*

(v) Lost edits. *Let the cost be the number of lost edits, with the solver keys as must-fire keys. Deciding whether some threshold repair c-repairs one complaint and loses no edit is strongly NP-hard, for every p and c, already for the deployed scoping of logs with orthonormal own keys; so, unless* P = NP*, the least number of lost edits has no polynomial-time approximation within any factor, already on instances on which it is* 0 *or* 1*. Under the certificate the same question is answered in one pass (Proposition B.72(c)).*

(vi) No least repair. *By Proposition B.29(d), the sets of un-fired edits that c-repair a complaint need not be closed under supersets, and they need not have a least element: for the scalar contributions $+1, -1, +1$ they are $\{1\}$, $\{3\}$ and $\{1, 2, 3\}$, for every $c < 1$ (Proposition B.12, consequence (iii)).*

Proof. (a) is Theorem A.5; $r([T]) = 0$.

(b)(i) Lemma A.9, last claim: the system is solvable for every target, in particular for $V = -dWK_q$. $\operatorname{rank}[K_E, K_q] = \operatorname{rank} K_E + A$ iff no nontrivial combination of the complaint keys lies in $\operatorname{col} K_E$; then $(W_E + X^\star)k_i = W_E k_i$. (ii) If $k_q \in \operatorname{col} K_{\text{prot}}$, every X with $XK_{\text{prot}} = 0$ has $Xk_q = 0$, so the residual $(dW + X)k_q$ is the damage, and the criterion fails as $c < 1$ and the damage is not zero. If $k_q \notin \operatorname{col} K_{\text{prot}}$, an admissible X exists (Lemma A.9(b)) and has residual 0. So a sacrifice set is feasible for the tolerance iff it is feasible in the sense of Sec. A.7, and the smallest one is OPT of Min-Sacrifice. For $r_i = 1$, $z = 0$ and $a = A$, $s^\star_Z$ is the least s with $s = T$ or $T - s \le d - A$, which is $\max(0, T + A - d)$ for $A \le d$. (iii) Proposition B.75(b),(f),(g). The set $\{X : XK_{\text{prot}} = 0\}$ is a subspace and each constraint $\|(dW + X)k_q\|_p \le c\|dWk_q\|_p$ is convex in X; for $p \in \{1, \infty\}$ it is expressed by linear inequalities.

(c)(i) Certified implies c-repaired (Lemma B.68(c)). (ii), (iii) and (vi) are the statements cited; in (vi), keeping the edits 1, 2, 3 or 3 or 1 alone leaves the residual 1, the damage, and keeping $\{2, 3\}$, $\{1, 2\}$ or none leaves 0; keeping $\{2\}$ leaves -1 and keeping $\{1, 3\}$ leaves 2. (iv) Proposition B.74(b). The least cost of a mask can exceed the least cost of a threshold repair by any factor (T lost edits against none in Proposition B.75(f)), and a threshold repair need not be a mask, so an algorithm for either problem solves or approximates the other only where the optima are shown to coincide.

(v) Let $(U = [3h]; C_1, \ldots, C_H)$ be an instance of X3C in which every element lies in some set, and take the $T = H + 2$ contributions of $\mathcal{I}(C, H)$ of Sec. A.4: the sets $\chi(C_j)$, the carrier and the balancer w_{b}; the damage is e_*. Let the log have $\psi_i = k_i = e_i$, so that $s_i = 1$, and let the complaint q have one position with the key $k_q := e_{\text{b}} + \frac{1}{2}\sum_{i \ne \text{b}} e_i$: its coordinate is 1 under the balancer and $\frac{1}{2}$ under every other edit. With the write vectors $u_i := w_i/\zeta_i(k_q)$ the contributions $\Delta_i k_q$ are those of $\mathcal{I}(C, H)$, and the scoped log is $\mathcal{S}^E$. A threshold repair un-fires on q a set S of edits, and by Lemma A.6 it c-repairs q iff $S = [T]$ or S consists of the carrier and an exact cover, for every p and $c \in [0, 1)$. An edit other than the balancer is un-fired on q and released by a threshold in $(\frac{1}{2}, 1]$; the balancer is un-fired on q only by a threshold above 1, which loses it. So a c-repairing threshold repair that loses no edit exists iff an exact cover exists, and otherwise the least number of lost edits is 1 (thresholds 1, and 1^+ for the balancer). The numbers of the instance are $\frac{1}{2}$ and integers of absolute value at most $\max(2, H)$, so the hardness is strong. An algorithm with any approximation factor returns a repair without lost edits whenever one exists. □

Status. (a), (b)(i), (c)(ii), (c)(iii) and (c)(vi) restate results of this report (Theorems A.5 and A.12, Sec. B.5.4). (b)(ii) and (c)(i),(iv) are one-line observations. (c)(v) is new, by the reduction of Lemma A.6; (b)(iii) is new as an observation. The column reads: under a tolerance, rollback and rescoping are always feasible and both are hard, with the lineage size of the complaint in the place of T in both; the optimum of rescoping is never larger; compensation at one key per complaint is a closed form while the complaint keys are independent of the protected keys, and beyond that its choice of what to give up is hard as well. Under a tolerance, feasible deletion sets and threshold repairs need not have a least element, and the least sacrifice of a compensation need not be unique (protected keys e_1, e_2 and the complaint key $e_1 + e_2$: the minimal sacrifice sets are $\{1\}$ and $\{2\}$).

B.10 Several edited modules

Published locate-and-edit configurations write one edit into several consecutive MLP modules. This subsection extends the certificate, the least repair and the calibration bound to such logs, and states what does not extend; Remark 3.5 of the paper points to it. The paper's claims and experiments are for one module. Proposition B.1, Theorem B.8 and Proposition B.58 are the case $M = 1$.

Definition B.78 (Scoped log over several modules). *Let the model have $M \ge 1$* edited modules *$\mu = 1, \ldots, M$, numbered in computation order; module μ is a linear map with the pre-edit weight $W_0^\mu \in \mathbb{R}^{n_\mu \times d_\mu}$. A* term *is a pair (i, μ) of an edit $i \in [T]$ and a module μ, the part of edit i that is written into module μ; I is the set of terms and I_μ the set of the terms of module μ. A* scoped log over the modules *is*

$$\mathcal{S} = \big((W_0^\mu)_{\mu \le M};\ (\psi_{i,\mu}, s_{i,\mu}, \mathcal{R}_{i,\mu}, \tau_{i,\mu}, g_{i,\mu}, v_{i,\mu})_{(i,\mu) \in I}\big),$$

with, for each term, a detector $\psi_{i,\mu} \in \mathbb{R}^{d_\mu}$, a normaliser $s_{i,\mu} > 0$ (the own coordinate of edit i at module μ), a role set $\mathcal{R}_{i,\mu} \subseteq$ Roles*, a threshold $\tau_{i,\mu} \in \Theta$, a response $g_{i,\mu} \colon \mathbb{R} \to \mathbb{R}$ and a value $v_{i,\mu} \in \mathbb{R}^{n_\mu}$. The* coordinate *of the term at a key $k \in \mathbb{R}^{d_\mu}$ is $\zeta_{i,\mu}(k) := \langle \psi_{i,\mu}, k \rangle / s_{i,\mu}$. When module μ is evaluated at a key k of a position with the role ϱ, the term (i, μ) is* released *iff $\varrho \in \mathcal{R}_{i,\mu}$ and $\zeta_{i,\mu}(k) \ge \tau_{i,\mu}$, and the module returns*

$$W_0^\mu k + \sum_{(i,\mu)\ \text{released}} g_{i,\mu}\big(\zeta_{i,\mu}(k)\big)\, v_{i,\mu}.$$

$\mathcal{M}[\mathcal{S}]$ is the model with these modules. The pre-edit model *is the scoped log without terms. The* deployed scoping *$\mathcal{S}^E$ of a log with the updates $u_{i,\mu}\psi_{i,\mu}^\top$ has all roles, all thresholds $-\infty$, the identity response*

and the values $s_{i,\mu}u_{i,\mu}$: its module μ returns $W_E^\mu k$ with $W_E^\mu = W_0^\mu + \sum_i u_{i,\mu}\psi_{i,\mu}^\top$, the deployed model, linear at every module.

Definition B.79 (Runs; assumptions (M1), (M2)). *A* call *is one evaluation of an edited module: it has a module, a position of the sequence that the model evaluates, the role of this position, and a key. The* run $\mathrm{run}_{\mathcal{S}}(q)$ *of $\mathcal{M}[\mathcal{S}]$ on a query q is the finite sequence of its calls $1, \dots, N$ in the order of execution, each with the output that the module returned, together with everything that the model emits: the logits of every decoding step, the generated tokens and the decision to stop.* $\mathrm{run}_0(q)$ *is the run of the pre-edit model. We assume, in one execution configuration:*

(M1) The log acts through the outputs of the edited modules only, and the model is deterministic. *For every $h \geq 1$, whether the run has an h-th call, the module, position, role and key of this call, and everything that is emitted before it (or, if there is no h-th call, until the run ends) are determined by q and by the outputs that the calls $1, \dots, h-1$ returned, by a rule that is the same for every scoped log over the pre-edit weights.*

(M2) Gate. *At a call, a module evaluates each of its terms (coordinate, release, response) from the parameters of this term and the key and role of the call alone; its output is determined by the key and by the released terms with their detectors, normalisers, responses and values, by a rule that is the same for every scoped log over the pre-edit weights; and where no term is released the output is $W_0^\mu k$ itself, as the pre-edit module computes it.*

Definition B.80 (Coordinates, firing sets, released sets, certified queries). *A call is* eligible *for a term (i,μ) if it is a call of module μ with a role in $\mathcal{R}_{i,\mu}$, and the term is* open *at a call, under $\mathcal{S}$, if the call is eligible and its key k has $\zeta_{i,\mu}(k) \geq \tau_{i,\mu}$. The* coordinate *$\zeta_{i,\mu}(q)$ of a query q is the largest $\zeta_{i,\mu}(k)$ over the keys of the eligible calls of the* pre-edit *run* $\mathrm{run}_0(q)$*, and $-\infty$ if there is none. As in Sec. B.1, $\tau_{i,\mu} > a$ means $\tau_{i,\mu} \geq a^+$, with $(-\infty)^+ := -\infty$, so that $\tau_{i,\mu} > -\infty$ always holds. The* firing set $\mathrm{Fire}_{\mathcal{S}}(q)$ *of q is the set of the terms that are open at some call of* $\mathrm{run}_0(q)$*: $(i,\mu) \in \mathrm{Fire}_{\mathcal{S}}(q)$ iff the term has an eligible call and $\zeta_{i,\mu}(q) \geq \tau_{i,\mu}$, that is, iff $\tau_{i,\mu} > \zeta_{i,\mu}(q)$ does not hold. The* released set $\mathrm{Rel}_{\mathcal{S}}(q)$ *is the set of the terms that are released at some call of* $\mathrm{run}_{\mathcal{S}}(q)$*. The query q is* certified *if $\mathrm{Fire}_{\mathcal{S}}(q) = \emptyset$, that is, if $\tau_{i,\mu} > \zeta_{i,\mu}(q)$ for every term. A pair $((i,\mu), q)$ is* covered *if $(i,\mu) \in \mathrm{Fire}_{\mathcal{S}}(q)$ and* released *if $(i,\mu) \in \mathrm{Rel}_{\mathcal{S}}(q)$. We write Fire_τ and Rel_τ when only the thresholds vary.*

The coordinate $\zeta_{i,\mu}(q)$ is a function of q, of the pre-edit weights and of the detector, normaliser and role set of the term. It does not depend on the thresholds, on the values or on the other terms. The keys of $\mathrm{run}_{\mathcal{S}}(q)$ do: this is the failure of (H1) in Example B.3.

Lemma B.81 (Two runs coincide up to the first discordant call). *Assume* (M1) *and* (M2). *Let $\mathcal{S}$ and $\mathcal{S}'$ be scoped logs over the same pre-edit weights, q a query and N the number of calls of* $\mathrm{run}_{\mathcal{S}}(q)$*. Call the h-th call of* $\mathrm{run}_{\mathcal{S}}(q)$ concordant *if, at its key and role, $\mathcal{S}'$ releases the same terms as $\mathcal{S}$, with the same detectors, normalisers, responses and values. Let $h^\star \leq N+1$ be the least index of a call that is not concordant, and $h^\star := N+1$ if there is none. Then* $\mathrm{run}_{\mathcal{S}'}(q)$ *has the same calls $1, \dots, h^\star - 1$ as* $\mathrm{run}_{\mathcal{S}}(q)$*, with the same modules, positions, roles, keys and outputs, and the same is emitted before call $h^\star$. If $h^\star \leq N$, the $h^\star$-th call of* $\mathrm{run}_{\mathcal{S}'}(q)$ *has the module, position, role and key of the $h^\star$-th call of* $\mathrm{run}_{\mathcal{S}}(q)$*. If $h^\star = N+1$, then* $\mathrm{run}_{\mathcal{S}'}(q) = \mathrm{run}_{\mathcal{S}}(q)$.

Proof. We show by induction on $h \leq h^\star$ the following claim $A(h)$: the run $\mathrm{run}_{\mathcal{S}'}(q)$ has the calls $1, \dots, h-1$ of $\mathrm{run}_{\mathcal{S}}(q)$, with the same modules, positions, roles, keys and outputs. $A(1)$ is empty. Assume $A(h)$ with $h \leq h^\star$. By (M1), whether there is an h-th call, its module, position, role and key, and what is emitted before it, are determined by q and the outputs of the calls $1, \dots, h-1$, which are the same in the two runs. So both runs have an h-th call or neither has, the two calls have the same module, position, role and key, and the same is emitted before them. If $h = h^\star \leq N$, this is the second claim. If $h < h^\star$, then $h \leq N$ and call h is concordant: by (M2) each log evaluates its terms from their parameters and the key and role of the call, so at the h-th call of its own run $\mathcal{S}'$ releases the terms that it releases at the key and role of the h-th call of $\mathrm{run}_{\mathcal{S}}(q)$, which are those that $\mathcal{S}$ releases there, with the same detectors, normalisers, responses and values. By (M2) the two outputs are equal, which is $A(h+1)$. If $h^\star = N+1$, then $A(N+1)$ holds, and by (M1) neither run has a further call and both emit the same until they end. □

Theorem B.82 (Certificate for several edited modules). *Assume* (M1) *and* (M2). *Let $\mathcal{S}$ be a scoped log over the modules $1, \dots, M$ and q a query.*

(a) Certificate. *If $\mathrm{Fire}_{\mathcal{S}}(q) = \emptyset$, that is, if no term is open at any call of the pre-edit run of q, then $\mathrm{run}_{\mathcal{S}}(q) = \mathrm{run}_0(q)$: the scoped model makes the same calls with the same keys and outputs, and it emits the same logits at every decoding step and the same tokens, for a greedy decoding of any length, bit for bit. In particular $\hat{y}(q;\mathcal{S}) = y_0(q)$.*

(b) Run-time form. *$\mathrm{Fire}_{\mathcal{S}}(q) = \emptyset$ iff $\mathrm{Rel}_{\mathcal{S}}(q) = \emptyset$: no term is open on the pre-edit run iff no term is released in the run of the scoped model.*

(c) First release. *If $\mathrm{Fire}_{\mathcal{S}}(q) \neq \emptyset$, let $h^\star$ be the first call of $\mathrm{run}_0(q)$ at which a term of $\mathcal{S}$ is open. Then $\mathrm{run}_{\mathcal{S}}(q)$ and $\mathrm{run}_0(q)$ have the same calls before $h^\star$, with the same outputs, and emit the same before call $h^\star$; call $h^\star$ has the same module, position, role and key in both runs; no term is released in $\mathrm{run}_{\mathcal{S}}(q)$ before call $h^\star$; and the terms released at call $h^\star$ of $\mathrm{run}_{\mathcal{S}}(q)$ are exactly those that are open at call $h^\star$ of $\mathrm{run}_0(q)$.*

(d) A subset of the modules does not certify. *There are a model with two edited modules, a scoped log and a query such that no term of module 1 is open on the pre-edit run and the answer differs from the pre-edit answer.*

Proof. By (M2) a module of the pre-edit model returns $W_0^\mu k$ at every call, so $\mathrm{run}_0(q)$ is the run of the scoped log $\mathcal{S}^\emptyset$ without terms. Apply Lemma B.81 to the pair $(\mathcal{S}^\emptyset, \mathcal{S})$. As $\mathcal{S}^\emptyset$ releases nothing, a call of $\mathrm{run}_0(q)$ is concordant iff $\mathcal{S}$ releases no term at its key and role, that is, iff no term of $\mathcal{S}$ is open at this call. So the index $h^\star$ of the lemma is the first call of $\mathrm{run}_0(q)$ at which a term of $\mathcal{S}$ is open.

(a) If $\mathrm{Fire}_{\mathcal{S}}(q) = \emptyset$, every call is concordant, and the lemma gives $\mathrm{run}_{\mathcal{S}}(q) = \mathrm{run}_0(q)$. The run contains the logits of every decoding step, the tokens and the decision to stop, so the answer is $y_0(q)$.

(c) The lemma gives the common calls before $h^\star$, what is emitted before call $h^\star$, and the common module, position, role and key of call $h^\star$. At the calls before $h^\star$ the key and the role are those of $\mathrm{run}_0(q)$, at which no term of $\mathcal{S}$ is open; so none is released. At call

$h^\star$ the scoped model evaluates each term from its parameters and this key and role (M2), so it releases the terms that are open at call $h^\star$ of $\mathrm{run}_0(q)$.

(b) If $\mathrm{Fire}_{\mathcal{S}}(q) = \emptyset$, then by (a) the calls of $\mathrm{run}_{\mathcal{S}}(q)$ are those of $\mathrm{run}_0(q)$, at which no term is open, so no term is released. If $\mathrm{Fire}_{\mathcal{S}}(q) \neq \emptyset$, a term is released at call $h^\star$ of $\mathrm{run}_{\mathcal{S}}(q)$ by (c).

(d) Two scalar modules with $W_0^1 = W_0^2 = 1$, the output of the first being the key of the second, one position; module 1 has no term and module 2 has one term with detector 1, normaliser 1, threshold 0, response 1 and value 10. At the key 2 the pre-edit output is 2 and the scoped output is 12. □

Remark B.83 (The architectural facts that are used, and those that are not). (i) *What* (M1) *asks.* Every key of an edited module, every logit and every token is computed from the query and from outputs of edited modules that were computed before it, and decoding is deterministic. This is true of every feed-forward network under every schedule: position by position with a key–value cache, layer by layer over the prompt, or with the whole sequence recomputed at every decoding step. (M1) takes the place of (H1)–(H3); it does not say that keys are independent of the log, which is false from the second module on. The induction of Lemma B.81 runs over the calls in the order of execution, that is, over the decoding steps, the modules in computation order and the positions, nested as the schedule nests them. (ii) *What is not used.* No property of attention is used, causal masking included; nor that the edited modules are consecutive, that an edit has a term at every module, or that the network is linear anywhere outside the gate. (iii) *What* (M2) *asks.* As (H4), for each module: the gate decides the release of a term at the key that this module receives, and it branches, returning the pre-edit product where nothing is released (Remark B.5(ii)). A design in which the gate of one module releases the terms of an edit at all modules is outside (M2) and is not treated here. (iv) *Positions.* With a key–value cache each pair of a module and a position is called once. The calls of $\mathrm{run}_0(q)$ are then the pairs (μ, t) with t a position of the pre-edit trace $x^0(q)$, and $\mathrm{Fire}_{\mathcal{S}}(q) = \emptyset$ reads: no term is released at any module and any eligible position of the pre-edit trace. If positions are recomputed in later passes, each pass is a call with its own key. With causal attention the keys of one pair in two passes are equal in exact arithmetic and can differ in the last bits in floating point; the certificate then ranges over all calls, or over one pass with a guard band larger than this discrepancy (Remark B.9(iii)). Without causal masking only the form with all calls is correct. (v) *What is observed.* By (b) the certificate of a query is read off the run of the scoped model itself; and by (c) this run also shows the first call at which a term is open on the pre-edit run, with the pre-edit key. After the first release, equality with the calls of the pre-edit run is no longer guaranteed. (vi) Under sampling with a common random tape the statements hold for each realisation of the random choices, as in Remark B.5(iii). Throughout, one deterministic execution configuration is fixed, and the calls are all the calls that are actually made, recomputations included.

Proposition B.84 (Frame rule on the run of the log; lineage). *Assume* (M1) *and* (M2). *Let $\mathcal{S}'$ differ from $\mathcal{S}$ only in the parameters of the terms of a set $J \subseteq I$, and let q be a query.*

(a) Frame rule. *If no term of J is released in $\mathrm{run}_{\mathcal{S}}(q)$, and no term of J is open under $\mathcal{S}'$ at a call of $\mathrm{run}_{\mathcal{S}}(q)$, then $\mathrm{run}_{\mathcal{S}'}(q) = \mathrm{run}_{\mathcal{S}}(q)$, bit for bit.*

(b) Raise-only repairs. *If $\mathcal{S}'$ only raises thresholds or shrinks role sets, the first condition of* (a) *suffices: $J \cap \mathrm{Rel}_{\mathcal{S}}(q) = \emptyset$ implies $\mathrm{run}_{\mathcal{S}'}(q) = \mathrm{run}_{\mathcal{S}}(q)$.*

(c) Lineage. *Silencing every term outside $\mathrm{Rel}_{\mathcal{S}}(q)$ leaves the run of q unchanged. Silencing the terms of $\mathrm{Fire}_{\mathcal{S}}(q)$ certifies q; and a scoped log that certifies q differs from $\mathcal{S}$ in the parameters of every term of $\mathrm{Fire}_{\mathcal{S}}(q)$.*

(d) What is monotone. *If $\tau' \geq \tau$, then $\mathrm{Fire}_{\tau'}(q) \subseteq \mathrm{Fire}_{\tau}(q)$ for every query q, and for every fixed run the set of the terms that are open at its calls shrinks.*

Proof. (a) Every call of $\mathrm{run}_{\mathcal{S}}(q)$ is concordant in the sense of Lemma B.81: a term outside J has the same parameters in both logs, so by (M2) it is released under $\mathcal{S}'$ at the key and role of the call iff it is released under $\mathcal{S}$, with the same detector, normaliser, response and value; a term of J is not released there under $\mathcal{S}$, by the first hypothesis, and not under $\mathcal{S}'$, by the second. The lemma gives the claim.

(b) A term of J that is not released at a call under $\mathcal{S}$ has a role set that does not contain the role of the call, or a coordinate below its threshold. Both remain true when the role set shrinks and the threshold rises, so the term is not open under $\mathcal{S}'$ at this call.

(c) The first claim is (b) with $J := I \setminus \mathrm{Rel}_{\mathcal{S}}(q)$. For the other two, whether a term is open at a call of $\mathrm{run}_0(q)$ depends on the parameters of this term alone, because $\mathrm{run}_0(q)$ does not depend on the log. So after the terms of $\mathrm{Fire}_{\mathcal{S}}(q)$ are silenced no term is open on $\mathrm{run}_0(q)$, and a term of $\mathrm{Fire}_{\mathcal{S}}(q)$ whose parameters are kept is still open there.

(d) The keys of a fixed run are fixed, and at a fixed key and role the test $\zeta_{i,\mu}(k) \geq \tau_{i,\mu}$ can only turn from true to false when the threshold rises. $\mathrm{run}_0(q)$ is one fixed run for all τ. □

Theorem B.85 (Least repair for several edited modules). *Assume* (M1) *and* (M2). *Let detectors, normalisers and role sets be fixed, let $\tau^0 \in \Theta^I$ be the current thresholds and $\mathcal{Q}^-$ a finite set of wrong-answer complaints, and put*

$$\tau^\star_{i,\mu} := \max\Big(\tau^0_{i,\mu},\ \max_{q \in \mathcal{Q}^-} \zeta_{i,\mu}(q)^+\Big) \qquad ((i,\mu) \in I).$$

(a) Existence and uniqueness. *A threshold vector $\tau \geq \tau^0$ certifies every $q \in \mathcal{Q}^-$ iff $\tau \geq \tau^\star$. The set of these vectors is closed under coordinatewise minimum and maximum, and $\tau^\star$ is its least element: the unique least repair of $\mathcal{Q}^-$ under the certificate criterion.*

(b) Order independence; raise-only. *The update U_q with $U_q(\tau)_{i,\mu} := \max\big(\tau_{i,\mu}, \zeta_{i,\mu}(q)^+\big)$ is inflationary and idempotent, and two updates commute: every sequence of updates U_q, $q \in \mathcal{Q}^-$, in which each complaint of $\mathcal{Q}^-$ occurs at least once turns τ^0 into $\tau^\star$. U_q changes exactly the thresholds of the terms of $\mathrm{Fire}_{\tau}(q)$, and none if q is certified.*

(c) The complaints are certified, and certified queries stay certified. *Under $\tau^\star$, and under every $\tau \geq \tau^\star$, each $q \in \mathcal{Q}^-$ is certified: its run is the pre-edit run, bit for bit, and it is answered with $y_0(q)$. Every query that is certified under some τ is certified under every $\tau' \geq \tau$, with the same run.*

(d) Scope maximality. *For every $\tau \geq \tau^0$ that certifies Q^- and every term, the scope $\{k : \zeta_{i,\mu}(k) \geq \tau_{i,\mu}\}$ under τ is contained in the scope under $\tau^\star$, which is contained in the scope under τ^0. So $\tau^\star$ minimises at once every threshold, the set of changed terms and the set of pairs $((i,\mu), q')$ that lose coverage.*

(e) Computation at run time. *Without a pre-edit run, $U_q(\tau)$ is the result of the following loop of repeated runs of the scoped model (one stored run does not suffice): run $\mathcal{M}[\tau]$ on q; stop if no term is released; otherwise raise the threshold of each term that is released at the first call with a release to $\zeta_{i,\mu}(k)^+$, k the key of this call, and repeat. The loop makes at most one run per call of $\mathrm{run}_0(q)$ at which a term is open under the initial thresholds, and one more.*

(f) One threshold per edit. *If the thresholds of the terms of an edit are required to be equal, $\tau_{i,\mu} = \tau_i$ for all μ, then* (a)–(d) *hold with $\tau_i^\star := \max\big(\tau_i^0, \max_{q\in Q^-} \max_\mu \zeta_{i,\mu}(q)^+\big)$.*

Proof. (a) By Definition B.80, q is certified under τ iff $\tau_{i,\mu} \geq \zeta_{i,\mu}(q)^+$ for every term: a conjunction of lower bounds, one per coordinate, which do not depend on τ because the coordinates are taken on the pre-edit run. Together with $\tau \geq \tau^0$ they read $\tau \geq \tau^\star$. The set $\{\tau : \tau \geq \tau^\star\}$ is closed under coordinatewise minimum and maximum and has the least element $\tau^\star$.

(b) As in Theorem B.8(b): the maximum is associative, commutative and idempotent, and U_q changes $\tau_{i,\mu}$ iff $\zeta_{i,\mu}(q)^+ > \tau_{i,\mu}$, that is, iff the term has an eligible call and $\zeta_{i,\mu}(q) \geq \tau_{i,\mu}$, iff it belongs to $\mathrm{Fire}_\tau(q)$.

(c) By (a) the complaints are certified, and Theorem B.82(a) gives their runs and answers. If q' is certified under τ and $\tau' \geq \tau$, then $\mathrm{Fire}_{\tau'}(q') \subseteq \mathrm{Fire}_\tau(q') = \emptyset$ by Proposition B.84(d), and under both vectors the run of q' is $\mathrm{run}_0(q')$.

(d) The scope of a term is antitone in its threshold, and $\tau \geq \tau^\star \geq \tau^0$ by (a). If $\tau^\star_{i,\mu} \neq \tau^0_{i,\mu}$, then $\tau_{i,\mu} \geq \tau^\star_{i,\mu} > \tau^0_{i,\mu}$. A pair that is covered under τ^0 and not under $\tau^\star$ has $\zeta_{i,\mu}(q') < \tau^\star_{i,\mu} \leq \tau_{i,\mu}$ and is not covered under τ.

(e) Let $\tau^{(0)} := \tau$ and let $\tau^{(r)}$ be the thresholds after r rounds. We show $\tau \leq \tau^{(r)} \leq U_q(\tau)$ by induction. By Theorem B.82(b),(c), applied to $\tau^{(r)}$, no term is released in the run iff q is certified under $\tau^{(r)}$; otherwise the first call with a release is the first call h_r of $\mathrm{run}_0(q)$ at which a term is open under $\tau^{(r)}$, its key k is the pre-edit key, and the released terms are the terms that are open there. Such a term is eligible at call h_r of $\mathrm{run}_0(q)$, so $\tau^{(r)}_{i,\mu} \leq \zeta_{i,\mu}(k) \leq \zeta_{i,\mu}(q)$: the new threshold $\zeta_{i,\mu}(k)^+$ is larger than the old one and at most $\zeta_{i,\mu}(q)^+$. After the round no term is open at call h_r of $\mathrm{run}_0(q)$, and none at an earlier call, since thresholds only rise; so $h_{r+1} > h_r$, and each h_r is a call at which a term is open under τ. Hence the loop stops, after at most as many rounds with a release as there are such calls, at a vector τ' with $\tau \leq \tau' \leq U_q(\tau)$ that certifies q. By (a), applied to the complaint set $\{q\}$ and the initial thresholds τ, $\tau' \geq U_q(\tau)$.

(f) The vectors with $\tau_{i,\mu} = \tau_i$ form a set that is closed under coordinatewise minimum and maximum. For such a vector the bounds of (a) read $\tau_i \geq \zeta_{i,\mu}(q)^+$ for all μ and q, and (a)–(d) follow as before, with the edits in the place of the terms and $\max_\mu \zeta_{i,\mu}(q)$ in the place of the coordinate. □

Proposition B.86 (What is lost against the one-module theorem). *For one edited module, under* (H1)*, and queries whose trace is the prompt, $\mathrm{Rel}_\tau(q) = \mathrm{Fire}_\tau(q)$ for all τ, and Theorem B.8*(c) *holds with "released" in the place of "covered". For two edited modules, at one position and with responses equal to* 1*, each of the following fails.*

(i) The firing set is the lineage of the deployed answer. *$\mathrm{Fire}_{\tau^0}(q)$ and $\mathrm{Rel}_{\tau^0}(q)$ are incomparable in general: neither contains the other, also for one query. The least repair of a complaint raises the terms of $\mathrm{Fire}_{\tau^0}(q)$: it can leave a term that is released in the run of the complaint as it is, and it can raise a term that is not released in this run.*

(ii) The least repair loses the fewest releases. *There are τ^0, a complaint, a pair $((i,\mu), q')$ and a certifying $\tau' \geq \tau^\star$ such that the pair is released under τ^0 and under τ', covered under $\tau^\star$, and not released under $\tau^\star$.*

(iii) Only queries with a raised term in their firing set change. *There are τ^0, a complaint and a query q' whose answer changes under $\tau^\star$ although no raised term belongs to $\mathrm{Fire}_{\tau^0}(q')$. What remains is Proposition B.84*(b)*: a query whose run changes has a raised term in $\mathrm{Rel}_{\tau^0}(q')$.*

(iv) A raise-only repair releases no new term. *There are τ^0, a complaint and a query q' with $\mathrm{Rel}_{\tau^\star}(q') \not\subseteq \mathrm{Rel}_{\tau^0}(q')$.*

(v) A least repair exists when pairs must stay released. *There are τ^0, a complaint and a pair $((i,\mu), q')$ that is released under τ^0 such that the vectors $\tau \geq \tau^0$ that certify the complaint and keep the pair released have two incomparable minimal elements and no least one.*

Proof. For one module, (H1) says that the keys of the prompt positions do not depend on the log. If the trace is the prompt, the calls of $\mathrm{run}_\tau(q)$ therefore have the keys of $\mathrm{run}_0(q)$, and a term is released iff it is open on the pre-edit run. (i) In Example B.87 the term e_2 is released in the run of the complaint and is not in its firing set, so the least repair does not change it. In Example B.3, at the key 2, e_2 is in the firing set (the pre-edit key of the second module is $2 \geq 0$) and is not released (the key in the run is -3); the least repair raises both thresholds above 2, and raising that of e_1 alone gives the output 12. For one query in which neither set contains the other, add to Example B.87 a further term e_5 of module 2 with the coordinate $-x$, the threshold -1 and the value 1: the complaint then has the firing set $\{e_1, e_5\}$ and the released set $\{e_1, e_2\}$. (ii) and (iii) are Example B.87, (iv) is Example B.88 and (v) is Example B.89. □

Example B.87 (Released is not covered; bystanders). *One position, with the role* last*. Module* 1 *has keys in $\mathbb{R}^3$ and $W_0^1 k = k_{(1)}$, the first entry of k; module* 2 *is scalar with $W_0^2 = 1$, and its key is the output x of module* 1*. All terms have the normaliser* 1*, the response* 1 *and the threshold* 1*: e_1, in module* 1*, has the coordinate $k_{(2)}$ and the value* 5*; e_3, in module* 1*, has the coordinate $k_{(3)}$ and the value -5; e_2, in module* 2*, has the coordinate x and the value* 10*.*

The complaint q has the key $(0,1,0)$. Its pre-edit run has $x = 0$ and the output 0*, and $\mathrm{Fire}_{\tau^0}(q) = \{e_1\}$. In its run under τ^0, e_1 is released, $x = 5$, e_2 is released, and the output is* 15*: $\mathrm{Rel}_{\tau^0}(q) = \{e_1, e_2\}$. The least repair $\tau^\star$ raises the threshold of e_1 to 1^+ and changes nothing else; under it the run of q is the pre-edit run.*

A released pair. *The query q' has the key $(2,1,1)$, the pre-edit value $x=2$ and the pre-edit output 2. Under τ^0, e_1 and e_3 are released, $x=2$, e_2 is released, and the output is 12. Under $\tau^\star$ only e_3 is released, $x=-3$, and e_2 is not released: the output is -3, although the pair (e_2,q') is covered, its pre-edit coordinate being $2\geq 1$. Under the larger vector τ' that also raises e_3 to 1^+, which certifies q as well, $x=2$ and e_2 is released, with the output 12.*

A value. *Changing the value of e_2 from 10 to 20 changes the output of q under τ^0 from 15 to 25, although $e_2\notin \mathrm{Fire}_{\tau^0}(q)$.*

A bystander. *A second complaint, with the key $(\frac{3}{2},0,0)$, has the firing set $\{e_2\}$, and its least repair raises the threshold of e_2 to $(\frac{3}{2})^+$ and nothing else. The query q'' with the key $(-\frac{19}{5},1,0)$ has $\mathrm{Fire}_{\tau^0}(q'')=\{e_1\}$. Under τ^0, e_1 is released, $x=\frac{6}{5}$, e_2 is released, and the output is $\frac{56}{5}$; after the repair e_2 is not released and the output is $\frac{6}{5}$. The answer of q'' changes, the only raised term is e_2, and $e_2\notin \mathrm{Fire}_{\tau^0}(q'')$, while $e_2\in \mathrm{Rel}_{\tau^0}(q'')$.*

Example B.88 (The least repair of one complaint releases a term on another query). *One position. Module 1 has keys in $\mathbb{R}^2$ and $W_0^1 k=k_{(1)}$; module 2 is scalar with $W_0^2=1$, and its key is the output x of module 1. Normalisers and responses are 1. The term e_1 of module 1 has the coordinate $k_{(2)}$, the threshold 1 and the value 5; the term e_2 of module 2 has the coordinate $-x$, the threshold -3 (it is released where $x\leq 3$) and the value 10. The complaint has the key $(9,1)$: its pre-edit output is 9, its firing set is $\{e_1\}$, and under τ^0 its output is 14. The least repair raises e_1 to 1^+. The query q' with the key $(2,1)$ has the pre-edit output 2. Under τ^0, e_1 is released, $x=7$, e_2 is not released, and the output is 7: $\mathrm{Rel}_{\tau^0}(q')=\{e_1\}$. Under $\tau^\star$, $x=2$, e_2 is released, and the output is 12: $\mathrm{Rel}_{\tau^\star}(q')=\{e_2\}$. The repair of the complaint releases a term on q' that was not released before, and moves the output of q' away from its pre-edit value.*

Example B.89 (Two minimal repairs when a pair must stay released). *One position. Module 1 has keys in $\mathbb{R}^4$ and $W_0^1 k=k_{(1)}$; module 2 is scalar with $W_0^2=1$, and its key is the output x of module 1. Normalisers, responses and thresholds are 1. In module 1, e_1 has the coordinate $k_{(2)}$ and the value 2, e_3 the coordinate $k_{(3)}$ and the value -2, e_4 the coordinate $k_{(4)}$ and the value -2; in module 2, e_2 has the coordinate x and the value 10. The complaint has the key $(0,1,0,0)$ and the firing set $\{e_1\}$. The query q' has the key $(3,1,1,1)$; under τ^0 all three terms of module 1 are released, $x=1$, and e_2 is released. Let $\tau\geq\tau^0$ certify the complaint, so that $\tau_{e_1}>1$. In the run of q', e_1 is not released, and $x=3-2\,\mathbf{1}[\tau_{e_3}=1]-2\,\mathbf{1}[\tau_{e_4}=1]$. The term e_2 is released iff $x\geq\tau_{e_2}$; as $\tau_{e_2}\geq 1$, this requires $\tau_{e_3}>1$ or $\tau_{e_4}>1$. So the admissible vectors are those with $\tau_{e_1}>1$, with $\tau_{e_3}>1$ or $\tau_{e_4}>1$, and with $\tau_{e_2}\leq x$. Their minimal elements are $(1^+,1^+,1,1)$ and $(1^+,1,1^+,1)$, in the order (e_1,e_3,e_4,e_2); they are incomparable, and their coordinatewise minimum $(1^+,1,1,1)$ gives $x=-1$ and does not release e_2.*

Reading: certified against resolved. For the complaints nothing is lost: the least repair exists, is unique and does not depend on the order of the complaints, it only raises thresholds, and a certified complaint is resolved, exactly (Theorem B.85(a)–(c)). What is lost concerns the queries that are *not* certified. The pre-edit coordinates of such a query determine its run only up to the first release (Theorem B.82(c)); behind it the keys are no longer the pre-edit keys. So every statement of Theorem B.8(c) survives as a statement about coverage on the pre-edit run (Theorem B.85(d)) and fails as a statement about the runs of the repaired model (Proposition B.86): the least repair has the largest scopes but need not keep the most releases; a raise-only repair can release a term on another query; and the queries whose answer can change are those with a raised term in their released set, which is read off the runs of the log in service and not off stored pre-edit coordinates. In particular "least" refers to the certificate criterion only, as for one module (Remark B.9(iv)): the vectors $\tau\geq\tau^0$ under which a complaint is *resolved* need not have a least element. If two edits with the value 1 fire on a complaint whose answer changes iff the module output rises by 2, raising either threshold resolves it, and the least certifying repair raises both.

Corollary B.90 (Calibration for several edited modules). *Assume* (M1) *and* (M2). *Let every term have an eligible call in the pre-edit run of every query, let $\zeta(q):=(\zeta_{i,\mu}(q))_{(i,\mu)\in I}\in\mathbb{R}^I$ be the coordinate vector of a query, and let the coordinate vectors of the benign calibration queries $q_1,\dots,q_m$ and of a fresh benign query q_{m+1} be exchangeable. Fix $\alpha\in(0,1)$ and $\tau^0\in(0,\infty)^I$ without the queries, split the calibration queries without looking at them into $\mathcal{B}_1$ and $\mathcal{B}_2$ with $m_1\geq 1$ and m_2 queries, and let*

$$b_{i,\mu}:=\max_{q\in\mathcal{B}_1}\zeta_{i,\mu}(q),\qquad \mathrm{sc}_2(q):=\max_{(i,\mu)\in I}\frac{\zeta_{i,\mu}(q)-b_{i,\mu}}{\tau^0_{i,\mu}},$$

the largest margin over the modules, the edits and the eligible calls of the pre-edit run. Let $l_2:=\lfloor\alpha(m_2+1)\rfloor$, let $\hat\eta$ be the (m_2+1-l_2)-th smallest of the scores $\mathrm{sc}_2(q)$, $q\in\mathcal{B}_2$, or $+\infty$ if $l_2=0$, and $\hat\tau_{i,\mu}:=(b_{i,\mu}+\hat\eta\,\tau^0_{i,\mu})^+$.

(a) Bound. *$\mathbb{P}\big(q_{m+1}$ is not certified under $\hat\tau\big)\leq l_2/(m_2+1)\leq\alpha$, with equality in the first inequality if the scores sc_2 of the queries of $\mathcal{B}_2$ and of q_{m+1} are almost surely pairwise distinct. The rule assigns finite thresholds iff $m_2\geq 1/\alpha-1$, whatever T and M. Coordinates, baselines and scales are real numbers here; in floating point the gate must store the comparison $(\zeta_{i,\mu}-b_{i,\mu})/\tau^0_{i,\mu}>\hat\eta$ (Proposition B.47, Remark B.59): the comparison $\zeta_{i,\mu}>b_{i,\mu}+\hat\eta\,\tau^0_{i,\mu}$ as computed can release a term on a query whose score ties with the scale, and the bound then fails.*

(b) What the event is. *The following events coincide: q_{m+1} is not certified; some term is open at a call of its pre-edit run; some term is released in its run under $\hat\tau$. On the complement the run under $\hat\tau$ is the pre-edit run, bit for bit. Hence $\mathbb{P}\big(\mathrm{run}_{\hat\tau}(q_{m+1})\neq\mathrm{run}_0(q_{m+1})\big)\leq l_2/(m_2+1)$ and $\mathbb{P}\big(\hat y(q_{m+1};\hat\tau)\neq y_0(q_{m+1})\big)\leq l_2/(m_2+1)$.*

(c) Monotonicity. *The upper bounds of* (a) *and* (b) *and the implications of* (b) *hold for every threshold vector $\tau\geq\hat\tau$, whatever else it depends on; the equality of* (a) *concerns $\hat\tau$ itself.*

(d) The other rules. *Theorems B.45, B.46 and B.50, Proposition B.58*(a) *and Theorems B.61–B.63 hold with the terms I in the place of the edits $[T]$ and "not certified" in the place of "fires". In particular the least repair of the calibration queries leaves a fresh query uncertified with a probability of at most $\min\big(1,|I|/(m+1)\big)$, where $|I|\leq TM$ is the number of terms.*

Proof. The statements that are quoted are statements about a finite family of real coordinates per query and about the events that a coordinate reaches its threshold; their proofs use of the edits only

that (1) the coordinate vector of a query depends neither on the thresholds nor on the calibration queries, (2) an edit fires on a query iff its coordinate reaches its threshold, and (3) a query on which no edit fires has the pre-edit answer (LC). For the terms, (1) holds because $\zeta_{i,\mu}(q)$ is taken on the pre-edit run, (2) is Definition B.80, and (3) is Theorem B.82(a). This gives (d); in Theorem B.50(c) the number of holders is at most $\min(|I|, m+1)$.

(a) The term (i,μ) belongs to $\mathrm{Fire}_{\hat\tau}(q)$ iff $\zeta_{i,\mu}(q) > b_{i,\mu} + \hat\eta\,\tau^0_{i,\mu}$, so q is not certified iff $\mathrm{sc}_2(q) > \hat\eta$. The function sc_2 depends on the first block only, and a permutation of the queries of $\mathcal{B}_2$ and the fresh query leaves the joint law of all coordinate vectors invariant; so the scores sc_2 of these m_2+1 queries are exchangeable. If $l_2 = 0$ the event is empty. Otherwise (2), with m_2 in the place of m and $\kappa = m_2 + 1 - l_2$, gives the bound $l_2/(m_2+1)$, with equality for distinct scores, and $l_2 \le \alpha(m_2+1)$. The thresholds are finite iff $l_2 \ge 1$.

(b) Theorem B.82(b) gives the equality of the events and Theorem B.82(a) the run on the complement. So the event that the run differs from the pre-edit run is contained in the event of (a), and the event that the answer differs is contained in it as well.

(c) If $\tau \ge \hat\tau$, then $\mathrm{Fire}_\tau(q) \subseteq \mathrm{Fire}_{\hat\tau}(q)$ for every q (Proposition B.84(d)), so the event of (a) shrinks, and (b) holds for every threshold vector. □

Remark B.91 (What "false firing" means for several modules). (i) *One event.* For a query, "some term is open on the pre-edit run", "some term is released in the run of the scoped model" and "the query is not certified" are the same event (Theorem B.82(b)). The false-firing rate of several modules is its probability for a fresh benign query: the quantity that calibration bounds is the one that a deployment observes, the share of benign queries in whose run some term is released, and no pre-edit pass is needed to monitor it. (ii) *Not per term and not per module.* On a query that is not certified the released set and the firing set agree at the first release (Theorem B.82(c)) and can differ behind it, in both directions (Proposition B.86(i)). So the statements about a single edit and about the number of firing edits, Theorem B.50(a),(b) read for the terms, concern the firing set $\mathrm{Fire}_{\hat\tau}(q)$ of the pre-edit run. They bound neither the probability that a given term is released, nor the expected number of released terms, nor the rate at which a given module releases a term. Each bound on the event "some term" transfers. (iii) *The scores come from pre-edit runs.* If the calibration queries are scored on the runs of the log in service, the deployed log or a scoped log, the scores are still exchangeable, for a fixed log, and the rule is a valid test of the event that the score of the fresh query, taken on its run under the log in service, exceeds the scale. This event is no certificate: a fresh query that passes this test can release terms under the calibrated thresholds and change its answer. For one module and prompt positions the two scorings coincide, by (H1). Calibration thus needs the pre-edit weights of all edited modules, as a repair does. (iv) *The number of terms.* A rule whose sample size grows with the number of thresholds pays for the terms: the least repair of the calibration queries has the bound $\min(1, |I|/(m+1))$, and the bound with T in the place of $|I|$ is false. The sample sizes of the shared scale and of the two-block rule do not depend on $|I|$. With one threshold per edit (Theorem B.85(f)) all statements hold with the edits as index set and the coordinates $\max_\mu \zeta_{i,\mu}(q)$.

B.10.1 What does not extend. Four statements of the one-module theory do not extend. For each we give the form that fails, the smallest example that our checks found, and what remains. We did not search for restricted versions.

Proposition B.92 (Scope-Cut with coverage on the pre-edit run). *Let an instance consist of finite sets Q^- of complaints and Q^+, $Q^!$ of pairs (i,q') of an edit and a query, with weights, and of the coordinates $\zeta_{i,\mu}(\cdot)$ of these queries; let a pair (i,q') be* covered *under $\tau \in \Theta^I$ if every term of edit i belongs to $\mathrm{Fire}_\tau(q')$, and let $\mathcal{G}$ have an arc from $q \in Q^-$ to (i,q') whenever $\zeta_{i,\mu}(q') \le \zeta_{i,\mu}(q)$ for some term (i,μ) of edit i. Assume that every term of the edit of a pair has an eligible call in the pre-edit run of the query of the pair, that is, $\zeta_{i,\mu}(q') > -\infty$ for every pair (i,q') and every term (i,μ) of edit i; a complaint may have the coordinate $-\infty$. Then parts* (a)–(d) *of Theorem B.14 hold, with the thresholds $\tau^C_{i,\mu} := \min\{\zeta_{i,\mu}(q') : (i,q') \in (Q^+ \cup Q^!)\setminus C\}$: the maximum of Φ_λ over the feasible threshold vectors is the maximum weight of an admissible set and equals $\omega(Q^-) - \mathrm{mincut}(N_\lambda)$, and the source side of a minimum cut gives optimal thresholds. If the pairs are pairs $((i,\mu),q')$ of a term and a query, all of Sec. B.4 applies with the terms in the place of the edits.*

Proof. Let $U(\tau)$ be the set of the pairs that are not covered and $C(\tau) := \mathrm{Clean}(\tau) \cup U(\tau)$. We verify parts (a) and (b) of Theorem B.14 in this setting; parts (c) and (d) follow from them as there, by the same words. (a) Let q be certified under τ and let $\mathcal{G}$ have an arc from q to (i,q'), so that $\zeta_{i,\mu}(q') \le \zeta_{i,\mu}(q)$ for a term (i,μ). Then $\tau_{i,\mu} > \zeta_{i,\mu}(q) \ge \zeta_{i,\mu}(q')$, the term (i,μ) is not in $\mathrm{Fire}_\tau(q')$, and the pair is in $U(\tau)$. So $C(\tau)$ is closed; its weight is $\Phi_\lambda(\tau)$, and it is disjoint from $Q^!$ iff τ is feasible. (b) Let C be closed, $q \in C \cap Q^-$ and (i,μ) a term. No pair (i,q') outside C receives an arc from q, so $\zeta_{i,\mu}(q') > \zeta_{i,\mu}(q)$ for each of them, and $\tau^C_{i,\mu} > \zeta_{i,\mu}(q)$, the minimum over the empty set being $+\infty$: q is certified under τ^C. A pair (i,q') outside C has $\zeta_{i,\mu}(q') \ge \tau^C_{i,\mu}$ for every term of edit i and is covered; so $U(\tau^C) \subseteq C$, and $\Phi_\lambda(\tau^C) \ge \omega_\lambda(C)$. If C is admissible, the pairs of $Q^!$ lie outside C and are covered. □

Proposition B.93 (Keeping pairs released is NP-hard). *The following problem is NP-hard: given a model with two edited modules that are composed at one position (the output of module 1 is the key of module 2) with rational pre-edit weights, a scoped log with responses equal to 1, a wrong-answer complaint q and pairs $((i_h,\mu_h),q_h)$, decide whether some $\tau \in \Theta^I$ certifies q and has $(i_h,\mu_h) \in \mathrm{Rel}_\tau(q_h)$ for every h. It is in NP if the inputs are rationals in binary and the model is evaluated in polynomial time in exact arithmetic, with numbers of polynomial bit length. With "covered" in the place of "released", and for one edited module and prompt positions, the problem is decided by one comparison per pair.*

Proof. *Reduction from satisfiability.* Let a formula in conjunctive normal form have the variables $x_1,\dots,x_L$ and the clauses $C_1,\dots,C_H$, no clause containing a variable twice. Module 1 has keys $(\xi,\beta) \in \mathbb{Q}^L \times \mathbb{Q}^H$ and $W^1_0(\xi,\beta) = \beta$; module 2 has keys in $\mathbb{Q}^H$, the outputs of module 1, and W^2_0 is the identity. All normalisers and responses are 1. Module 1 has a term e_j for each variable, with the coordinate ξ_j and the value $v_j \in \mathbb{Q}^H$ whose h-th entry is 1 if x_j occurs in C_h, -1 if its negation does, and 0 otherwise. Module 2 has a term e'_h for each clause, with the coordinate β'_h, the h-th

entry of its key, and an arbitrary value. The complaint q has the key $(0, -\mathbf{1})$, with $\mathbf{1}$ the vector of ones. For each clause the query q_h has the key (ξ^h, β^h), where $\xi^h_j = 1$ if x_j or its negation occurs in C_h and $\xi^h_j = 0$ otherwise, and β^h is $-\mathbf{1}$ except for its h-th entry, which is the number of negated variables of C_h minus 1; the pair is (e'_h, q_h).

Let τ certify q. The pre-edit coordinates of q are 0 for the terms e_j and -1 for the terms e'_h, so $\tau_{e_j} > 0$ and $\tau_{e'_h} > -1$. In the run of q_h the term e_j has the coordinate $\xi^h_j \in \{0, 1\}$; it is released iff x_j occurs in C_h, negated or not, and $\tau_{e_j} \leq 1$. Call x_j true if $\tau_{e_j} \leq 1$. The h-th entry of the output of module 1 is then the number of negated variables of C_h, minus 1, plus the number of true variables that occur unnegated, minus the number of true variables that occur negated: the number of true literals of C_h, minus 1. The term e'_h is released iff this integer is at least $\tau_{e'_h} > -1$, which requires a true literal. So if τ also releases every pair, the assignment satisfies the formula. Conversely, for a satisfying assignment let $\tau_{e_j} := 1$ if x_j is true and 1^+ otherwise, and $\tau_{e'_h} := (-1)^+$: then q is certified, and in the run of q_h the h-th entry is at least 0 and e'_h is released. The instance has $L + H$ terms and $H + 1$ queries.

Membership. Let τ be admissible, and for each term let a be the least of the coordinates that the term takes at its eligible calls in $\mathrm{run}_0(q)$ and in the runs $\mathrm{run}_\tau(q_h)$ and that are at least its threshold; replace the threshold by a, or by $+\infty$ if there is no such coordinate. At every call of these runs each term is open under the new thresholds iff it is open under τ, so by Lemma B.81 the runs are unchanged, and the new vector is admissible. Its entries are numbers that the model computes, so it is a certificate of polynomial size that is verified by $H + 1$ runs.

One module, or coverage. A vector certifies q and covers the pairs iff $\zeta_{i,\mu}(q) < \tau_{i,\mu} \leq \zeta_{i,\mu}(q_h)$ for the terms and pairs concerned, which is possible iff $\zeta_{i_h,\mu_h}(q) < \zeta_{i_h,\mu_h}(q_h)$ for every pair; for one module and prompt positions released pairs are covered pairs (Proposition B.86). □

The frame rule on stored coordinates. The frame rule holds in the form of Proposition B.1(b), on the run of the log itself (Proposition B.84(a),(b)). It fails when the lineage is taken from stored pre-edit coordinates: "if no term of J is open on $\mathrm{run}_0(q)$, under $\mathcal{S}$ and under $\mathcal{S}'$, then q is answered alike" is false. The smallest example is Example B.4: two scalar modules, two terms and one key; changing the value of a term that is not in the firing set changes the output, because the term is released in the run. One term is not enough: a term that is not open on the pre-edit run is alone in its log and is never released (Theorem B.82(b)). For one module the two forms coincide on prompt positions and differ once tokens are generated (Example B.2); with several modules they differ at one position. What remains, for changes that only raise thresholds or shrink role sets: the queries that a change of the terms of J can affect are those with a term of J in their released set (for other changes both conditions of Proposition B.84(a) are needed: one term with the threshold 2 and a query with the key 1 is affected when the threshold falls to 1). This set is read off the runs of the log in service, and it has to be read again after a repair, which can enlarge the released sets of other queries (Example B.88).

Monotonicity of firing in the thresholds. The firing set $\mathrm{Fire}_\tau(q)$ is antitone in τ, and so is the set of the terms that are open on any fixed run (Proposition B.84(d)). The released set $\mathrm{Rel}_\tau(q)$ is not. The smallest example is Example B.3: two scalar modules, two terms and one key; raising the threshold of e_1 releases e_2. One term is not enough: if it is released under $\tau' \geq \tau$, it is open on the pre-edit run under τ', hence under τ, and is then released under τ (Theorem B.82(b)). So Proposition B.1(c) holds for a fixed run only, and the statement of the nested frontier that the firing set of no query grows along the chain of optima holds for firing sets and fails for released sets. "Certified queries stay certified" is not affected (Theorem B.85(c)).

The minimum cut on the scoped run. The problem Scope-Cut is stated in terms of the coordinates of the complaints and of the pairs. With coverage on the pre-edit run it is the same closure problem (Proposition B.92). With "released in the run of the repaired model" in the place of "covered", which is what a why-not complaint asks for, the structure is lost. The threshold vectors that certify a complaint and keep a pair released are not closed under coordinatewise minimum: the smallest example that we found is Example B.89, with two modules, four terms, one complaint and one pair, and two incomparable minimal repairs. And deciding whether such a vector exists is NP-hard already for one complaint (Proposition B.93), while for one module it takes one comparison per pair: no polynomial-time algorithm decides it, in particular no cut formulation that is built in polynomial time, unless P = NP.

The rebase lemma and the retrofit. Let a log over several modules record, for edit i and module μ, the solver key $k_{i,\mu}$ that the module received when the batch of the edit was solved, the target $z^\star_{i,\mu}$ and the update $u_{i,\mu}\psi^\top_{i,\mu}$ with $u_{i,\mu} = z^\star_{i,\mu} - (W^\mu_0 + \sum_{j<i} u_{j,\mu}\psi^\top_{j,\mu})k_{i,\mu}$. Lemma B.65, applied to module μ, gives $z^\star_{i,\mu} - W^\mu_0 k_{i,\mu} = u_{i,\mu} + \sum_{j<i}\langle\psi_{j,\mu}, k_{i,\mu}\rangle\, u_{j,\mu}$: as an identity between logged quantities the lemma holds at every module. What does not extend is its use, Theorem B.66(a). The scoped log with these values does not attain the targets of the modules $\mu \geq 2$, because the key that module μ receives in the run of the scoped log is not $k_{i,\mu}$: the logged key was computed behind the deployed outputs of the earlier modules, at the solver position and at the positions that it attends to, and these outputs contain the shortfall of the attainment and the leaks that a scoped log removes (Example B.94). If edit i is the only term released at that call, with the response 1, the values that attain the targets are $z^\star_{i,\mu} - W^\mu_0 k'$ with k' the key in the run of the scoped log itself. They are not computed from the log alone: they need a forward pass of the scoped model for every edit, module after module, they depend on the thresholds and values of the terms of the earlier modules, and they are stale as soon as such a threshold is repaired. The second identity of the lemma, too, describes a key that the deployed model no longer produces at a module $\mu \geq 2$ once later edits have been written. In our checks the retrofit did attain all targets in one configuration: one position, one edit per batch, own coordinates equal to 1, and solver keys collected after the earlier modules were updated; then the outputs of the earlier modules in the replay are those of the solver pass. Values for several modules remain open.

Example B.94 (The rebased values of several modules do not attain the targets). *(i)* One edit, one position. *Two scalar modules with* $W_0^1 = W_0^2 = 1$*, the output of the first being the key of the second. The edit has the solver key* $k_{1,1} = 1$ *and the target* $z^\star_{1,1} = 3$ *at module* 1*, so* $u_{1,1} = 2$*, and the read-out vector* $\psi_{1,1} = \frac{1}{2}$*, so* $s_{1,1} = \frac{1}{2}$*. The updated module* 1 *returns* $1 + \frac{1}{2} \cdot 2 = 2$ *at the solver key, which is the solver key* $k_{1,2} = 2$ *of module* 2*. With the target* $z^\star_{1,2} = 7$*,* $u_{1,2} = 5$ *and* $\psi_{1,2} = \frac{1}{2}$*, so* $s_{1,2} = 1$*, and the deployed model returns* 7*. The rebased values are* $v^{\mathrm{rb}}_{1,1} = 3 - 1 = 2$ *and* $v^{\mathrm{rb}}_{1,2} = 7 - 2 = 5$*. In the scoped log with these values and the response* 1*, with both terms released, module* 1 *returns its target* 3*; module* 2 *receives the key* 3*, where its term has the coordinate* $\frac{3}{2}$*, and returns* $3 + 5 = 8 \neq 7$*. The value that attains the target is* $7 - 3 = 4$*.*

(ii) Exact attainment, two positions. *Module* 1 *has keys in* $\mathbb{R}^2$ *and* $W_0^1 k = k_{(1)} + k_{(2)}$*; module* 2 *is scalar with* $W_0^2 = 1$*, and its key at the second position is the sum of the outputs of module* 1 *at the two positions. The prompt of the edit has the keys* $(1, \frac{1}{2})$ *and* $k_{1,1} = (0, 1)$ *at module* 1*; the target is* $z^\star_{1,1} = 3$*, so* $u_{1,1} = 2$*, and* $\psi_{1,1} = (0, 1)$*, so* $s_{1,1} = 1$*. The updated module* 1 *returns* $\frac{3}{2} + \frac{1}{2} \cdot 2 = \frac{5}{2}$ *at the first position, where the edit has the coordinate* $\frac{1}{2}$*, and* 3 *at the second:* $k_{1,2} = \frac{11}{2}$*. With* $z^\star_{1,2} = 10$*,* $u_{1,2} = \frac{9}{2}$ *and* $\psi_{1,2} = \frac{2}{11}$*. The rebased values are* 2 *and* $\frac{9}{2}$*. In the scoped log with a threshold in* $(\frac{1}{2}, 1]$ *for the term of module* 1*, the term is released at the second position only, module* 1 *returns* $\frac{3}{2}$ *and* 3*, and module* 2 *receives the key* $\frac{9}{2}$*, where its term has the coordinate* $\frac{9}{11}$*. With a threshold above* $\frac{9}{11}$ *the term of module* 2 *is not released on the own request of the edit and the module returns* $\frac{9}{2}$*; with a threshold of at most* $\frac{9}{11}$ *it returns* 9*. The target* 10 *is missed in both cases.*

B.11 Complaint operators beyond the least repair

Sec. 4 of the paper uses one complaint operator, the least repair of reported wrong-answer complaints (Theorem B.8), with its certificate. Algorithm 1 composes it with two further operators, a repair of the detectors from complaints and a repair for why-not complaints, in one loop over the complaints, whose coordinates are taken on the pre-edit trace (Def. 2.3). These two are extensions: they are specified and implemented, and the paper makes no claim about them. In the algorithm j indexes detectors (one per edit until a why-not repair adds one), and ξ is the guard band of Algorithm 1.

Wrong-answer complaint (line 3). Raise is the update of Theorem B.8(b) with a guard band (Remark B.9). A complaint that no threshold separates from a pair of $Q^!$ is refused and counted; the thresholds are the least repair of the accepted complaints, up to the band, whatever their order. Thresholds only rise, so certified queries stay certified and a bound on false firing stays valid (Proposition B.1(c)).

Detector repair from complaints (lines 11–12). A least repair is scope-maximal (Theorem B.8(c)) and does not anticipate the next complaint. At the end of a batch, the reported keys are removed from each detector of the log: the detector is replaced by the closest vector with coordinate zero on these keys and s_j on the own key, a projection in closed form (Proposition B.37); a detector is kept if its own key lies in the span of these keys, and it moves far if its own key is close to it. Optimal detectors are NP-hard to approximate already for one edit (Theorem B.38); this is the realisable case. As coordinates change, the thresholds are recalibrated and the recorded complaints raised again. The recalibration uses a *fresh* calibration sample, which no earlier step has used: complaints are selected by thresholds that were fitted on the earlier sample, so the repaired detectors depend on it, whereas Proposition B.58 needs detectors that do not depend on the calibration queries; that the samples are disjoint from the complaints is not enough. The bound on false firing is claimed for the thresholds of the fresh sample only.

Why-not complaint (lines 5–10). Lowering τ_j to the coordinate of q' covers (i, q'); it is done if no certified complaint fires again (no arc to the pair in the digraph of Theorem B.14). Otherwise edit i gets a further detector with the same role set and value: the normalised key of q' with the cosine as coordinate, the gate feature of NeuralDB [22], at a fixed threshold $\tau' < 1$. The edit fires where one of its detectors does, and a granted pair stays assigned to the detector that covers it, which the guard of Algorithm 1 caps: with assigned coverage, threshold repair remains the problem of Theorem B.14 (Proposition B.39), while free coverage makes it inapproximable (Theorem B.40); Theorem B.8 and Proposition B.58 hold with detectors in place of edits. Until recalibrated, lowered thresholds and added detectors are outside the bound and are counted; a covered pair need not be resolved, and another edit may be selected.

Pilot outcomes. In pilots on the two development logs of the paper both lines missed the criteria that were registered for them. Detector repair from reported complaints lowered the rate of false firing at equal coverage of rephrasings, but not by the registered factor on every log and split. Thresholds fitted to complaints and why-not pairs by the minimum cut of Theorem B.14 fired, out of sample and at equal coverage, as often as thresholds optimised edit by edit (Proposition B.18) and more often than one shared threshold. The registered experiments with complaint streams on top of the calibrated method were not run. The paper reports this and claims, about complaints, the least repair with its certificate only.

Algorithm 1 Complaint operators beyond the least repair (extensions)

Input: calibrated scoped log $\mathcal{S}$ with thresholds τ; α, ξ and the guard keys as in Algorithm 1; must-cover pairs $Q^!$, each held by a guard key; batches of complaints, and for each batch a fresh sample $\mathcal{B}'$ of benign queries that no earlier step has used; $Q^- = \emptyset$; threshold τ' of added detectors

1 **for each** batch **do**
2 **for each** complaint of the batch **do**
3 **if** it is a wrong-answer complaint q **then** $Q^- \leftarrow Q^- \cup \{q\}$; Raise$(q)$
4 **else** ▷ *why-not complaint* (i, q')
5 **if** edit i fires on q' **then** $Q^! \leftarrow Q^! \cup \{(i, q')\}$; **continue**
6 $j \leftarrow$ the detector of edit i with the least $\tau_j - \zeta_j(q')$; $a \leftarrow \zeta_j(q') - \xi$
7 **if** $\zeta_j(q) < a$ for every certified $q \in Q^-$ **then** $\tau_j \leftarrow a$ ▷ *lower*
8 **else if** no certified $q \in Q^-$ fires on the cosine detector ψ' of the key of q' at the threshold τ' **then** give edit i the further detector (ψ', τ')
9 **else** count (i, q') as refused; **continue**
10 $Q^! \leftarrow Q^! \cup \{(i, q')\}$
11 $\psi_j \leftarrow$ the closest $\psi \perp$ keys of Q^- with $\langle \psi, k_j \rangle = s_j$, for each log detector j
12 $\tau \leftarrow$ Algorithm 1 on $\mathcal{S}$, α, ξ, the guard keys with those of $Q^!$, and the fresh sample $\mathcal{B}'$ of the batch; Raise(q) for every $q \in Q^-$

Raise(q): $\tau_j \leftarrow \max\big(\tau_j,\ (\zeta_j(q) + \xi)^+\big)$ for every detector j; if a pair of $Q^!$ would lose its coverage, leave τ unchanged and count q as refused

Table 7: Compared methods by deployment class: what each needs beyond the pre-edit weight W_0. Gates are analogues built from the log under repair, in the style of the cited editors. Cal.: Algorithm 1 on m benign queries. [a] also keys of the logged requests; [b] also the logged requests and unrelated items.

Needs	Methods and how their thresholds are set
W_E only	LINEAR: the deployed log
log: factors, own keys	RESCOPE-W (repair; all positions, logged values): log-only thresholds[a], least repair of n complaints, or Cal. RESCOPE (migration; last prompt token, rebased values, top-1): Cal.; variant with near-scope calibration[b]
log, n complaints	COMP-B (minimum compensation, answer keys of the edits protected); RE-EDIT (needs the editor); RB-UNION, RB-TOT (greedy rollback); blocklist
solver keys, values	COSINE [22], BALL [29], ROUTER [68], Z-SCORE [46], EXACT (cosine ≥ 0.999): published threshold; Cal.
and benign key statistics	cosine family (centred, whitened, null-repaired cosine; whitened coordinate): Cal.
content of the requests, two encoders	retrieval on image and question (three fusion rules): Cal.; lookup (same image and a shared word; text logs: the subject string)
edit targets	DUAL: the deployed answer if it equals a logged target, else the pre-edit answer; two generations

C EXPERIMENTAL SUPPLEMENT

C.1 Compared Methods by Deployment Class

Table 7 lists the compared methods of Sec. 5 by what each of them needs.

C.2 Calibration Validity under Workload Shift

Figure 1 belongs to Sec. 5.6 of the paper.

C.3 Coverage against Far and Near-Scope Firing

Figure 2 belongs to Sec. 5.3 of the paper: it traces what one common threshold on a detector buys, without calibration and without guards.

C.4 Deployment Classes: All Rows

Table 8 is Table 2 of the paper with the five rows that left it for space: RESCOPE-W with two-block thresholds from benign queries (near-scope cells at gate level, [g]), the cosine detector on the solver keys under the same calibration, the cosine at its published threshold, the lookup on the requests, and the answer-match baseline (DUAL: the deployed answer if it equals a logged target, else the pre-edit answer).

C.5 Equal Evidence on Prefix Logs; Capacity; Revocation and Cost

Prefix logs. Table 9 gives the six prefix logs behind the prefix part of H1; they are sub-streams of the two development streams, not independent logs. On the prefixes of 10 and 20 edits a rollback reaches at most 3.6 % of changed held-out answers by deleting nearly every edit (Rel at most 8 %); on the prefixes of 50 edits the rollback with the fewest changed answers leaves 27.5 | 36.0 % (Qwen | LLaVA) at Rel 33 | 43.

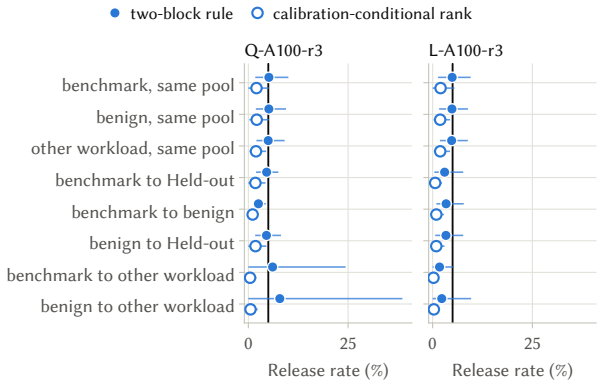


Figure 1: Calibration validity on the two development logs: realised release rate of RESCOPE on the test queries over 500 calibration draws of $m = 200$ queries, mean with the 5th to 95th percentile, per pair of a calibration and a test source; vertical line: the level α. Filled: two-block rule; hollow: calibration-conditional rank ($\delta = 0.1$). Thm. 3.6 applies to the pairs of one pool; the others measure what a workload shift costs. Off the scale and not drawn: calibrated on the out-of-benchmark workload and tested on benchmark or benign queries, the rule releases an edit on 90 to 98 % of them (conditional rank: 76 to 91 %).

Capacity of linear restoration. The check of Thm. 3.2 on the development logs evaluates the formula of the theorem on numerical ranks of captured key matrices (double precision; relative tolerance 10^{-9}, the ranks unchanged over two decades of it) and compares it, on a grid of numbers of complaints (step 5 around the transition), with the number of edits that the minimum-norm restoring weight preserves; the two agree at every grid point. With the complaints taken in three seeded orders, all T edits remain preservable up to 25, 30, 35 complaints on Qwen and 35 on LLaVA (largest value of the grid; from the rank after every complaint: 29, 33, 35 and 37, 35, 35), and from 30, 35, 40 and 40 complaints on the keys of the complaints span the key space, so that only W_0 restores them at all positions (after every complaint: 30, 34, 36 and 38, 36, 36). Two predictions were registered before the measurement. The first, the largest number of complaints up to which all edits stay preservable, was 23, 24, 34 and 35 (rounded down to the grid: 20, 20, 30 and 35); it is missed on Qwen and met on LLaVA. The second, the number of complaints from which on only W_0 restores them, was 25, 25, 36 and 36 (on the grid: 25, 25, 40 and 40) against the measured 30, 35, 40 and 40: it is missed on Qwen in two of the three orders and met on LLaVA. The hypothesis of the theorem (independent write vectors, a nonsingular coordinate matrix) holds on Qwen and holds on LLaVA. For the same complaints the thresholds-only repair, started from the log-only thresholds, silences at most 0 edits on their own requests or solver keys.

Lineage. Of the 431 | 504 queries on which RESCOPE releases an edit under the default calibration, 93.3 | 91.5 % return to the logits of the pre-edit model bit for bit after masking one edit, and none needs more than 3 | 2 masks; a rollback of the linear log for one complaint deletes a median of 3 to 6 edits. Before any revocation,

Table 8: Deployment classes on the two development logs at the default budget (%); columns as in Table 2 of the paper; Cap: capped edits of a calibrated row (mean of 20 draws), for which the bound is void.

		Q-A100-r3									L-A100-r3								
Needs	Method	Rel	T-Gen	I-Gen	Gen	Rls	D1	D2	Near	Cap	Rel	T-Gen	I-Gen	Gen	Rls	D1	D2	Near	Cap
–	deployed log	78	9	22	13	–	54.2	15.9	–	–	88	29	66	41	–	62.3	39.7	–	–
log	ReScope-W, log-only thresholds	78	7	26	14	18.5	1.1	1.9	21	–	92	28	68	41	26.2	2.8	1.3	50	–
log, n	Comp-B, all complaints	78	10	25	14	–	49.0	18.7	–	–	86	32	66	41	–	59.7	37.2	–	–
log, $\mathcal{B}$	ReScope-W, two-block thresholds	78	5	24	–	6.5	0.4	0.9	13[g]	2.1	93	29	69	–	3.8	0.9	0.0	36[g]	0.1
	ReScope	100	56	68	44	3.9	3.9	9.3	37/27/41	0.0	98	38	90	57	3.1	3.1	3.8	42/22/38	0.2
	ReScope, log-derived negatives	100	21	57	26	2.5	2.5	2.8	11/13/9	0.0	98	15	70	33	2.3	2.3	1.3	11/7/10	1.1
	cosine on solver keys	99	13	49	–	4.2	4.2	–	9/20/12	3.9	97	12	52	–	4.3	4.3	–	34/23/43	36.8
log, $\mathcal{B}$, stat.	whitened gate, selected for H2	100	60	79	49	6.9	6.9	7.5	42/44/47	0.0	98	38	93	53	3.8	3.8	5.1	48/38/44	0.0
	whitened gate, selected for H3	100	45	75	42	5.4	5.4	6.5	28/28/29	0.0	98	29	93	49	3.6	3.6	5.1	46/39/39	0.0
log	cosine, published threshold	100	37	69	–	48.2	46.9	–	100	–	98	22	93	–	41.4	41.2	–	97	–
requests	lookup on the requests	100	57	0	–	0.0	0.0	–	6	–	98	57	1	–	0.0	0.0	–	4	–
requests, enc., $\mathcal{B}$	retrieval, two encoders	100	86	95	63	4.5	4.5	3.7	28/48/50	0.0	98	89	97	76	6.7	6.7	2.6	54/53/46	0.3
edit targets	answer match (Dual)	64	8	18	13	0.0	0.0	0.0	2	–	72	28	51	41	0.4	0.4	0.0	17	–

Table 9: Prefix logs after 10 reported complaints (%): D1 on Held-out and Rel of the deployed log, of ReScope-W (least repair from its log-only thresholds) and of three linear repairs of the same complaints; H1: the registered rule (D1 at most 10 % and Rel within 2 points of the deployed log); last column: D1 at most 10 % and Rel not more than 2 points below the deployed log.

		Deployed		ReScope-W		Comp-B		RB-Union		RB-Tot		H1	Rel $\geq$ dep. $-$ 2
Stream	T	D1	Rel	D1	Rel	D1	Rel	D1	Rel	D1	Rel	(registered)	(D1 $\leq$ 10)
Q-A100-r3	10	23.9	80	0.6	80	19.9	80	3.7	3	0.0	0	met	yes
Q-A100-r3	20	29.9	75	1.1	75	28.7	75	11.3	15	2.8	2	met	yes
Q-A100-r3	50	43.0	82	0.9	78	36.6	80	27.5	33	28.0	33	not met	no
L-A100-r3	10	26.4	90	0.5	90	29.2	90	6.8	7	2.3	0	met	yes
L-A100-r3	20	35.4	90	0.9	90	37.2	90	11.3	12	3.6	8	met	yes
L-A100-r3	50	50.5	86	1.8	92	51.4	86	36.0	43	37.1	39	not met	yes

all 497 | 484 tested sequences without a release have the logits of the pre-edit model bit for bit. Table 10 gives the size of the pre-edit firing set of benign Held-out queries, the edits that fire on the teacher-forced pre-edit trace of a query (for ReScope this is its lineage; for ReScope-W, with all positions eligible, the lineage on the log's own run can differ, Sec. 2).

Revocation and cost. Table 11 gives the revocation experiment for every number of revoked edits and the cost measurements in full. Timings depend on the load of the shared card (92 % median utilisation during the measurement); only the paper's own gates (ReScope, ReScope-W) and the answer-match baseline were timed, not the gates of the other methods (Sec. 5.7). The gate of ReScope-W evaluates every position and costs more than that of ReScope: its rows give the all-positions form with the thresholds of the ReScope calibration and ReScope-W with its log-only thresholds; guard keys are not evaluated in any timed gate. The table prints the primary latency pass; in the second pass of the same experiment on the development log of Qwen the gate of ReScope takes 1.84 ms (10.5 % end to end) and the all-positions form 8.14 ms (14.5 %). The scoped store reaches the size of the edited module at 1,726 edits on Qwen.

C.6 Unseen Logs in Full

Tables 12 and 13 hold every row behind Table 3(a) of the paper, one table per editor family; the two families of a backbone share requests and solver keys, so the rows of gates on the solver keys are equal at equal T, and request-keyed rows agree up to the rebased values, which differ slightly between the families. Table 14 lists the registered read-outs log by log: on the unseen logs they are replications, not the confirmatory family. After a Holm correction within each read-out over the unseen logs and the baselines, 18 of 22 comparisons of H2 are rejected, all against ReScope, and 27 of 33 of H3, 25 against ReScope and 2 in its favour (one of the 25 is at the limit of the bootstrap resolution: an independent recomputation with more resamples finds 27 or one fewer). On the out-of-benchmark workload the release of ReScope exceeds the level on the first family of LLaVA (14.9 and 6.7 % at the two unseen lengths).

C.7 Ablations, Near-Scope Cells per Draw, Exact Tests

Table 15 changes one factor of ReScope at a time on the development logs; behavioural cells exist for the variants that were run at behaviour, the others are gate level (–). A variant that changes only what is released (gain response, all firing edits) equals the default at gate level by construction. Table 16 lists the near-scope cells of Table 2 draw by draw, with the wrong-target rate; Table 17 gives the exact McNemar tests of ReScope against the other rows.

C.8 Related Work That Left the Body

These comparisons belong to Sec. 6 of the paper. Table 18, which that section cites, sets the closest works side by side. *Unlearning.* Removing information from a model is treated in model management as view maintenance under deletions of training data [61, 71] and evaluated on database tasks [43]; the terms of an edit log have

Table 10: Size of the pre-edit firing set of benign HELD-OUT queries: share of queries on which no edit fires, and median, mean and maximum of the size over the other queries.

	Q-A100-r3				L-A100-r3			
Scoped log	size 0 (%)	median	mean	max	size 0 (%)	median	mean	max
deployed scoping (no thresholds; all positions)	0.0	100	100.00	100	0.0	100	100.00	100
REScope-W, log-only thresholds (all positions)	81.5	1	2.19	9	73.8	1	1.24	4
REScope, two-block thresholds (last prompt position; 20 draws pooled)	95.4	1	1.07	3	96.9	1	1.04	2

Table 11: Top: revocation of m edits (means of three seeded sets, the same sets on both logs). Lgt: tested sequences on which no revoked edit was released and whose logits change, of those tested; Ans: answer flags that change on them; Kept: Rel of the edits that are not revoked (%); D1 on HELD-OUT afterwards (%). Bottom: cost. numbers: stored numbers; MB in double precision; %par: share of the backbone's parameter count; gate: per forward pass, of REScope in the row of a log and, in the two rows below it, of the all-positions form of REScope-W with the thresholds of the REScope calibration and of REScope-W with its log-only thresholds; end to end: overhead of a forward pass with its spread over sequences; synthetic logs have the dimensions of the named backbone.

	Q-A100-r3								L-A100-r3							
	scoped log: mask				linear log: subtract				scoped log: mask				linear log: subtract			
m	Lgt	Ans	Kept	D1	Lgt	Ans	Kept	D1	Lgt	Ans	Kept	D1	Lgt	Ans	Kept	D1
1	0/762	0	100.0	5.0	762/762	22	79.1	54.4	0/766	0	98.0	5.3	766/766	14	88.2	62.3
5	0/749	0	100.0	4.7	749/749	43	77.9	53.5	0/751	0	98.6	5.1	751/751	35	88.1	61.8
10	0/733	0	100.0	4.7	733/733	59	78.5	52.3	0/735	0	98.2	4.8	735/735	50	89.3	59.8
20	0/699	0	100.0	4.2	699/699	75	80.4	51.1	0/701	0	98.3	4.1	701/701	71	91.7	59.4
50	0/603	0	100.0	3.1	603/603	120	82.7	43.7	0/601	0	98.0	2.5	601/601	115	90.7	51.2

Log	T	numbers	MB	%par	gate (ms)	end to end (%) [10th, 90th pct.]	cal. (s)	repair (ms)	revoke (ms)
Q-A100-r3	100	1,305,900	10.0	0.03	1.05	5.0 [−12, +35]	0.03	0.48	0.05
all positions, thresholds of the REScope calibration					6.73	14.4 [−15, +52]			
REScope-W, log-only thresholds					4.21	9.8 [−14, +50]			
L-A100-r3	100	1,510,700	11.5	0.02	1.08	4.9 [−4, +30]	0.03	0.47	0.05
all positions, thresholds of the REScope calibration					6.25	11.9 [−3, +45]			
REScope-W, log-only thresholds					4.53	8.3 [−2, +39]			
synthetic, Qwen	10^3	13,059,000	100	0.35	0.76	–	0.03	1.3	0.05
synthetic, Qwen	10^4	130,590,000	996	3.48	1.28	–	0.22	14.4	0.14
synthetic, Qwen	10^5	1,305,900,000	9963	34.78	8.71	–	2.32	178.4	0.36
synthetic, LLaVA	10^3	15,107,000	115	0.21	0.82	–	0.03	1.3	0.06
synthetic, LLaVA	10^4	151,070,000	1153	2.14	1.37	–	0.22	17.3	0.13
synthetic, LLaVA	10^5	1,510,700,000	11526	21.39	8.54	–	2.11	173.2	0.37

no training data behind them. *Stealth edits* [63] bound the firing of an implanted detector on other inputs. *Selective risk.* The concurrent TANSAR bounds the selective risk of one threshold from labelled queries [1]; our bound is on firing and needs no labels (Thm. 3.6). *Explanations.* Unexpected and missing answers of a query [14, 66] are explained by interventions on the input [60, 70]. *Query refinement* adjusts predicate parameters to meet constraints on aggregates or group sizes [3, 45]; our constraints are certified complaints. *Model lakes* ask for records of transformations such as model editing [55]. *Repair by editing.* A workshop paper proposes to repair a model against constraints by editing facts [52]; we repair the log of applied edits. *Gated writes.* ALOE [47] and CoachHooK [46] gate a write by a learned address with its own threshold and by a z-score threshold. With a log, an edit is deleted by subtraction [73]. *Conformal gates.* Conformal thresholds with a marginal guarantee gate neural operators in query processing [33].

Table 12: First editor family (AlphaEdit-style logs; %): the development logs ($T = 100$) and the unseen logs. Rel, T-Gen, I-Gen: exact match on all edits; subscript >: later edits only (stream positions from 500 on Qwen and from 100 on LLaVA; –: the log has none); Rls, D1: Held-out; Near: firing on near-scope negatives, kinds pooled (gate level); Cap: capped edits (mean of 20 draws). ReScope-W with benign queries was not run on unseen logs.

Log	Method	Rel	T-Gen	I-Gen	T-Gen $_{>}$	I-Gen $_{>}$	Rls	D1	Near	Cap
Q-A100-r3	deployed log	78	9	22	–	–	–	54.2	–	–
	ReScope-W, log-only	78	7	26	–	–	18.5	1.1	–	–
	ReScope	100	56	68	–	–	3.9	3.9	34	0.0
	ReScope, log-derived negatives	100	21	57	–	–	2.5	2.5	10	0.0
	whitened gate (H2)	100	60	79	–	–	6.9	6.9	39	0.0
	whitened gate (H3)	100	45	75	–	–	5.5	5.5	28	0.0
	cosine on solver keys	99	13	49	–	–	4.2	4.2	14	3.9
	retrieval	100	86	95	–	–	4.5	4.5	37	0.0
	lookup	100	57	0	–	–	0.0	0.0	4	–
Q-A200	deployed log	66	5	23	–	–	–	77.4	–	–
	ReScope-W, log-only	74	4	25	–	–	16.8	0.9	–	–
	ReScope	100	40	61	–	–	4.0	4.0	23	0.0
	ReScope, log-derived negatives	100	18	49	–	–	2.4	2.4	8	0.4
	whitened gate (H2)	100	30	65	–	–	3.4	3.4	26	0.0
	whitened gate (H3)	100	27	66	–	–	3.8	3.8	23	0.0
	cosine on solver keys	99	8	44	–	–	4.0	4.0	12	16.8
	retrieval	100	76	91	–	–	4.7	4.7	33	0.0
	lookup	100	44	0	–	–	0.0	0.0	4	–
Q-A500	deployed log	55	4	16	–	–	–	89.0	–	–
	ReScope-W, log-only	69	4	21	–	–	13.8	1.1	–	–
	ReScope	100	25	57	–	–	1.8	1.8	12	0.4
	ReScope, log-derived negatives	99	16	43	–	–	2.7	2.7	6	6.5
	whitened gate (H2)	100	23	60	–	–	2.8	2.8	15	0.0
	whitened gate (H3)	100	18	58	–	–	3.3	3.3	13	0.0
	cosine on solver keys	99	6	38	–	–	4.2	4.2	9	72.6
	retrieval	100	74	89	–	–	7.0	7.0	27	1.5
	lookup	100	49	0	–	–	0.0	0.0	5	–
L-A100-r3	deployed log	88	29	66	–	–	–	62.3	–	–
	ReScope-W, log-only	92	28	68	–	–	–	2.8	–	–
	ReScope	98	38	90	–	–	3.1	3.1	30	0.2
	ReScope, log-derived negatives	98	15	70	–	–	3.5	2.3	8	1.1
	whitened gate (H2)	98	38	93	–	–	3.6	3.8	37	0.0
	whitened gate (H3)	98	29	93	–	–	3.3	3.6	36	0.0
	cosine on solver keys	97	12	52	–	–	4.9	4.3	29	36.8
	retrieval	98	89	97	–	–	7.2	6.7	44	0.3
	lookup	98	57	1	–	–	0.0	0.0	3	–
L-A200	deployed log	70	22	51	16	43	–	76.1	–	–
	ReScope-W, log-only	89	26	66	23	64	–	3.4	–	–
	ReScope	98	33	85	34	83	2.8	2.9	26	0.3
	ReScope, log-derived negatives	98	19	68	24	67	4.2	4.0	6	4.3
	whitened gate (H2)	98	28	87	30	86	3.7	3.5	33	0.0
	whitened gate (H3)	98	23	89	24	88	3.3	3.4	31	0.0
	cosine on solver keys	98	11	47	12	44	5.8	6.5	28	89.8
	retrieval	98	83	96	84	96	5.8	4.2	36	0.5
	lookup	98	45	1	32	0	0.0	0.0	3	–
L-A500	deployed log	41	11	33	10	33	–	91.9	–	–
	ReScope-W, log-only	89	20	63	19	62	–	10.4	–	–
	ReScope	99	18	79	18	79	1.9	1.8	18	1.4
	ReScope, log-derived negatives	99	15	63	16	64	2.4	2.8	4	35.1
	whitened gate (H2)	99	18	82	17	82	3.1	3.3	21	0.0
	whitened gate (H3)	99	14	83	13	83	2.3	2.9	19	0.0
	cosine on solver keys	98	8	37	9	36	3.9	3.7	21	289.7
	retrieval	99	72	91	73	91	5.9	4.9	27	5.3
	lookup	99	43	2	39	2	0.0	0.0	3	–

Table 13: Second editor family (MEMIT-style logs; %), all unseen; columns as in Table 12.

Log	Method	Rel	T-Gen	I-Gen	T-Gen $_{>}$	I-Gen $_{>}$	Rls	D1	Near	Cap
Q-M100	deployed log	53	11	19	–	–	–	89.5	–	–
	ReScope-W, log-only	26	5	12	–	–	25.6	7.7	–	–
	ReScope	100	18	52	–	–	4.2	4.2	16	5.3
	ReScope, log-derived negatives	99	8	38	–	–	2.3	2.3	5	4.4
	whitened gate (H2)	100	60	79	–	–	6.9	6.9	39	0.0
	whitened gate (H3)	100	45	75	–	–	5.5	5.5	28	0.0
	cosine on solver keys	99	13	49	–	–	4.2	4.2	14	3.9
	retrieval	100	86	95	–	–	4.5	4.5	37	0.0
	lookup	100	57	0	–	–	0.0	0.0	4	–
Q-M500	deployed log	25	6	10	–	–	–	96.6	–	–
	ReScope-W, log-only	11	1	3	–	–	45.6	11.2	–	–
	ReScope	100	11	43	–	–	1.8	1.8	9	19.0
	ReScope, log-derived negatives	98	8	33	–	–	2.9	2.9	5	39.3
	whitened gate (H2)	100	23	60	–	–	2.8	2.8	15	0.0
	whitened gate (H3)	100	18	58	–	–	3.3	3.3	13	0.0
	cosine on solver keys	99	6	38	–	–	4.2	4.2	9	72.6
	retrieval	100	74	89	–	–	7.0	7.0	27	1.5
	lookup	100	49	0	–	–	0.0	0.0	5	–
Q-M1000	deployed log	15	5	7	3	6	–	98.9	–	–
	ReScope-W, log-only	7	1	2	0	1	44.1	9.0	–	–
	ReScope	100	7	34	8	34	1.5	1.5	9	33.7
	ReScope, log-derived negatives	97	5	27	5	27	4.1	4.1	5	87.7
	whitened gate (H2)	100	14	47	14	46	4.5	4.5	13	0.0
	whitened gate (H3)	100	12	46	11	43	4.7	4.7	12	0.0
	cosine on solver keys	99	3	28	3	25	3.0	2.6	13	224.6
	retrieval	100	59	77	52	73	5.8	5.8	23	3.9
	lookup	100	42	0	36	0	0.0	0.0	5	–
Q-M2000	deployed log	10	4	5	3	4	–	97.6	–	–
	ReScope-W, log-only	5	0	1	0	1	31.6	4.7	–	–
	ReScope	100	5	31	5	31	2.8	2.8	8	59.8
	ReScope, log-derived negatives	98	3	22	3	23	3.0	3.0	4	220.1
	whitened gate (H2)	100	10	42	10	41	5.8	5.8	11	0.0
	whitened gate (H3)	100	8	39	7	39	6.2	6.2	9	0.0
	cosine on solver keys	99	2	22	2	22	3.4	3.2	12	591.4
	retrieval	100	54	74	51	72	8.0	8.0	19	24.7
	lookup	100	42	0	39	0	0.0	0.0	5	–
L-M100	deployed log	78	26	65	–	–	–	77.8	–	–
	ReScope-W, log-only	81	17	62	–	–	–	17.5	–	–
	ReScope	98	11	59	–	–	4.1	2.4	35	18.9
	ReScope, log-derived negatives	96	4	54	–	–	4.4	3.0	8	24.1
	whitened gate (H2)	98	38	93	–	–	3.6	3.8	37	0.0
	whitened gate (H3)	98	29	93	–	–	3.3	3.6	36	0.0
	cosine on solver keys	97	12	52	–	–	4.9	4.3	29	36.8
	retrieval	98	89	97	–	–	7.2	6.7	44	0.3
	lookup	98	57	1	–	–	0.0	0.0	3	–
L-M500	deployed log	46	13	38	13	38	–	87.0	–	–
	ReScope-W, log-only	55	10	39	9	35	–	14.7	–	–
	ReScope	98	8	56	8	59	2.8	3.1	20	90.6
	ReScope, log-derived negatives	95	4	39	5	42	3.3	3.0	6	170.6
	whitened gate (H2)	99	18	82	17	82	3.1	3.3	21	0.0
	whitened gate (H3)	99	14	83	13	83	2.3	2.9	19	0.0
	cosine on solver keys	98	8	37	9	36	3.9	3.7	21	289.7
	retrieval	99	72	91	73	91	5.9	4.9	27	5.3
	lookup	99	43	2	39	2	0.0	0.0	3	–
L-M1000	deployed log	37	11	30	11	29	–	86.1	–	–
	ReScope-W, log-only	48	7	34	7	33	–	11.7	–	–
	ReScope	99	5	55	6	57	3.3	2.8	19	148.6
	ReScope, log-derived negatives	97	4	39	4	40	3.9	4.7	5	410.9
	whitened gate (H2)	99	8	72	8	72	3.4	2.8	21	0.0
	whitened gate (H3)	99	8	77	8	77	3.0	2.8	18	0.0
	cosine on solver keys	98	7	33	7	32	3.9	4.9	23	618.8
	retrieval	99	58	79	58	79	6.9	3.8	26	12.5
	lookup	99	44	1	42	1	0.0	0.0	4	–

Table 14: Registered read-outs per log (points with intervals; paired bootstrap crossed over calibration draws and edits; 3 draws, one on the logs with 1000 edits or more): H2, mean of T-Gen and I-Gen (exact) of ReScope minus the gate selected for H2; H3, the same under the near-scope constraint, ReScope minus the gate selected for H3 (pair 1), with log-derived negatives on both sides (pair 2), and against retrieval. First row of each backbone: the development log (confirmatory); the others are replications. Under the raised scale the guard caps are kept: on average at most 219 edits of ReScope and 0 of the gate sit at their cap in pair 1, 380 and 280 in pair 2, 93 of the retrieval rule (largest values over the logs, attained at the largest T); the constraint is reachable in every draw.

Log	H2, all edits	H2, later edits	H3, pair 1	H3, pair 2	H3, retrieval
Q-A100-r3	−7.0 [−12.0, −2.0]	–	−1.3 [−7.7, +5.3]	+2.0 [−3.0, +7.0]	−22.3 [−36.0, −9.0]
Q-A200	+2.8 [−4.0, +10.8]	–	+1.2 [−3.5, +5.8]	+4.8 [+0.8, +9.2]	−15.5 [−23.5, −7.5]
Q-A500	−0.4 [−3.0, +2.1]	–	+5.1 [+2.1, +8.0]	+3.1 [−0.6, +7.2]	−18.9 [−25.5, −12.1]
Q-M100	−34.5 [−41.5, −27.5]	–	−11.3 [−20.0, −3.7]	−9.3 [−15.3, −4.0]	−32.3 [−46.7, −18.3]
Q-M500	−14.6 [−17.5, −11.6]	–	−7.9 [−11.8, −4.3]	−5.7 [−9.7, −1.8]	−32.1 [−39.8, −25.0]
Q-M1000	−10.0 [−11.7, −8.4]	−8.6 [−10.7, −6.5]	−7.8 [−10.0, −5.5]	−3.1 [−4.6, −1.7]	−37.4 [−40.8, −33.9]
Q-M2000	−8.1 [−9.1, −7.1]	−7.3 [−8.5, −6.2]	−5.0 [−6.4, −3.6]	−2.9 [−3.9, −1.9]	−36.6 [−39.2, −34.1]
L-A100-r3	−1.3 [−8.0, +4.7]	–	−4.0 [−9.7, +1.7]	−0.3 [−6.0, +5.3]	−12.7 [−25.0, −0.3]
L-A200	+1.3 [−5.1, +8.5]	+0.7 [−6.7, +9.0]	−4.2 [−8.7, 0.0]	+4.0 [−0.5, +8.7]	−12.5 [−20.5, −4.2]
L-A500	−1.1 [−4.4, +2.1]	−1.3 [−5.1, +2.4]	−2.7 [−6.0, +0.7]	+5.8 [+2.0, +9.9]	−15.5 [−20.7, −10.3]
L-M100	−31.0 [−40.2, −22.2]	–	−23.3 [−32.3, −14.7]	−14.7 [−23.3, −6.3]	−32.0 [−43.3, −21.0]
L-M500	−17.8 [−21.2, −14.7]	−16.0 [−19.3, −12.5]	−14.9 [−18.9, −11.1]	−11.2 [−14.8, −7.5]	−27.7 [−32.5, −23.3]
L-M1000	−10.2 [−11.8, −8.7]	−8.9 [−10.4, −7.3]	−9.0 [−11.0, −6.9]	−4.2 [−6.0, −2.4]	−24.6 [−27.9, −21.5]

Table 15: Ablations of ReScope on the development logs (%). FF, Near: release on Held-out and on near-scope negatives (gate level, 20 draws); Cov_T, Cov_I: text and image rephrasings on which the own edit is released; Rel, T-Gen, I-Gen, D1: behaviour (3 draws).

	Q-A100-r3								L-A100-r3							
Variant	FF	Near	Cov_T	Cov_I	Rel	T-Gen	I-Gen	D1	FF	Near	Cov_T	Cov_I	Rel	T-Gen	I-Gen	D1
default (two-block rule, last position, replace, rebased)	4.7	39	62	73	100	56	68	3.9	3.1	34	43	91	98	38	90	3.1
one shared scale	4.6	29	60	73	100	60	72	4.8	3.4	24	33	90	98	34	89	4.3
log-only thresholds	0.0	1	5	28	100	6	28	0.0	1.3	18	20	84	98	20	83	1.3
per-edit running maxima	32.6	86	74	81	100	71	79	29.6	34.0	54	57	94	98	54	94	34.0
calibration-conditional rank	0.9	17	35	60	–	–	–	–	0.8	21	25	79	–	–	–	–
all positions eligible	5.9	13	14	43	100	16	43	0.4	3.0	36	38	87	93	37	85	2.6
linear response	4.7	39	62	73	78	8	26	2.1	3.1	34	43	91	92	29	69	1.4
gain response	4.7	39	62	73	–	–	–	–	3.1	34	43	91	–	–	–	–
logged instead of rebased values	4.7	39	62	73	99	56	67	3.9	3.1	34	43	91	99	39	91	3.1
guards off	4.7	39	62	73	100	56	68	3.9	3.1	34	43	91	98	38	90	3.1
all firing edits released	4.7	39	64	75	–	–	–	–	3.1	34	45	92	–	–	–	–
calibration sample contaminated, 1 %	4.7	39	62	72	–	–	–	–	2.7	32	42	90	–	–	–	–
contaminated, 5 %	2.3	25	47	64	–	–	–	–	1.0	23	28	82	–	–	–	–
contaminated, 20 %	0.0	4	10	34	100	8	27	0.1	0.0	4	2	45	98	3	36	0.0

Table 16: Release on near-scope negatives at the 3 behavioural calibration draws (%, in the order of the draws) per kind, and WT: near-scope negatives answered with the target of the released edit (first draw).

	Q-A100-r3				L-A100-r3			
Method	HN-I	HN-Q	Cross	WT	HN-I	HN-Q	Cross	WT
ReScope	37/24/38	23/18/29	52/39/55	10.2	26/11/23	45/26/42	55/29/49	27.5
ReScope, log-derived negatives	9/12/7	7/9/6	16/17/14	3.6	6/4/5	11/7/10	15/10/15	6.6
cosine on solver keys	5/14/6	9/24/14	12/21/15	–	21/12/31	40/27/49	39/29/48	–
whitened gate (H2)	34/35/38	38/41/45	54/56/59	12.1	21/14/18	59/49/54	64/51/60	30.9
whitened gate (H3)	19/19/20	29/29/30	36/36/38	9.0	20/15/15	57/50/51	62/52/53	30.8
retrieval	41/58/60	7/20/27	36/68/63	7.4	64/64/54	23/23/19	75/71/67	21.4

Table 17: Exact McNemar tests on the development logs, first calibration draw: b, c: rows on which only ReScope, only the other method shows the outcome (D1: the pre-edit answer is not reproduced; Rel: the request is answered with the target); p: Holm-adjusted over backbones, rows and draws.

	Q-A100-r3						L-A100-r3					
	D1 on Held-out			Rel			D1 on Held-out			Rel		
ReScope against	b	c	p	b	c	p	b	c	p	b	c	p
deployed log	10	239	$< 10^{-55}$	22	0	$< 10^{-4}$	6	273	$< 10^{-69}$	12	2	1.000
ReScope-W, log-only	21	3	0.026	22	0	$< 10^{-4}$	19	7	1.000	8	2	1.000
ReScope, log-derived negatives	18	5	0.775	0	0	1.000	22	11	1.000	0	0	1.000
cosine on solver keys	17	8	1.000	2	0	1.000	19	21	1.000	1	0	1.000
whitened gate (H2)	13	19	1.000	0	0	1.000	18	15	1.000	0	0	1.000
whitened gate (H3)	14	14	1.000	0	0	1.000	17	17	1.000	0	0	1.000
cosine, published threshold	0	195	$< 10^{-56}$	0	0	1.000	12	180	$< 10^{-36}$	0	0	1.000
lookup	23	0	$< 10^{-4}$	0	0	1.000	25	0	$< 10^{-5}$	0	0	1.000
retrieval	16	8	1.000	0	0	1.000	19	37	1.000	0	0	1.000
answer match	23	0	$< 10^{-4}$	36	0	$< 10^{-8}$	24	1	$< 10^{-3}$	27	1	$< 10^{-4}$
exact match	23	0	$< 10^{-4}$	1	0	1.000	25	0	$< 10^{-5}$	0	0	1.000
pre-edit model	23	0	$< 10^{-4}$	100	0	$< 10^{-27}$	25	0	$< 10^{-5}$	97	0	$< 10^{-26}$

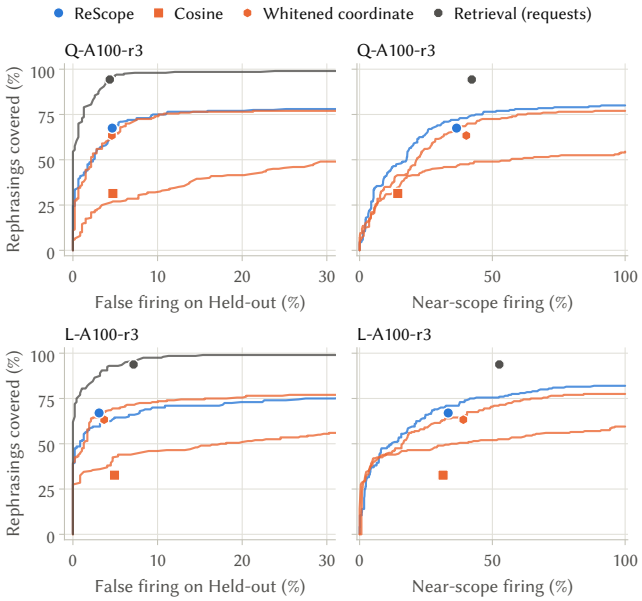


Figure 2: Coverage of rephrasings (text and image) against false firing on HELD-OUT (left) and against the mean firing on near-scope negatives (right) as one common threshold moves, per detector, on the two development logs. Markers: the calibrated operating points at the default level (mean of 20 draws; retrieval: its fused score has no single threshold on the right). Up and to the left is better; a detector that is ahead on the left and behind on the right buys coverage with near-scope firing.

Table 18: Closest works. *: Sec. 5 of the paper compares an analogue built from the log, not the published system. §: one global threshold in use. †: property of the repair (Sec. 3), not of the cited work. ‡: marginal bound of Thm. 3.6; every gate of Sec. 5 receives this calibration.

Work	Log as input	Per-edit scope	Com-plaints	Exact locality	Firing bound
NeuralDB* [22]	no	one§	no	yes	no
ALOE [47]	no	yes	no	yes	no
CoachHooK* [46]	no	no	no	yes	no
Compensation [21]	yes†	no	yes†	no†	–
Rollback [73]	yes	no	yes†	no†	–
QFix [69] (query logs)	yes	yes	yes	–	–
What-if queries [12]	yes	given	no	–	–
Query repairs [65]	no	–	yes	–	–
RESCOPE-W, RESCOPE (the paper)	yes	yes	yes	yes	cal.‡

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